MultiQT notes

November 11, 2026

1 Notation

Unless stated otherwise, fix one disease, location and forecast horizon.

Forecasts. The index t labels a forecast round (reference week), and y_t is the outcome for its target week. Use s for an earlier round. At quantile levels $\mathcal{A} = \{\alpha_1 < \cdots < \alpha_m\} \subset (0,1)$, b_t^α is the base forecast and q_t^α the recalibrated forecast. The vectors $b_t, q_t \in \mathbb{R}^m$ collect all levels. Values are hospital counts unless a transformation is shown.

Coverage and loss. Write $\mathbf{1}\{\cdot\}$ for an indicator and $x_+ = \max(x, 0)$. Coverage at level α is $\mathrm{cov}_t^\alpha = \mathbf{1}\{y_t \le q_t^\alpha\}$, with target α. Quantile loss, with prediction z first and observation r second, is

$$\rho_\alpha(z, r) = \alpha(r - z)_+ + (1 - \alpha)(z - r)_+. \tag{1}$$

Ordering and states. Forecasts must be nondecreasing across levels. Write

$$\mathcal{K} = \{q \in \mathbb{R}^m : q^{\alpha_1} \le \cdots \le q^{\alpha_m}\},$$

and assume $b_t \in \mathcal{K}$. Projection onto a nonempty closed convex set C means the nearest point in Euclidean distance:

$$\Pi_C(x) = \arg\min_{z \in C} \|z - x\|^2.$$

A tilde marks hidden accumulated state; the untilded state is its projected version. Standard offsets use $\tilde{\theta}_t, \theta_t$; studentized offsets use $\tilde{u}_t, u_t$. Superscripts $\mathrm{std}, \mathrm{stu}$ distinguish these methods, and η or η_t is the step size.

The methodological recursions forecast, observe y_t, then update. Delayed reports and revisions are introduced in Section 3.2; implementation, weighting and evaluation symbols are defined where they are used.

2 Methodological notes

2.1 Studentized score

Standard MultiQT tracks additive residual corrections. Studentization tracks scaled residuals, so changing the scale also changes the accumulated correction in outcome units. A falling scale can shrink a positive correction after a peak without erasing the hidden state. The quantile coverage target is unchanged; this construction alone does not establish calibration guarantees.

2.1.1 Standard MultiQT and score tracking

Starting with zero hidden state, MultiQT [1] projects the forecast but updates the unprojected state:

$$\theta_t = \Pi_{\mathcal{K}-b_t}(\tilde{\theta}_t), \qquad q_t^{\text{std}} = b_t + \theta_t, \tag{2}$$

$$\text{cov}_t^{\text{std},\alpha} = \mathbf{1}\{y_t \le q_t^{\text{std},\alpha}\}, \qquad \tilde{\theta}_{t+1}^{\alpha} = \tilde{\theta}_t^{\alpha} - \eta(\text{cov}_t^{\text{std},\alpha} - \alpha). \tag{3}$$

For a score $S_t(y)$, let $r_t = S_t(y_t)$ be its observed value, c_t its threshold and $\delta = 1 - \alpha$ the target miscoverage. The scalar score tracker of Angelopoulos et al. [2] uses

$$\mathcal{P}_t = \{y : S_t(y) \le c_t\}, \qquad \text{err}_t = \mathbf{1}\{r_t > c_t\}, \tag{4}$$

and updates

$$c_{t+1} = c_t + \eta(\text{err}_t - \delta) = c_t - \eta(\mathbf{1}\{r_t \le c_t\} - \alpha). \tag{5}$$

This is a quantile-loss subgradient step. In the threshold argument,

$$\partial_z \rho_\alpha(z, r) = \begin{cases} \{-\alpha\}, & z < r, \\ [-\alpha, 1-\alpha], & z = r, \\ \{1-\alpha\}, & z > r, \end{cases} \tag{6}$$

and the selected subgradient, including ties, is

$$g_\alpha(z; r) = \mathbf{1}\{r \le z\} - \alpha \;\in\; \partial_z \rho_\alpha(z, r). \tag{7}$$

Loss depends on distance from the threshold; the update depends only on which side contains the observation.

2.1.2 Residual coordinates

The signed residual score recovers standard MultiQT:

$$S_t^{\text{std},\alpha}(y) = y - b_t^{\alpha}, \qquad r_t^{\text{std},\alpha} = y_t - b_t^{\alpha}. \tag{8}$$

Its threshold θ_t^{α} defines the forecast endpoint and coverage:

$$\{y : S_t^{\text{std},\alpha}(y) \le \theta_t^{\alpha}\} = (-\infty, b_t^{\alpha} + \theta_t^{\alpha}] = (-\infty, q_t^{\text{std},\alpha}], \tag{9}$$

$$\text{cov}_t^{\text{std},\alpha} = \mathbf{1}\{r_t^{\text{std},\alpha} \le \theta_t^{\alpha}\}. \tag{10}$$

Thus

$$\tilde{\theta}_{t+1}^{\alpha} = \tilde{\theta}_t^{\alpha} - \eta(\mathbf{1}\{r_t^{\text{std},\alpha} \le \theta_t^{\alpha}\} - \alpha), \tag{11}$$

with the same loss in residual coordinates:

$$\ell_t^{\text{std}}(\theta) = \sum_{\alpha \in \mathcal{A}} \rho_\alpha(b_t^{\alpha} + \theta^{\alpha}, y_t) = \sum_{\alpha \in \mathcal{A}} \rho_\alpha(\theta^{\alpha}, r_t^{\text{std},\alpha}). \tag{12}$$

For one level there is no ordering constraint, so this is scalar quantile tracking. For several levels, evaluate the subgradient at the projected state and apply it to the hidden state.

2.1.3 Studentized residuals and their objective

Choose finite, positive scales σ_t^α from information available at the forecast cutoff and hold them fixed when differentiating. Offsets u_t^α are in studentized units, whereas θ_t^α is in outcome units. Define

$$S_t^{\mathrm{stu},\alpha}(y) = \frac{y - b_t^\alpha}{\sigma_t^\alpha}, \qquad r_t^{\mathrm{stu},\alpha} = \frac{y_t - b_t^\alpha}{\sigma_t^\alpha}. \tag{13}$$

The score is increasing, so inversion preserves the coverage event:

$$q_t^{\mathrm{stu},\alpha} = b_t^\alpha + \sigma_t^\alpha u_t^\alpha, \qquad \mathrm{cov}_t^{\mathrm{stu},\alpha} = \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u_t^\alpha\}. \tag{14}$$

Quantile loss of the standardized scores is a weighted outcome-unit loss:

$$\ell_t^{\mathrm{stu}}(u) = \sum_{\alpha\in\mathcal{A}} \rho_\alpha(u^\alpha, r_t^{\mathrm{stu},\alpha}) = \sum_{\alpha\in\mathcal{A}} \frac{\rho_\alpha(b_t^\alpha + \sigma_t^\alpha u^\alpha, y_t)}{\sigma_t^\alpha}. \tag{15}$$

Its selected subgradient is

$$g_t^{\mathrm{stu},\alpha}(u_t) = \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u_t^\alpha\} - \alpha = \mathrm{cov}_t^{\mathrm{stu},\alpha} - \alpha. \tag{16}$$

Indeed, away from ties the inverse scale in the loss cancels the scale from differentiation of the forecast:

$$\frac{\partial}{\partial u^\alpha} \frac{\rho_\alpha(b_t^\alpha + \sigma_t^\alpha u^\alpha, y_t)}{\sigma_t^\alpha} = \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u^\alpha\} - \alpha. \tag{17}$$

At ties use (7). Starting from $\tilde{u}_1 = 0$, the update is

$$\tilde{u}_{t+1}^\alpha = \tilde{u}_t^\alpha - \eta g_t^{\mathrm{stu},\alpha}(u_t). \tag{18}$$

The projected threshold determines coverage; only the hidden threshold accumulates increments. This is not gradient descent on the unweighted outcome-unit loss in u: that gradient has an additional factor σ_t^α. The coding section also tests median-centred forecasts and nonzero initial states.

2.1.4 Ordering and projection

Ordering always concerns the returned forecasts. For standard residuals,

$$C_t^{\mathrm{std}} = \mathcal{K} - b_t, \qquad b_t + \theta \in \mathcal{K} \iff \theta \in C_t^{\mathrm{std}}, \tag{19}$$

$$\theta^{\alpha_i} - \theta^{\alpha_{i+1}} \le b_t^{\alpha_{i+1}} - b_t^{\alpha_i}, \qquad i = 1, \dots, m-1. \tag{20}$$

For studentized residuals, collect the scales in $A_t = \mathrm{diag}(\sigma_t^{\alpha_1}, \dots, \sigma_t^{\alpha_m})$. Then

$$C_t^{\mathrm{stu}} = \{u : b_t + A_t u \in \mathcal{K}\} = A_t^{-1}(\mathcal{K} - b_t). \tag{21}$$

This is a nonempty closed convex polyhedron containing zero, generally not a cone. Unequal scales do not preserve $\mathcal{K}$, so it cannot generally be written as $\mathcal{K} - A_t^{-1} b_t$. The unique Euclidean projection gives

$$u_t = \Pi_{C_t^{\mathrm{stu}}}(\tilde{u}_t), \qquad q_t^{\mathrm{stu}} = b_t + A_t u_t. \tag{22}$$

Changing variables to $q = b_t + A_t u$ yields weighted isotonic regression:

$$q_t^{\mathrm{stu}} = \arg\min_{q\in\mathcal{K}} \sum_{\alpha\in\mathcal{A}} \frac{(q^\alpha - b_t^\alpha - \sigma_t^\alpha \tilde{u}_t^\alpha)^2}{(\sigma_t^\alpha)^2}, \tag{23}$$

For positive-definite W, write $\Pi_{\mathcal{K}}^W(v) = \arg\min_{q\in\mathcal{K}} (q - v)^\top W (q - v)$ for weighted projection. Then the equivalent projection identity is

$$b_t + A_t \Pi_{A_t^{-1}(\mathcal{K} - b_t)}(\tilde{u}_t) = \Pi_{\mathcal{K}}^{A_t^{-2}}(b_t + A_t \tilde{u}_t). \tag{24}$$

Weights $1/(\sigma_t^\alpha)^2$ follow from Euclidean projection in score coordinates; ordinary residual coordinates give equal weights.

2.1.5 Common and constant scales

For a scale σ_t common to all levels, $A_t = \sigma_t I$, where I is the identity matrix. Positive scalar multiplication preserves $\mathcal{K}$:

$$C_t^{\text{stu}} = \mathcal{K} - b_t/\sigma_t, \qquad q_t^{\text{stu}} = \Pi_{\mathcal{K}}(b_t + \sigma_t \tilde{u}_t). \tag{25}$$

The projection is ordinary isotonic regression. This is standard MultiQT on b_t/σ_t and y_t/σ_t, with the returned forecast multiplied by σ_t. For constant σ, it is also equivalent to standard MultiQT on the original inputs after matching initialization and converting state and rate:

$$\tilde{\theta}_t = \sigma \tilde{u}_t, \qquad \eta_{\text{std}} = \sigma \eta_{\text{stu}}. \tag{26}$$

For varying σ_t, changing the scale rescales the accumulated state; changing only the current rate does not reproduce that effect.

2.2 Questions and ideas

Which property of the base forecast should another score recalibrate? Increasing, invertible scores preserve the quantile coverage event; changing the target to PIT or highest-density-region calibration would require a separate formulation.

3 Coding Updates

The tracker transform is $\psi(x) = x$ or $\log(1 + x)$, with transformed base $\beta_t^\alpha = \psi(b_t^\alpha)$. Offsets θ_t are in these tracker units. The count divisor c_{scale} used for scoring is population/100,000 by default.

3.1 Data

Script 00 downloads CMU–TimeSeries hub forecasts and dated NHSN reports, and generates our own base forecasts where the required inputs are available. Script 01 prepares cases (one disease, location and forecast horizon) from the selected base forecasts and reports. Cases retain generation cutoffs, revisions, feedback schedules for delays 0–4, populations and filled-forecast flags.

Reference weeks are Saturdays (`reference_date`); the target week of y_t is `target_end_date` in forecasts and `target_date` in prepared cases. The information cutoff ι_t is normally the Wednesday of the reference week, or the actual generation date where known. Write $y_s^{[v]}$ for the latest report for the target week of row s, dated on or before cutoff v (archive field `version`). A valid report is finite and nonnegative. "Latest counts" means the latest reports in the downloaded archive, through 30 October 2026; further revisions remain possible.

Hub forecasts: the active design. The `hub_first_months` design uses submitted CMU–TimeSeries forecasts for both diseases from reference date 23 November 2024. The stored quantiles are used subject to the cleaning and filling below; we do not refit them using latest counts. Feedback uses the report available at the relevant cutoff, while evaluation uses latest counts.

Target dates determine the split: tuning through 31 May 2025, then evaluation from June 2025 through September 2026. A season means the *full October–September year*, including summer. Overall scores use all available evaluation weeks; before/after-peak scores restrict these to the corresponding peak windows. Flu has long summer gaps in submitted forecasts, rather than an explicit winter-only evaluation rule. The flu winter 2024/25 peak belongs to tuning. Plotted target weeks after September 2026 are outside the evaluation period and are not scored.

Own `windowed_seasonal` forecasts: the longer-history alternative. The `seasons` design starts on reference date 14 October 2023, tunes on 2023/24 and evaluates on 2024/25 and 2025/26, using the same October–September definition. Our COVID forecasts have been generated with unchanged, pinned Delphi code and packages. Flu still uses hub forecasts: we have not generated the corresponding own flu series. Earlier flu hub coverage is incomplete, especially at horizon -1; this is not a matched comparison of two locally generated base models. This alternative has not supplied the performance results below.

Why own flu forecasts are blocked. The COVID training archive uses NHSN counts alone. Flu adds historical ILI+ and adjusted FluSurv-NET training rows, requiring eight missing CSVs from the non-public Delphi `aux_data/flusion_data/` folder: `flu_burden.csv`, `us_pop.csv`, and six WHO/NREVSS laboratory files (national, HHS-region and state series, before 2015/16 and from clinical laboratories). The first two support adjustments to FluSurv-NET hospitalization rates; the laboratory series support ILI+, combining influenza-like illness with influenza test positivity. Script 00 skips flu when these files are absent. This is a missing-input blocker, not an observed failure of the flu model. COVID does not require these inputs; the underlying surveillance data are not necessarily all private.

How earlier reports are reconstructed. Report versions start with a retrospective bulk load on 19 November 2024, so the original reports available during 2023/24 cannot be recovered. For our earlier forecasts, we keep only past target weeks eligible at the generation date, but use their latest revised counts. Future target weeks are excluded; later revisions of past weeks are used in hindsight. Feedback for target weeks through 27 April 2024 likewise uses latest counts, assigned an assumed release on the following Wednesday and flagged `synthetic_final`. This is not an exact replay of real-time information, and cannot assess the benefit of waiting for revisions in that earlier period. From 20 November 2024, our base forecasts use reports available at their generation dates. The active hub design does not use this synthetic earlier feedback.

Other limitations of the longer-history experiment. These are not flu-specific. The generated COVID base has zero width at horizon -1: its target is already a predictor, so all quantiles coincide before recalibration. The default regression solver could stall in this case; using the production solver `fn` resolved that problem. Target weeks 4 May–2 November 2024 have incomplete reporting and are excluded from learning and scoring; our forecast schedule pauses across this gap. Stateful methods carry accumulated corrections and weights forward when rounds resume, rather than restarting. The missing report versions and this reporting gap also affect COVID, despite its available training inputs.

Cleaning and filling.

1. **Missing forecasts.** Fill gaps of at most six rounds from the most recent earlier forecast for the same target: horizon h at reference date r uses $h + k$ at $r - 7k$ days, with the smallest available $k > 0$. Horizon 3 has no source. Filled rows are flagged and scored; unfillable rounds remain absent.
2. **Broken submissions.** Remove the known Delphi bugs: COVID 23 November 2024; COVID $h = -1$ on 25 April 2026; US flu 14 and 21 December 2024 and 4 January 2025. Also remove all-zero submissions across every location and quantile of a disease–round–horizon (flu $h = -1$, 22 November 2025).
3. **Missing reports.** If unpublished or currently invalid, stateful feedback waits and re-runs omit it; neither substitutes an older valid report. The active cohort has no stored missing values, although a report can be unavailable at an earlier cutoff. Targets 4 May–2 November 2024 and forecast rows without finalized truth are excluded from learning and scoring.
4. **Report errors.** The main workflow rejects nonfinite or negative reports only. Finite, nonnegative errors can pass: Pennsylvania COVID for 16 November 2024 changed from 6,124 to 1,005 in successive releases. The screen below and withholding experiment are confined to `agent/`; neither is adopted in `R/`.

Disease	Missing reference weeks	Unfilled target weeks, $h = 2$
COVID	25 January–1 February 2025	15 February 2025
Both	25 October–15 November 2025	15–29 November 2025
Flu	25 January 2025; 18 April 2026	None
Flu	7 June–11 October 2025	21 June–25 October 2025
Flu	6 June–3 October 2026	20 June–17 October 2026

The COVID $h = 2$ weight plots (page 85) show the first two gaps; only the first missing round can use an earlier $h = 3$ forecast. US flu also lacks target 4 January 2025 at $h = 2$ after successive broken submissions. Lighter bars denote filled rows. Lines break at gaps longer than 28 days and can bridge shorter missing intervals. Updates occur only at retained rounds: pending feedback waits across unfilled rounds, without resetting the state.

Locations. Every disease–location–horizon case learns independently. Detailed plots show the US, California, Texas, New York and Pennsylvania; this is the selected list, not the four largest states by population. Scores cover all 53 locations. Tuning and pooled summaries use the 52 non-US locations; the US is evaluated separately with their selected parameters. Forecasts are never averaged across locations.

Experimental report screen. The experiment is in `agent/scripts/lib/report_screen.R`, with corrected results in `agent/outputs/report_screen_notes_audit/README.md`. At cutoff v, select the latest version before checking validity. In this screen, u indexes target weeks. Write $z_u^{[v]} = \log(1 + y_u^{[v]})$ and $d_u^{[v]} = z_u^{[v]} - z_{u-1}^{[v]}$. Use up to eight preceding consecutive-week changes, without bridging missing weeks. Given at least six valid changes and a valid previous target week,

$$\mu_u^{[v]} = \operatorname{median}_j d_j^{[v]}, \qquad s_u^{[v]} = 1.4826 \operatorname{median}_j |d_j^{[v]} - \mu_u^{[v]}|,$$
$$\widehat{z}_u^{[v]} = z_{u-1}^{[v]} + \mu_u^{[v]}. \tag{27}$$

Flag unusual movement when

$$|z_u^{[v]} - \widehat{z}_u^{[v]}| > \max\{\log 2, 4s_u^{[v]}\} \quad \text{and} \quad |y_u^{[v]} - (e^{\widehat{z}_u^{[v]}} - 1)| \geq 10. \tag{28}$$

With insufficient history, the report is unassessed. A separate revision flag requires successive valid stored values with a $(y + 1)$ ratio exceeding two in either direction and a count difference of at least ten. Flags identify unusual movements or disagreements, not proven errors; they do not change scoring truth.

3.2 Feedback

Feedback uses the latest report at each cutoff as defined in Section 3.1, independently of the recalibration method. Retrospective initialization is described in Section 3.3.

Stateful learning. Let ν_s be the first valid publication date for row s. Wait d calendar weeks, giving maturity date $m_s^{(d)}$, then consume the newest report when valid at the next retained forecast cutoff with $t > s$. Write $\tau_d(s)$ for that round and $\mathcal{R}_t^{(d)}$ for the rows consumed before forecasting at t:

$$m_s^{(d)} = \nu_s + 7d \text{ days}, \tag{29}$$
$$\tau_d(s) = \min\{t > s : \iota_t \geq m_s^{(d)},\ y_s^{[\iota_t]} \text{ valid}\}, \qquad \mathcal{R}_t^{(d)} = \{s : \tau_d(s) = t\}. \tag{30}$$

An invalid newest report postpones learning; an older version is never substituted. Use the report at actual consumption time ι_t, even if a missing forecast delayed it beyond maturity. Each row is learned at most once; subsequent revisions are ignored. The report used for a consumed row is

$$\bar{y}_s^{(d)} = y_s^{[\iota_{\tau_d(s)}]}, \qquad \tau_d(s) \text{ exists}. \tag{31}$$

Plain MultiQT on counts starts at $\tilde{\theta}_0 = 0$ and, once active, updates against previously returned forecasts before computing q_t:

$$\tilde{\theta}_t^{\alpha} = \tilde{\theta}_{t-1}^{\alpha} - \eta_t \sum_{s \in \mathcal{R}_t^{(d)}} (\mathbf{1}\{y_s^{[\iota_t]} \leq q_s^{\alpha}\} - \alpha), \tag{32}$$
$$\theta_t = \Pi_{\mathcal{K} - b_t}(\tilde{\theta}_t), \qquad q_t = b_t + \theta_t. \tag{33}$$

The hidden state is unprojected; an empty release set gives no update. The `log1p` tracker applies the same rule in ψ coordinates.

Weekly re-run. For forecasts and states, a superscript $[t]$ identifies the re-run at reference week t: $q_s^{[t]}$ is its reconstructed forecast for row s. This differs from the report cutoff date v in $y_s^{[v]}$. At reference week t, reconstruct plain MultiQT from zero using reports current at ι_t, retaining the release positions $j = \tau_d(s)$ and delay d, but omitting invalid current reports:

$$\tilde{\theta}_j^{\alpha,[t]} = \tilde{\theta}_{j-1}^{\alpha,[t]} - \eta_j^{[t]} \sum_{\substack{s \in \mathcal{R}_j^{(d)} \\ y_s^{[\iota_t]} \text{ valid}}} (\mathbf{1}\{y_s^{[\iota_t]} \le q_s^{\alpha,[t]}\} - \alpha), \tag{34}$$

$$\theta_j^{[t]} = \Pi_{\mathcal{K}-b_j}(\tilde{\theta}_j^{[t]}), \qquad q_j^{[t]} = b_j + \theta_j^{[t]}. \tag{35}$$

Only $q_t^{[t]}$ is returned; earlier returned forecasts are unchanged. All re-runs keep the stateful S_0 and L_0 fixed (Section 3.3). Studentized methods use (22), reconstruct their scale history and return the base until a required scale exists. An invalid revised report removes an update, not an expert: the declared score bank remains fixed (Table 3).

Experimental withholding. The separate screen experiment retains $m_s^{(d)}$ but waits until the newest eligible report clears the screen. Movement flags are reassessed at every cutoff; a fresh large-revision flag blocks only the first cutoff at which that version is current, since unchanged values are not stored as new versions. Unassessed reports pass. Dated manual reviews could override flags, but none were supplied. Unresolved flags can prevent a row from being learned; older reports are never substituted. Only stateful delays 0 and 1 were tested, excluding re-runs and earlier A–E mixtures. Tuned parameters stay fixed, but S_0 and L_0 are recomputed from the changed release schedule: the comparison combines postponed updates with changed initialization.

3.3 Tuning

Reference scales use the first 20 consumed rows in release order, $I_{20}^{(d)}$, which need not span 20 calendar weeks. Parameter selection uses the tuning period; these are separate operations.

Reference scales and retrospective initialization. Write Q_p for an empirical p-quantile. The initial error scale S_0, distinct from the score function S_t, pools absolute base errors in tracker units:

$$S_0 = \max\Big\{\varepsilon_S, Q_{0.9}\Big(\Big\{|\psi(b_s^\alpha) - \psi(\bar{y}_s^{(d)})| : s \in I_{20}^{(d)},\ \alpha \in \mathcal{A}\Big\}\Big)\Big\}, \qquad \varepsilon_S = 10^{-6}. \tag{36}$$

It sets rates $\eta = \kappa S_0$, with rate multiplier κ, and studentized scale bounds $[.05S_0, S_0]$; studentized step sizes are selected in standardized units. Script 02 labels the multiplier m, distinct from $m = |\mathcal{A}|$ here. The reference loss L_0 is defined in Section 3.5.3.4.

With the default `replay_warmup=TRUE`, both constants use the eventual first 20 releases and apply from the start. Forecasts before the 20th release therefore use retrospective scale information. In the active data these rows have tuning target dates, but some reports arrive after early evaluation cutoffs: tuning-period membership does not imply availability before evaluation. With replay disabled, learning waits for warmup; fewer than 20 releases leaves learning inactive. The constants stay fixed across weekly re-runs; states, weights and time-varying scales are reconstructed. Evaluation never resets the tracker, so its initial evaluated correction is generally nonzero.

Cases and selection. Tune COVID before flu, and `log1p` before `identity` within each disease. For each pair, run stateful delay 0, delay 1 and weekly re-run at delay 0. These 12 configurations have two stages each (24 total). Each starts from defaults; configurations without

tuning rows stop. The hub design tunes horizons $-1, \ldots, 3$; the seasons design tunes $0, \ldots, 3$, with -1 reusing their choices.

Candidates replay the full history with common S_0, L_0 within each case, tracker and feedback design. Only tuning target weeks are scored, including initialization rows when valid. Scores use latest archived truth, which need not have been available before evaluation. Pool case-level means equally over non-US locations and horizons. The criterion is log quantile loss for `log1p` and quantile loss per 100,000 for `identity`, without L_0 normalization.

Stage 1 selects tracker rates separately for `fixed`, `leaky`, `absolute_slope`, `studentized`, `studentized_base`, `median_constant`, `studentized_full` and `studentized_level_rms`. Let $\overline{\mathcal{L}}_j(\vartheta)$ be the mean loss of method j at candidate ϑ over tuning cases, and $\overline{\mathcal{L}}_0$ the original's mean loss. Retain candidates satisfying

$$\overline{\mathcal{L}}_j(\vartheta) \leq \min_{\vartheta'} \overline{\mathcal{L}}_j(\vartheta') + .01\,\overline{\mathcal{L}}_0,$$

then select the smallest mean calibration error. This elbow rule allows one percentage point above the minimum on the loss-change plot; it does not estimate curvature. Stage 2 holds those rates fixed and minimizes loss over H separately for `weighted`, `weighted_common`, `shared`, `shared_common`, `restart_weighted`, `restart_grid` and `score_learned`. Both stages break ties by the smaller parameter value. This single pass avoids a Cartesian search; earlier rate choices are not revisited and need not give the best joint combination.

Stage	Setting	Candidates
1	`fixed_multiplier`	.001, .0025, .005, .01, .02, .05, .1, .2, .4
1	`slope_min_multiplier`	.002, .004, .01, .02, .05, .1
1	`median_rate`	.003, .01, .03, .1, .3, 1
1	`leaky_multiplier`	.001, .002, .004, .008, .02, .05, .1, .2
2	`H`	2, 4, 8, 16, 32

Multipliers scale S_0; the slope multiplier controls its rate floor. `median_rate` is standardized. H counts processed weight updates and also sets sharing $\gamma = 1/(2H)$ where enabled.

Fixed and inherited settings. Learned mixtures use adaptive λ from $\lambda_{\text{start}} = 2.72$; $\lambda = 0$ gives equal weights in `exp_weights()`. Recent-start count is 8; outer and Decline rates are $.001S_0$ and $.05S_0$; target-signal threshold/width are 0/.05. Other constants remain as specified with each method. `studentized_zero` and `median_constant_zero` inherit their corresponding normal-start rates; `score_gated` inherits H from `score_learned`, and outer variants retain the selected settings of their inner mixtures.

Tuning plots. The current `multiqt_tuning.pdf` pairs calibration error with percentage loss change $100(\overline{\mathcal{L}}_j/\overline{\mathcal{L}}_0 - 1)$. Solid/dashed lines use the different log/identity criteria. Filled rate points meet the tolerance; open points exceed it; large circles mark selected values. A horizontal CE line marks the original. Pooled pages by disease and feedback design precede pages for California, Texas, New York and Pennsylvania, when available. State pages average over horizons but circle the pooled selection. H curves condition on selected rates. The `on_grid_edge` flag marks an endpoint selection, not evidence of an interior optimum. Embedded plots at page 47 retain their recorded earlier procedure.

3.4 Evaluation

Peaks and evaluation windows. Let $y_t^\star$ be the latest archived count for the target of row t. `find_peaks()` removes nonfinite counts and the incomplete-reporting interval. Write x_i for the

retained latest counts in date order, z_i for their smoothed values and P_i for peak prominence:

$$z_i = (x_{i-1} + x_i + x_{i+1})/3, \qquad P_i = z_i - \max\left\{ \min_{l_i \le j \le i} z_j, \min_{i \le j \le r_i} z_j \right\}.$$

Here l_i, r_i are the nearest strictly higher smoothed positions on either side, or the series ends; X below is the largest retained count on a forecast target date. Endpoints retain the unsmoothed value. Candidates are maxima within eight retained observations on each side, earliest at ties, satisfying

$$P_i > 0, \qquad P_i \ge .30 z_i, \qquad P_i \ge .10 X.$$

Place each peak at the largest raw count within one retained observation of its candidate, earliest at ties; keep dates within the forecast target range. These windows count observations, not calendar weeks across gaps. Peaks are hindsight labels, used for prediction only by the target oracle.

Let T_t be the target date, $p(t)$ its nearest retained peak date (earlier at ties), and $u_t^{\text{peak}} = (T_t - p(t))/7$ the distance in weeks. For evaluation rows E with finite truth, score the windows

$$\begin{aligned} W_{\text{overall}} &= E, \\ W_{\text{before}} &= \{t \in E : -7 \le u_t^{\text{peak}} \le 0\}, \\ W_{\text{after}} &= \{t \in E : 1 \le u_t^{\text{peak}} \le 8\}. \end{aligned} \tag{37}$$

Rows count once; the peak week belongs to "up to peak". Overall also includes weeks outside these windows. Missing rows and period boundaries shorten windows without discarding peaks. Period membership follows the target date; its nearest peak may lie in another period. Seasonal scores restrict these sets; the main plots combine both evaluation seasons.

Loss and calibration error. A case fixes location, horizon and feedback design within a disease. For a selected window W of $N = |W|$ rows, average over rows and all 23 quantile levels:

$$\mathcal{L}^{\text{nat}}(W) = \frac{1}{mN} \sum_{t \in W} \sum_{\alpha \in \mathcal{A}} \rho_\alpha(q_t^\alpha / c_{\text{scale}}, y_t^\star / c_{\text{scale}}), \tag{38}$$

$$\mathcal{L}^{\log}(W) = \frac{1}{mN} \sum_{t \in W} \sum_{\alpha \in \mathcal{A}} \rho_\alpha(\log(1 + \max\{q_t^\alpha, 0\}), \log(1 + y_t^\star)), \tag{39}$$

$$\text{CE}(W) = \frac{1}{m} \sum_{\alpha \in \mathcal{A}} \left| \frac{1}{N} \sum_{t \in W} \mathbf{1}\{y_t^\star \le q_t^\alpha\} - \alpha \right|. \tag{40}$$

Columns are `quantile_loss`, `log_quantile_loss` and `coverage_error`; tuning uses `loss`, `log_loss` and `coverage_error`. CE averages weeks before taking the absolute residual. All feedback designs use the same latest truth; empty windows give missing scores. Evaluation losses are never divided by L_0. Scripts 04/06 display the criterion of the tracker; script 02 displays loss per 100,000.

Quantile-level diagnostics. `multiqt_by_level.pdf` repeats the fixed-rate trade-off at each level:

$$\begin{aligned} \mathcal{L}_\alpha^{\text{nat}}(W) &= \frac{1}{N} \sum_{t \in W} \rho_\alpha(q_t^\alpha / c_{\text{scale}}, y_t^\star / c_{\text{scale}}), \\ \text{CE}_\alpha(W) &= \left| \frac{1}{N} \sum_{t \in W} \mathbf{1}\{y_t^\star \le q_t^\alpha\} - \alpha \right|. \end{aligned} \tag{41}$$

Pages fix disease and level, with location rows, horizon columns and phase symbols. There are 23 COVID pages then 23 flu pages, in increasing α order; .1, .5, .9 are loaded at page 41.

All-location trade-offs. Write $\mathcal{L}_{j,c}$ and $\mathrm{CE}_{j,c}$ for the mean quantile loss and calibration error of method j in case c; $j = 0$ is the original. A set $\mathcal{C}$ contains J cases. In `multiqt_peak_reasoning_tradeoff.pdf`, columns are horizons $-1, \ldots, 3$; blocks are delay 0, delay 1 and re-run, each showing overall, up to peak and after peak. Lower left is better. The first page per disease shows individual non-US cases and large method means

$$\left(J^{-1} \sum_c \mathcal{L}_{j,c},\ J^{-1} \sum_c \mathrm{CE}_{j,c} \right).$$

These average location scores, not forecasts or pooled coverage indicators. The first row of `multiqt_error_tradeoff.pdf` uses the same aggregation for fixed rates. Feedback designs have separately selected parameters, so their differences include retuning. Axes vary by panel; all-location limits use the 1st–99th percentiles and mean points, with loss capped near four times the base. Extremes can be hidden or marked `>`. Original/common-weight/outer methods use open circles/triangles/squares or diamonds; exact forecast duplicates are hidden.

Evaluation tree. Script 06 splits pooled disease cases by feedback design, then horizon. For each window it reports

$$R_j(\mathcal{C}) = 100 \left[1 - \frac{\sum_{c \in \mathcal{C}} N_c \mathcal{L}_{j,c}}{\sum_{c \in \mathcal{C}} N_c \mathcal{L}_{0,c}} \right], \tag{42}$$

$$P_j(\mathcal{C}) = J^{-1} \sum_{c \in \mathcal{C}} \mathbf{1}\{\mathcal{L}_{j,c} < \mathcal{L}_{0,c}\}. \tag{43}$$

`reduction` is a ratio of week-weighted losses (positive is better); `share_improved` weights cases equally, counts ties as not improved, and is printed for at least five cases. CE panels also average case scores with week weights. Each page displays the ten methods with lowest pooled after-peak loss: a descriptive evaluation ranking, not tuning. Overlapping horizons/designs are not independent replicates. Tree panels share axes; clipping is marked `<` or `>`. By contrast, script 02 phase-rate plots show loss change $100(\mathcal{L}_j/\mathcal{L}_0 - 1)$, where negative is better.

Forecast curves. `multiqt_method_forecasts.pdf` shows levels $.1, \ldots, .9$ and weekly loss per 100,000 at a fixed horizon; gray marks tuning, dotted lines peaks. `multiqt_season_forecasts.pdf` adds coverage and loss trajectories for selected fixed rates. `multiqt_base_forecast_paths.pdf` shows full count timelines, median paths around all retained peaks and the three earlier indicators (Section 4.6). It spans tuning and evaluation; its flu winter 2024/25 peak is within tuning, not held-out evidence. Weight timelines and archived A–E soft-switch timelines explain mixtures, rather than providing additional performance scores. Gap dates are listed in Section 3.1.

3.5 Methods

The methods below use the preceding data, feedback, tuning and evaluation conventions unless stated otherwise.

3.5.1 Single trackers

These methods maintain one MultiQT tracker for each case.

Fixed rate: `fixed`. Use (32)–(33) with $\eta = \kappa S_0$ in tracker units. Tune κ separately (default .05). Corrections persist across rounds.

Leaky tracker: `leaky`. Add a common decaying shift to persistent MultiQT offsets, as in (63)–(64). The separately tuned method also belongs to the score bank.

Slope-dependent rate: `absolute_slope`. Let Δ_t be elapsed calendar weeks since the previous reference date, at least one. Use the delayed MultiQT update with rate

$$\eta_t = \min\left\{\kappa_{\max} S_0,\ \kappa_{\min} S_0 + \left|\frac{\beta_t^{.5} - \beta_{t-1}^{.5}}{\Delta_t}\right|\right\}. \tag{44}$$

The first slope is zero. Tune the floor multiplier (default .004) and keep the cap at .08. The slope compares base medians at the same horizon across reference dates, per elapsed calendar week; it uses no future outcomes. Unlike studentization, it changes only new increments.

Persistent restart control: `restart_persistent`. Keep the first eligible weekly-start expert with $\kappa = .05$ (Section 3.5.3.5). It learns only from forecasts made since its start. Its rate is not separately tuned, so this control for restart mixtures need not reproduce `fixed`.

3.5.2 Studentized scores

Studentized variants and median initialization. In tracker units, use centre $a_t^\alpha = \beta_t^\alpha$ for the full base or $a_t^\alpha = \beta_t^{.5}$ for its median. After processing available feedback, return

$$q_t = \psi^{-1}\left[\Pi_{\mathcal{K}}^{A_t^{-2}}(a_t + A_t \tilde{u}_t)\right]. \tag{45}$$

For a common scale, the projection is the ordinary isotonic projection. After warmup, the delayed update is applied to the hidden state,

$$\tilde{u}_t^\alpha = \tilde{u}_{t-1}^\alpha + \eta \sum_{s \in \mathcal{R}_t^{(d)}} \left(\alpha - \mathbf{1}\{\bar{y}_s^{(d)} \le q_s^\alpha\}\right). \tag{46}$$

Compare each report with its historical forecast q_s, reconstructed inside a re-run when applicable. Project only the forecast; the hidden state persists across peaks and seasons. The standardized step has default $\eta = .1$ and is tuned for each normal-start or full-base method. Zero-start variants inherit the corresponding normal-start rate.

Scales from recent errors or the current forecast. Let $\mathcal{H}_t^{(d)}$ contain the last five consumed rows, or fewer if unavailable, and set $e_s^\alpha = \psi(\bar{y}_s^{(d)}) - \beta_s^\alpha$. These need not be consecutive calendar

weeks. A hat marks a raw scale estimate, and Φ is the standard normal distribution function. The default scales are

$$\widehat{\sigma}_{t,\mathrm{med}} = \left[\frac{1}{|\mathcal{H}_t^{(d)}|}\sum_{s\in\mathcal{H}_t^{(d)}}(e_s^{.5})^2\right]^{1/2}, \tag{47}$$

$$\widehat{\sigma}^{\alpha}_{t,\mathrm{res}} = \left[\frac{1}{|\mathcal{H}_t^{(d)}|}\sum_{s\in\mathcal{H}_t^{(d)}}(e_s^{\alpha})^2\right]^{1/2}. \tag{48}$$

Both require five consumed rows. RMS includes bias because the errors are not centred. The optional robust alternative is median $|e_s^{\alpha}|/\Phi^{-1}(.75)$.

The common scale inferred from the current base is

$$\widehat{\sigma}_{t,\mathrm{base}} = \frac{\beta_t^{.9} - \beta_t^{.1}}{\Phi^{-1}(.9) - \Phi^{-1}(.1)}. \tag{49}$$

The denominator is approximately 2.563. The earlier `studentized_levels` method used the level-specific base scales

$$\widehat{\sigma}^{\alpha}_{t,\mathrm{pair}} = \frac{\beta_t^{1-a_\alpha} - \beta_t^{a_\alpha}}{\Phi^{-1}(1-a_\alpha) - \Phi^{-1}(a_\alpha)}. \tag{50}$$

Here $a_\alpha = \min(\alpha, 1-\alpha)$, replacing $a_{.5}$ by the nearest lower grid level. This construction requires a quantile grid symmetric about .5. Base-width scales respond immediately to changes in the current forecast; residual scales respond only after the corresponding reports arrive.

Floor raw scales at 10^{-6}, then clip to $[.05S_0, S_0]$ once S_0 exists; denote these bounded scales by σ. The bounds limit how much the scale suppresses or amplifies accumulated corrections. They are implementation defaults, not requirements of studentization.

Table 1: Studentized methods. Scales refer to the bounded versions above. Initial states are specified before the first update.

Method ID	Centre and scale	Initialization and purpose
`studentized`	$a_t^{\alpha} = \beta_t^{.5}$; common $\sigma_{t,\mathrm{med}}$	$\tilde{u}^{\alpha}_{\mathrm{init}} = \Phi^{-1}(\alpha)$. Reconstruct the distribution around the base median using recent errors.
`studentized_base`	$a_t^{\alpha} = \beta_t^{.5}$; common $\sigma_{t,\mathrm{base}}$	Normal initial state. Reconstruct the distribution from the median and the central width of the current base forecast.
`median_constant`	$a_t^{\alpha} = \beta_t^{.5}$; common constant σ	Normal initial state; freeze $\sigma_{t,\mathrm{med}}$ at the first reference week after both required warmups. Separates replacing the base by its median from subsequently changing the scale.
`studentized_zero`	$a_t^{\alpha} = \beta_t^{.5}$; common $\sigma_{t,\mathrm{med}}$	$\tilde{u}_{\mathrm{init}} = 0$. Start all levels at the median; use the rate selected for `studentized`.
`median_constant_zero`	$a_t^{\alpha} = \beta_t^{.5}$; common constant σ	Zero initial state; use the scale and rate of `median_constant`.
`studentized_full`	$a_t^{\alpha} = \beta_t^{\alpha}$; common $\sigma_{t,\mathrm{base}}$	$\tilde{u}_{\mathrm{init}} = 0$. Keep the full base and rescale its accumulated corrections together.

Continued on next page

Table 1 (continued)

Method ID	Centre and scale	Initialization and purpose
`studentized_level_rms`	$a_t^\alpha = \beta_t^\alpha$; $\sigma_{t,\mathrm{res}}^\alpha$	Zero initial state. Set the scale at each level from recent errors at that level; weighted isotonic projection.

The paired-width method `studentized_levels`, removed on 11 November 2026, is retained here only to interpret earlier plots. The scale of `median_constant` is fixed within each reconstructed history but can change between re-runs; S_0 and L_0 cannot. Zero-start variants compare initialization at the same selected rate. Their first active forecast may already include a feedback update.

The residual-median methods also allow $t_5^{-1}(\alpha)$ initialization; the base-width median method always uses normal quantiles. It retains the median and bounded central width but imposes a symmetric normal shape in tracker units, so it need not reproduce the original forecast. Until a required residual scale exists, return the original full base. Feedback consumed while waiting is not replayed; later feedback on those rounds is compared with the original forecasts.

Studentization rescales the accumulated correction, rather than only the newest increment. For full-base `log1p` methods without crossings,

$$q_t^\alpha = (1 + b_t^\alpha)\exp(\sigma_t^\alpha \tilde{u}_t^\alpha) - 1, \tag{51}$$

so the count correction is non-additive. With `identity`, it is additive with a changing multiplier. Quantile-specific scales retain the coverage target, but no calibration guarantee is established here for those variants.

3.5.3 Weighted experts

An expert is a complete forecasting procedure; a bank is the set being combined. Weights learn from quantile losses separately for each location–horizon case. A fixed learning rate still changes the weights; an adaptive rate changes their responsiveness. The formulas first assume a fixed bank with all experts eligible.

3.5.3.1 Mixtures and feedback losses For K experts, let $f_{k,t}^\alpha$ be expert k's forecast and w_k^α its weight. Suppressing time indices for one forecast, form the mixture $\hat{q}$ and repair any crossings:

$$\hat{q}^\alpha = \sum_{k=1}^K w_k^\alpha f_k^\alpha, \qquad w_k^\alpha \geq 0, \quad \sum_{k=1}^K w_k^\alpha = 1, \qquad q = \Pi_{\mathcal{K}}(\hat{q}). \tag{52}$$

Per-quantile weights can introduce crossings, repaired by isotonic projection. Common weights replace w_k^α by w_k and preserve order when every expert is ordered. All mixtures average counts, including for `log1p` trackers.

Let n count processed feedback updates, with forecast row $s(n)$ and report Y_n. Score the expert forecasts for that row, using L for unnormalized losses and ℓ for losses normalized by L_0:

$$L_{k,n}^\alpha = \rho_\alpha\left(f_{k,s(n)}^\alpha / c_{\mathrm{scale}}, Y_n / c_{\mathrm{scale}}\right), \qquad L_{k,n} = \frac{1}{m}\sum_{\alpha\in\mathcal{A}} L_{k,n}^\alpha, \tag{53}$$

$$\ell_{k,n}^\alpha = L_{k,n}^\alpha / L_0, \qquad \ell_{k,n} = \frac{1}{m}\sum_{\alpha\in\mathcal{A}} \ell_{k,n}^\alpha. \tag{54}$$

The same L_0 applies to all experts and levels within a case and feedback design (Section 3.5.3.4). Score the originally computed forecasts in stateful learning and the reconstructed forecasts in a re-run.

3.5.3.2 Exponential updates and optional sharing Let $w^{\alpha}_{k,n}$ and λ^{α}_{n} be the weight and rate before update n. Before optional sharing, update separately at each level to obtain $\bar{w}^{\alpha}_{k,n+1}$:

$$\bar{w}^{\alpha}_{k,n+1} = \frac{w^{\alpha}_{k,n} \exp(-\lambda^{\alpha}_{n} \ell^{\alpha}_{k,n})}{\sum_{j=1}^{K} w^{\alpha}_{j,n} \exp(-\lambda^{\alpha}_{n} \ell^{\alpha}_{j,n})}. \tag{55}$$

With common weights, there is one update using the mean loss:

$$\bar{w}_{k,n+1} = \frac{w_{k,n} \exp(-\lambda_{n} \ell_{k,n})}{\sum_{j=1}^{K} w_{j,n} \exp(-\lambda_{n} \ell_{j,n})}. \tag{56}$$

Lower loss increases relative weight; larger λ strengthens the response. Weights start uniformly, and $\lambda = 0$ keeps equal weights.

When sharing is enabled with fraction γ, follow the exponential step by

$$w^{\alpha}_{k,n+1} = (1-\gamma)\bar{w}^{\alpha}_{k,n+1} + \gamma/K. \tag{57}$$

Omit α for common weights. Sharing helps downweighted experts recover: use $\gamma = 1/(2H)$ when enabled and zero otherwise. Table 2 distinguishes sharing from common weights across levels.

3.5.3.3 Adaptive learning rates The lookback H counts processed updates. The first H updates use starting rate λ_{start}. Record the weighted variance of each update using its weights before the exponential step and sharing:

$$\begin{aligned} M^{\alpha}_{n} &= \sum_{k=1}^{K} w^{\alpha}_{k,n} \left(\ell^{\alpha}_{k,n} - \sum_{j=1}^{K} w^{\alpha}_{j,n} \ell^{\alpha}_{j,n} \right)^{2}, \\ M_{n} &= \sum_{k=1}^{K} w_{k,n} \left(\ell_{k,n} - \sum_{j=1}^{K} w_{j,n} \ell_{j,n} \right)^{2}. \end{aligned} \tag{58}$$

Treat variances at most $10^{-12} \sum_{k} w^{\alpha}_{k,n} (\ell^{\alpha}_{k,n})^{2}$ as zero, and analogously for common weights, to avoid large rates from numerical rounding. For per-quantile weighting, the rate is

$$\lambda^{\alpha}_{n} = \begin{cases} \lambda_{\text{start}}, & 1 \le n \le H, \\ \sqrt{\dfrac{\log(2KH)+1}{\sum_{r=n-H}^{n-1} M^{\alpha}_{r}}}, & n > H. \end{cases} \tag{59}$$

Only the previous H variances enter; a zero sum retains the previous finite rate. Greater loss differences reduce the rate, whereas a large error shared by all experts does not increase the variance.

Per-quantile weighting maintains a vector, variance history and rate for each level. Common weighting uses one of each: average losses across levels *before* computing M_r, then omit α in (59). This is not an average of separately learned weights or per-quantile variances.

Later starts and implementation. Only restart experts that forecast the released row are eligible. Normalize their weights for the update, then restore their total mass; other weights stay unchanged. Use the eligible count in the rate formula. A new expert receives $1/K$ of the enlarged bank, preserving existing relative weights. Updates with fewer than two eligible experts do not count toward H.

3.5.3.4 Expert loss normalization Using the quantile loss in (1), define one reference loss per case and delay:

$$L_0 = \max\left\{\varepsilon_L, \frac{1}{20|\mathcal{A}|}\sum_{s\in I_{20}^{(d)}}\sum_{\alpha\in\mathcal{A}}\rho_\alpha\left(\frac{b_s^\alpha}{c_{\text{scale}}}, \frac{\bar{y}_s^{(d)}}{c_{\text{scale}}}\right)\right\}, \qquad \varepsilon_L = 10^{-6}. \tag{60}$$

Losses are relative to the initial mean base loss: $\ell = .8$ means 80% of that reference, not necessarily a 20% improvement in the current week. Values can exceed one. Weighting uses scaled counts even for `log1p` trackers.

Initial and fixed-rate weight updates. Dividing by L_0 preserves rankings and makes weight changes depend on relative loss gaps. A change of count units scales both L and L_0 equally, leaving weights unchanged when the numerical floor is inactive. This matters throughout fixed-rate runs and during the first H adaptive-run updates at λ_{start}; these are distinct from the 20 rows used to estimate L_0.

For equal initial weights, losses 80 and 100, $L_0 = 100$ and $\lambda_{\text{start}} = 1$ give weights about .55 and .45 before sharing. The same numerical rate on raw losses would put almost all weight on the first expert.

Effect on adaptive learning rates. After H updates, the rule in (59) already adapts to the variation between expert losses. Use superscripts raw, norm for calculations without and with L_0 normalization. Conditional on the same pre-update weight vectors at updates $n-H,\ldots,n$ and a positive variance sum, normalization gives, separately at each level,

$$\begin{aligned} M_r^{\alpha,\text{norm}} &= M_r^{\alpha,\text{raw}}/L_0^2, & \lambda_n^{\alpha,\text{norm}} &= L_0\lambda_n^{\alpha,\text{raw}}, \\ \lambda_n^{\alpha,\text{norm}}\ell_{k,n}^\alpha &= \lambda_n^{\alpha,\text{raw}}L_{k,n}^\alpha. \end{aligned} \tag{61}$$

The identity also holds for common weights. The numerical zero-variance threshold scales by L_0^{-2} too. Thus the loss and rate changes cancel in the exponential update, conditional on the same weight history. Runs with the same numerical starting rate can still differ: their first H updates change later weights and variances.

Equivalent runs require matching settings and $\lambda_{\text{start}}^{\text{raw}} = \lambda_{\text{start}}^{\text{norm}}/L_0$, including the same scaling of retained rates when the variance sum is zero. Keep normalization throughout a run so incoming losses, stored variances and rates remain in consistent units. After adaptation, L_0 is not an additional reduction in update strength.

Effect across quantile levels and re-runs. A single L_0 does not equalize losses across levels. Per-quantile weights use the loss and adaptive rate of each level; common weights use mean losses. Re-runs reconstruct losses but retain the stateful L_0. Evaluation losses are not normalized by L_0, and normalization alone guarantees neither accuracy nor calibration.

3.5.3.5 Expert combinations implemented in the code Table 2 groups the current combinations. Every learned mixture uses the adaptive variance-based rate above.

Method IDs	Weighting and bank
`uniform`	Equal weights on 11 persistent rate experts.
`weighted`, `shared`	Same rate bank; per-quantile weights, without/with sharing.
`weighted_common`, `shared_common`	Same rate bank; common weights, without/with sharing.
`restart_uniform`	Equal weights on all weekly starts, rate $.05S_0$.
`restart_weighted`	Same starts; adaptive λ, sharing, per-quantile weights.
`restart_grid`	Weekly starts at $.01S_0, .05S_0, .2S_0$; adaptive λ, sharing, per-quantile weights.
`restart_recent`	Equal weights on the eight most recent starts, rate $.05S_0$.
`score_equal`	Equal weights on the declared score bank.
`score_learned`	Same score bank; adaptive λ, sharing, per-quantile weights.
`score_gated`	Mix `score_learned` with the original using the target-specific gate $G^T_{t,h}$.

Table 2: Current expert combinations.

The five `_outer` variants of `shared`, `shared_common`, `restart_uniform`, `restart_weighted` and `restart_grid` add persistent MultiQT at $.001S_0$ to their inner mixture, retaining its settings and learning from their own final forecasts. Rate and H selection follow Section 3.3.

Rate and weekly-start banks. The rate bank uses

$$\kappa \in \{0, .01, .015, .02, .03, .05, .075, .1, .15, .2, .3\};$$

all experts start together and the zero-rate member is the original. Weekly-start banks add a fresh copy at every retained reference week when initialization permits; with warmup replay, starts begin in the first round. Each copy starts with zero correction and learns only from feedback on forecasts made since that start. Older copies keep their accumulated corrections. There is no permanent zero-rate expert.

For `restart_recent`, $\mathcal{B}_t$ contains the most recent $K_{\text{recent}} = 8$ copies, or all copies if fewer have started. Then

$$q^\alpha_t = |\mathcal{B}_t|^{-1} \sum_{k \in \mathcal{B}_t} f^\alpha_{k,t}. \tag{62}$$

This controls the age of corrections without detecting peaks. Count starts, not elapsed weeks across gaps. A new expert receives weight before feedback, so a new color band alone does not show lower loss.

The six score-based experts and fixed membership. The bank compares ways to retain or rebuild the base and correct its errors. Write $S^\alpha_{k,t}(y)$ for member k's residual score. For the scores below, $r = \psi(y)$; project and invert as in (45). Fixed, leaky and studentized experts use separately selected rates; Decline uses $.05S_0$ and the original does not update. This compares complete procedures, not scores at a common rate.

Membership is fixed across reference weeks and re-runs, even when forecasts coincide. Exclude Decline at $h = -1$, where it equals the original by construction. Do not deduplicate complete forecast histories: later forecasts or revisions must not change earlier membership.

Table 3: The declared score bank. Scales are the bounded scales defined earlier; Decline is omitted at $h = -1$ under the fixed-membership correction.

Member	Score or forecast	Purpose
`original`	$f^{\alpha}_{k,t} = b^{\alpha}_{t}$	Preserve the base as an option.
`fixed`	$S^{\alpha}_{k,t}(y) = r - \beta^{\alpha}_{t}$	Persistent additive correction; $\eta = \kappa S_0$.
`leaky`	Fixed MultiQT plus v_t in (63)	Short-lived common directional correction alongside persistent offsets.
`studentized_base`	$S^{\alpha}_{k,t}(y) = (r - \beta^{5}_{t})/\sigma_{t,\text{base}}$	Rebuild around the median with current central width and normal initial thresholds.
`studentized_full`	$S^{\alpha}_{k,t}(y) = (r - \beta^{\alpha}_{t})/\sigma_{t,\text{base}}$	Keep the full base; rescale accumulated corrections with its central width.
`decline`	$S^{\alpha}_{k,t}(y) = r - \beta^{\alpha}_{t}$, updated only for $c^{\text{ctx}}_{s,h} = 1$	Learn and apply a persistent correction in declining forecast context; return the original otherwise.

Initialization and scales follow Table 1. Leaky and Decline change state updates, rather than defining new static residual scores. `studentized_level_rms` remains a standalone method; `studentized_levels` appears only in earlier results.

The leaky expert. Besides its persistent MultiQT offsets, this expert has one common shift v_t on the log scale, initially zero. Write $\text{clip}_{[a,b]}(x) = \min\{b, \max\{a, x\}\}$. After warmup, its update is

$$v_t = \text{clip}_{[-.2,.2]}\left[2^{-\Delta_t} v_{t-1} + .2 \sum_{s \in \mathcal{R}^{(d)}_t} \frac{1}{m} \sum_{\alpha \in \mathcal{A}} \left(\alpha - \mathbf{1}\{\bar{y}^{(d)}_s \le f^{\alpha}_{\text{leaky},s}\}\right)\right], \tag{63}$$

The shift halves per calendar week, including gaps. Sum reports within a batch, then cap the shift. Errors in the same direction reinforce it; opposite errors across levels can cancel. It therefore targets a common directional error. Ordinary offsets persist and update at each level.

For `log1p`, add v_t after projection and before inversion. For `identity`, apply it to the ordinary projected count forecast $q^{\alpha}_{\text{plain},t}$ as

$$f^{\alpha}_{\text{leaky},t} = (1 + (q^{\alpha}_{\text{plain},t})_+)e^{v_t} - 1 + \min(q^{\alpha}_{\text{plain},t}, 0). \tag{64}$$

This gives the same log-shift for nonnegative counts and extends it to negative forecasts. These trackers do not clip final forecasts at zero.

Target-specific signal and Decline. Let $Z(t, h; v)$ be $\log(1 + \text{base median})$ at reference date t and horizon h, if the median is finite, nonnegative, available by v and not filled; otherwise it is missing. Write $Z_{t,h} = Z(t, h; \iota_t)$. The signal compares the target horizon with the largest available earlier median in the same forecast path. It defines context $c^{\text{ctx}}_{t,h} \in \{0, 1\}$ and gate

$G^T_{t,h} \in [0,1]$, the weight of the original in `score_gated`:

$$D_{t,h} = \max_{\substack{r=-1,\dots,h-1 \\ Z_{t,r} \text{ available}}} Z_{t,r} - Z_{t,h},$$

$$c^{\text{ctx}}_{t,h} = \mathbf{1}\{D_{t,h} > \log(1+\delta_G)\}, \qquad G^T_{t,h} = \left[1 + \exp\left\{-\frac{D_{t,h} - \log(1+\delta_G)}{\log(1+w_T)}\right\}\right]^{-1}. \tag{65}$$

The target median and at least one earlier median must be available by the cutoff at the exact reference date, excluding filled forecasts. Without this information, including at $h = -1$, set the context and gate to zero. The fixed settings are $\delta_G = 0$ and $w_T = .05$. The signal threshold δ_G is distinct from miscoverage δ; w_T controls the gate's transition width. When $D_{t,h} = 0$, the hard context is zero but the gate is .5. The signal describes a decline in the base forecast path; it is not a calibrated probability of being after the true peak.

Decline starts with zero hidden offsets and uses

$$\tilde{\theta}^\alpha_t = \tilde{\theta}^\alpha_{t-1} + .05 S_0 \sum_{s \in \mathcal{R}^{(d)}_t} c^{\text{ctx}}_{s,h} \left(\alpha - \mathbf{1}\{\bar{y}^{(d)}_s \le f^\alpha_{\text{decline},s}\}\right), \tag{66}$$

$$f_{\text{decline},t} = \begin{cases} \psi^{-1}\left[\Pi_{\mathcal{K}}(\beta_t + \tilde{\theta}_t)\right], & c^{\text{ctx}}_{t,h} = 1, \\ b_t, & c^{\text{ctx}}_{t,h} = 0. \end{cases} \tag{67}$$

The historical context of row s determines whether its feedback updates this state; the current context determines whether to apply the correction. Offsets persist between declining periods. The same warmup and re-run conventions apply as for Fixed MultiQT.

Equal, learned and gated score mixtures. `score_equal` averages the declared experts. For `score_learned`, learn one vector per quantile level using the normalized losses, adaptive rate and sharing above, then repair crossings. The gated mixture is

$$q^\alpha_{\text{score_gated},t} = (1 - G^T_{t,h}) q^\alpha_{\text{score_learned},t} + G^T_{t,h} b^\alpha_t. \tag{68}$$

The gate changes the returned forecast, retaining the weights, H and training losses of `score_learned`. It introduces no second weight vector. A missing signal returns the learned forecast; using the same gate across levels preserves order.

Earlier A–E experiments. The results in Section 4 use the earlier seven-member bank: original, fixed, leaky, studentized base width, studentized full base width, studentized level RMS and studentized paired level scales. They predate the six-member bank and the revised tuning procedure. A (`score_single`) learned one common vector. B (`score_switch`) used two softly updated vectors with a gate from forecast-path, nowcast and forecast-revision signals. C (`score_target_switch`) used the target-specific gate instead. D (`score_context`) trained each of two vectors only on its historical context; E (`score_fast_decline`) increased the learning rate during current declining context. Their hard-selection and no-sharing variants are also historical comparisons. These experiments used common weights, $H = 4$ and sharing 1/8 where enabled. Their adaptive rates used uncentered second moments, and the weekly-start mixtures used a fixed tuned λ. Their plots do not establish results for the current bank or for the new variance-based weighting. The earlier formulas are preserved in the archived notes accompanying those plots.

3.5.4 Reference benchmarks

The references are the original forecast, a hindsight control and two production configurations of Delphi.

Original forecast: `original`. Return $q_t = b_t$ without recalibration or learning. This is the reference for the reported loss changes and reductions.

Target oracle: `oracle_target`. This hindsight control returns

$$q_t^{\text{oracle}} = \begin{cases} q_t^{\text{fixed}}, & T_t \le p(t), \\ b_t, & T_t > p(t). \end{cases} \tag{69}$$

This bypasses correction for postpeak targets, including forecasts made before the peak. Fixed MultiQT still learns from its own forecasts. The control tests bypassing correction with perfect phase information; it is not deployable. Setting the rate to zero would instead freeze the offset, giving a different control.

Delphi benchmarks: separate scales and initialization. Both run for both diseases and all feedback designs, independently of the identity/log setting. States start at zero without S_0 or warmup; constants are untuned. For transform χ and projected forecast ξ_t in those units, the released coverage-error sum is

$$\epsilon_t^\alpha = \sum_{s \in \mathcal{R}_t^{(d)}} \left(\alpha - \mathbf{1}\{\chi(\bar{y}_s^{(d)}) \le \xi_s^\alpha\} \right). \tag{70}$$

In a weekly re-run, use its current reports and reconstructed forecasts, as in Section 3.2.

Adaptive fast and slow offsets: `ref_op`. Let $\mathcal{F}_t^{(d)} = \bigcup_{j \le t} \mathcal{R}_j^{(d)}$ contain all consumed rows, and $\mathcal{F}_{t,20}^{(d)}$ the last 20 in release order, or fewer if unavailable. The COVID production configuration uses $\chi(x) = \sqrt{x}$ on raw counts, with rates

$$\begin{aligned} \eta_t^F &= \max\left\{10^{-3},\ .03Q_{.9}\left(\left\{|\sqrt{\bar{y}_s^{(d)}} - \sqrt{b_s^\alpha}| : s \in \mathcal{F}_{t,20}^{(d)},\ \alpha \in \mathcal{A}\right\}\right)\right\}, \\ \eta_t^S &= .003Q_{.9}\left(\left\{|\sqrt{\bar{y}_s^{(d)}} - \sqrt{b_s^\alpha}| : s \in \mathcal{F}_t^{(d)},\ \alpha \in \mathcal{A}\right\}\right). \end{aligned} \tag{71}$$

Both rates are zero before the first release, pool absolute base errors over levels, and are recomputed as feedback arrives. The fast and slow hidden states $\tilde{\theta}_t^F, \tilde{\theta}_t^S$ and forecast are

$$\begin{aligned} \tilde{\theta}_t^{F,\alpha} &= .9\tilde{\theta}_{t-1}^{F,\alpha} + \eta_t^F \epsilon_t^\alpha, & \tilde{\theta}_t^{S,\alpha} &= \tilde{\theta}_{t-1}^{S,\alpha} + \eta_t^S \epsilon_t^\alpha, \\ \xi_t &= \Pi_{\mathcal{K}}(\sqrt{b_t} + \tilde{\theta}_t^F + \tilde{\theta}_t^S), & q_t^\alpha &= (\max\{\xi_t^\alpha, 0\})^2. \end{aligned} \tag{72}$$

The fast correction loses 10% per processed reference week, even without feedback; gaps do not compound decay. The slow correction persists. Re-runs reconstruct both rates and states. These rolling scales are separate from the frozen S_0.

Constant square-root rate: `sqrt_constant`. The flu production benchmark uses $\chi(x) = \sqrt{x/c_{\text{pop}}}$, where c_{pop} always equals population/100,000. Its update and forecast are

$$\begin{aligned} \tilde{\theta}_t^\alpha &= \tilde{\theta}_{t-1}^\alpha + .018\epsilon_t^\alpha, \\ \xi_t &= \Pi_{\mathcal{K}}\left(\sqrt{b_t/c_{\text{pop}}} + \tilde{\theta}_t\right), & q_t^\alpha &= c_{\text{pop}}(\max\{\xi_t^\alpha, 0\})^2. \end{aligned} \tag{73}$$

The rate .018 is in square-root counts-per-100,000 units. There is no adaptive rate, decay or second state. Delphi selected this constant using flu seasons 2024/25 and 2025/26, which overlap our evaluation seasons. Although we do not retune it, this is not an independently selected held-out benchmark for those seasons.

Zero boundary and plots. Square-root methods return zero when $\xi_t^\alpha < 0$ but learn from untruncated ξ_s^α. At zero outcomes, learning coverage can differ from coverage of the returned forecast. Both appear under "Single trackers" in script 04 and in the evaluation tree of script 06.

4 Results and where to see them

The performance comparisons describe the archived run of 10–11 November 2026: the seven-member score bank and A–E, before the six-member bank, variance-based weighting, simplified tuning and fixed-membership correction. The revised methods require new scores and plots. The updated forecast-path diagnostic is separate (Section 4.6).

The run uses `hub_first_months`, identity tracking and evaluation target weeks from June 2025 through September 2026. Unless specified otherwise, reductions pool 52 non-US locations, horizons $-1, \ldots, 3$ and three feedback designs, with week-weighted pooling as in (42). Positive percentages mean lower quantile loss per 100,000 than the original. The US is separate; longer-history and `log1p` evaluation remain separate work.

4.1 Do the full quantiles add value beyond the median?

Retaining the base width matters more than retaining the detailed shape for COVID; for flu, the full quantile curve performs better. The pooled overall reductions are:

Method	COVID	Flu
Median, constant sigma	−27.62	−34.31
Studentized median (residual RMS)	−11.89	−28.13
Studentized median (base width)	−.61	−9.60
Studentized full base (base width)	−1.28	−3.68
Studentized full base (level scales)	−1.10	−4.38
Studentized full base (level RMS)	.08	−.59

`studentized_base` replaces the detailed shape while retaining the median and CMU central width. The constant-scale and residual-RMS median methods also replace that width, worsening both diseases. `median_constant` is equivalent to raw median-residual tracking up to fixed rescaling of its rate and initial offsets. Full-base studentization retains β_t^α and rescales the correction (Table 1). These tuned comparisons do not support using the median alone, but do not isolate all shape and width information: initialization and S_0 still use the full base.

Plots: `multiqt_peak_reasoning_tradeoff.pdf`, pp. 1 and 7, "Overall" rows, compares loss and CE by horizon and feedback design. For forecast and weekly-loss curves, use the files in `multiqt_method_forecasts/`: `covid_horizon2_delay0.pdf` and `flu_horizon2_delay0.pdf`, pp. 6–7, showing the US median/full-base groups. These are loaded from notes pages 79 and 81.

4.2 What the studentized forecast curves show

At $h = 2$, delay 0, US COVID full-base curves remain close to the original and Fixed MultiQT, retaining the lag around the summer 2025 wave. Constant-scale median reconstruction keeps broad tails during low incidence, including negative lower and elevated upper quantiles, with additional loss. *Plots:* COVID forecast PDF, p. 7 (full base) and p. 6 (median), forecast and weekly-loss rows.

For US flu around January 2026, full-base width and level-scale methods raise central quantiles and add loss late in the wave; level RMS stays closer to the original. Median residual RMS produces a pronounced negative lower tail after the peak. These curves illustrate how a changing scale can amplify an accumulated correction. *Plots:* flu forecast PDF, pp. 6–7, January–March 2026 forecast and weekly-loss rows.

4.3 COVID, flu, the US and the selected four states

The selected states are California, Texas, New York and Pennsylvania, not the four largest by population. In the initial rate grid at $h = 2$, delay 0, after peaks, COVID $\kappa = .10$ lowers CE but raises loss in the US, Texas, New York and Pennsylvania; California worsens on both. New York improves both at $\kappa = .01$. For flu at $\kappa = .03$, US and Texas loss is nearly unchanged while CE improves; the other three states worsen on both. Better coverage therefore need not mean lower loss. *Plots:* `multiqt_error_tradeoff.pdf`, pp. 1 (COVID) and 4 (flu), $h = 2$, US/state rows, after-peak symbols; `multiqt_phase_rate_tradeoffs.pdf`, pp. 2–6, $h = 2$, after-peak curves. Script 02 labels κ as m.

For tuned methods after peaks, median/base-width reconstruction improves COVID loss in the US, Texas and New York but worsens California and Pennsylvania. Full-base level RMS improves flu loss in the US and Texas but worsens the other three states. *Plots:* `multiqt_peak_reasoning_tradeoff.pdf`, pp. 2–6 (COVID) and 8–12 (flu), after-peak rows across horizons/designs.

Across non-US locations, A–E give only small overall COVID gains and do not remove the pooled deterioration after peaks. No tested online method improves pooled flu loss overall. Keeping the eight most recent starts improves substantially over retaining all starts, with overall COVID gains in California and Texas but no after-peak gain there. *Plots:* `multiqt_peak_reasoning_tradeoff.pdf`, pp. 1 and 7 (all methods), pp. 3–4 (California/Texas starts); `multiqt_evaluation_tree.pdf`, pp. 1 and 7, root overall/after-peak panels. At $h = 2$, delay 0, COVID forecast pp. 10 and 17 and `multiqt_weight_timelines/covid_horizon2_delay0.pdf`, pp. 6 and 9, show the corresponding California/Texas curves and weights. The tree selects ten methods by after-peak loss; use the trade-off plot for the full comparison.

4.4 Delayed feedback and weekly re-runs

Waiting one additional week can help, but re-running is not consistently best. For context weighting D, the reductions are:

	Delay 0	Delay 1	Re-run
COVID: overall / after peak	$+.71\%/+.65\%$	$+.13\%/-2.09\%$	$-.06\%/-2.48\%$
Flu: overall / after peak	$-2.02\%/-5.79\%$	$-1.10\%/-3.94\%$	$-2.75\%/-7.88\%$

Delay 1 trades later learning for a later report, without guaranteeing better accuracy. Re-runs reconstruct tracker states from the report versions available then. Reports and tuned settings differ across designs, so these comparisons isolate neither lag nor report stability. Longer delays remain to be evaluated.

Plots: `multiqt_peak_reasoning_tradeoff.pdf`, pp. 1 and 7, D (blended), three feedback blocks; `multiqt_evaluation_tree.pdf`, p. 1, COVID D in the delay-0, delay-1 and re-run branches. Flu D is outside the ten methods on tree p. 7. For fixed rates, see `multiqt_phase_rate_tradeoffs.pdf`, pp. 1, 7 and 13, $h = 2$.

4.5 Do quantile-specific expert weights help?

For rate-grid mixtures, common weights lower pooled overall and after-peak loss in every disease–feedback configuration, with or without sharing and with an outer correction. Overall reductions are:

Rate-grid mixture	COVID: per-level $\to$ common	Flu: per-level $\to$ common
No sharing	$-5.00 \to -3.19$	$-2.83 \to -1.23$
Sharing	$-11.42 \to -8.59$	$-11.89 \to -8.89$
Sharing + outer	$-11.74 \to -8.91$	$-12.17 \to -9.16$

The selected H and outer rate match within pairs. Before the flu peak, per-quantile weights without sharing give a small advantage at delay 0 and under re-run. Lower loss also need not mean lower CE. *Plots:* `multiqt_peak_reasoning_tradeoff.pdf`, pp. 1 and 7, matched methods in all three windows; COVID forecast p. 2 and weight p. 1 show US curves and weights at $h = 2$, delay 0.

A–E use common weights only. Learned weekly starts have no matched common-weight control, so their results do not establish a benefit from per-quantile weighting. Equal weighting of all or recent starts changes the weighting rule too. *Plots:* COVID weight p. 2 (A–E) and p. 3 (starts), with forecast/loss pp. 4 and 3.

4.6 Base forecast paths and the three indicators

This diagnostic was updated on 11 November 2026; it does not rerun the method comparisons above. `multiqt_base_forecast_paths.pdf`, pp. 1–2, is loaded immediately below: COVID at page 25, flu at page 26. Each location has a full latest-count timeline, all peaks retained by script 01, and blue median curves for horizons $-1, \ldots, 3$. Curves use reference weeks $-7, \ldots, 8$ around each peak and are trimmed to the same target-date windows. The three rows underneath show the earlier indicators at their reference dates, using information available by the corresponding cutoff. Peaks and latest counts are retrospective; neither enters these indicators.

How the indicators are computed. Use $Z(t, h; v)$ from the target-signal definition in (65), and threshold $\varepsilon = \log(1.05)$. Here date differences are calendar days: $t - 7$ means exactly one week earlier, not the previous retained round. Blue cells mean indicator 1, white 0 and gray unavailable. Each indicator is computed separately.

1. **Path: a decline within the current forecast curve.** The indicator is

$$I_t^{\text{path}} = \mathbf{1}\left\{ \max_{\substack{h=-1,\ldots,2 \\ Z_{t,h},\, Z_{t,h+1} \text{ available}}} (Z_{t,h} - Z_{t,h+1}) > \varepsilon \right\}.$$

   Horizons $0, 1, 2, 3$ must be available; include the -1 to 0 comparison only when the nowcast exists. This detects any sufficiently large adjacent decline, rather than comparing one target with all earlier horizons.

2. **Nowcast: a decline or slowing positive growth.** Set $z_0 = Z(t, -1; \iota_t)$, $z_1 = Z(t - 7, -1; \iota_{t-7})$ and $z_2 = Z(t - 14, -1; \iota_{t-7})$, and let $g = z_0 - z_1$ and $g_{\text{prev}} = z_1 - z_2$. Then

$$I_t^{\text{nowcast}} = \mathbf{1}\{g < -\varepsilon \text{ or } (g_{\text{prev}} > \varepsilon \text{ and } g_{\text{prev}} - g > \varepsilon)\}.$$

   Thus slowing growth can trigger the indicator before the nowcast falls. It requires three consecutive reference weeks with usable nowcasts.

3. **Revision: downward forecast revisions that become more negative.** For older horizons $j = 0, \ldots, 3$, compare each nowcast with forecasts for the *same target week*:

$$r_j = z_0 - Z(t - 7(j + 1), j; \iota_t),$$
$$r_j^{\text{prev}} = z_1 - Z(t - 7(j + 2), j; \iota_{t-7}).$$

   Let J contain horizons with both comparisons available; require $|J| \geq 2$. The indicator is

$$I_t^{\text{revision}} = \mathbf{1}\Big\{ \text{median}_{j \in J}\, r_j < -\varepsilon \text{ and } \text{median}_{j \in J}(r_j - r_j^{\text{prev}}) < -\varepsilon \Big\}.$$

   The prior comparison uses its prior cutoff. These are revisions between *forecasts*, not revisions of reported hospital counts.

The logarithm is unnecessary for the current zero-threshold binary $c^{\mathrm{ctx}}_{t,h}$, but it defines the relative-change scale of these three indicators and their nonzero threshold. They are diagnostic warnings, not probabilities of having passed a true peak; the current expert methods are unchanged.

Why indicators are unavailable. A required median must exist at the exact reference date and horizon, be finite and nonnegative, be available by the cutoff and not be filled from an earlier forecast. Flu has no submitted forecasts for long summer intervals, although reported counts continue. Missing or removed rounds also interrupt the history: the nowcast indicator needs three consecutive nowcasts, and revision needs at least two matched older-horizon comparisons in both the current and prior round. Consequently, these two indicators can remain gray after forecasts resume, while Path is already available. For example, the unusable COVID reference weeks 25 January and 1 February 2025 leave Nowcast unavailable through 15 February and Revision through 22 February. Insufficient history at the start also gives gray cells. At the right edge, longer horizons can be absent from the prepared data because their finalized outcomes are not yet available; the indicators themselves do not use these outcomes. A gray cell does not mean indicator 0.

COVID | Base forecast paths and three indicators

cmu_hub | counts per 100k residents | All peaks from script 01 | Indicators: score > log(1.05)

Blue curves: one reference week, horizons -1 to 3; darker = later within each peak window. Open point: h = -1.

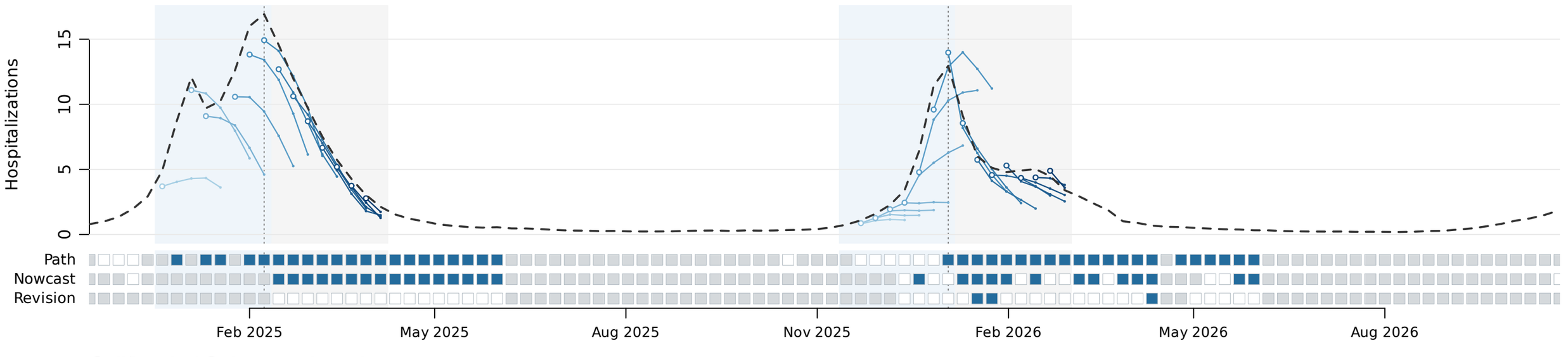
Timeline x-axis: target date. Indicator x-axis: reference date, using its cutoff. Same calendar scale; gray cells mean unavailable, not zero.
Blue curves use reference weeks -7 to 8 and are clipped to the same target-date windows. Counts and peak dates are retrospective.
Path: fall across horizons. Nowcast: fall or slowing growth. Revision: downward same-target revisions that become more negative.

FLU | Base forecast paths and three indicators

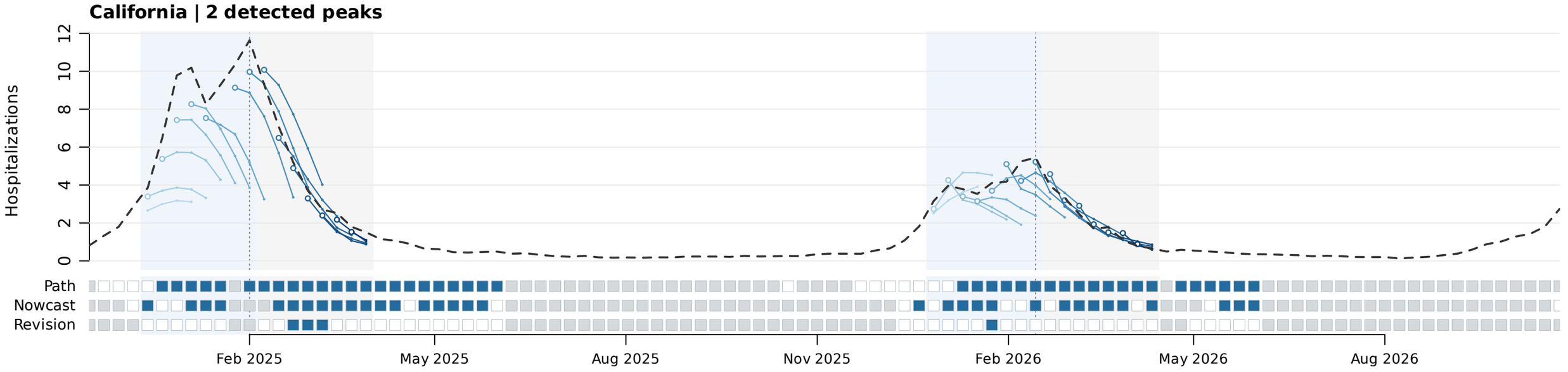
cmu_hub | counts per 100k residents | All peaks from script 01 | Indicators: score > log(1.05)

Blue curves: one reference week, horizons -1 to 3; darker = later within each peak window. Open point: h = -1.

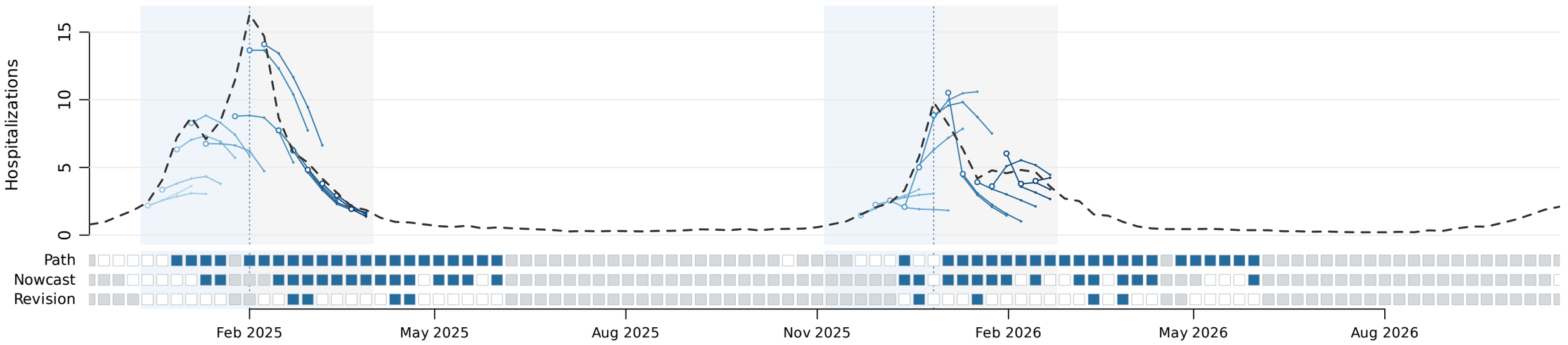
United States | 2 detected peaks



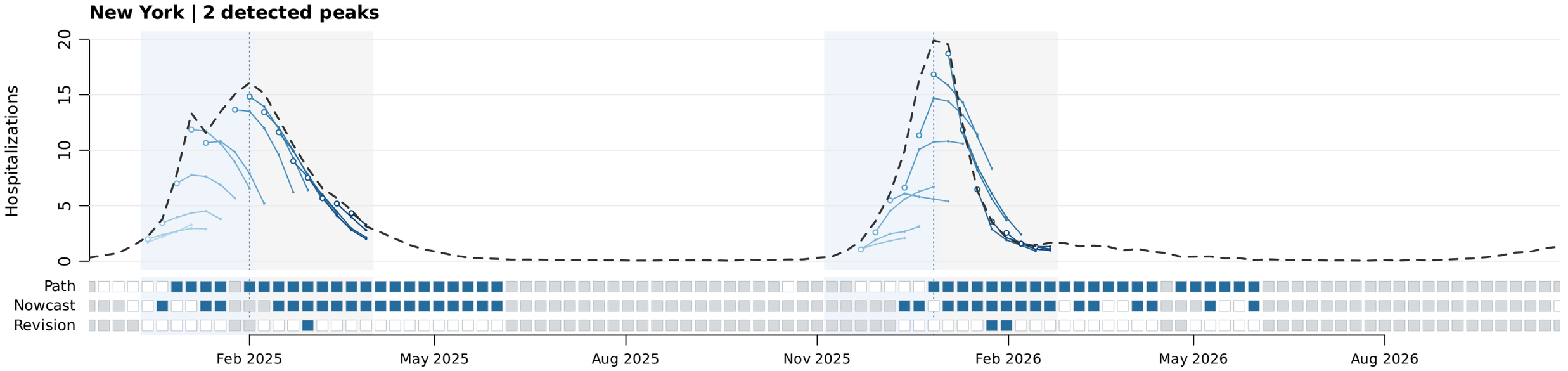
California | 2 detected peaks



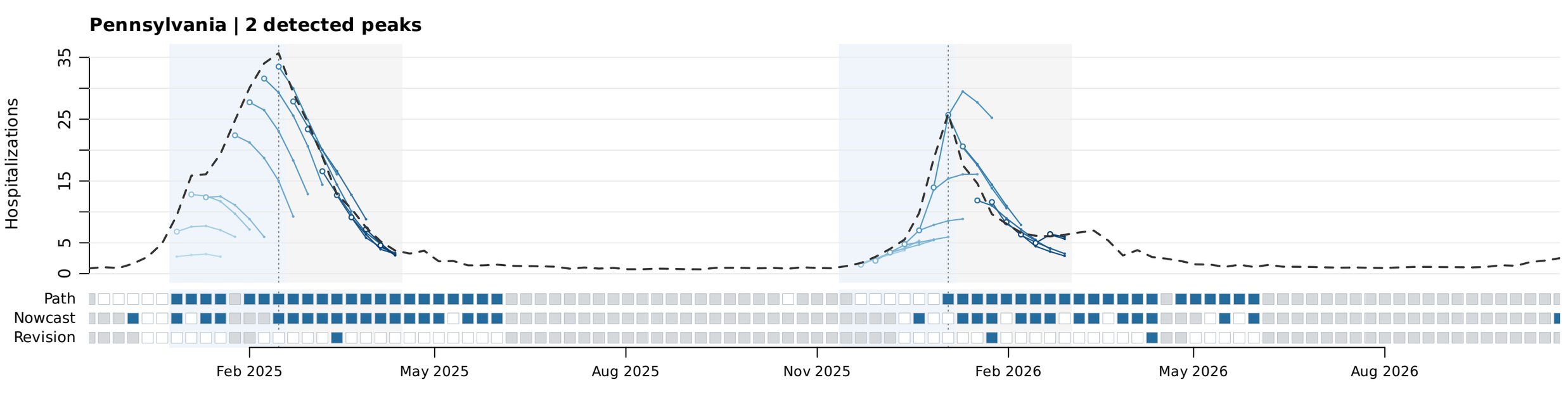
Texas | 2 detected peaks



New York | 2 detected peaks



Pennsylvania | 2 detected peaks



Timeline x-axis: target date. Indicator x-axis: reference date, using its cutoff. Same calendar scale; gray cells mean unavailable, not zero.
Blue curves use reference weeks -7 to 8 and are clipped to the same target-date windows. Counts and peak dates are retrospective.
Path: fall across horizons. Nowcast: fall or slowing growth. Revision: downward same-target revisions that become more negative.

Reading the plot references. Except for the updated forecast-path diagnostic above, filenames and page numbers refer to the earlier `output/pdf/` outputs, preserved under `agent/outputs/notes/plots_history_20261111/` with a source manifest. Forecast and weight files use the `multiqt_method_forecasts/` and `multiqt_weight_timelines/` subdirectories. The all-location trade-off markers are unweighted location means; tables and trees use week-weighted pooling. Check horizon/design panels as well as pooled results.

Loaded figures.

Original source PDF	Source pages	Notes start
`multiqt_base_forecast_paths.pdf` (updated)	1–2	25
`multiqt_error_tradeoff.pdf`	1, 4	30
`multiqt_phase_rate_tradeoffs.pdf`	1–7, 13	32
`multiqt_season_forecasts.pdf`	1	40
`multiqt_by_level.pdf`	4, 12, 20, 27, 35, 43	41
`multiqt_tuning.pdf`	1–6	47
`multiqt_peak_reasoning_tradeoff.pdf`	1–12	53
`multiqt_evaluation_tree.pdf`	1–12	65
`multiqt_method_forecasts.pdf`	2, 4	77
COVID $h = 2$, delay 0: studentized curves	6, 7	79
Flu $h = 2$, delay 0: studentized curves	6, 7, 13, 14	81
`multiqt_weight_timelines.pdf`	1, 2	85
`multiqt_weight_timelines.pdf`	3, 6, 9	87
`multiqt_method_forecasts.pdf`	3, 10, 17	90
Flu $h = 2$, delay 0: weights	3, 6, 9	93
Flu $h = 2$, delay 0: forecasts	3, 10, 17	96
`multiqt_soft_switch_timelines.pdf`	1	99

The singleton forecast, weight and switch PDFs are COVID, $h = 2$, delay 0; flu uses `flu_horizon2_delay0.pdf` in the matching subdirectory. Tuning pages show COVID delay 0, delay 1 and re-run, then the same flu designs, under the earlier procedure. The season and by-level pages are exact extracts from `notes_before_update.pdf` in the archive. These frozen figures are not replaced by new workflow outputs; updating the performance comparisons requires replacing scores, figures and page references together. The updated forecast-path figure is stored separately in `agent/outputs/notes/plots_diagnostics_20261111/`. The figures displayed here use copies in `agent/outputs/notes/plots_display_20261111/`; only captions are edited, and the archived figures and plotted data are preserved.

References

[1] Tiffany Ding, Isaac Gibbs, and Ryan J. Tibshirani. *Calibrated Multi-Level Quantile Forecasting.* arXiv:2512.23671v1, 2025. MultiQT paper.

[2] Anastasios N. Angelopoulos, Emmanuel J. Candès, and Ryan J. Tibshirani. *Conformal PID Control for Time Series Prediction.* Advances in Neural Information Processing Systems 36, 2023. NeurIPS paper.

[3] CMU Delphi. *Delphi hub forecasting: calibration experiments and tracker code.* Pinned source revision `9abebcfb`, inspected 9–10 November 2026. E09: constant learning rate and revision delay; E15: median shift and width change; tracker implementation.

[4] Tilmann Gneiting, Adrian E. Raftery, Anton H. Westveld III, and Tom Goldman. *Calibrated Probabilistic Forecasting Using Ensemble Model Output Statistics and Minimum CRPS Estimation*. Monthly Weather Review 133(5):1098–1118, 2005. Paper.

[5] Isaac Gibbs and Emmanuel J. Candès. *Conformal Inference for Online Prediction with Arbitrary Distribution Shifts*. Journal of Machine Learning Research 25(162):1–36, 2024. Paper and implementation.

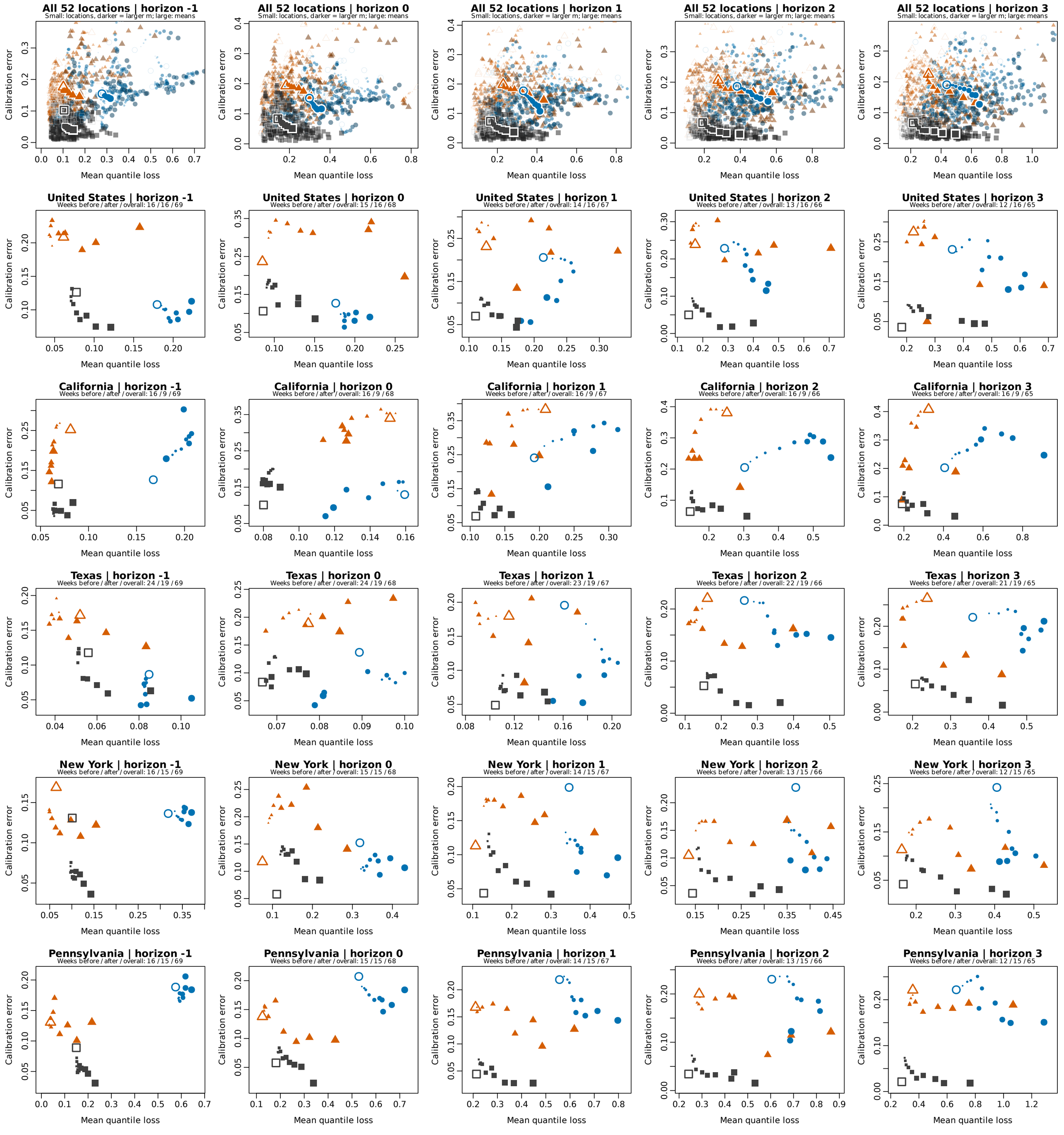
[6] Jonathan Berrisch and Florian Ziel. *CRPS Learning*. Journal of Econometrics 237, article 105221, 2023. Journal article; Author manuscript.

[7] Project experiment records, October 2026. Earlier rate-bank diagnosis: `agent/outputs/postpeak_diagnosis/REPORT.md`. Experiments and plan of 28 October: `agent/outputs/peak_antiwindup/README.md` and `agent/docs/peak_correction_todos.md`. Diverse-bank experiment of 30 October: `agent/outputs/expert_bank_comparison/README.md`, with saved forecasts and weights in `forecasts.rds` in the same directory. These are historical exploratory records, not the current evaluation protocol.

A Loaded plot pages

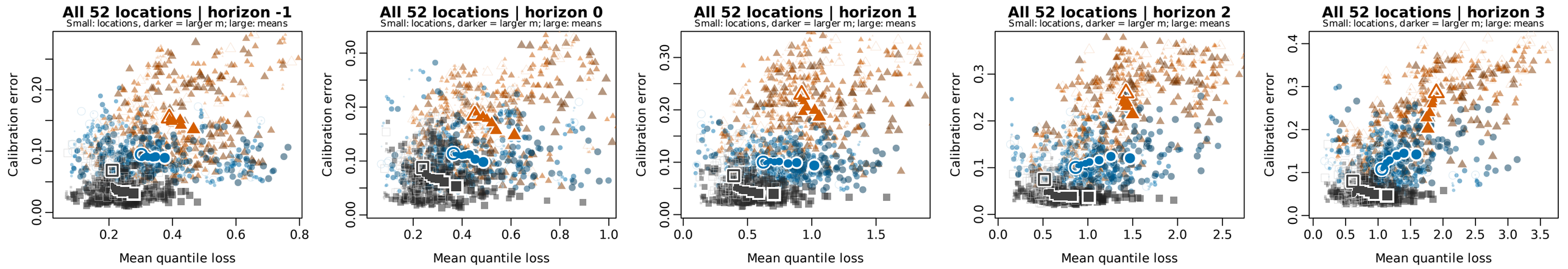
COVID | delay 0 | Fixed-rate MultiQT: calibration error vs mean quantile loss, mean over quantile levels

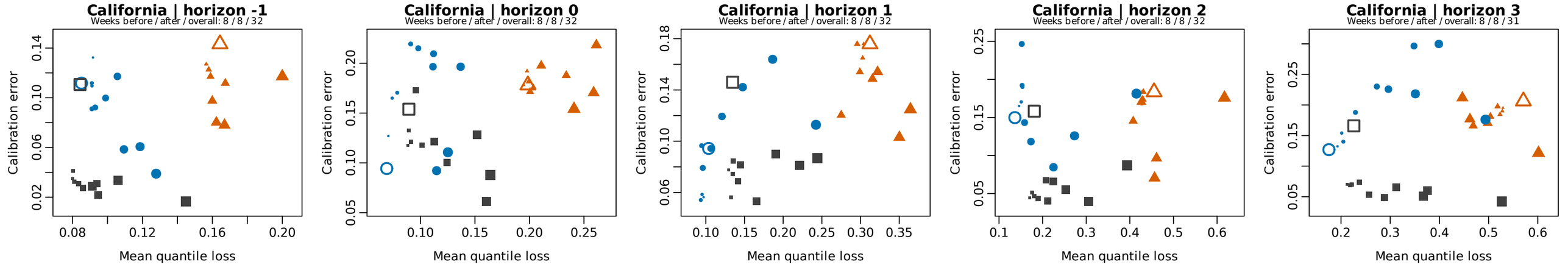
Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 | Scored against latest reported counts in counts per 100k residents | Lower left is better.

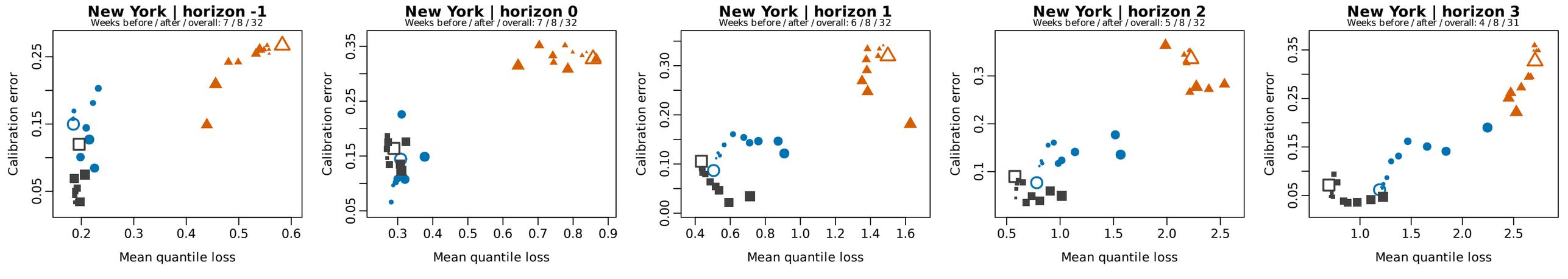


FLU | delay 0 | Fixed-rate MultiQT: calibration error vs mean quantile loss, mean over quantile levels

Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 | Scored against latest reported counts in counts per 100k residents | Lower left is better.



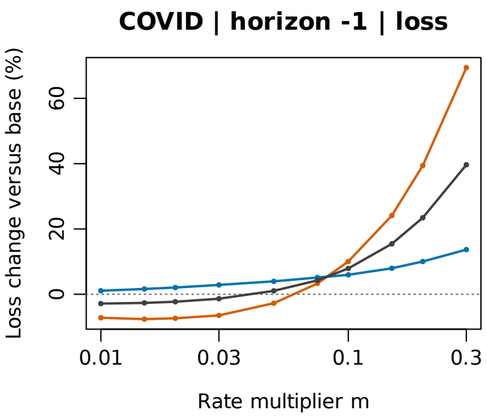


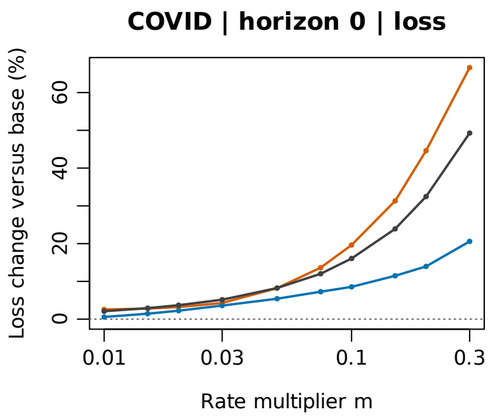


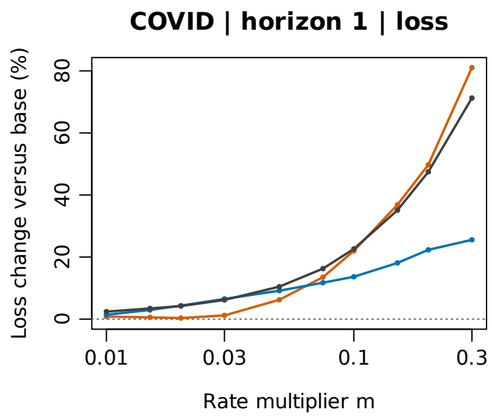
Before peak (weeks -7 to 0)
After peak (weeks 1 to 8)
Overall (Jun 2025 to Sep 2026)
Open symbol: m = 0 (original forecasts)
m = 0.01
m = 0.03
m = 0.1
m = 0.3

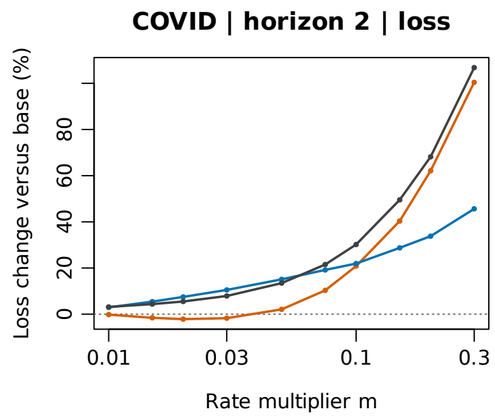
All 52 locations pooled (multipliers of the dots) | delay 0 | Fixed-rate MultiQT: loss and calibration before and after the peak

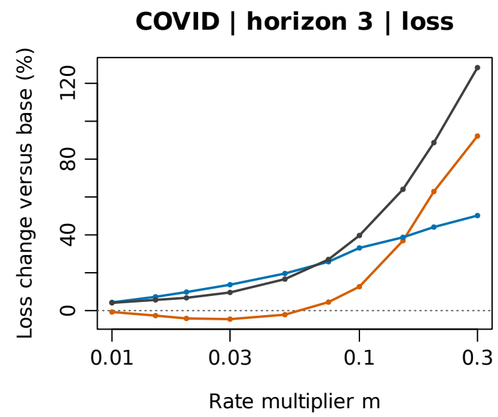
Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 | Scored against latest reported counts in counts per 100k residents

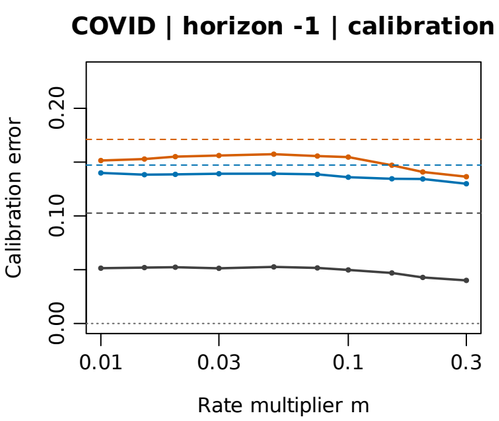


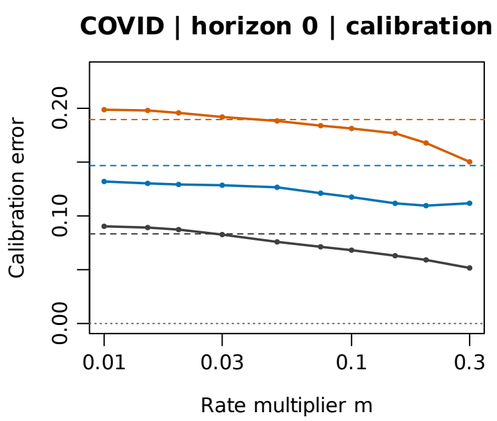


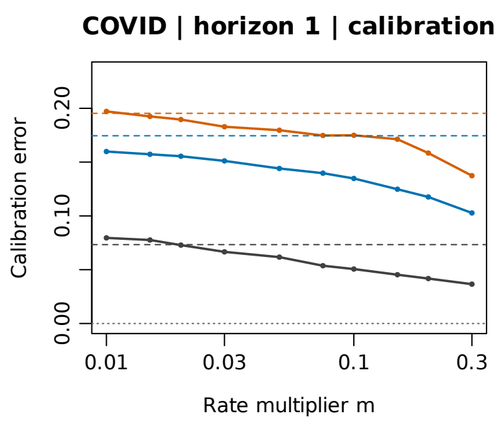


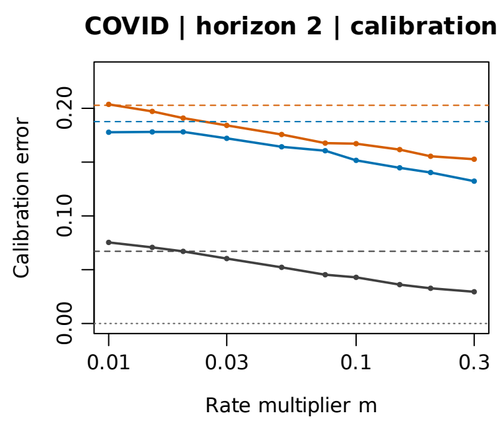


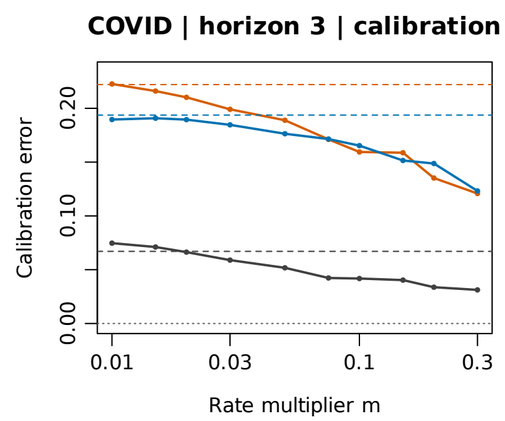


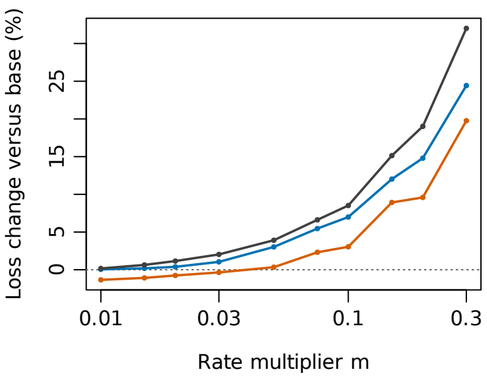


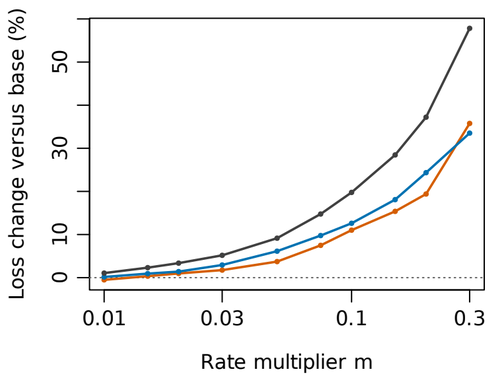


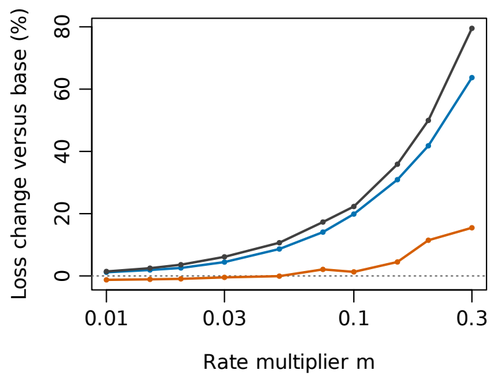


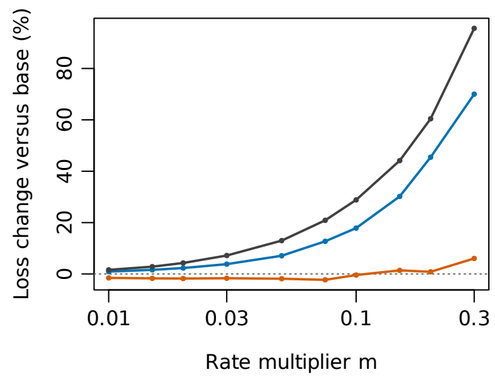


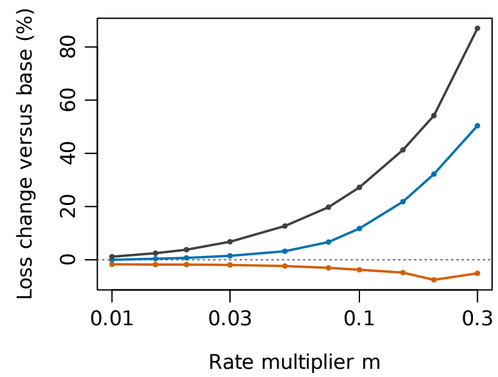


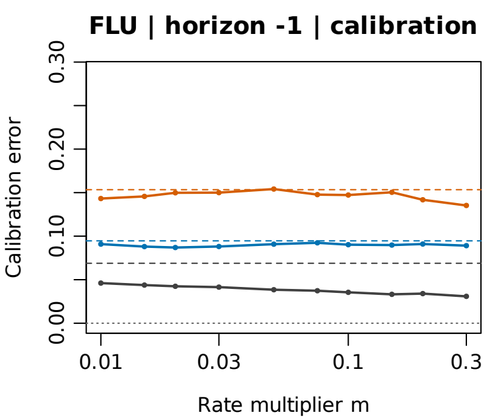


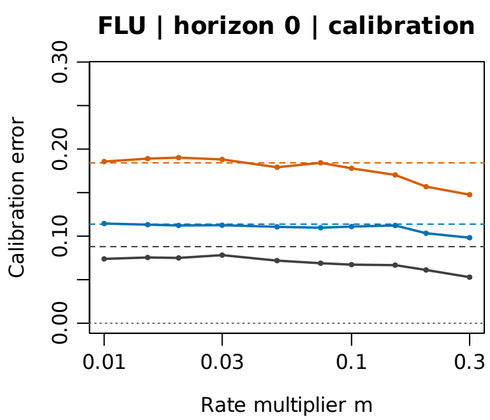


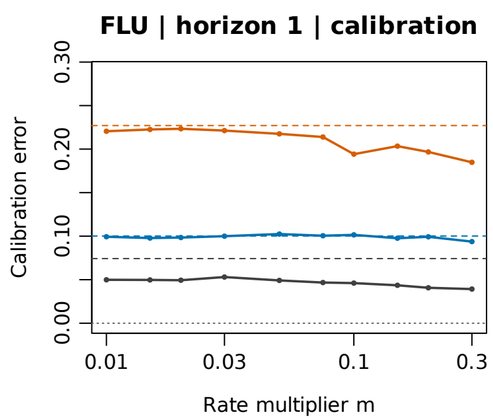


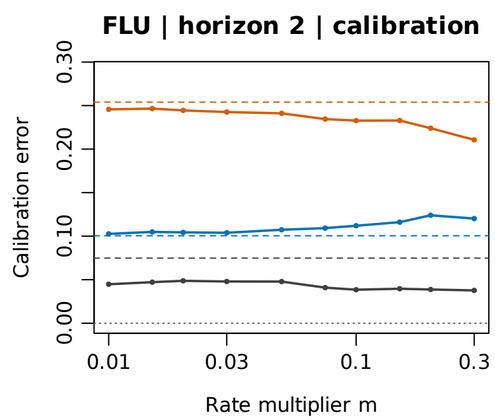


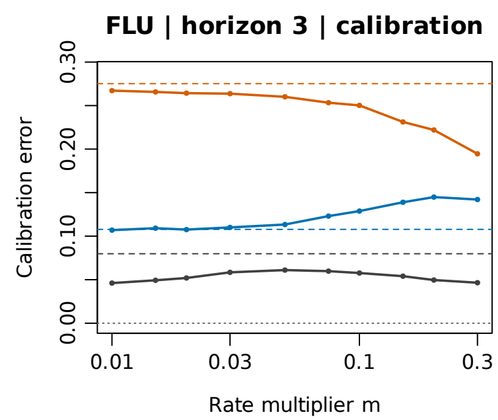










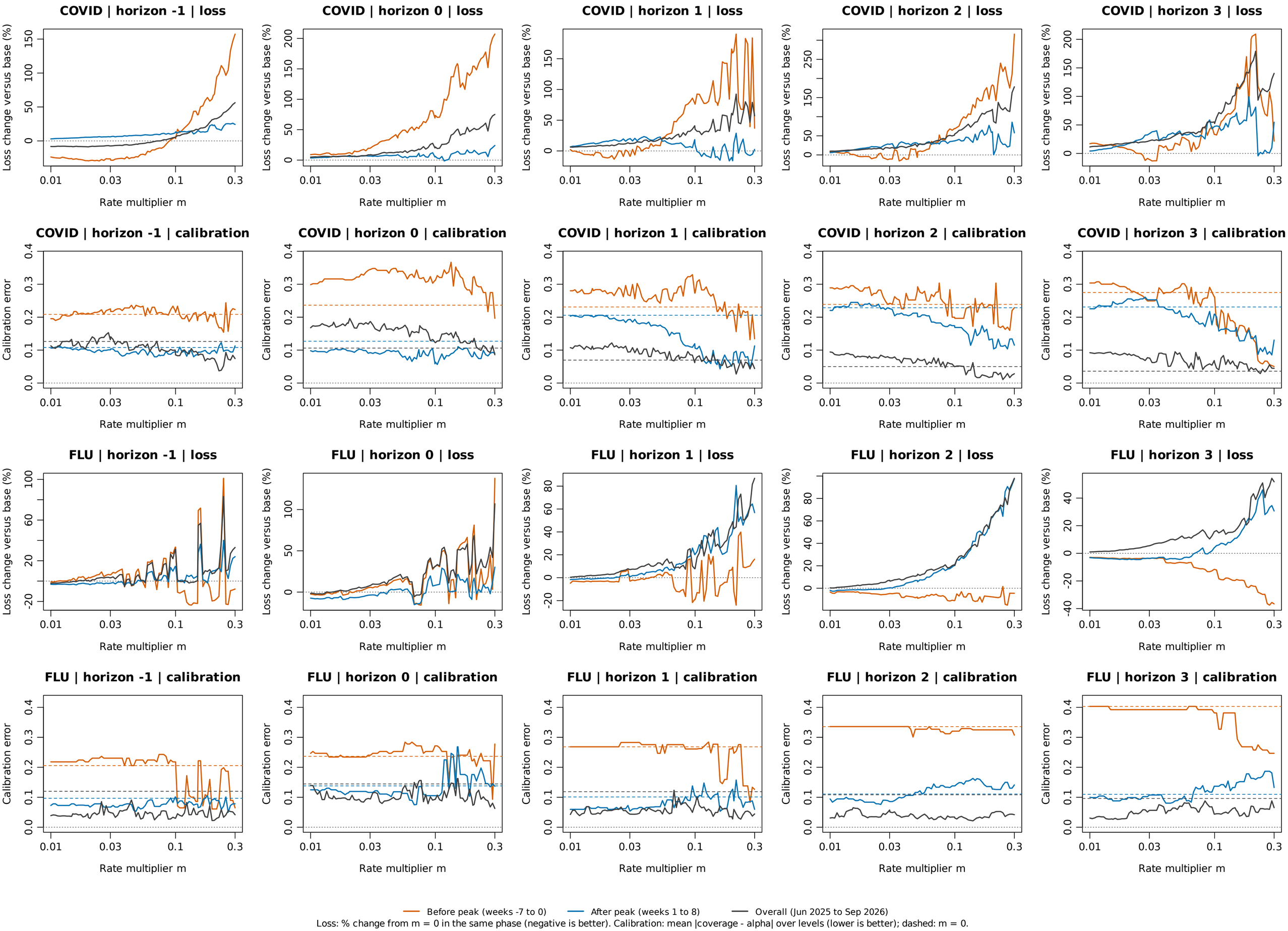


Before peak (weeks -7 to 0) After peak (weeks 1 to 8) Overall (Jun 2025 to Sep 2026)

Loss: % change from m = 0 in the same phase (negative is better). Calibration: mean |coverage - alpha| over levels (lower is better); dashed: m = 0. Pooled: each location weighted by its weeks.

United States | delay 0 | Fixed-rate MultiQT: loss and calibration before and after the peak

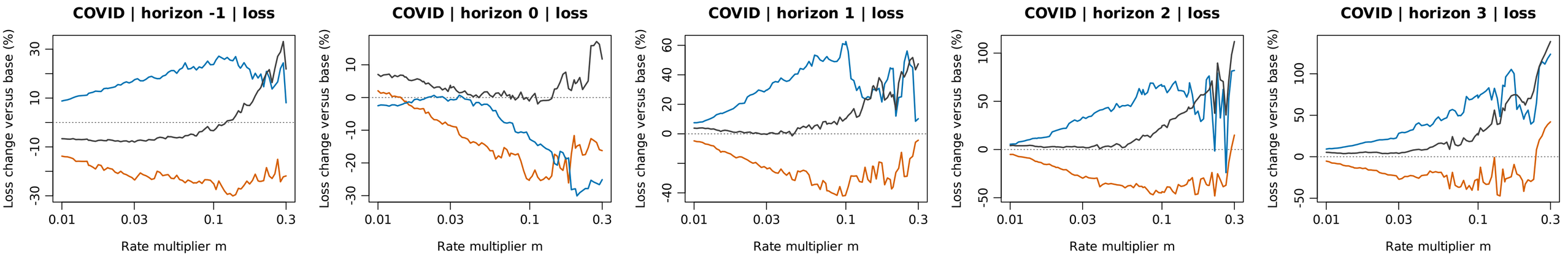
Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 | Scored against latest reported counts in counts per 100k residents

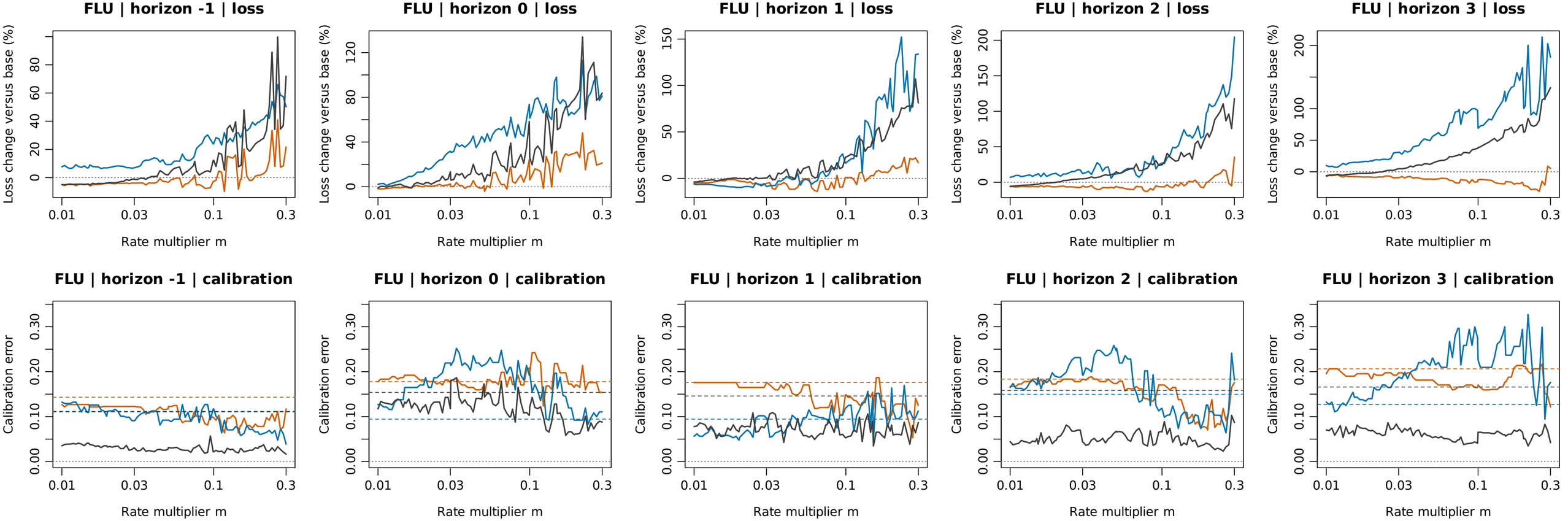


Loss: % change from m = 0 in the same phase (negative is better). Calibration: mean |coverage - alpha| over levels (lower is better); dashed: m = 0.

California | delay 0 | Fixed-rate MultiQT: loss and calibration before and after the peak

Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 | Scored against latest reported counts in counts per 100k residents



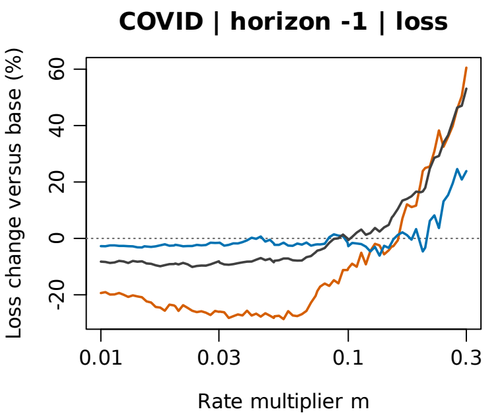


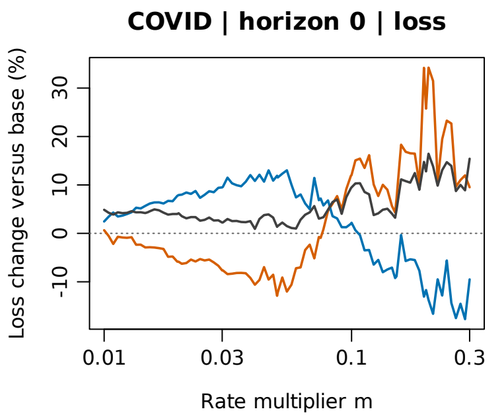
Before peak (weeks -7 to 0) | After peak (weeks 1 to 8) | Overall (Jun 2025 to Sep 2026)

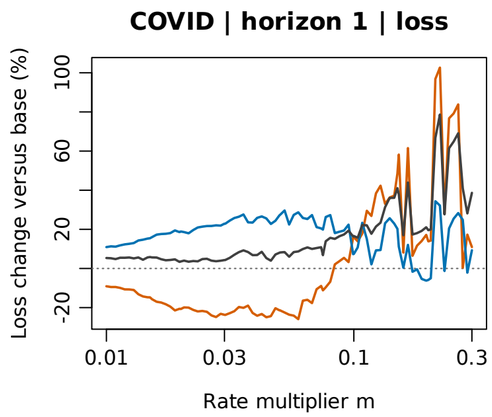
Loss: % change from m = 0 in the same phase (negative is better). Calibration: mean |coverage - alpha| over levels (lower is better); dashed: m = 0.

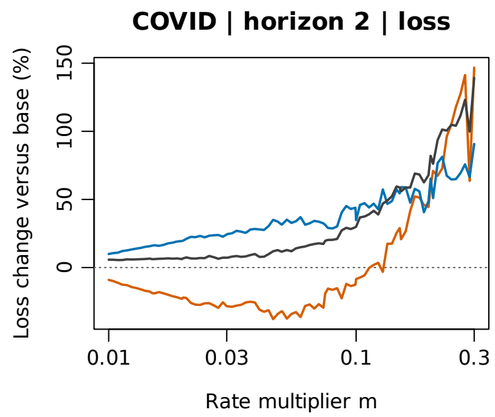
Texas | delay 0 | Fixed-rate MultiQT: loss and calibration before and after the peak

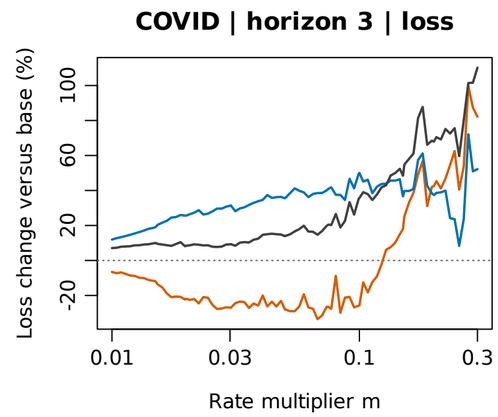
Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 | Scored against latest reported counts in counts per 100k residents

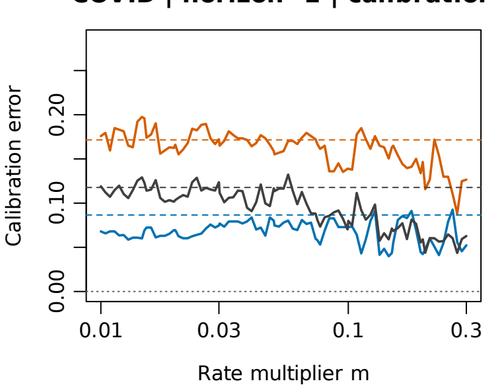


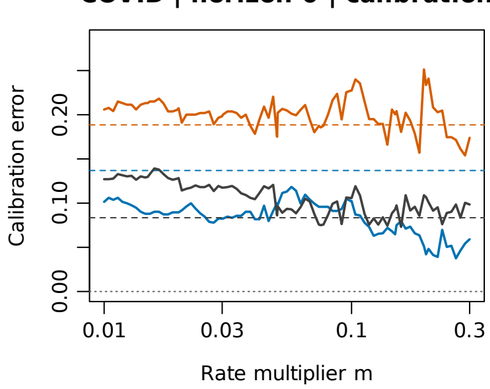


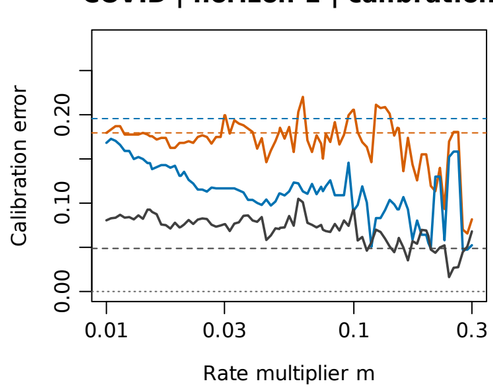


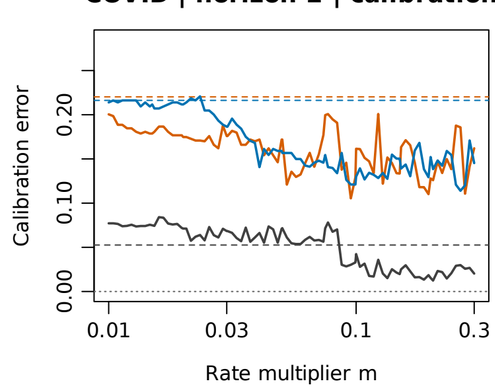


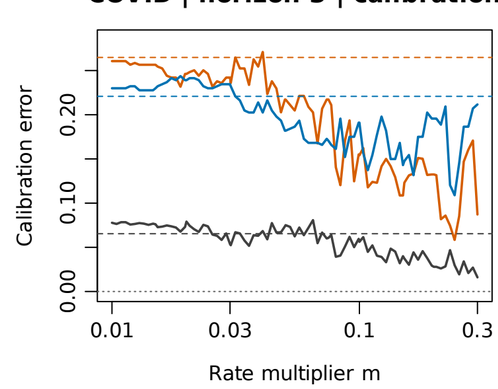


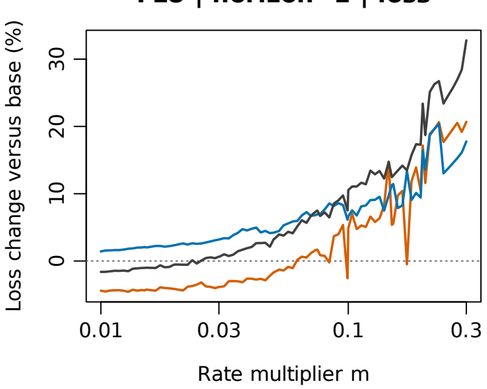


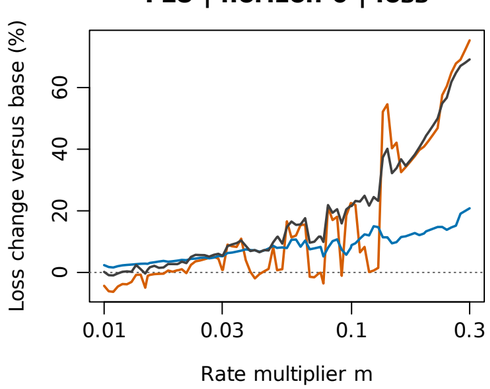


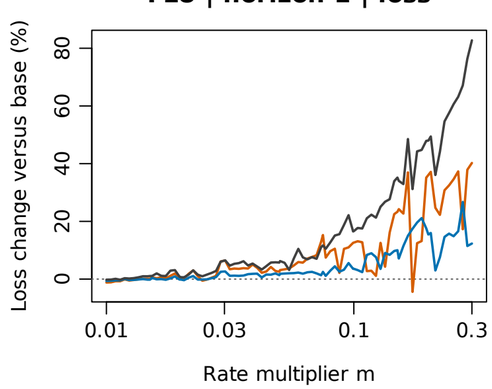


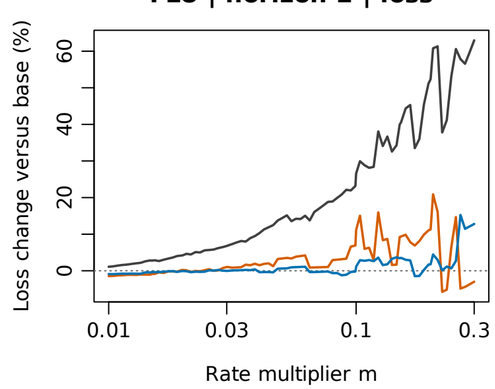


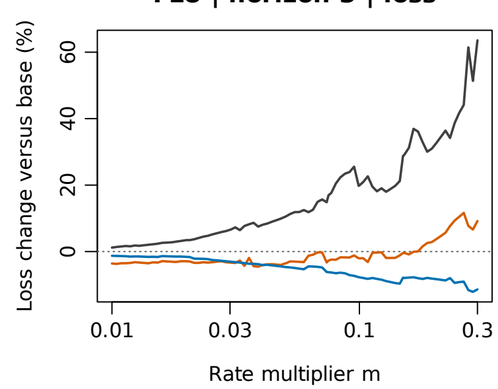


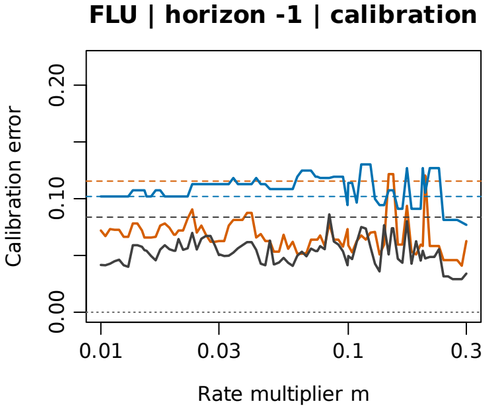


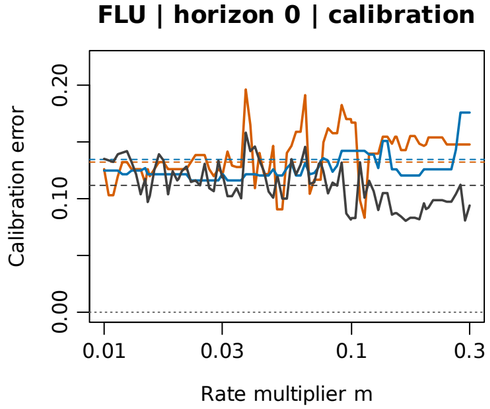


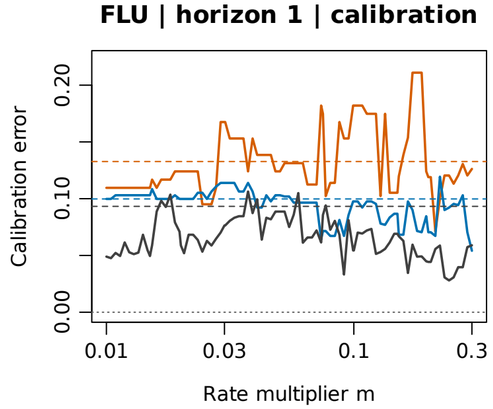


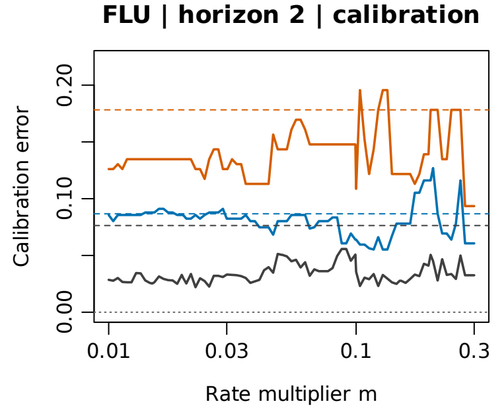


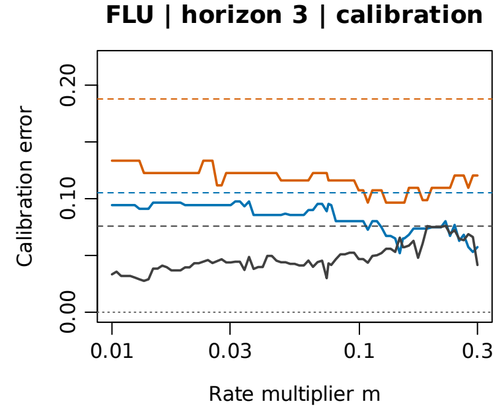










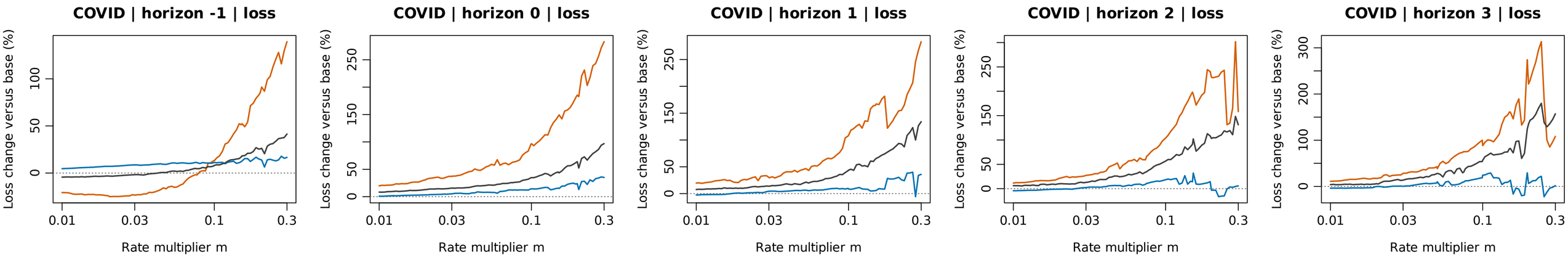


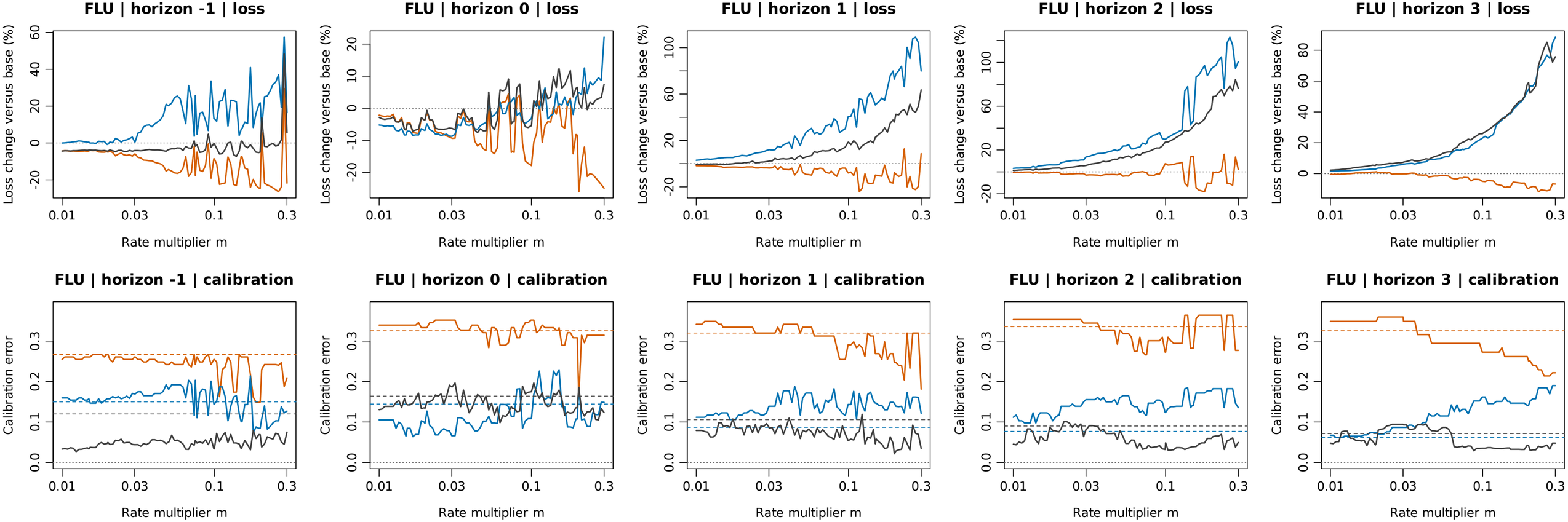
Before peak (weeks -7 to 0) | After peak (weeks 1 to 8) | Overall (Jun 2025 to Sep 2026)

Loss: % change from m = 0 in the same phase (negative is better). Calibration: mean |coverage - alpha| over levels (lower is better); dashed: m = 0.

New York | delay 0 | Fixed-rate MultiQT: loss and calibration before and after the peak

Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 | Scored against latest reported counts in counts per 100k residents



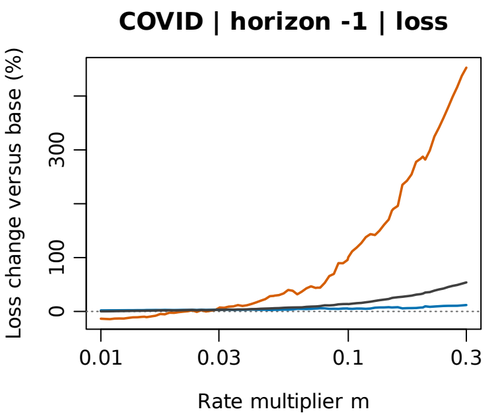


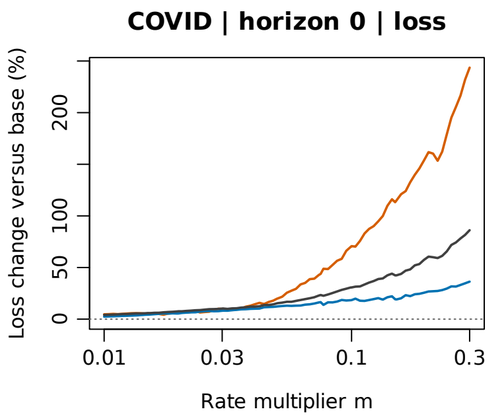
Before peak (weeks -7 to 0) | After peak (weeks 1 to 8) | Overall (Jun 2025 to Sep 2026)

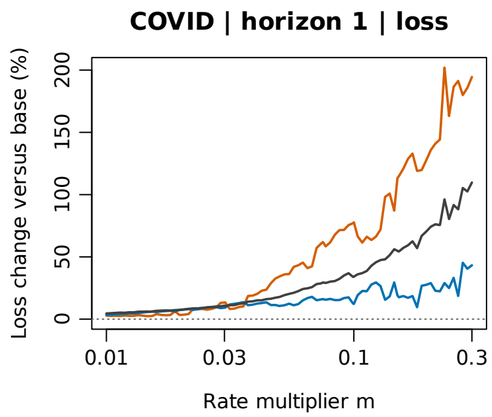
Loss: % change from m = 0 in the same phase (negative is better). Calibration: mean |coverage - alpha| over levels (lower is better); dashed: m = 0.

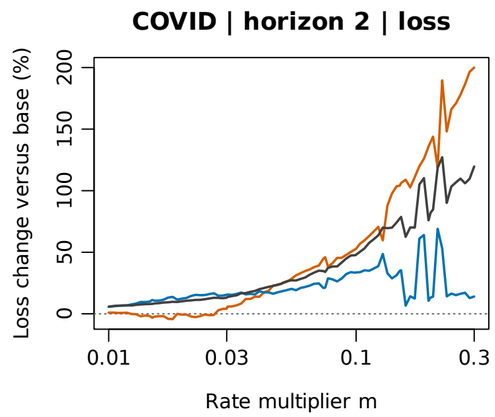
Pennsylvania | delay 0 | Fixed-rate MultiQT: loss and calibration before and after the peak

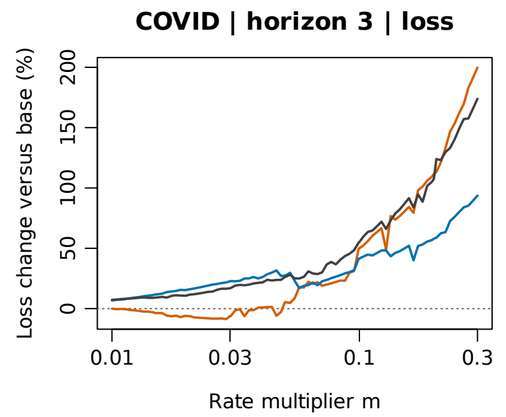
Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 | Scored against latest reported counts in counts per 100k residents

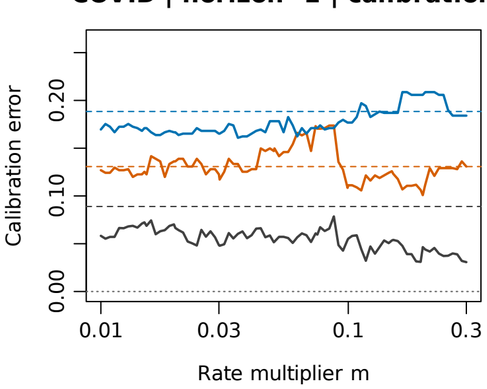


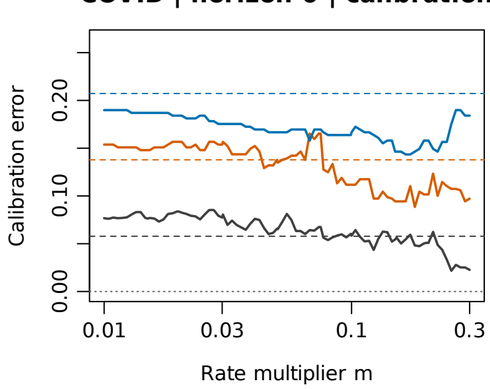


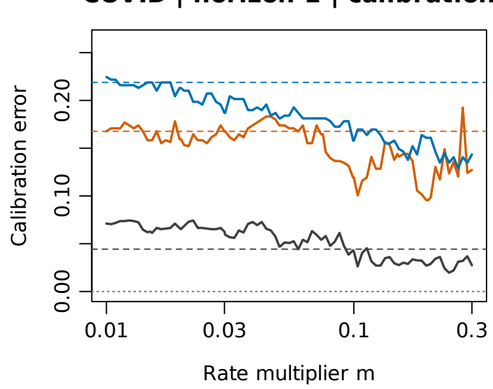


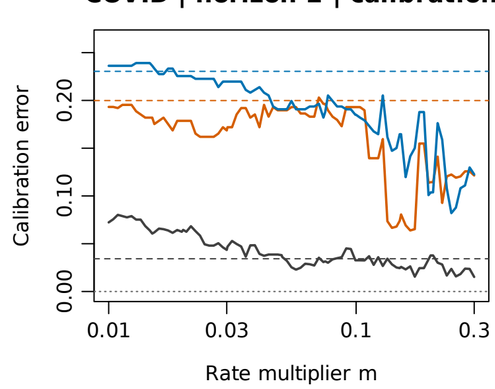


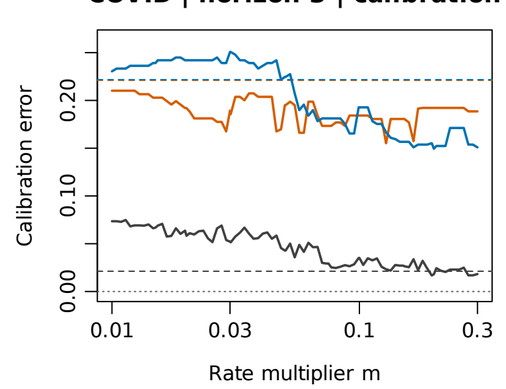


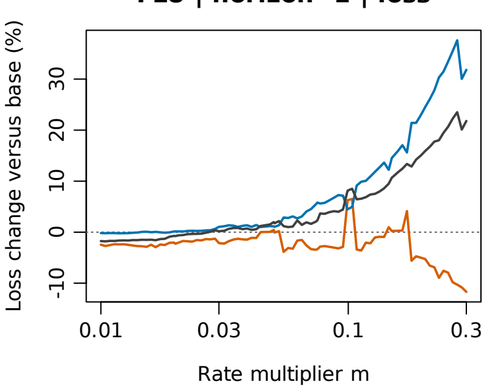


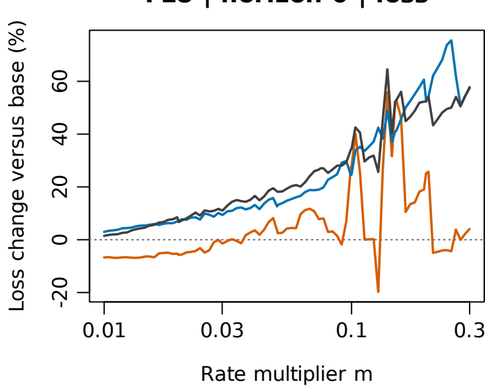


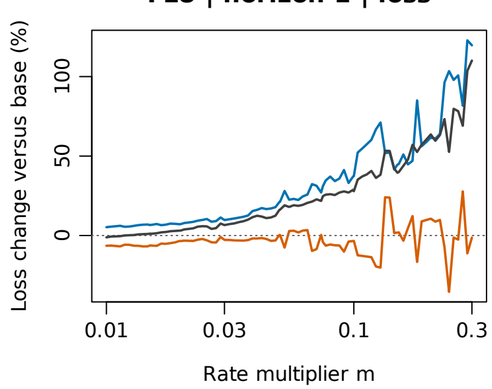


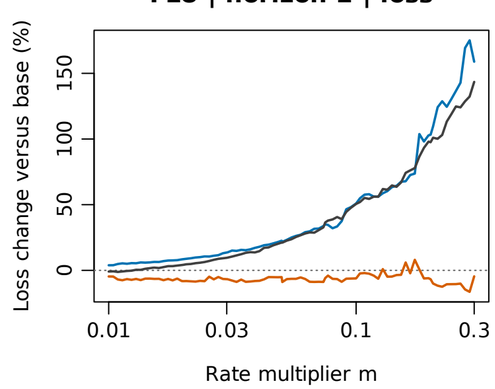


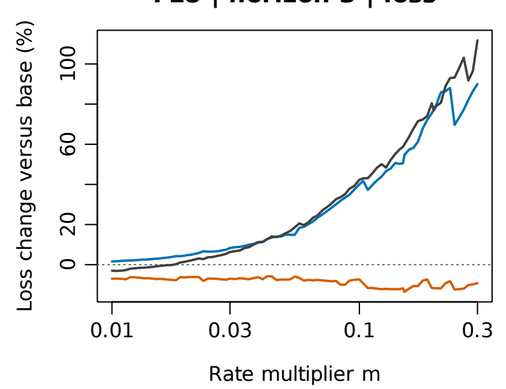


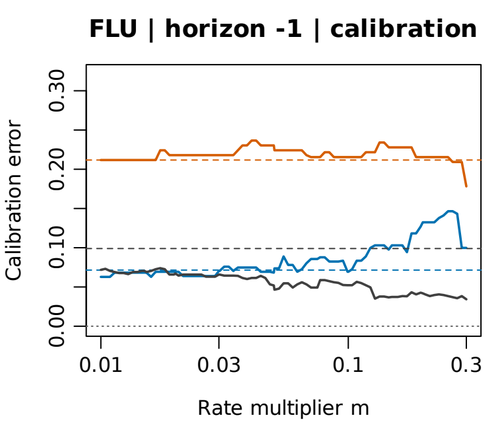


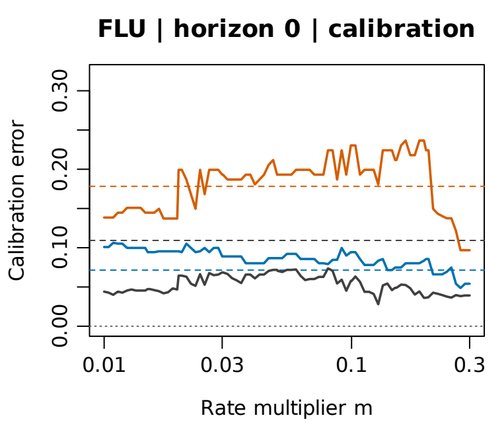


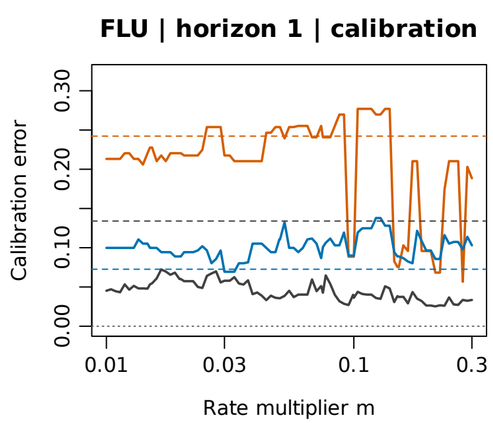


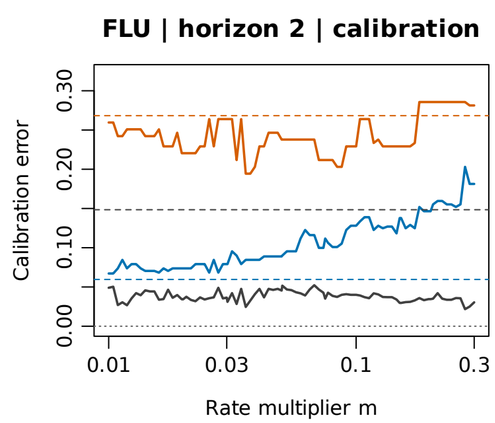


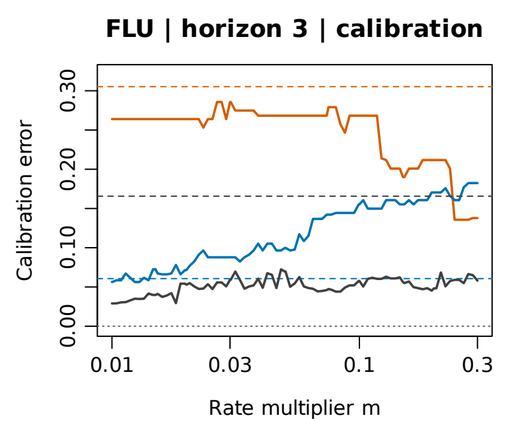










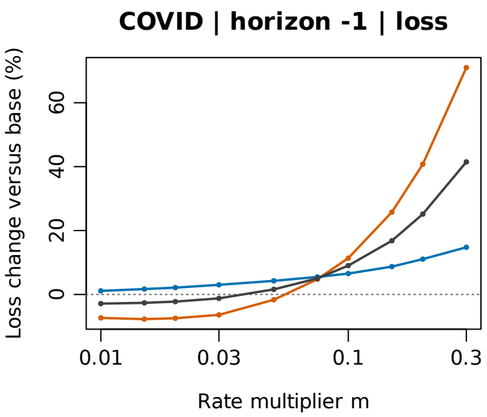


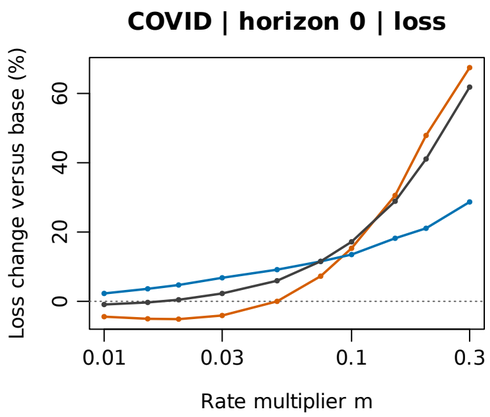
Before peak (weeks -7 to 0) | After peak (weeks 1 to 8) | Overall (Jun 2025 to Sep 2026)

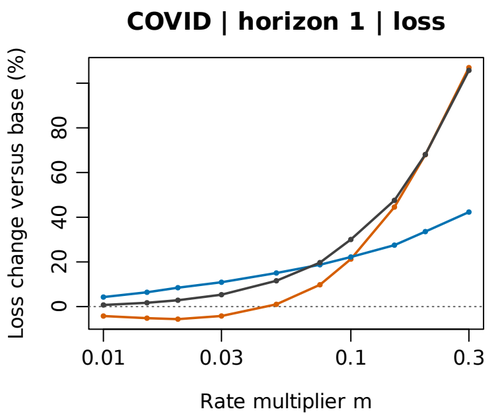
Loss: % change from m = 0 in the same phase (negative is better). Calibration: mean |coverage - alpha| over levels (lower is better); dashed: m = 0.

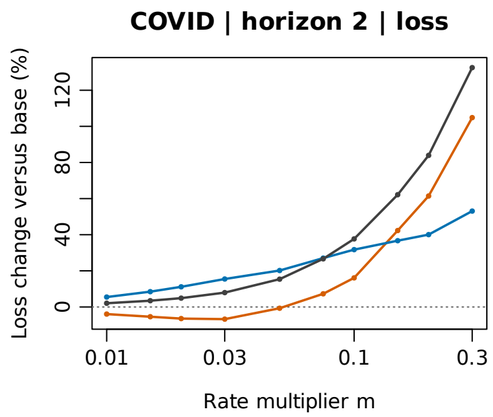
All 52 locations pooled (multipliers of the dots) | delay 1 | Fixed-rate MultiQT: loss and calibration before and after the peak

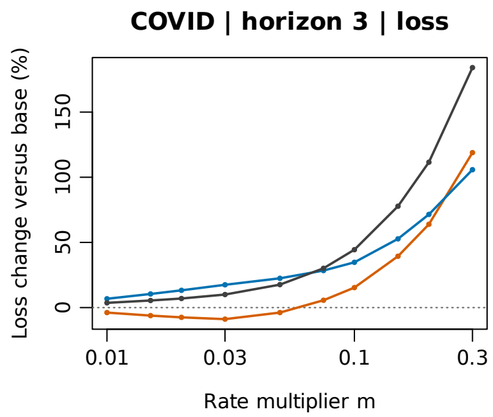
Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 1 | Scored against latest reported counts in counts per 100k residents

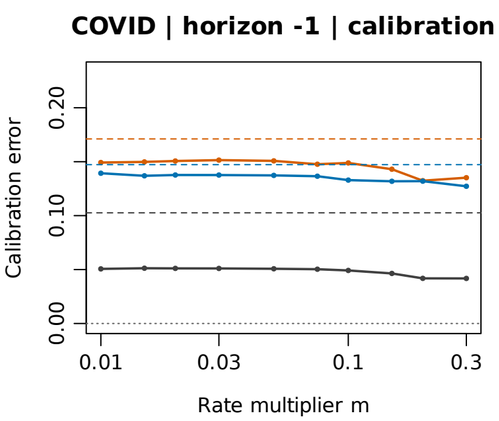


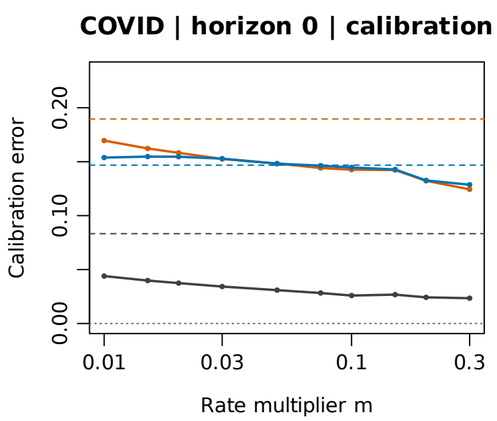


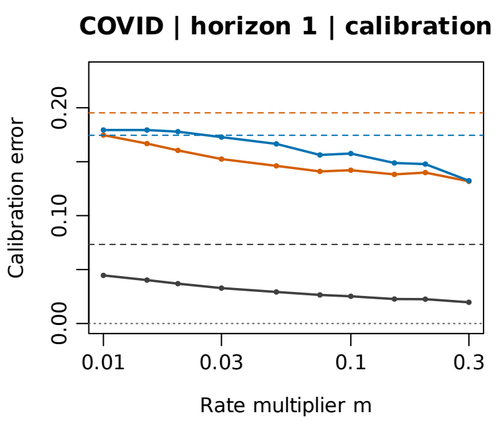


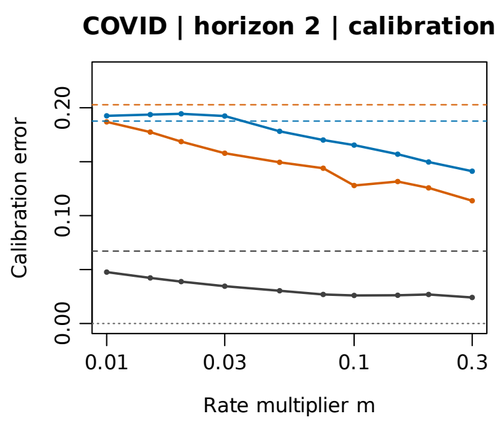


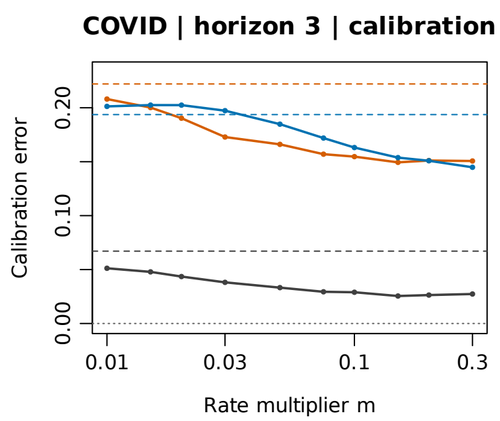


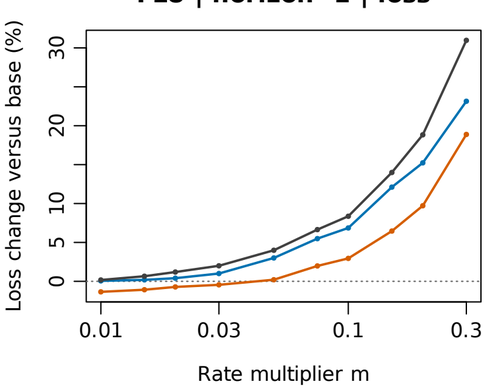


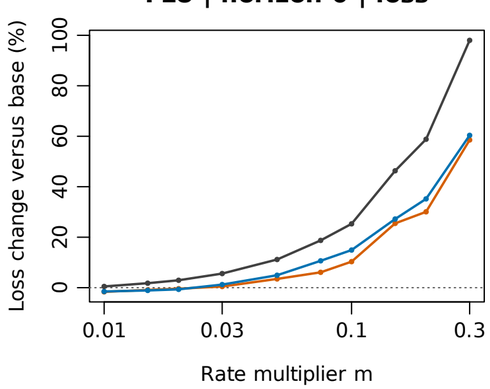


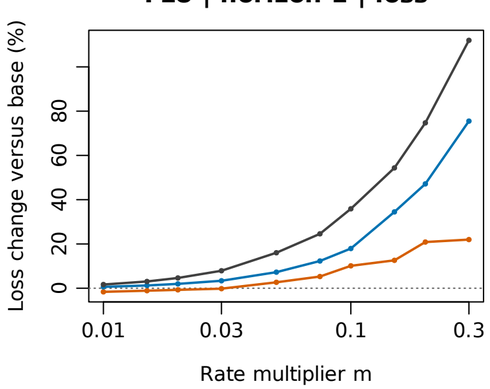


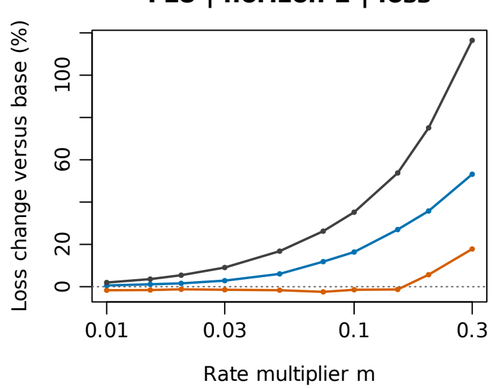


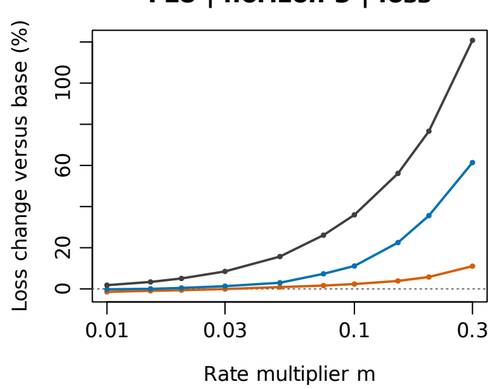


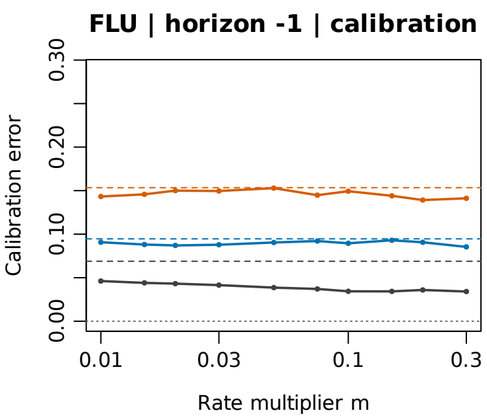


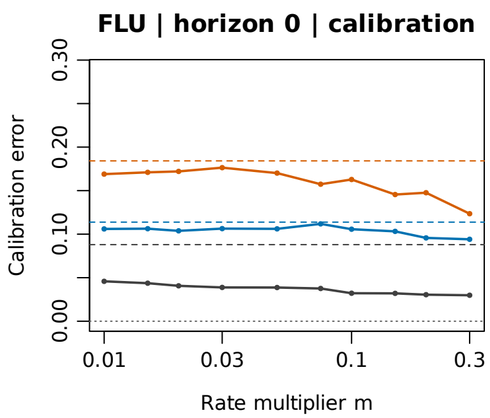


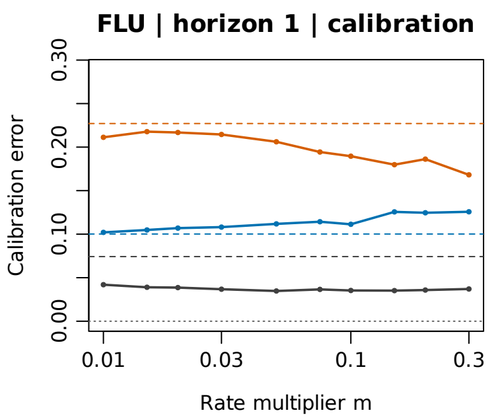


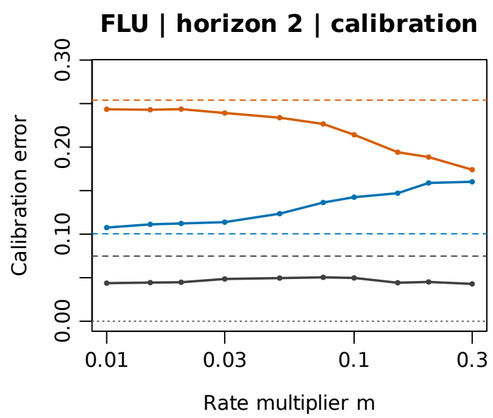


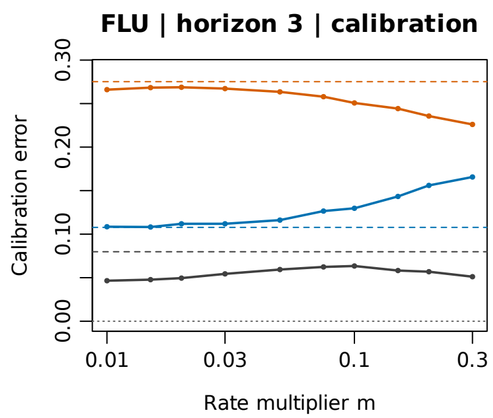










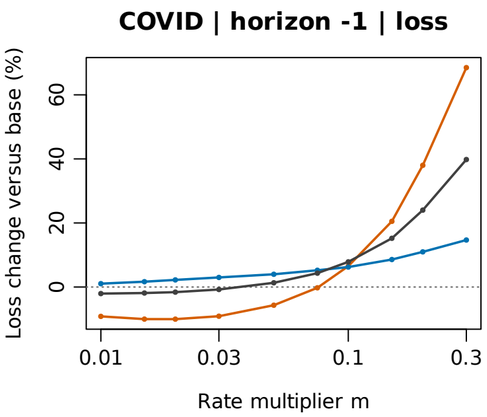


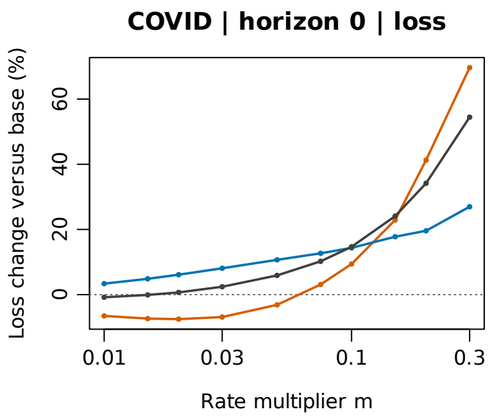
Before peak (weeks -7 to 0) | After peak (weeks 1 to 8) | Overall (Jun 2025 to Sep 2026)

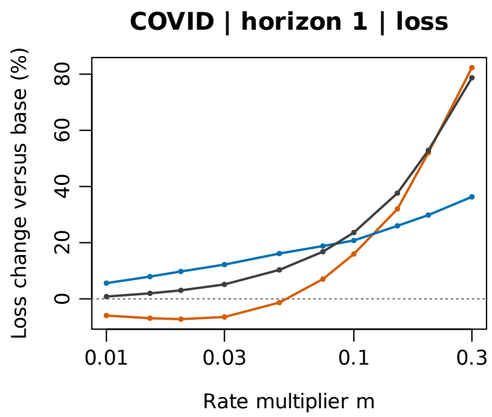
Loss: % change from m = 0 in the same phase (negative is better). Calibration: mean |coverage - alpha| over levels (lower is better); dashed: m = 0. Pooled: each location weighted by its weeks.

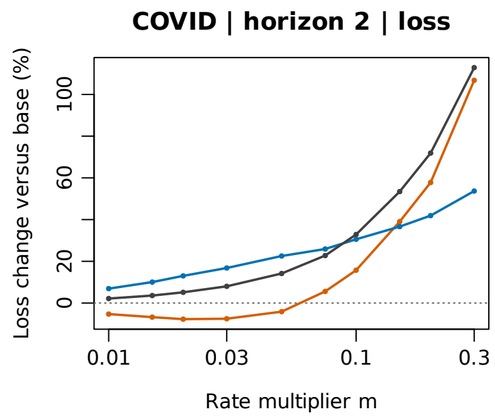
All 52 locations pooled (multipliers of the dots) | re-run | Fixed-rate MultiQT: loss and calibration before and after the peak

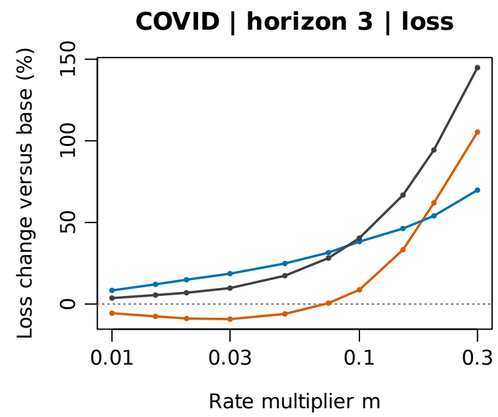
Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback re-run | Scored against latest reported counts in counts per 100k residents

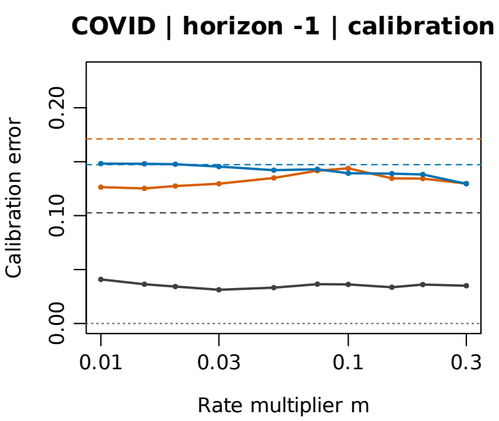


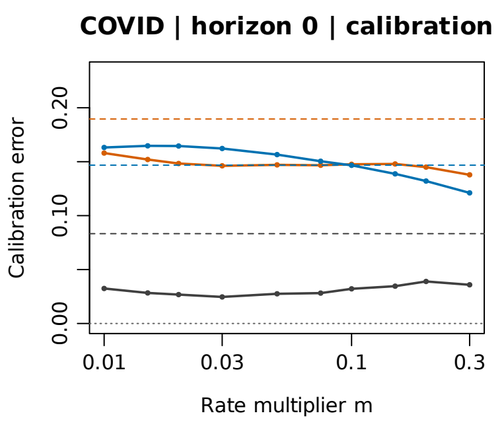


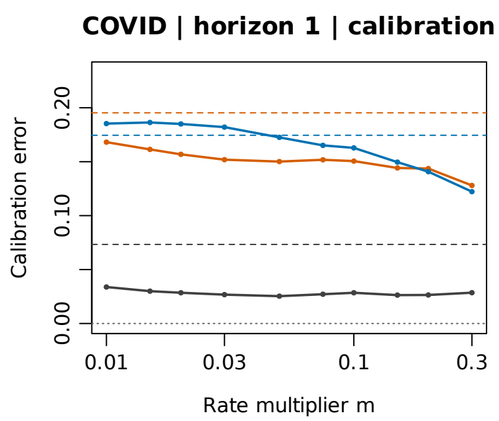


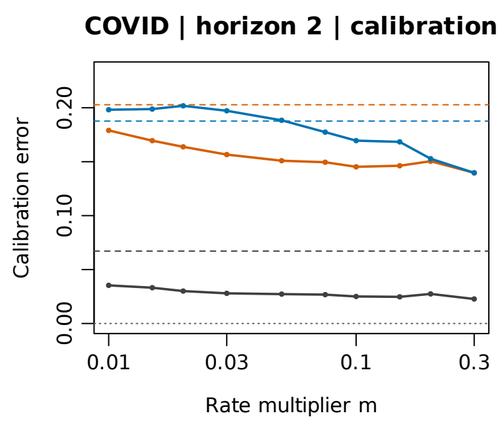


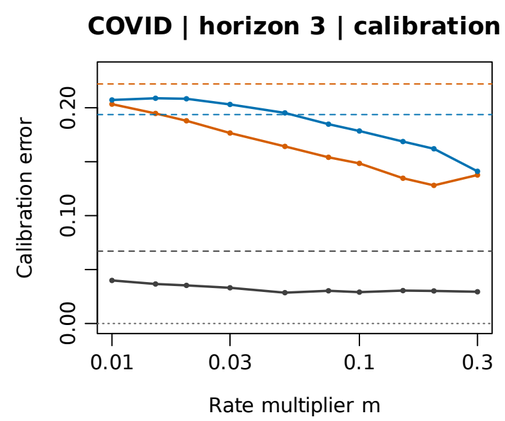


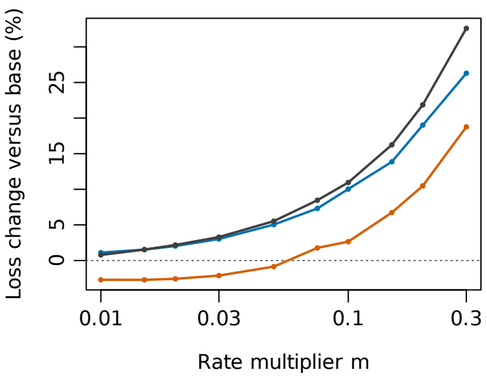


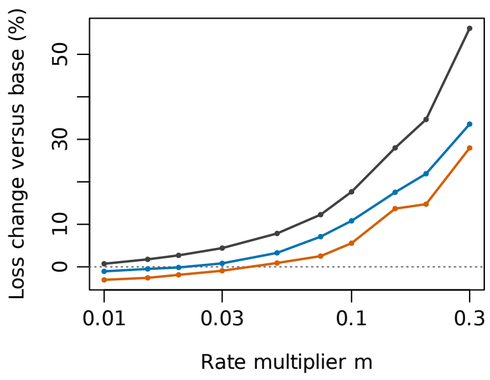


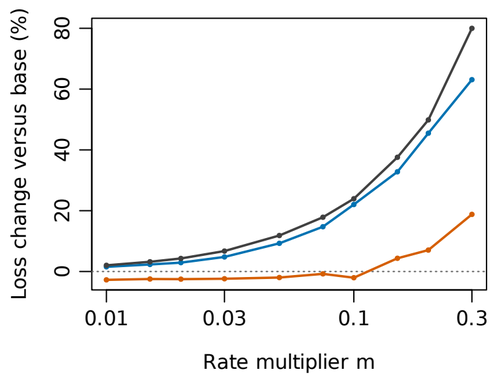


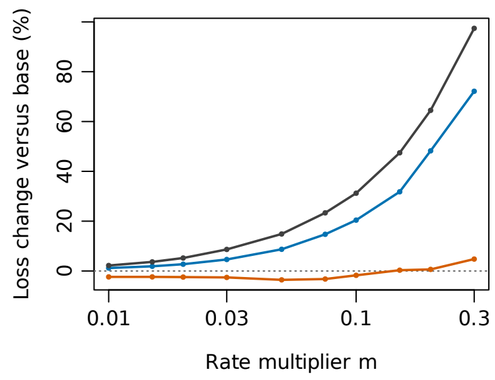


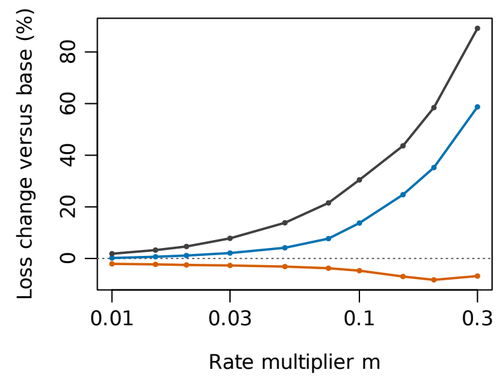


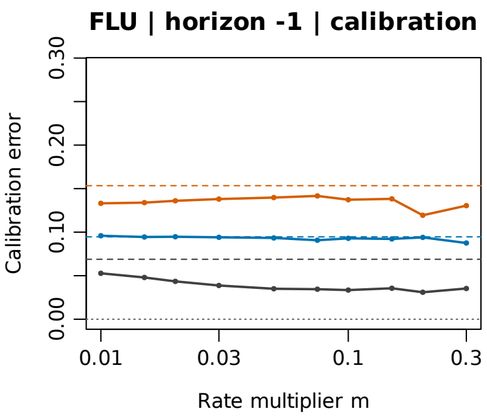


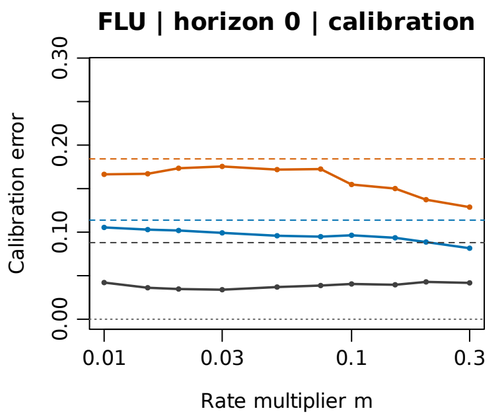


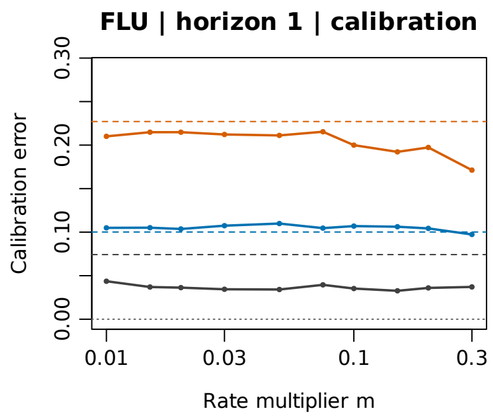


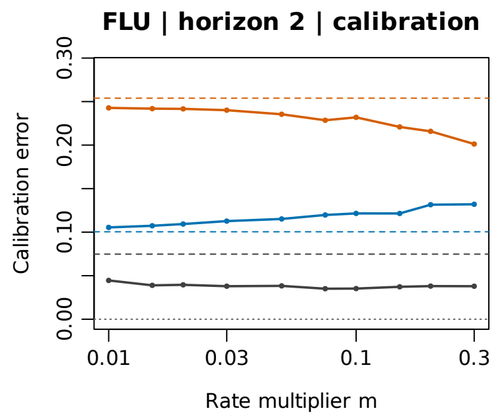


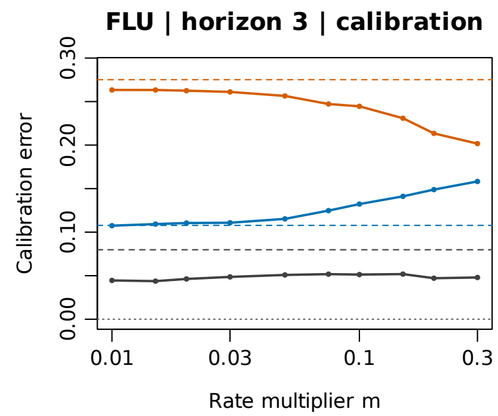










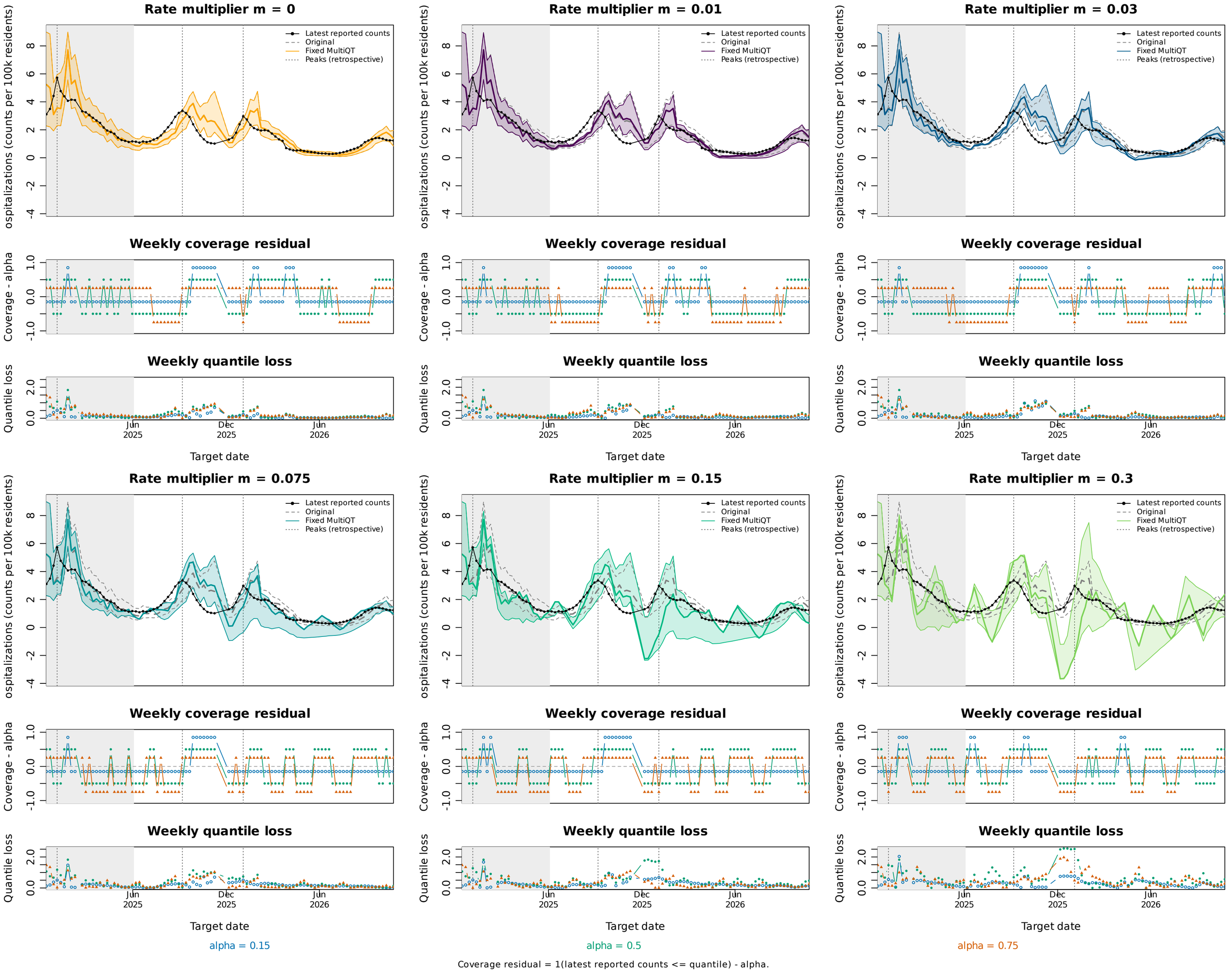


Before peak (weeks -7 to 0) | After peak (weeks 1 to 8) | Overall (Jun 2025 to Sep 2026)

Loss: % change from m = 0 in the same phase (negative is better). Calibration: mean |coverage - alpha| over levels (lower is better); dashed: m = 0. Pooled: each location weighted by its weeks.

covid | United States (US) | horizon 2 | delay 0 | 2024-12-14 to 2026-10-24

Count tracking | eta = m * S0, S0 = 17100 | Scores/plots: counts per 100k residents | Dotted lines: peaks (local maxima of the latest counts) | Gray: tuning period (not scored)

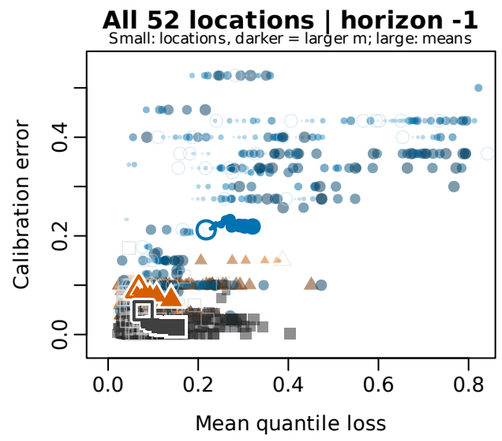


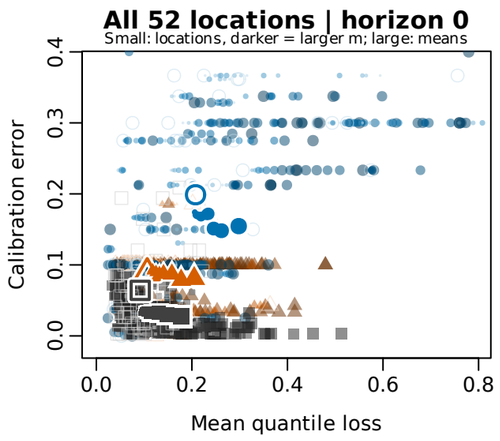
alpha = 0.15 | alpha = 0.5 | alpha = 0.75

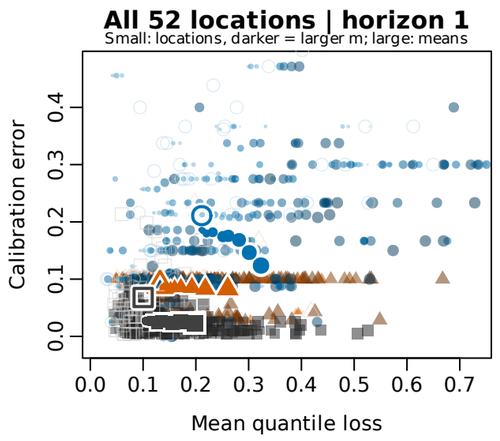
Coverage residual = 1(latest reported counts <= quantile) - alpha.

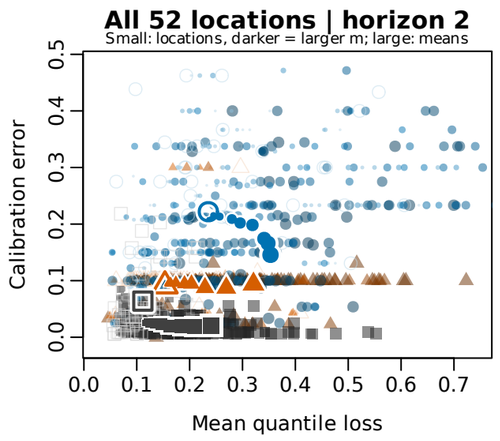
COVID | delay 0 | Fixed-rate MultiQT at quantile level alpha = 0.1: |coverage - alpha| vs quantile loss

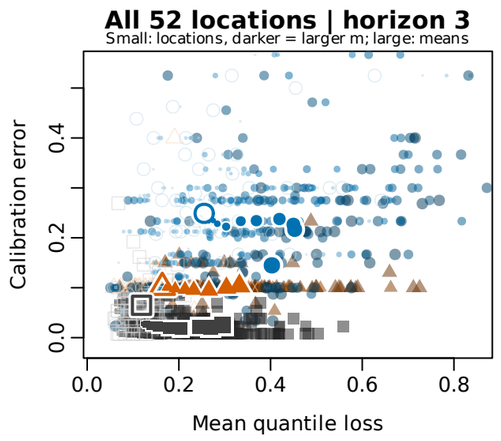
Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 | Scored against latest reported counts in counts per 100k residents | Lower left is better.

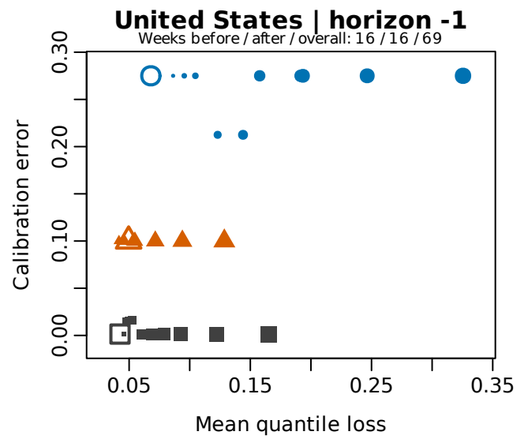


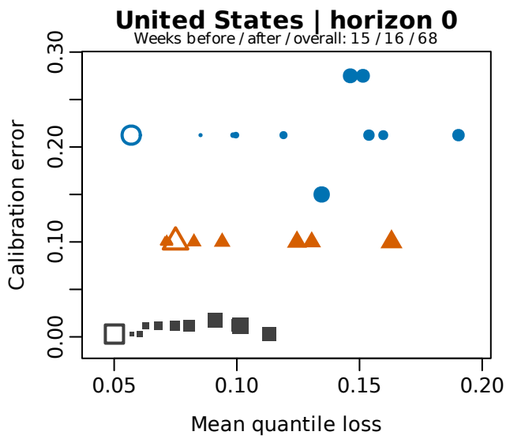


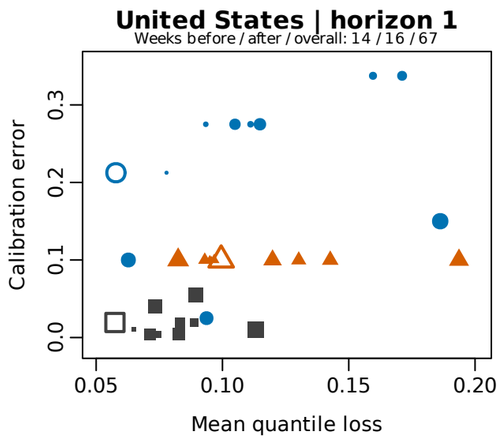


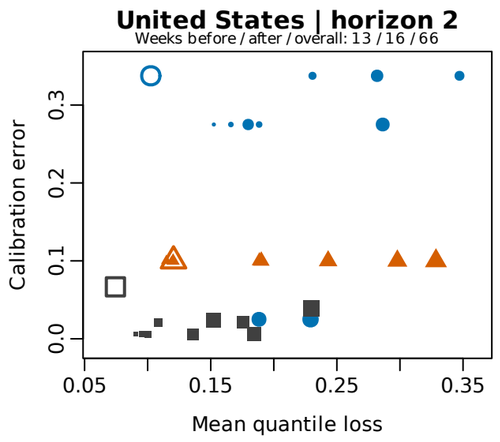


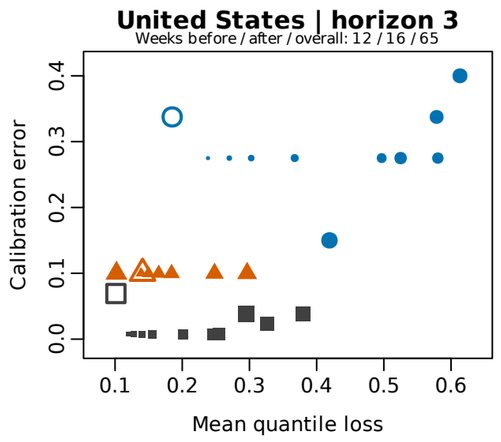


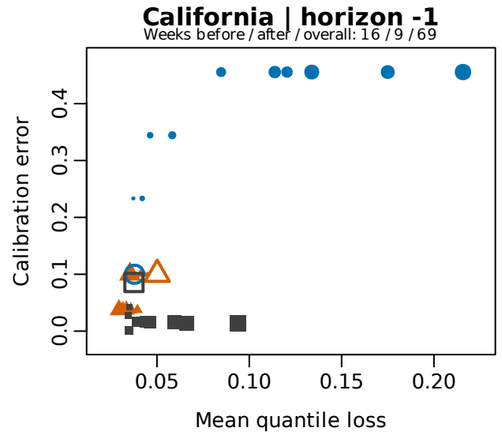


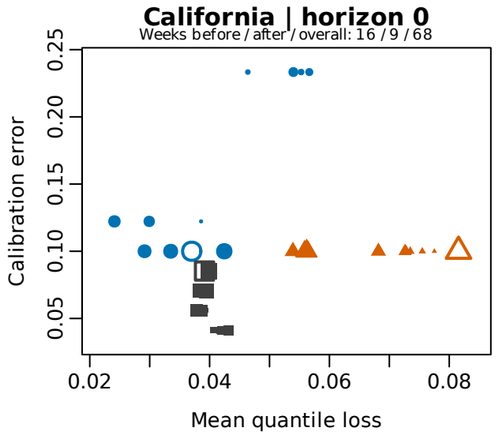


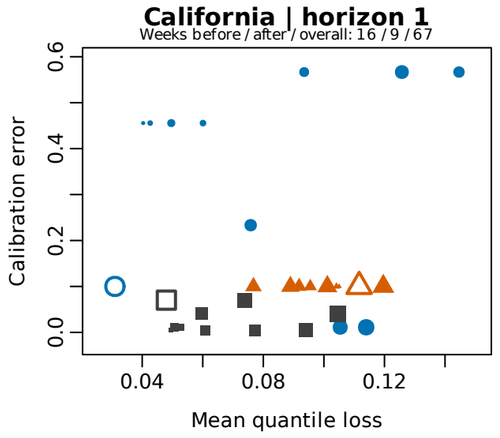


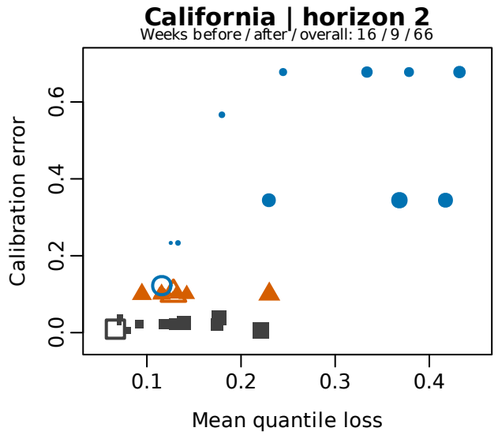


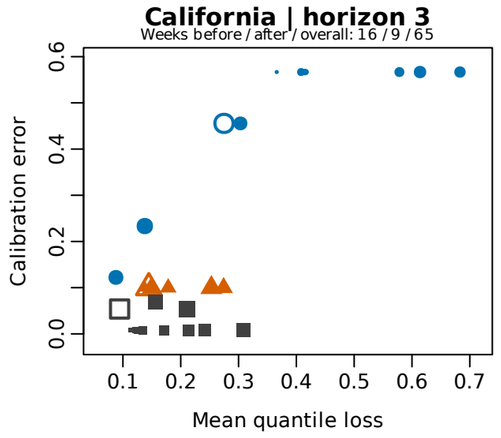


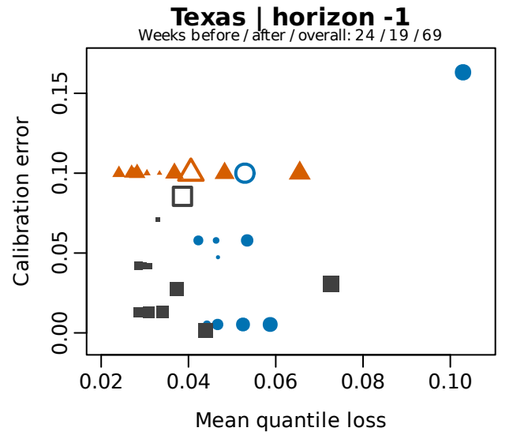


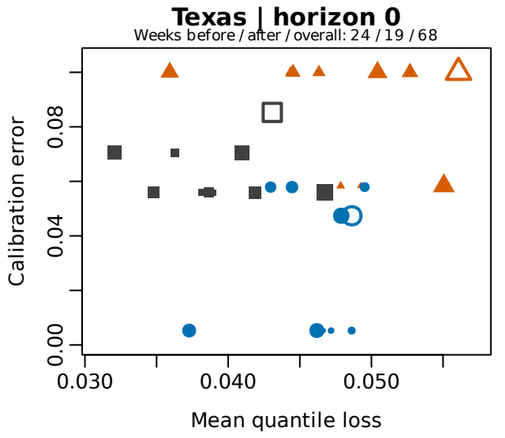


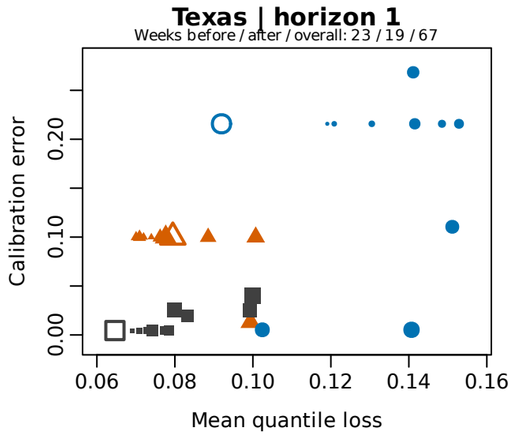


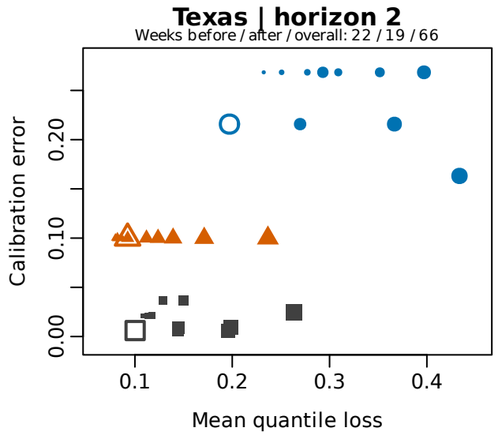


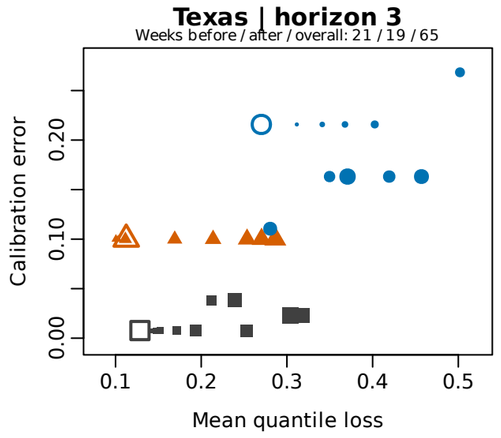


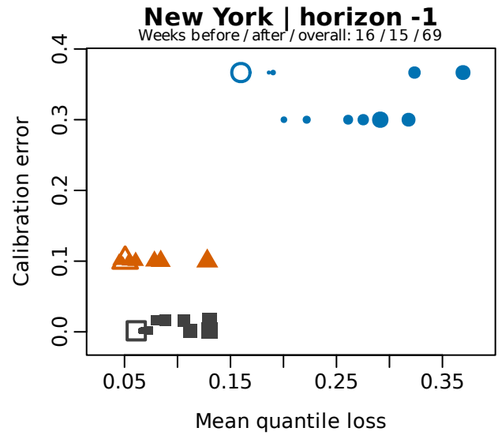


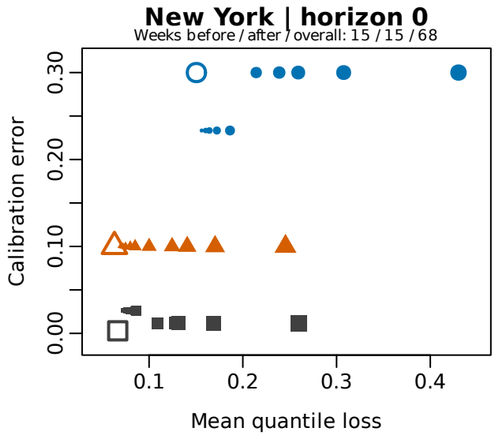


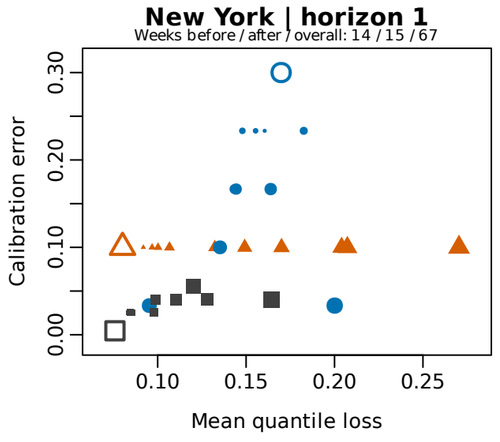


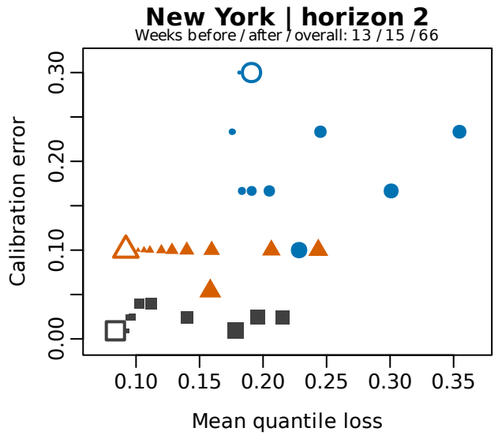


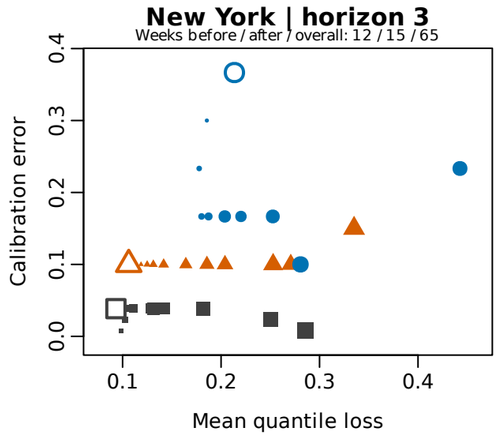


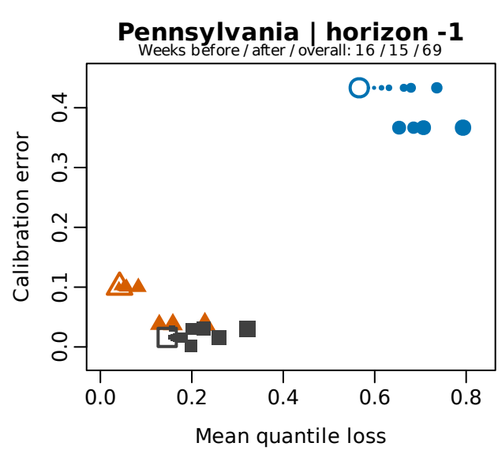


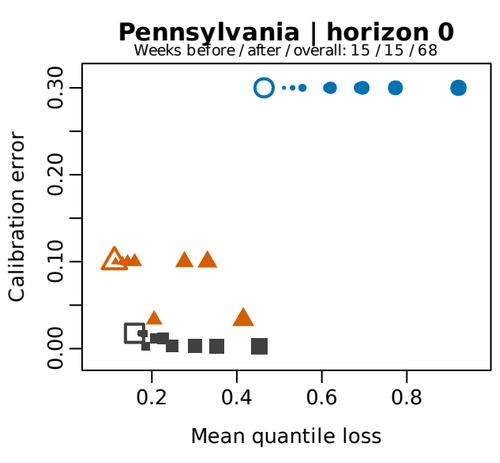


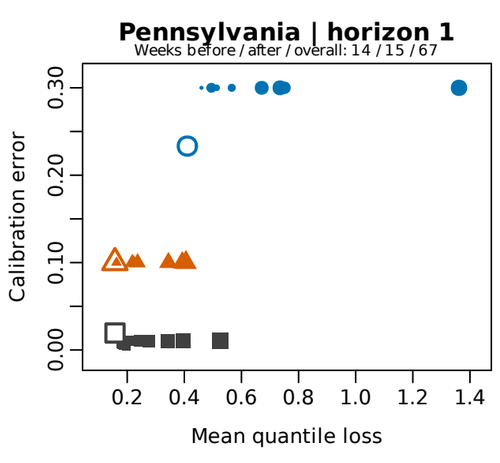


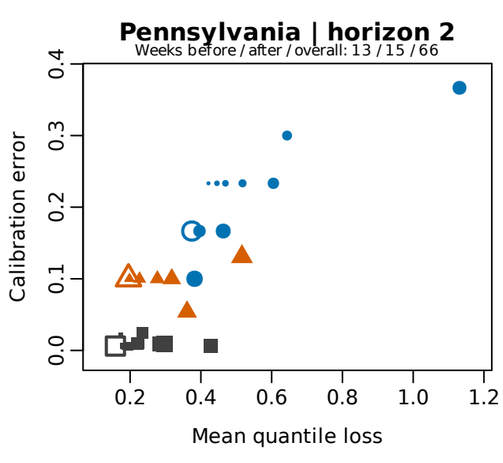


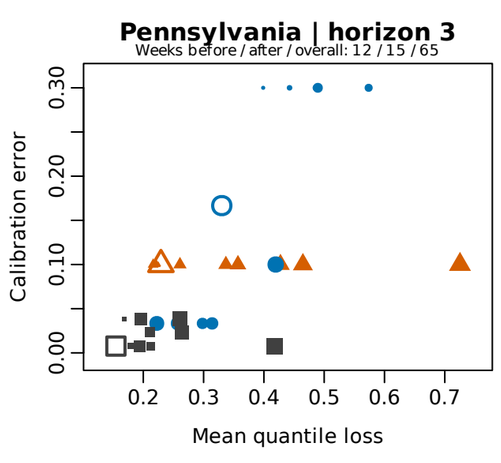








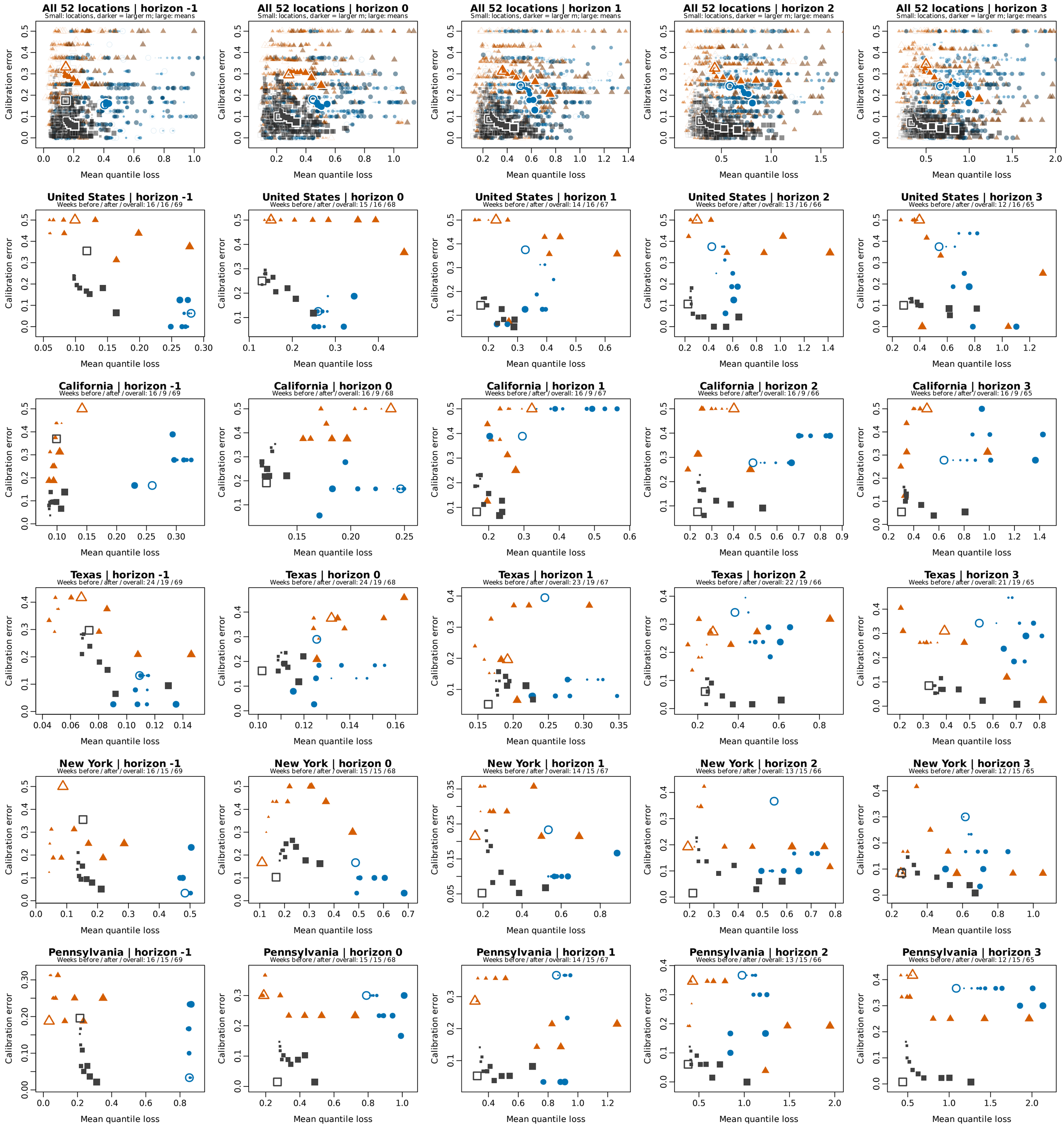




▲ Before peak (weeks -7 to 0)
● After peak (weeks 1 to 8)
■ Overall (Jun 2025 to Sep 2026)
○ Open symbol: m = 0 (original forecasts)
· m = 0.01
• m = 0.03
● m = 0.1
● m = 0.3

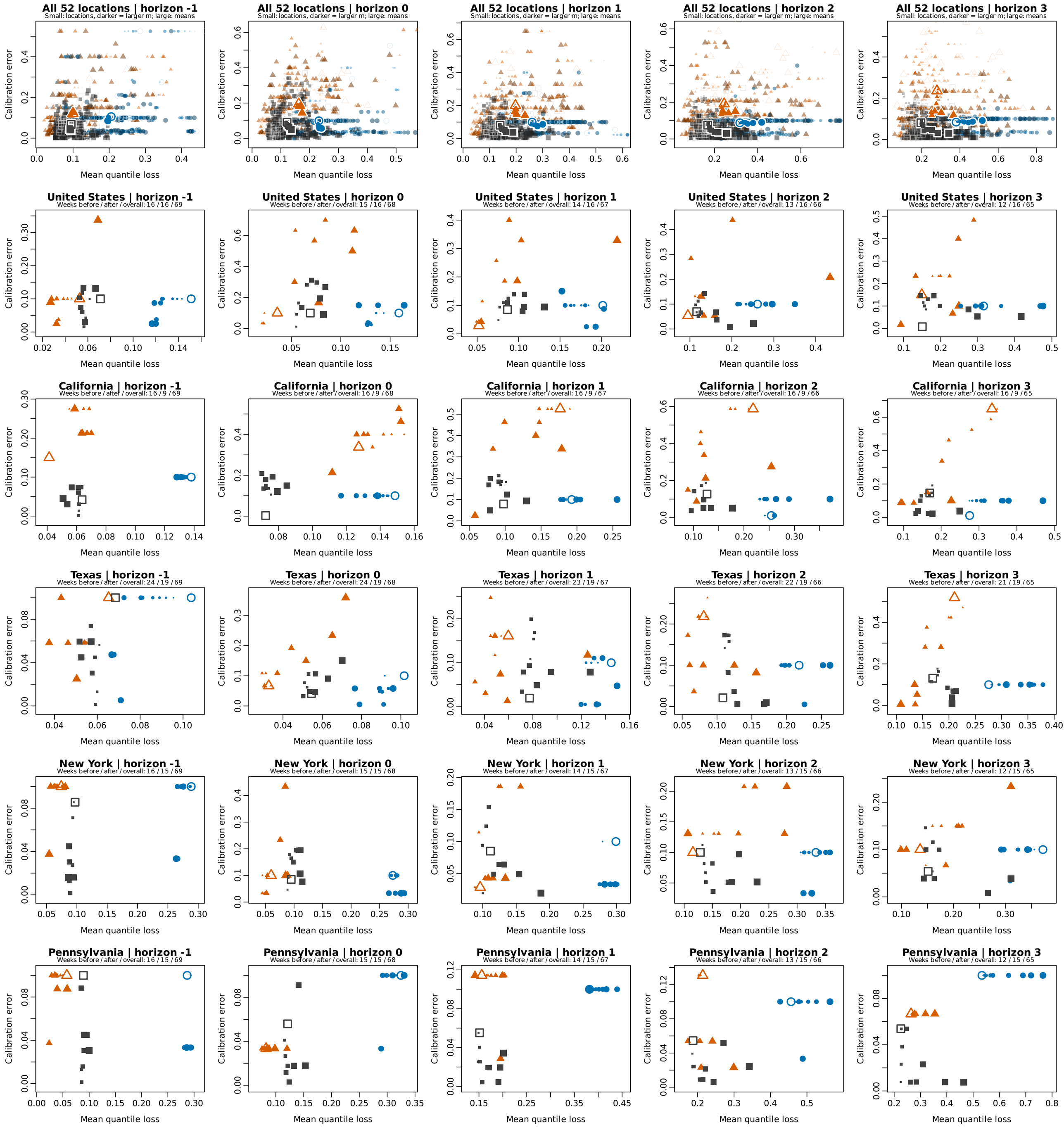
COVID | delay 0 | Fixed-rate MultiQT at quantile level alpha = 0.5: |coverage - alpha| vs quantile loss

Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 | Scored against latest reported counts in counts per 100k residents | Lower left is better.



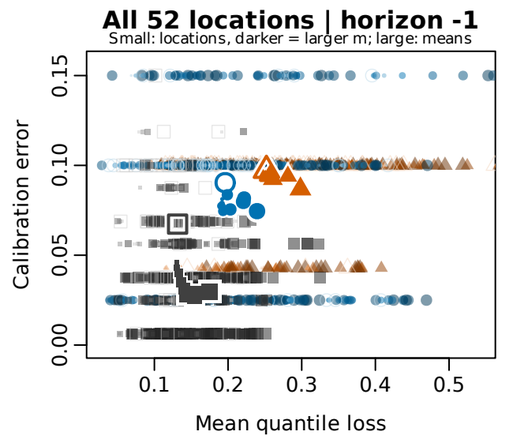
COVID | delay 0 | Fixed-rate MultiQT at quantile level alpha = 0.9: |coverage - alpha| vs quantile loss

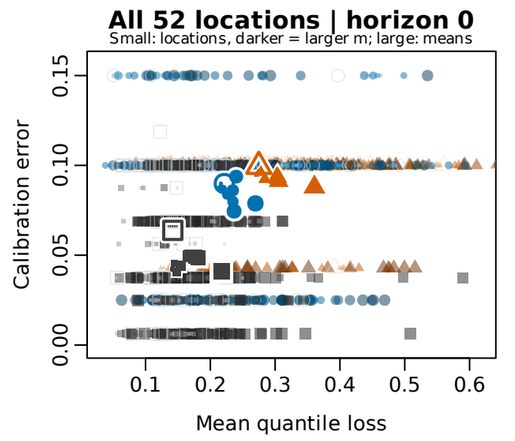
Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 | Scored against latest reported counts in counts per 100k residents | Lower left is better.

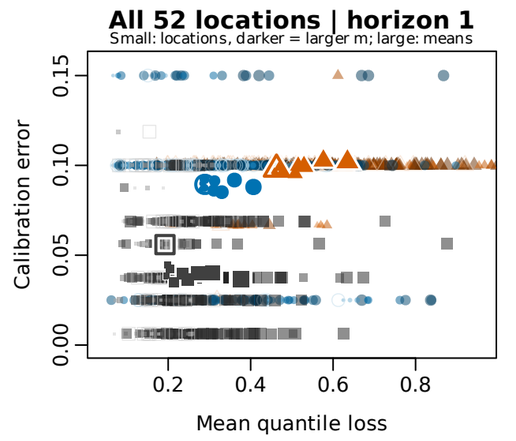


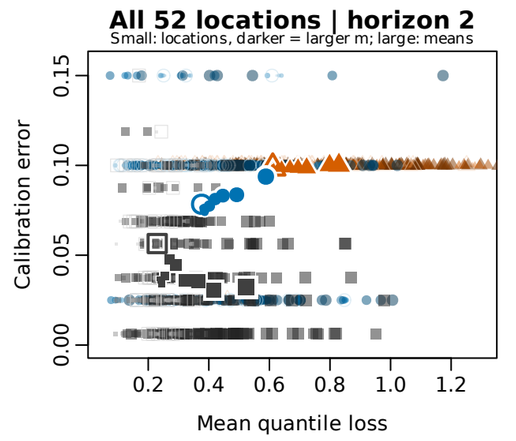
FLU | delay 0 | Fixed-rate MultiQT at quantile level alpha = 0.1: |coverage - alpha| vs quantile loss

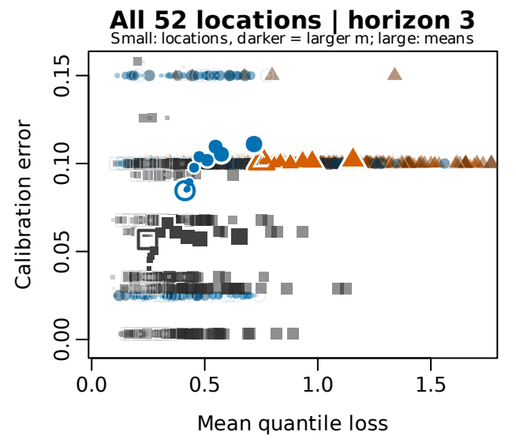
Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 | Scored against latest reported counts in counts per 100k residents | Lower left is better.

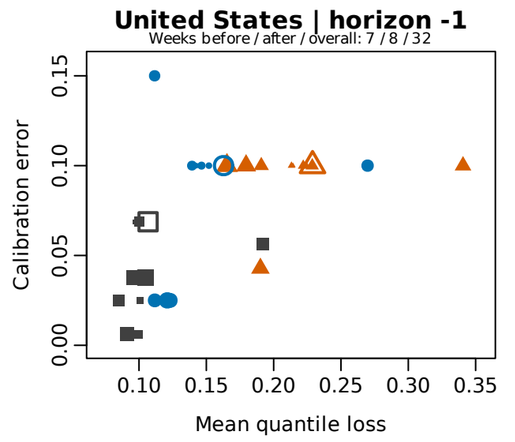


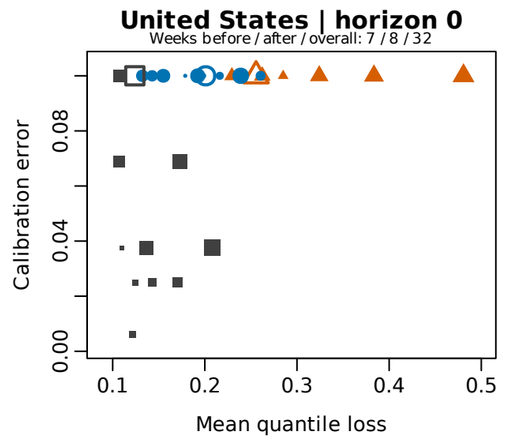


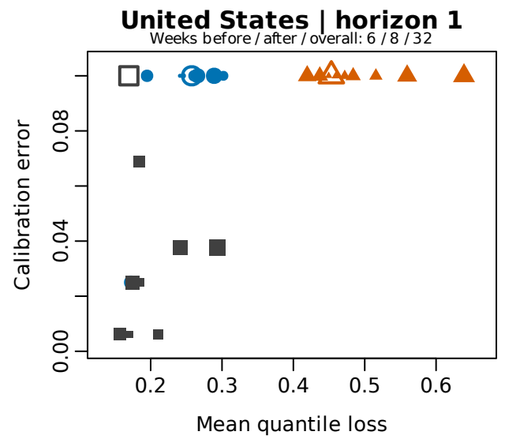


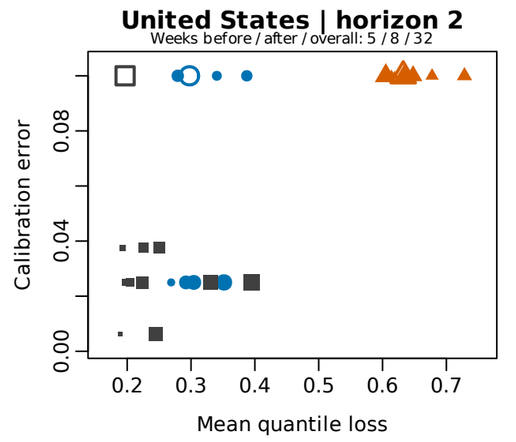


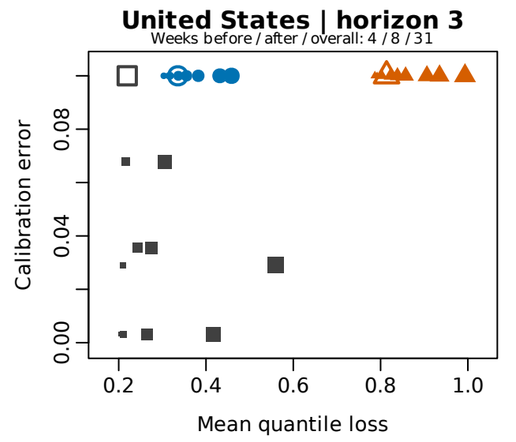


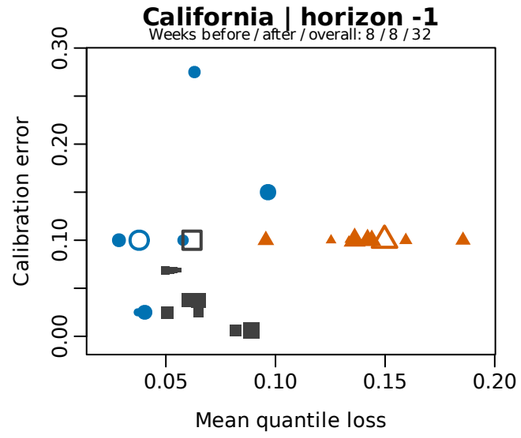


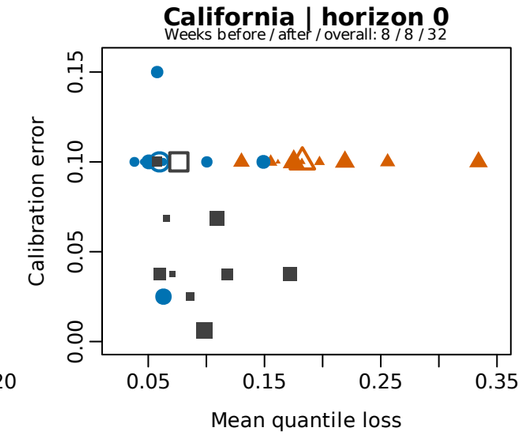


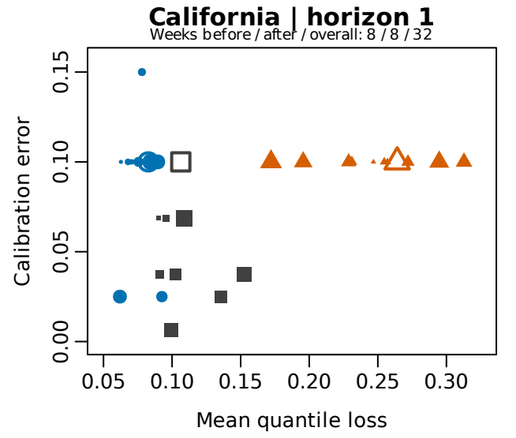


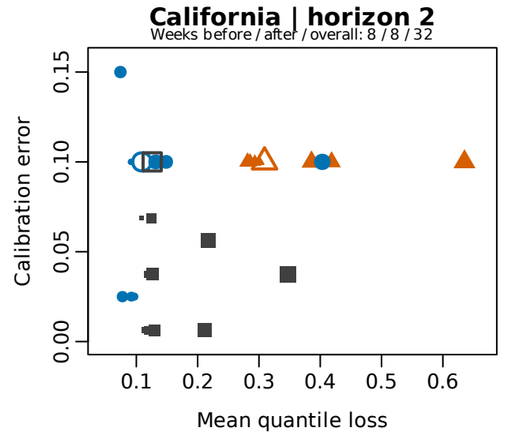


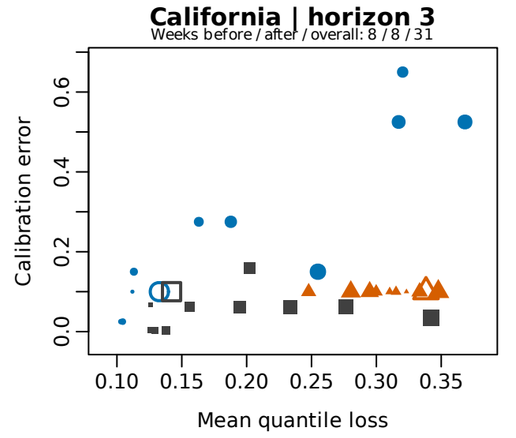


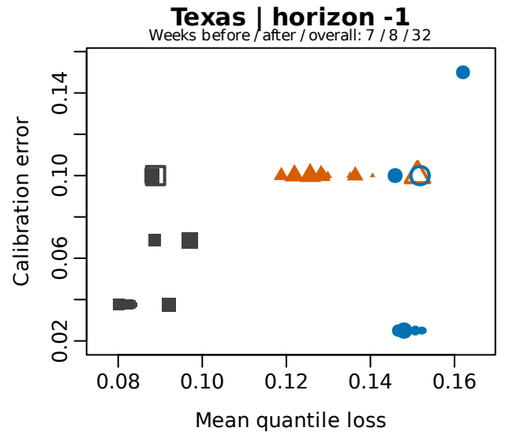


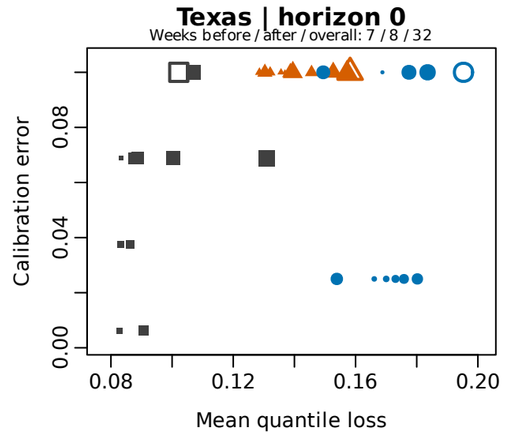


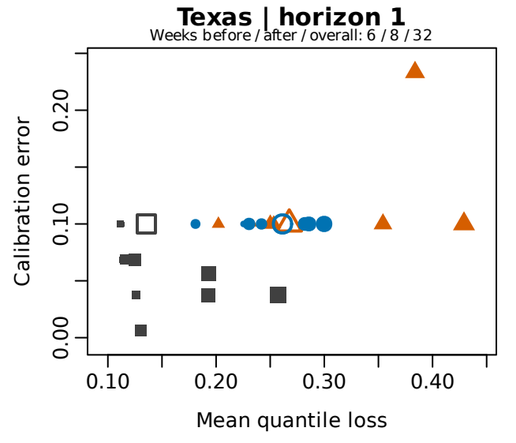


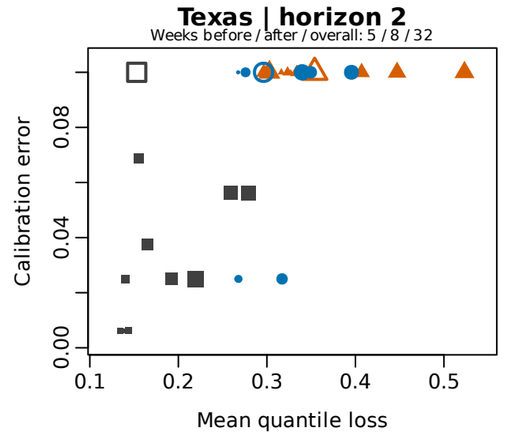


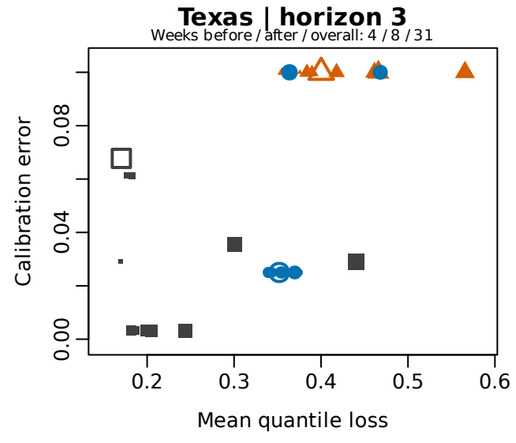


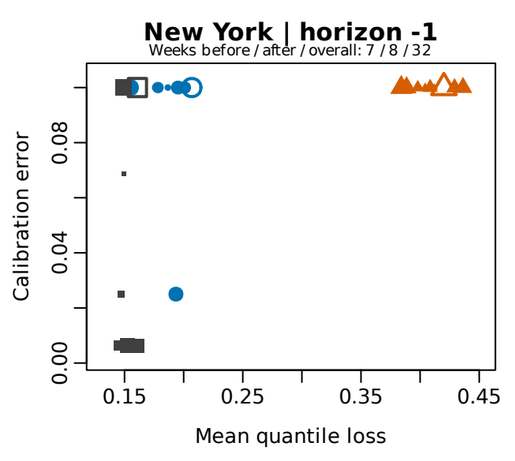


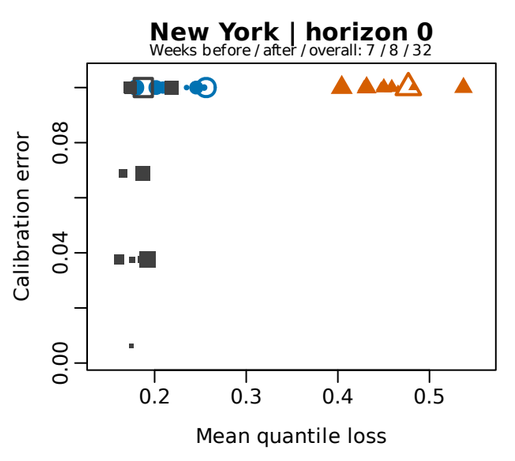


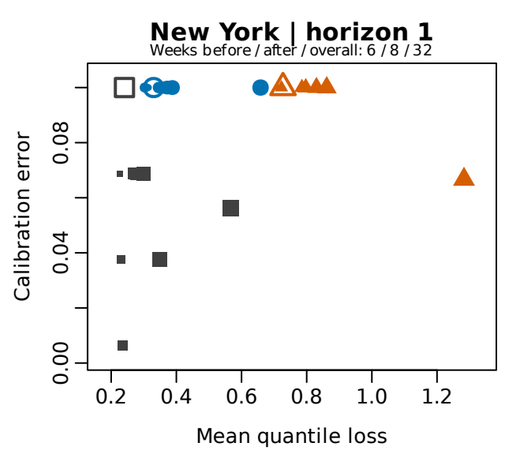


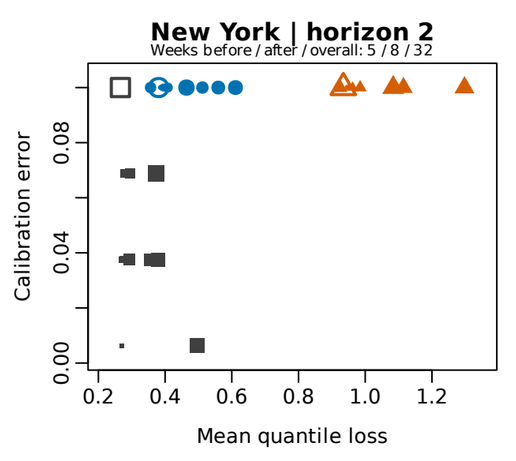


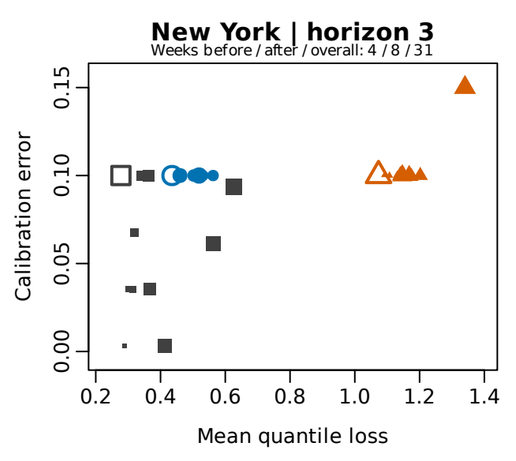


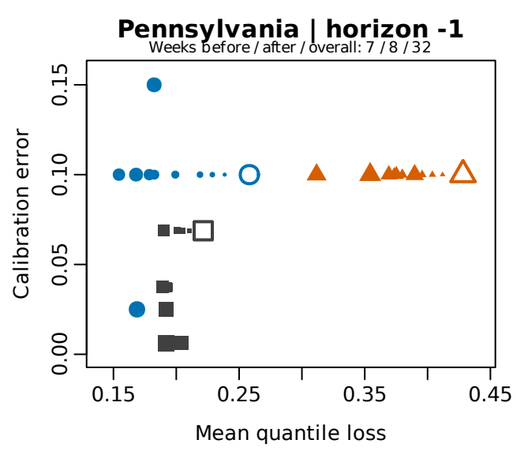


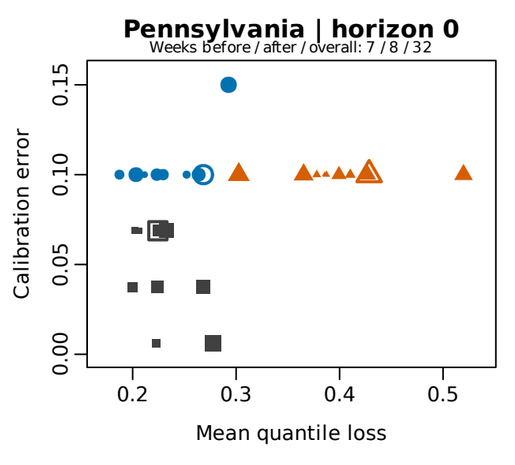


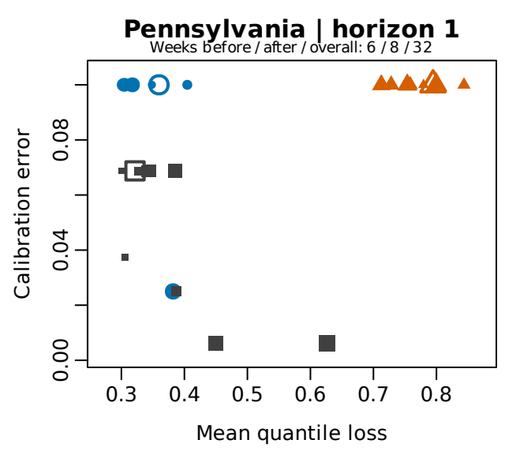


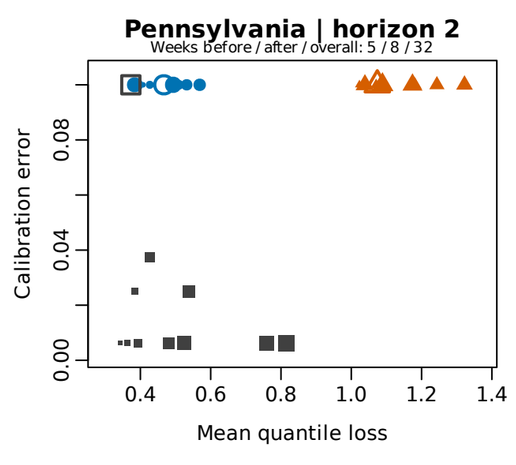


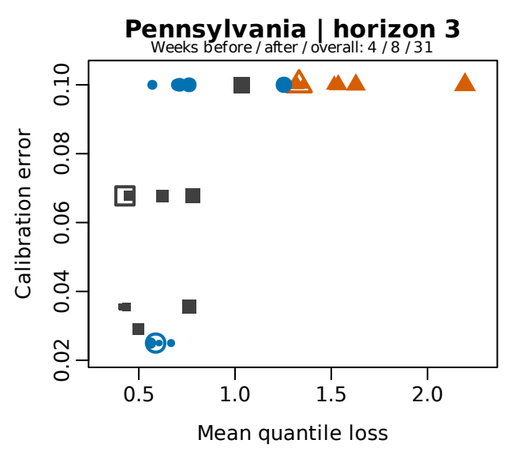








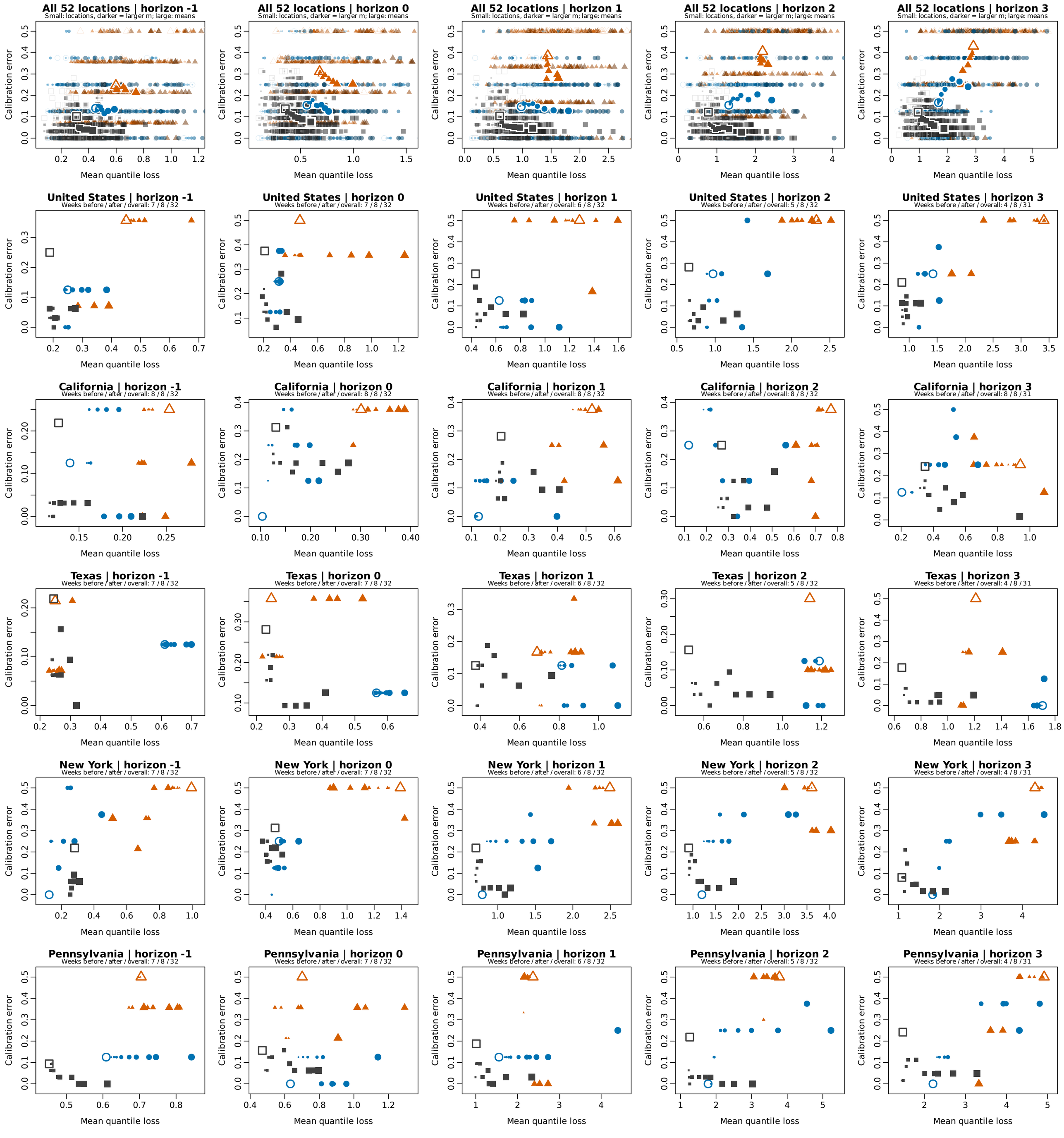




▲ Before peak (weeks -7 to 0)
● After peak (weeks 1 to 8)
■ Overall (Jun 2025 to Sep 2026)
○ Open symbol: m = 0 (original forecasts)
· m = 0.01
• m = 0.03
● m = 0.1
● m = 0.3

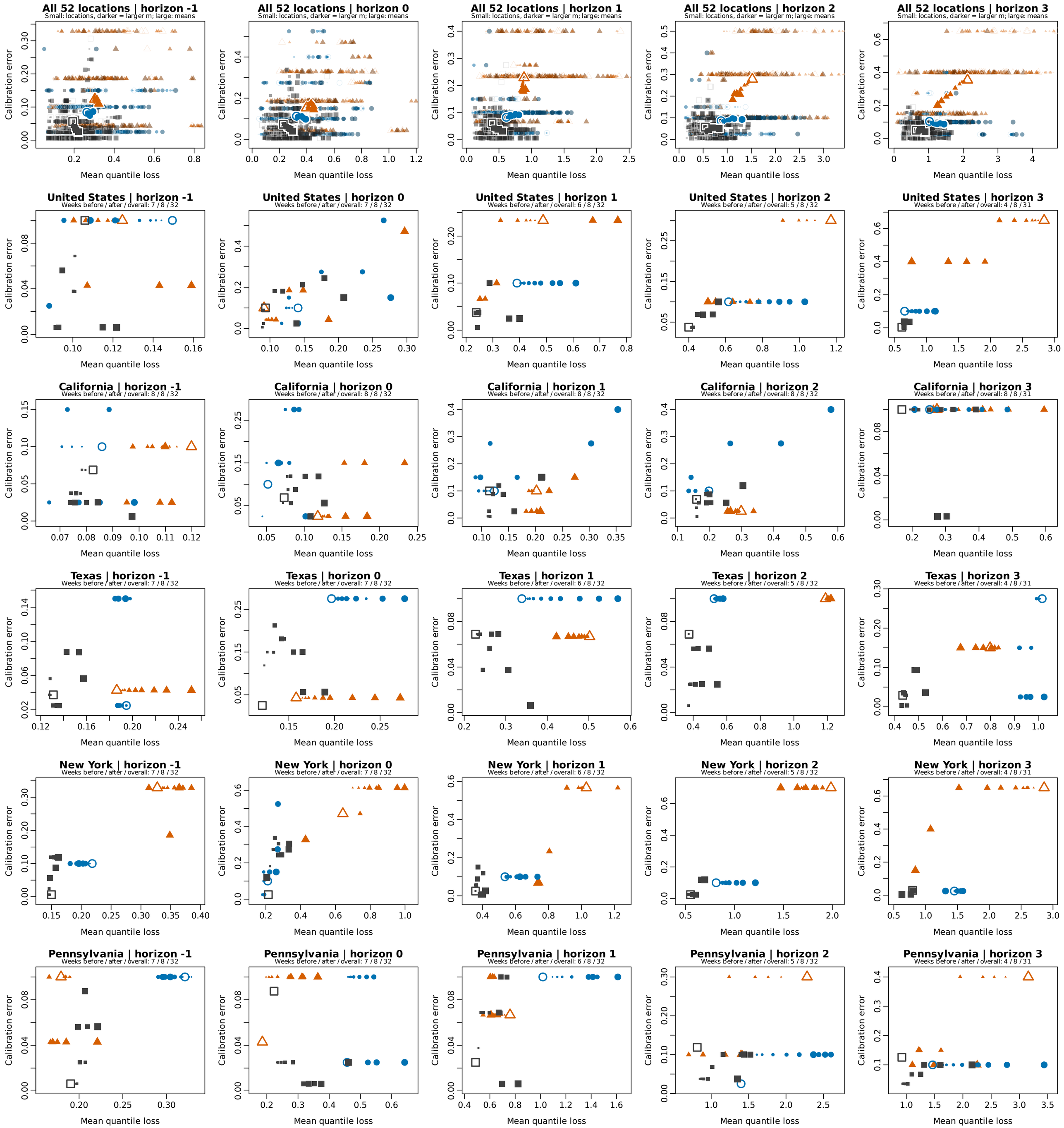
FLU | delay 0 | Fixed-rate MultiQT at quantile level alpha = 0.5: |coverage - alpha| vs quantile loss

Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 | Scored against latest reported counts in counts per 100k residents | Lower left is better.



FLU | delay 0 | Fixed-rate MultiQT at quantile level alpha = 0.9: |coverage - alpha| vs quantile loss

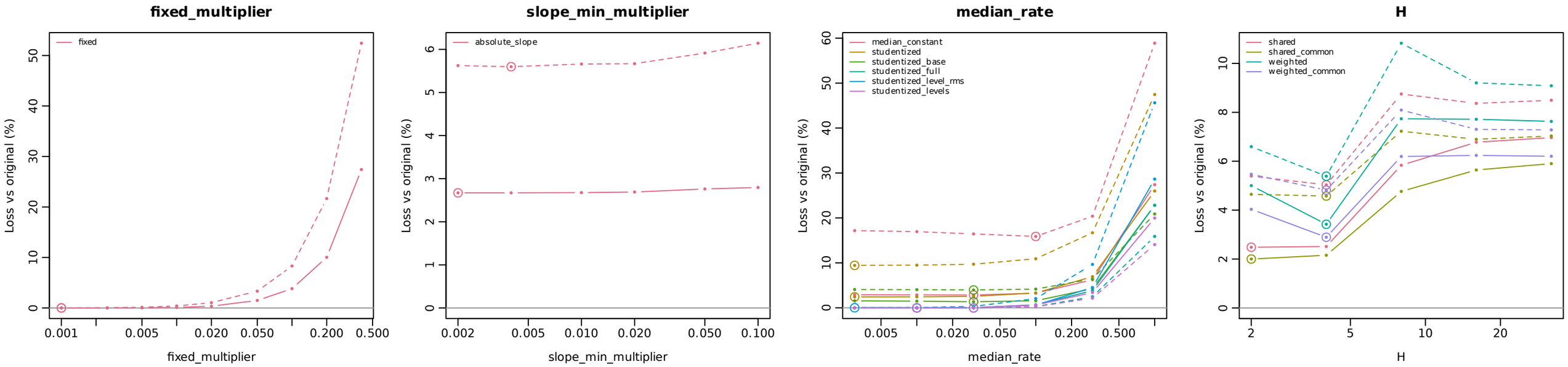
Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 | Scored against latest reported counts in counts per 100k residents | Lower left is better.



covid | design hub_first_months | feedback delay 0 | tuning period Nov 2024 to May 2025 | horizons -1 to 3 | 52 locations

Mean quantile loss relative to the original forecasts (mean over locations and horizons) on the scale of each tracker

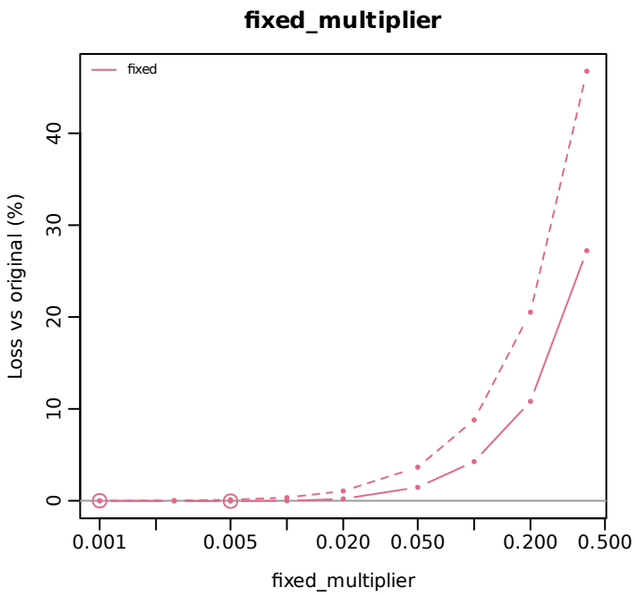
covid | design hub_first_months | feedback delay 1 | tuning period Nov 2024 to May 2025 | horizons -1 to 3 | 52 locations

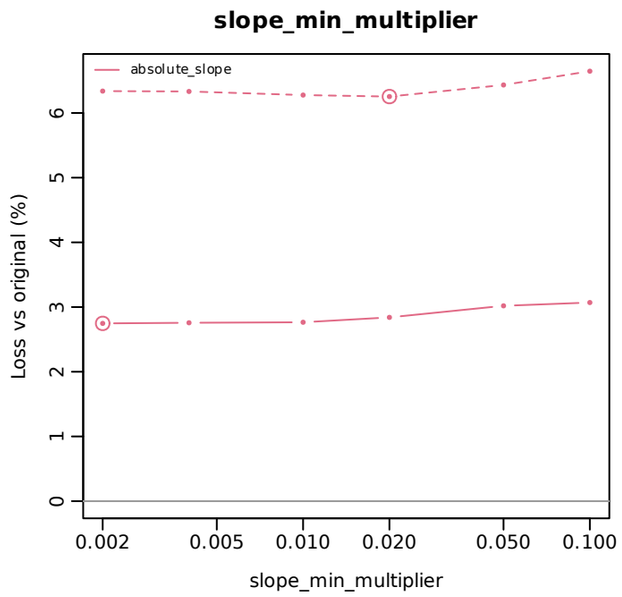


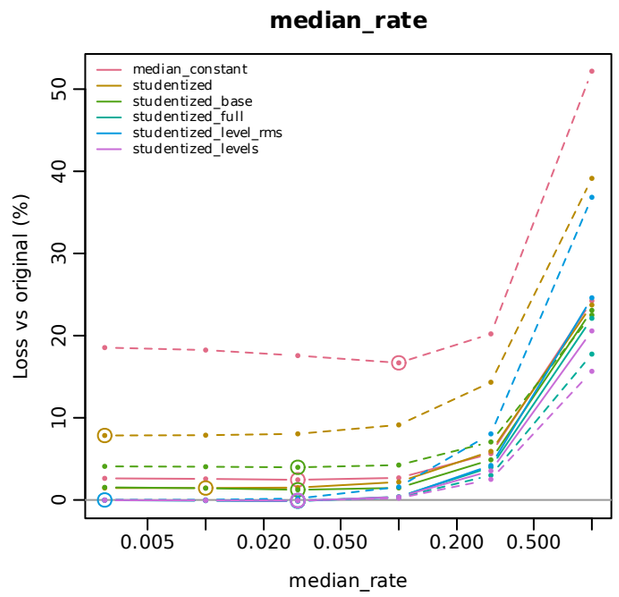
—— log1p tracker (loss of log(counts + 1)) - - - identity tracker (loss in counts per 100k residents) o chosen value

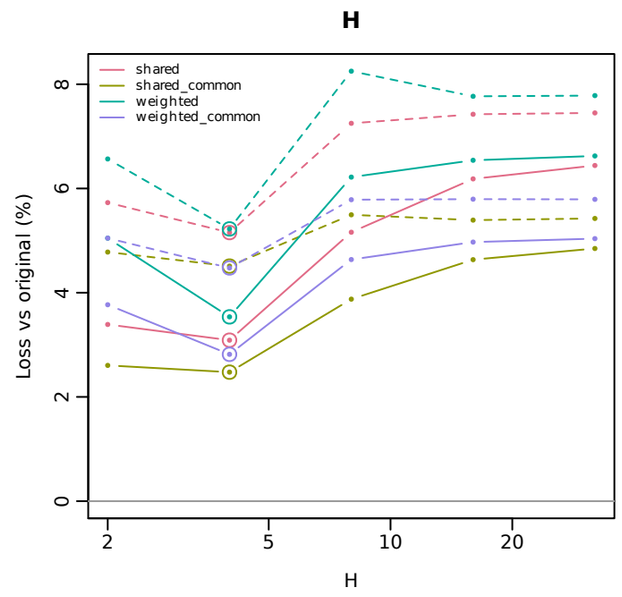
Mean quantile loss relative to the original forecasts (mean over locations and horizons) on the scale of each tracker

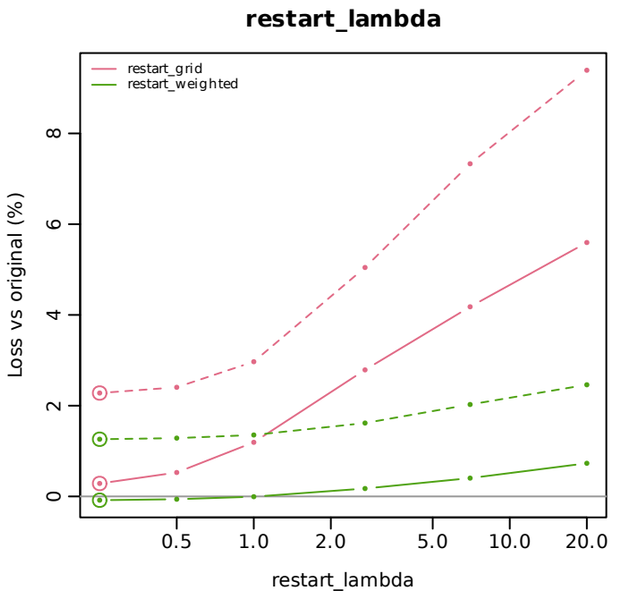
covid | design hub_first_months | feedback delay 0 + revisions (re-run) | tuning period Nov 2024 to May 2025 | horizons -1 to 3 | 52 locations

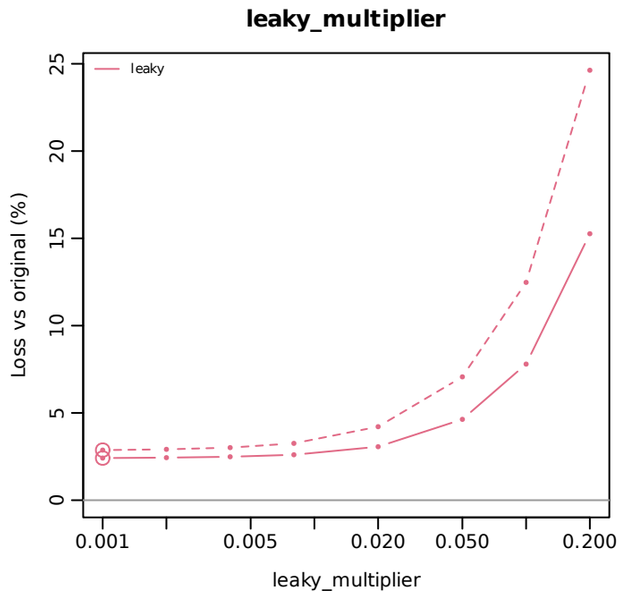


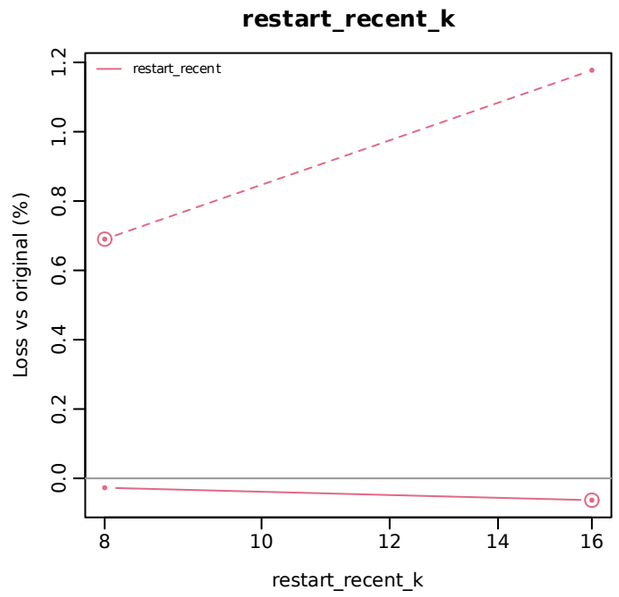


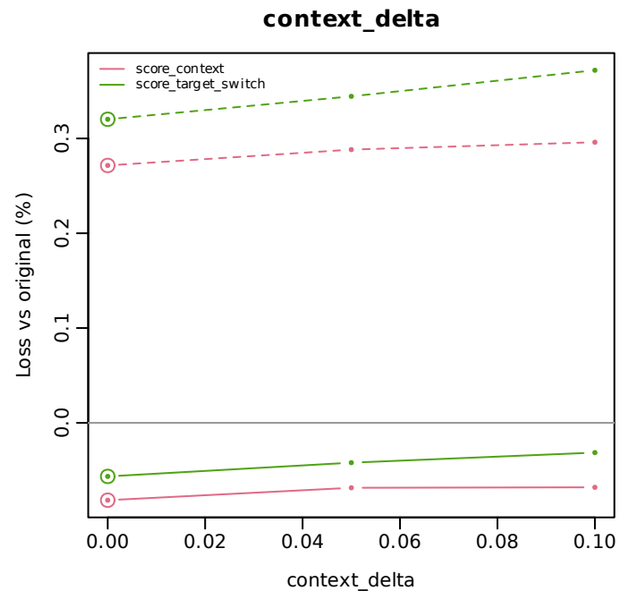


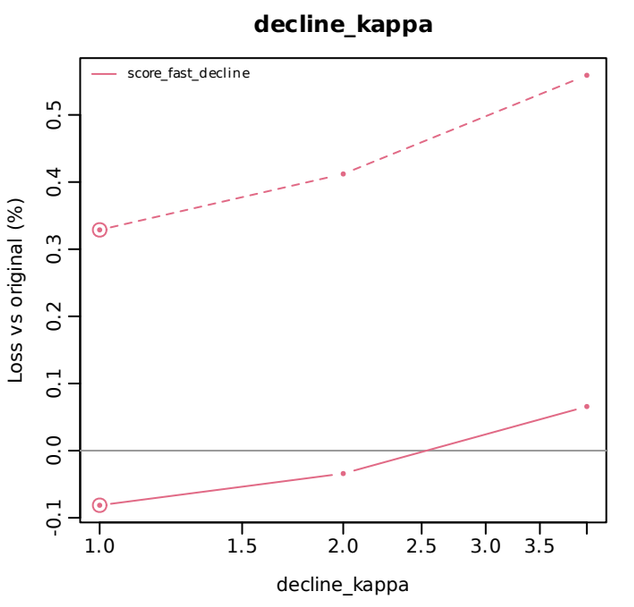


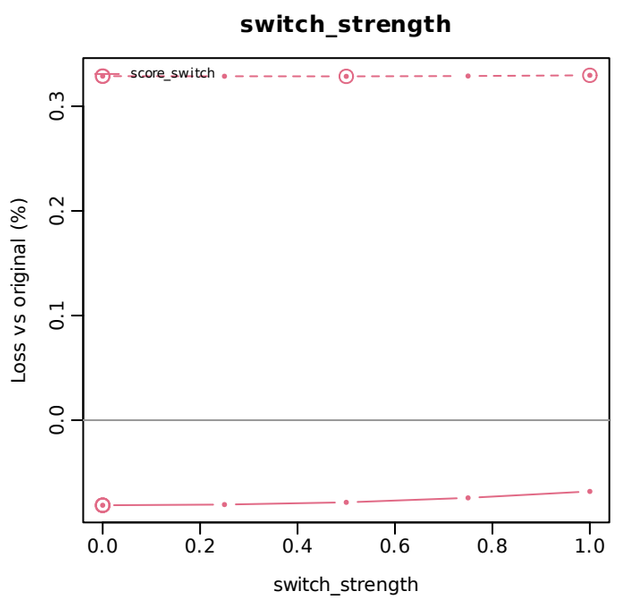


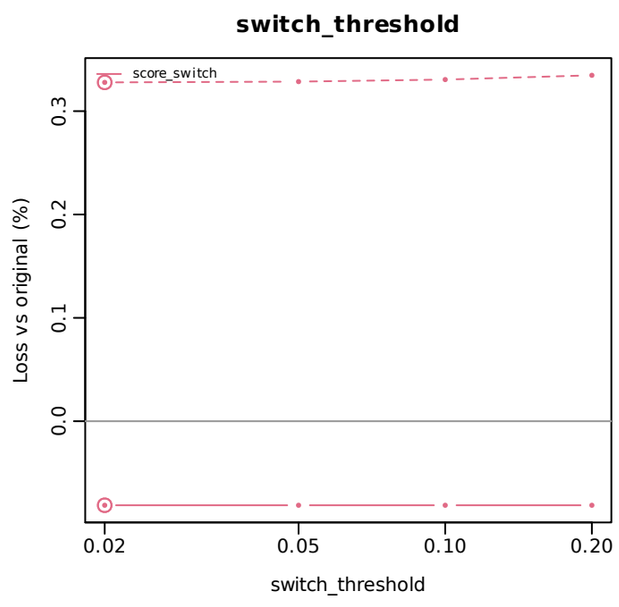


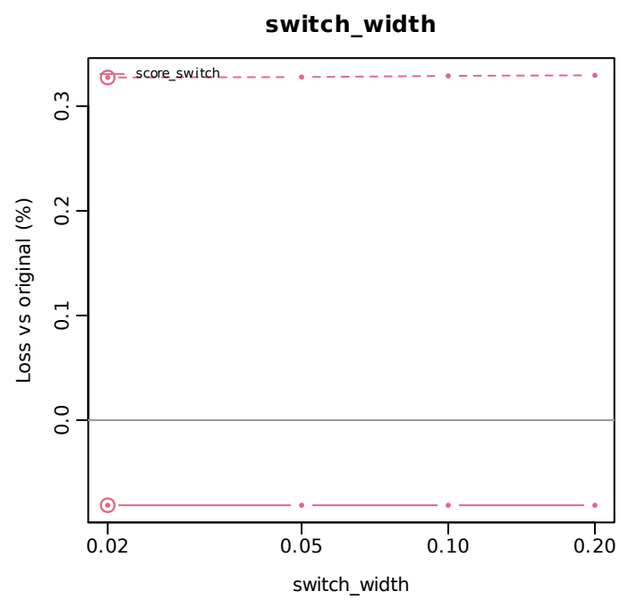


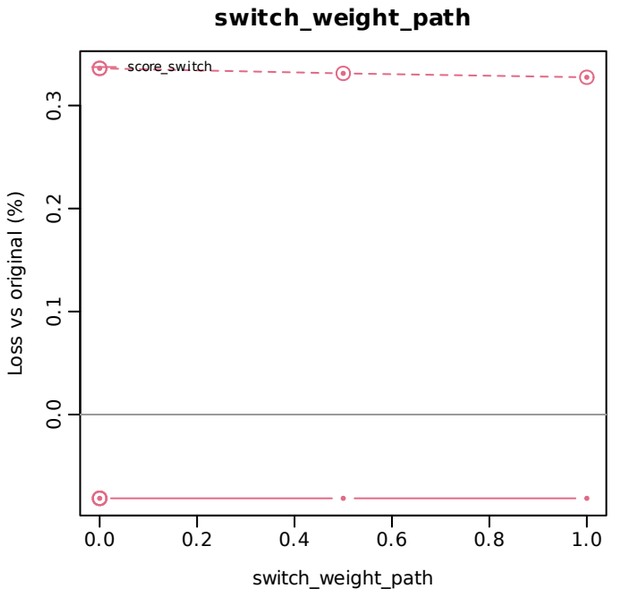


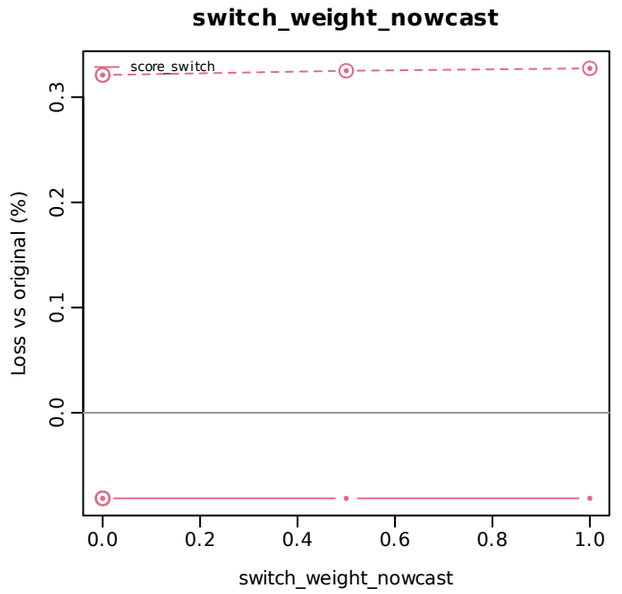


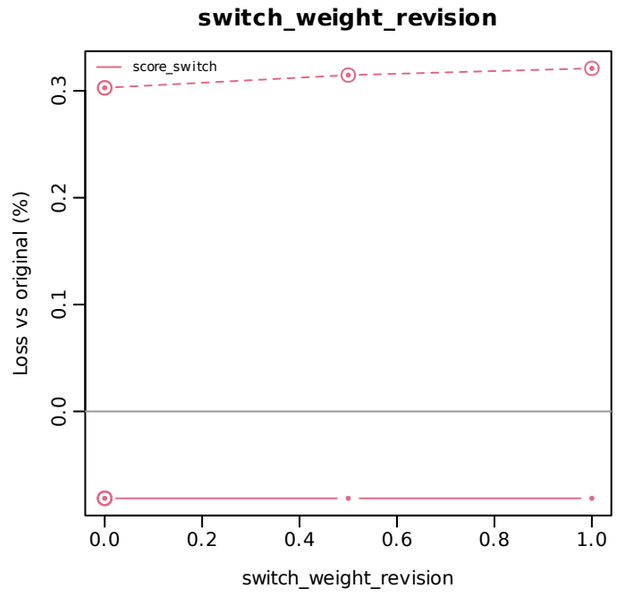


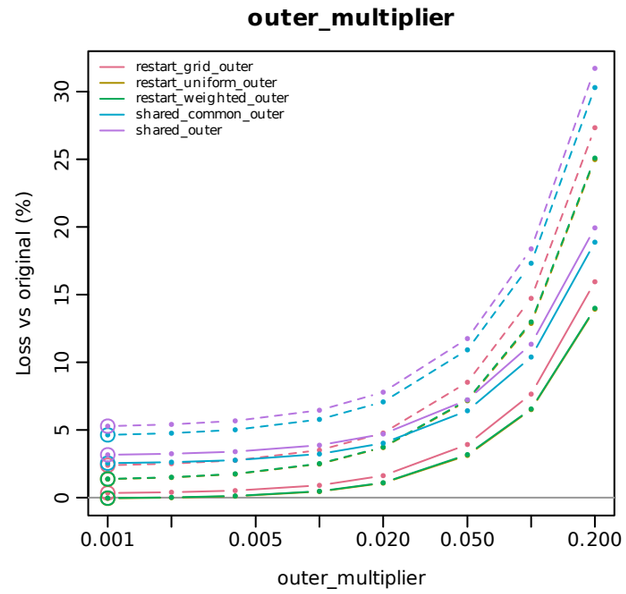








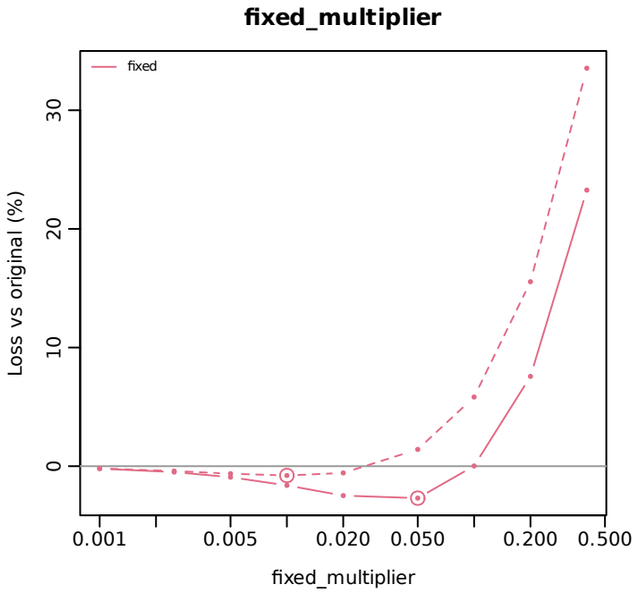


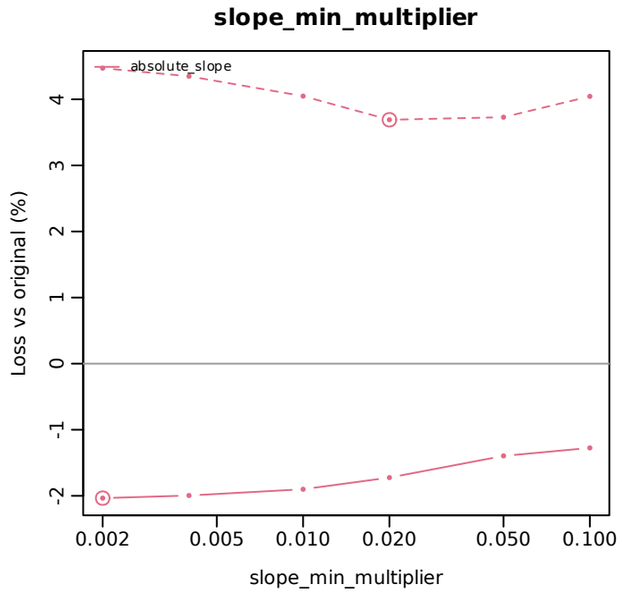


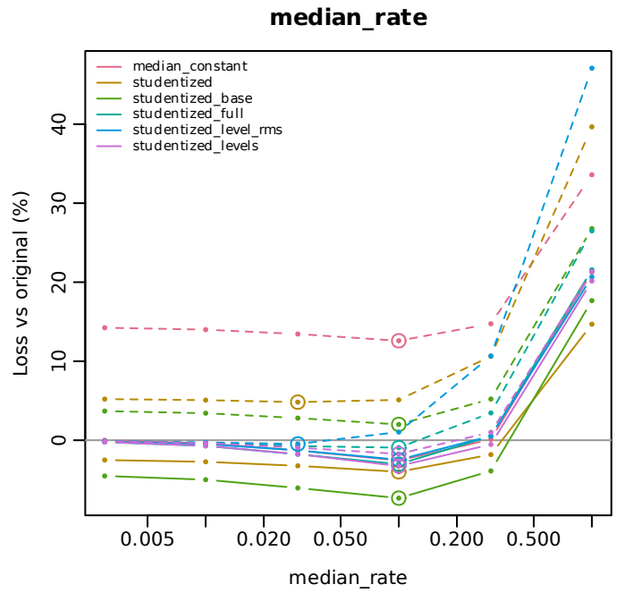
—— log1p tracker (loss of log(counts + 1)) - - - identity tracker (loss in counts per 100k residents) o chosen value

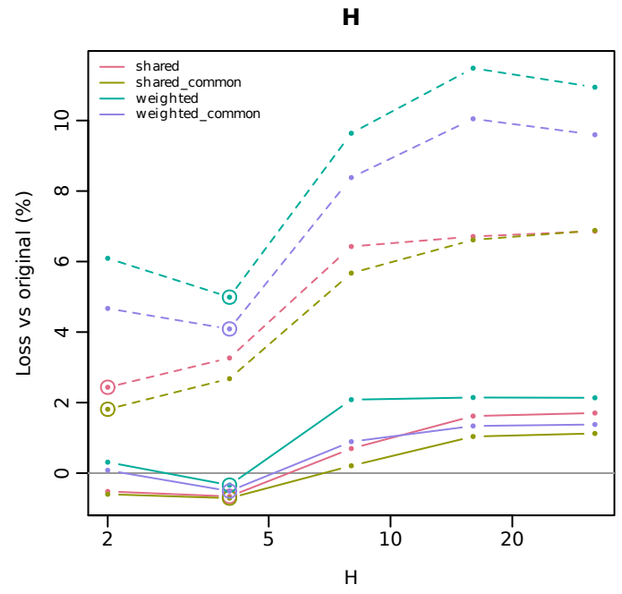
Mean quantile loss relative to the original forecasts (mean over locations and horizons) on the scale of each tracker

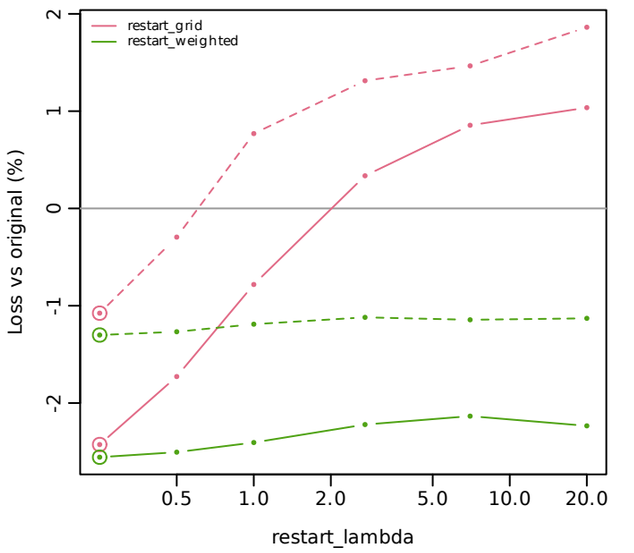
flu | design hub_first_months | feedback delay 0 | tuning period Nov 2024 to May 2025 | horizons -1 to 3 | 52 locations

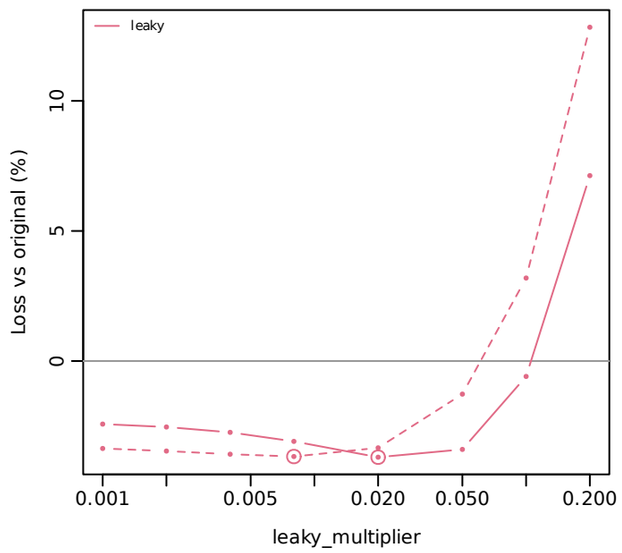


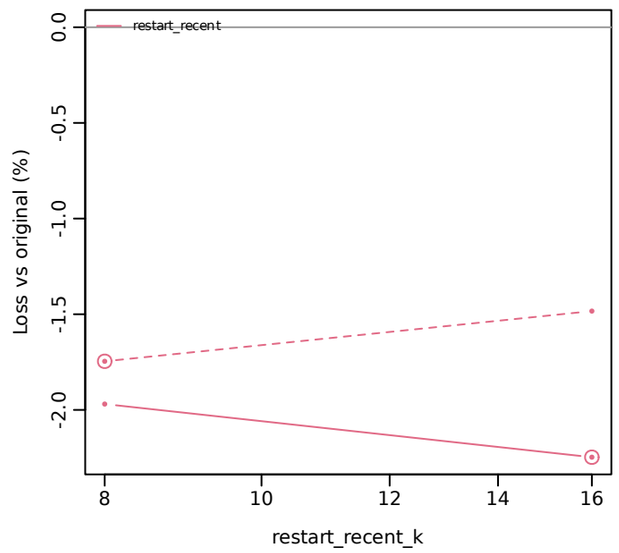


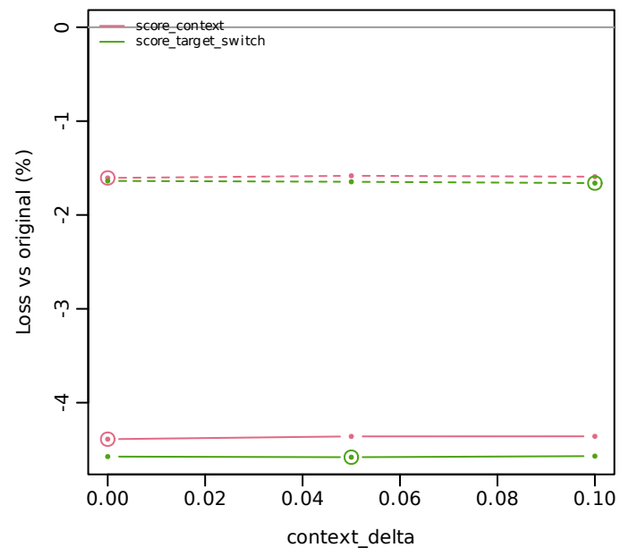


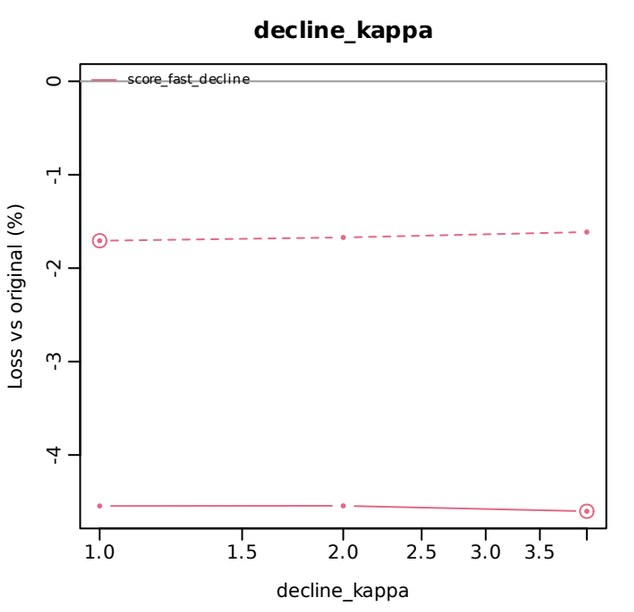


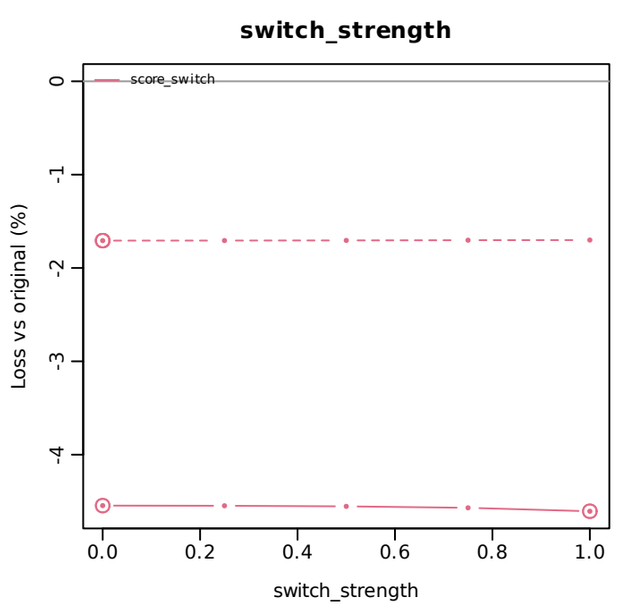


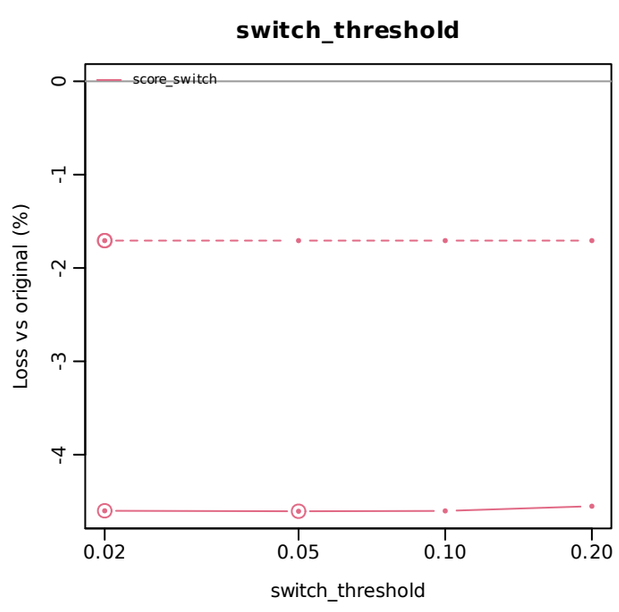


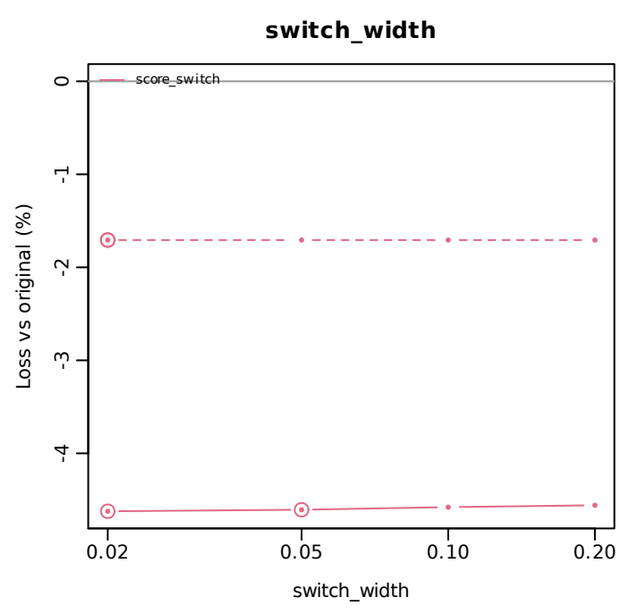


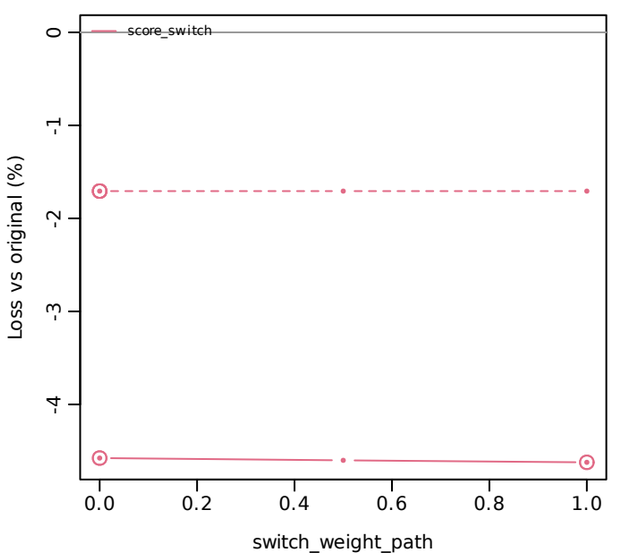


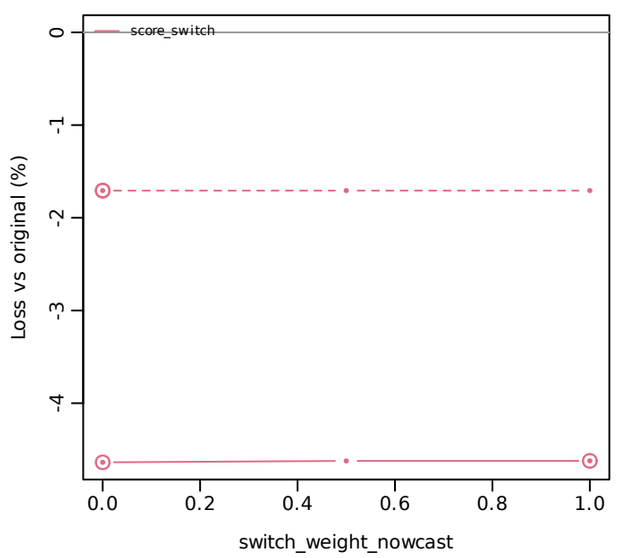


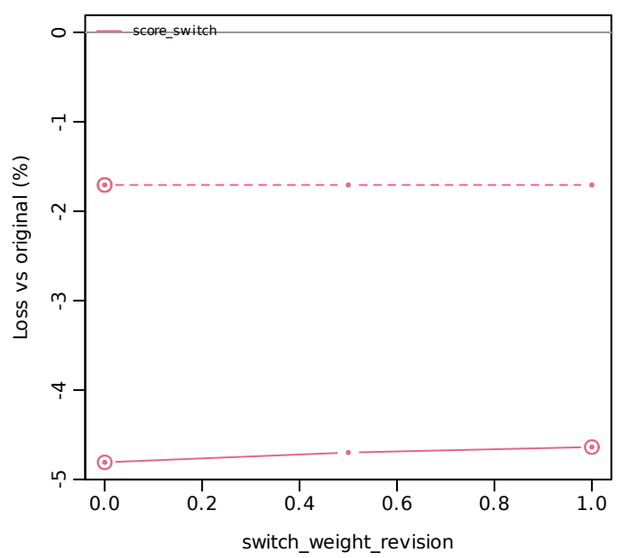


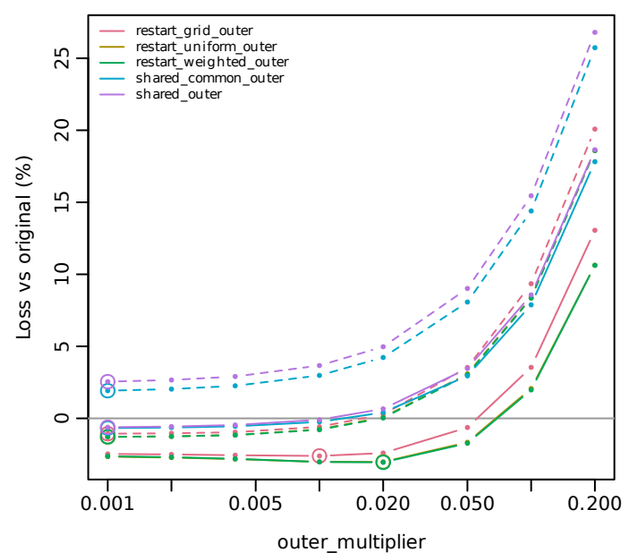












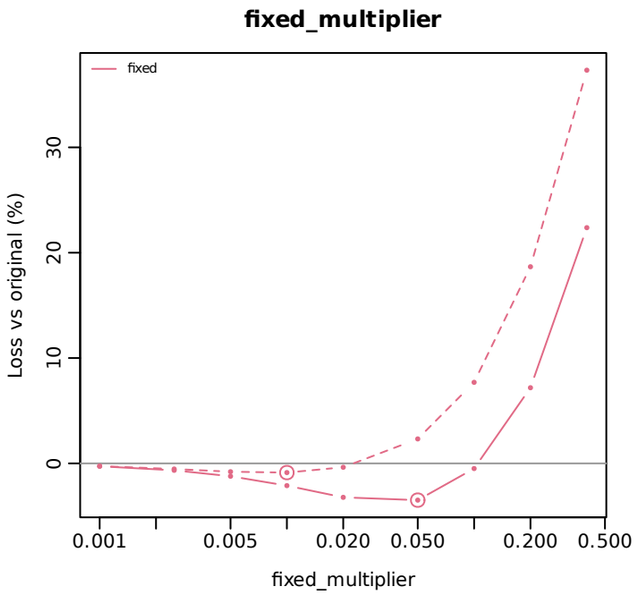
—— log1p tracker (loss of log(counts + 1)) - - - identity tracker (loss in counts per 100k residents) o chosen value

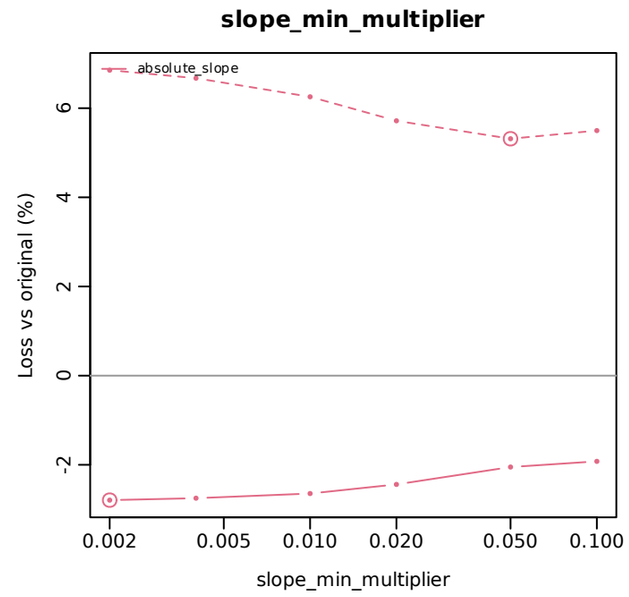
Mean quantile loss relative to the original forecasts (mean over locations and horizons) on the scale of each tracker

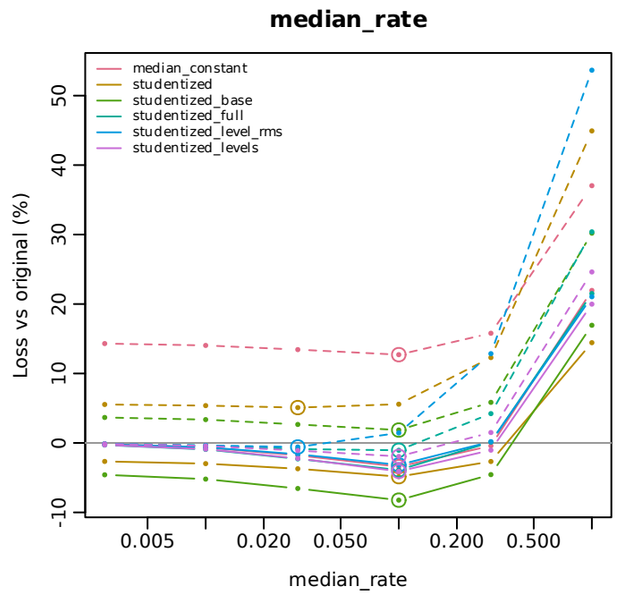
flu | design hub_first_months | feedback delay 1 | tuning period Nov 2024 to May 2025 | horizons -1 to 3 | 52 locations

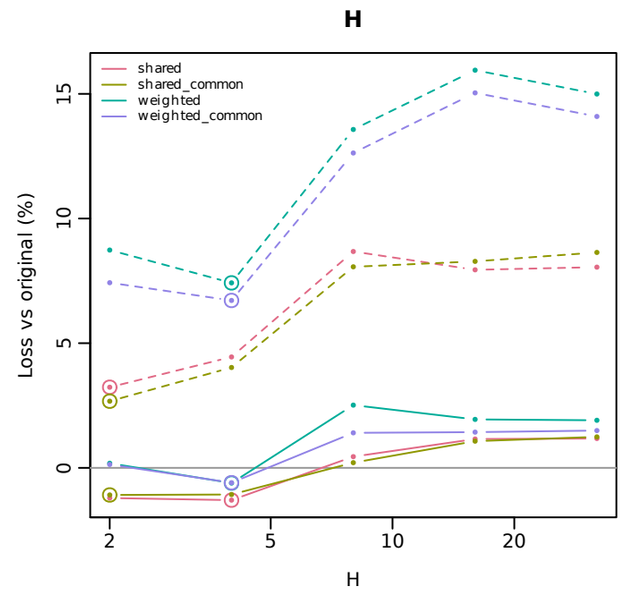
Mean quantile loss relative to the original forecasts (mean over locations and horizons) on the scale of each tracker

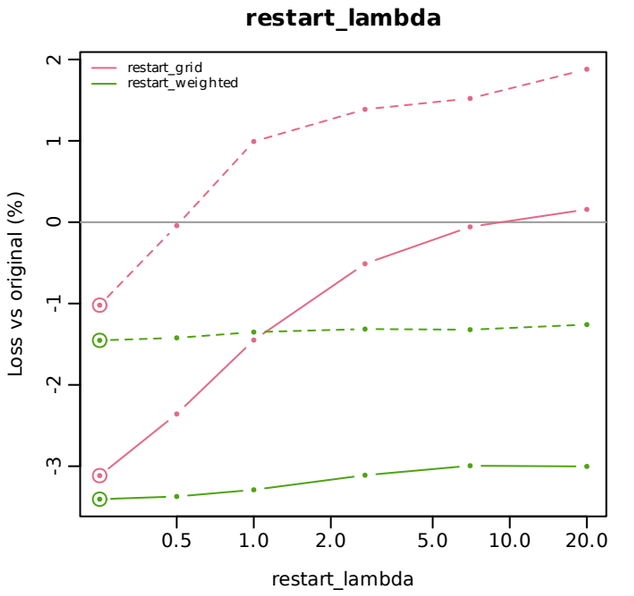
flu | design hub_first_months | feedback delay 0 + revisions (re-run) | tuning period Nov 2024 to May 2025 | horizons -1 to 3 | 52 locations

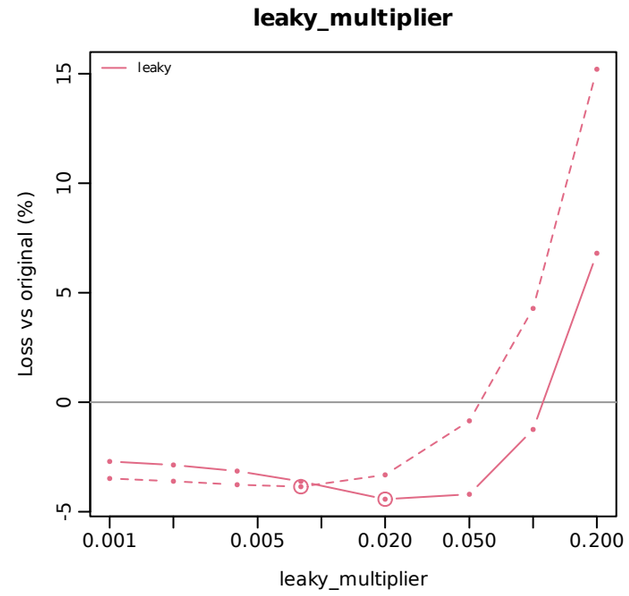


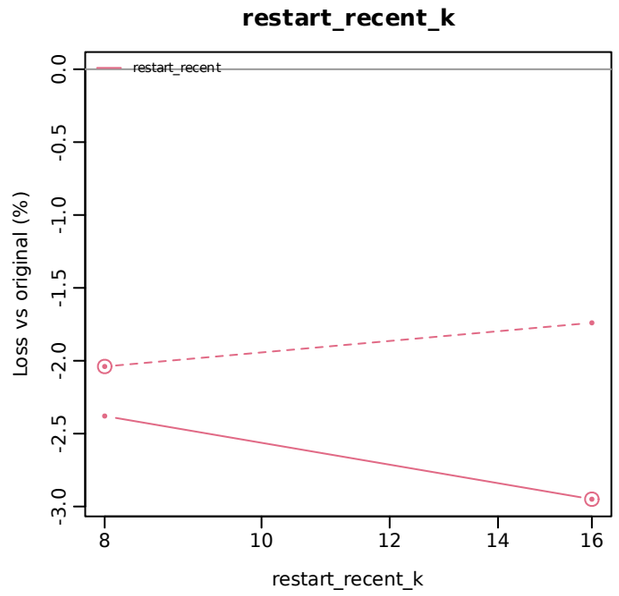


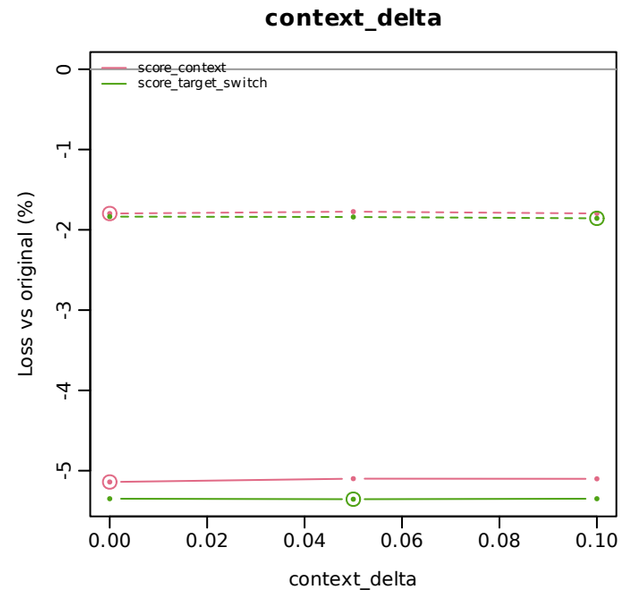


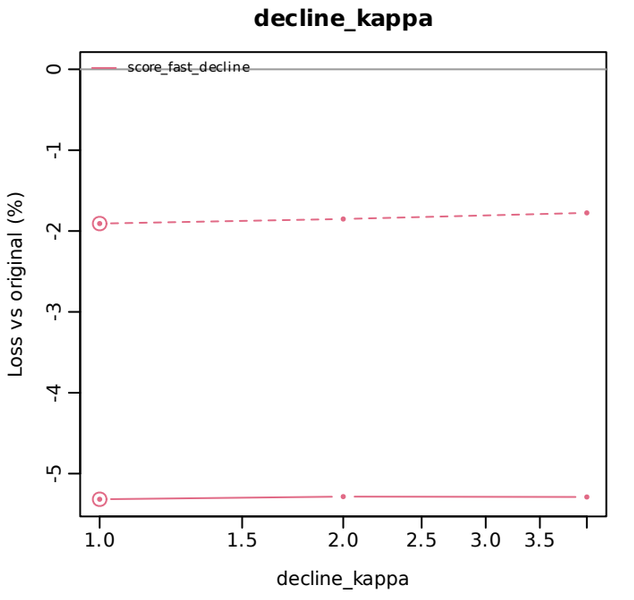


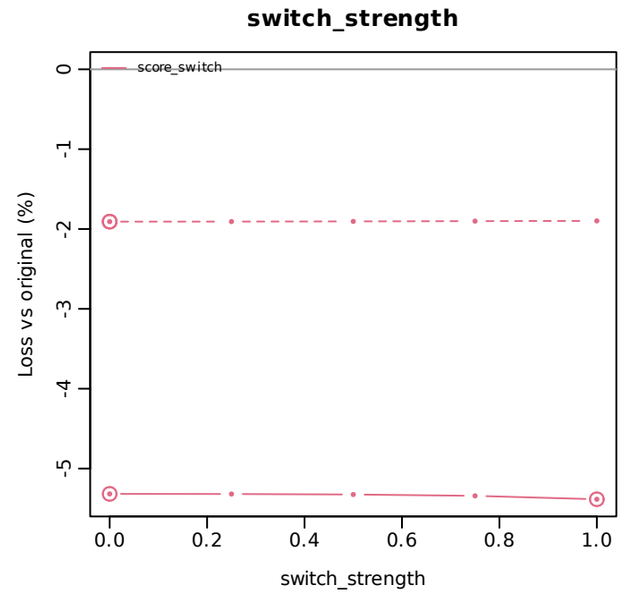


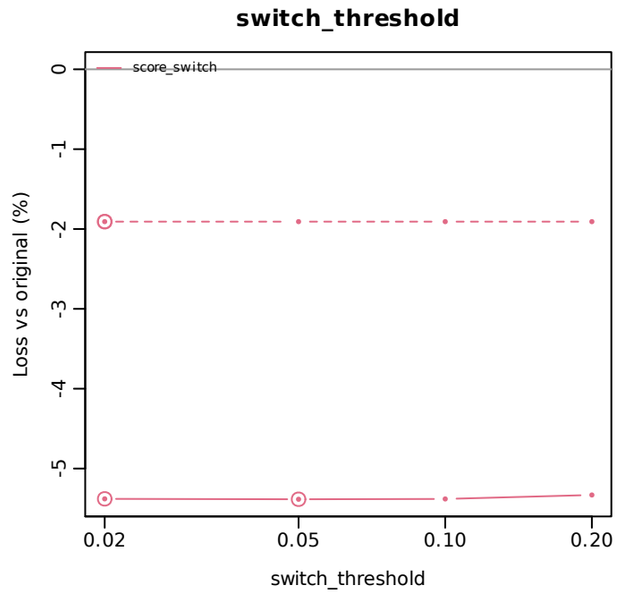


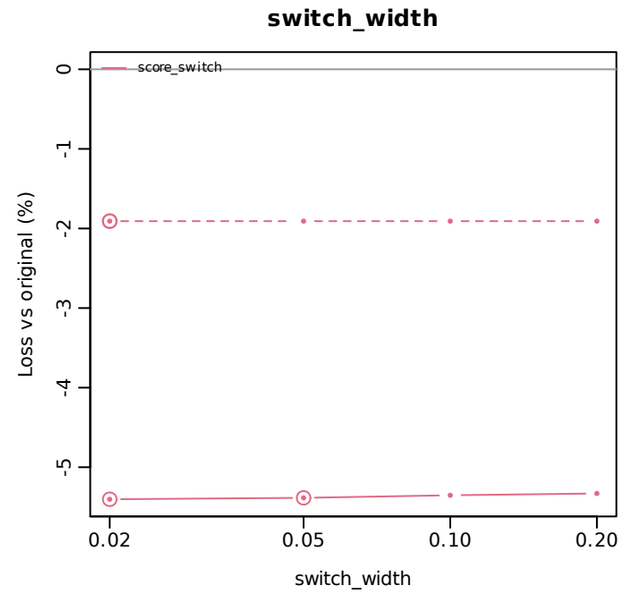


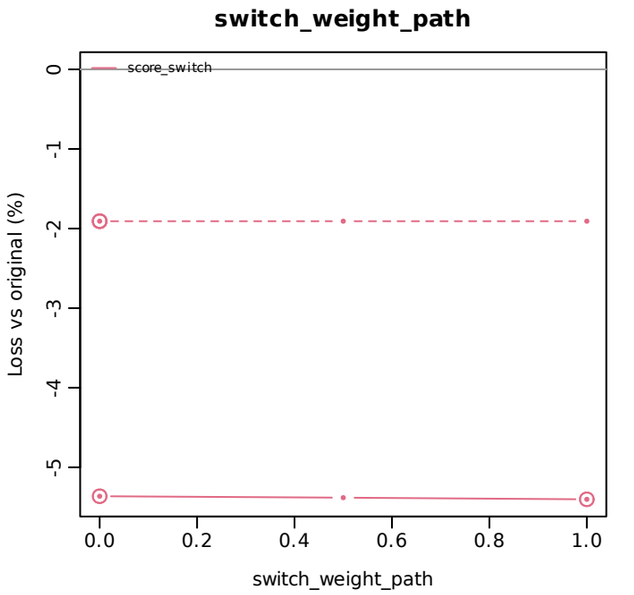


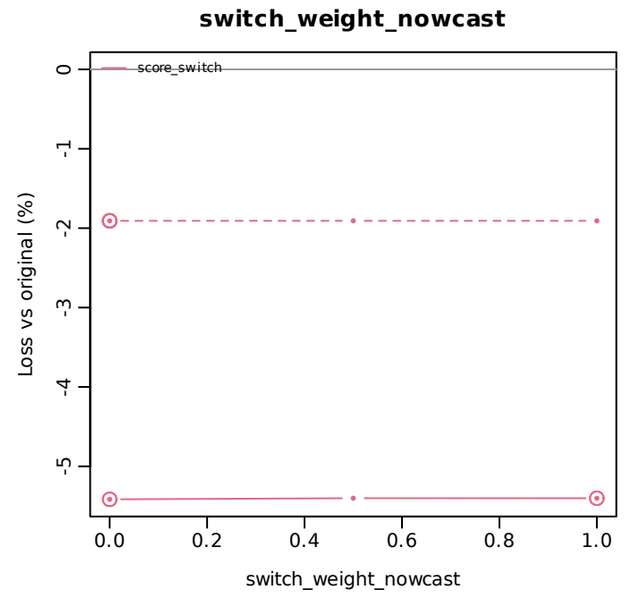


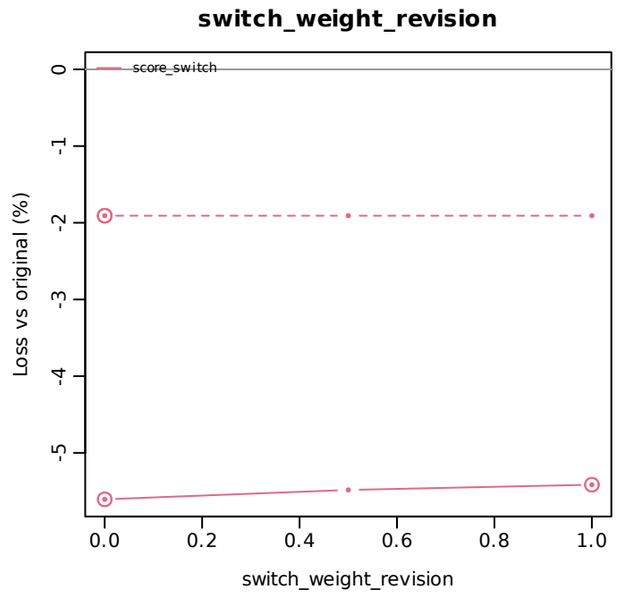


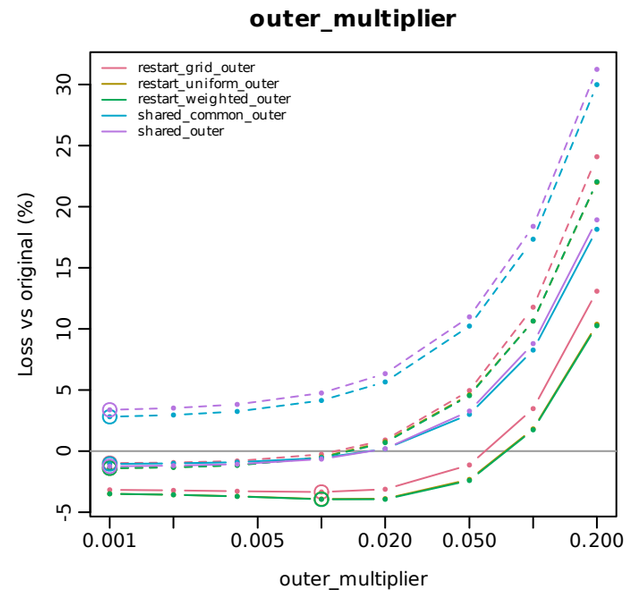












—— log1p tracker (loss of log(counts + 1)) - - - identity tracker (loss in counts per 100k residents) o chosen value

Mean quantile loss relative to the original forecasts (mean over locations and horizons) on the scale of each tracker

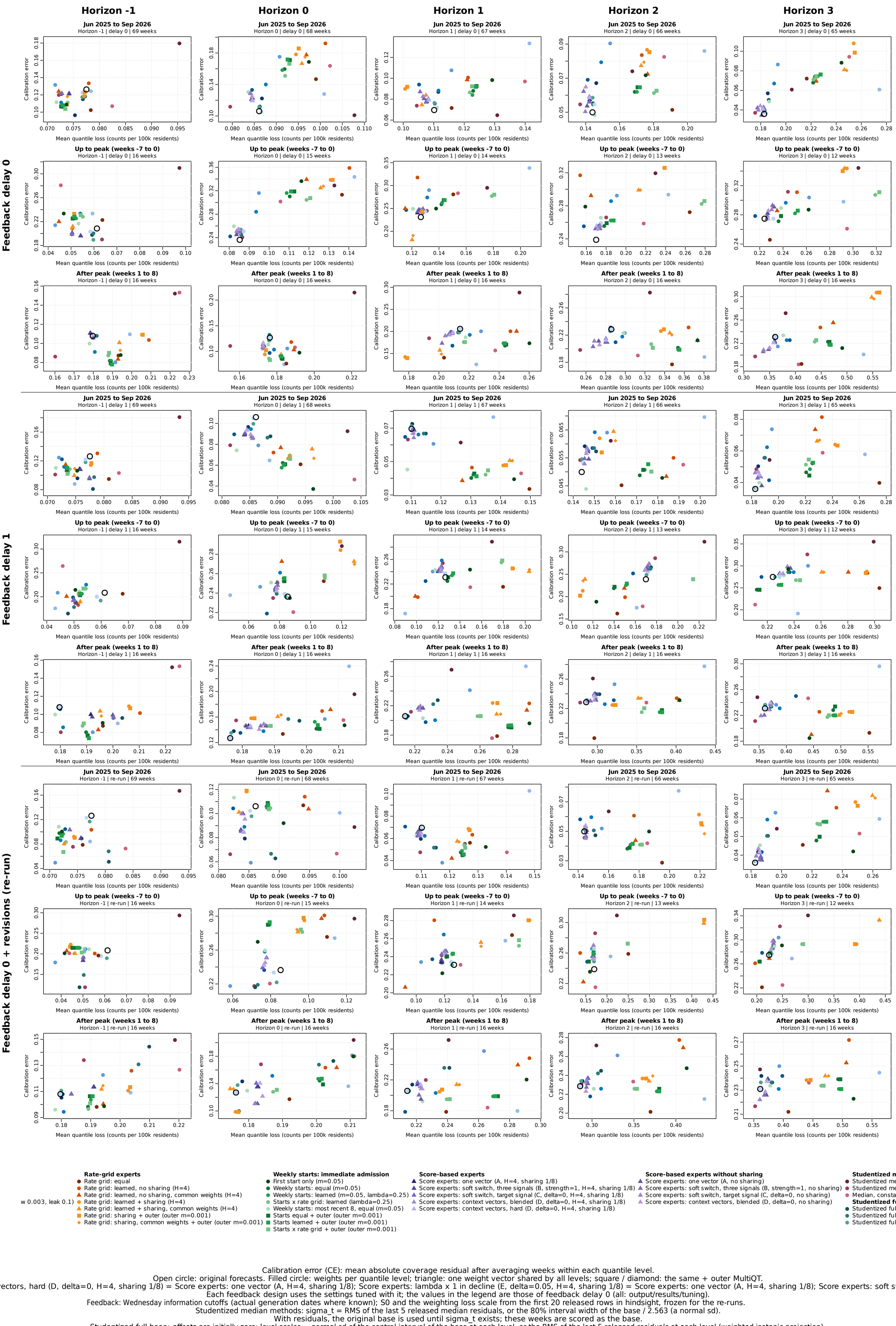
covid | all 52 locations | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.
Small: single locations; large: means over locations; axes: 1st to 99th percentile of both.

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
vectors, hard (D, delta=0, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: lambda x 1 in decline (E, delta=0.05, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: soft sw
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday information cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the 80% interval width of the base / 2.563 (a normal sd).
With residuals, the original base is used until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets are initially zero; level scales = normal sd of the central interval of the base at each level, or the RMS of the last 5 released residuals at each level (weighted isotonic projection)

covid | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.



Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
vectors, hard (D, delta=0, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: lambda x 1 in decline (E, delta=0.05, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: soft sw
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday information cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the 80% interval width of the base / 2.563 (a normal sd).
With residuals, the original base is used until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets are initially zero; level scales = normal sd of the central interval of the base at each level, or the RMS of the last 5 released residuals at each level (weighted isotonic projection)

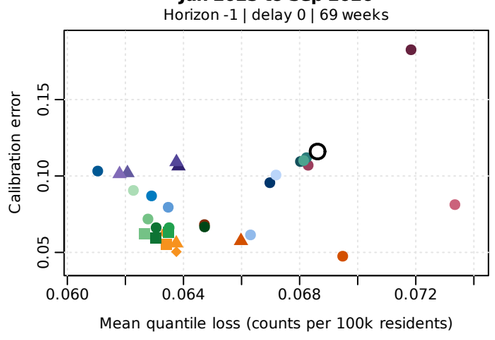
covid | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

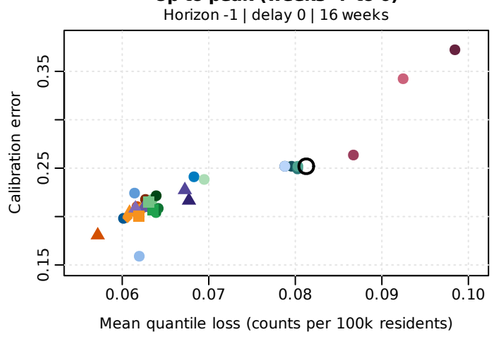
Horizon -1

Feedback delay 0

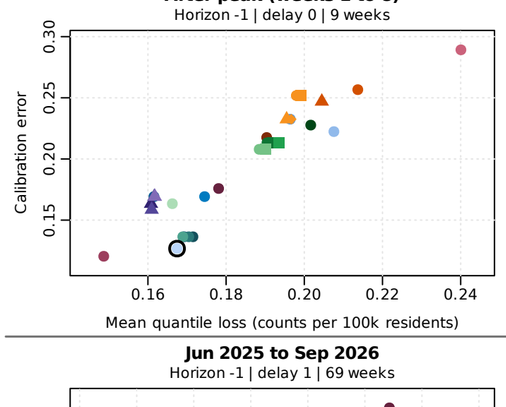
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 16 weeks

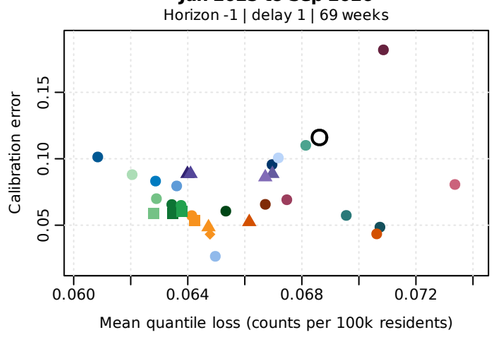


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 9 weeks

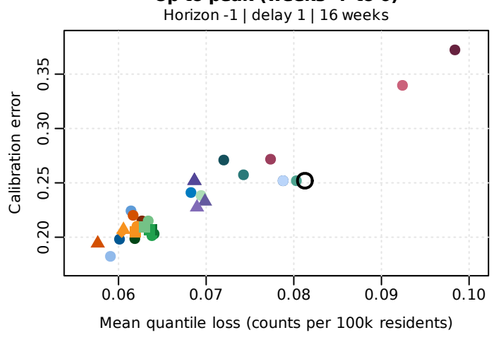


Feedback delay 1

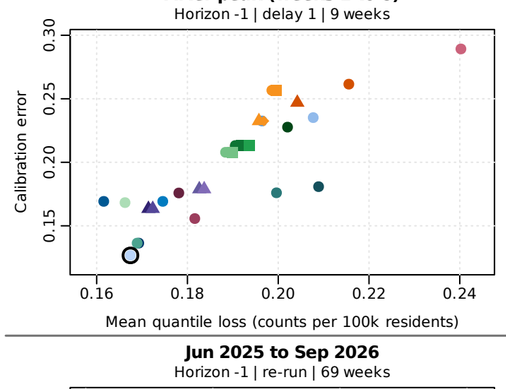
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 16 weeks

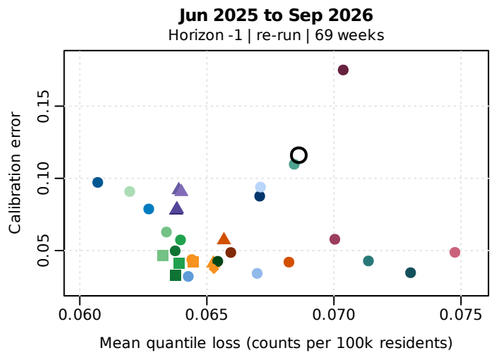


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 9 weeks

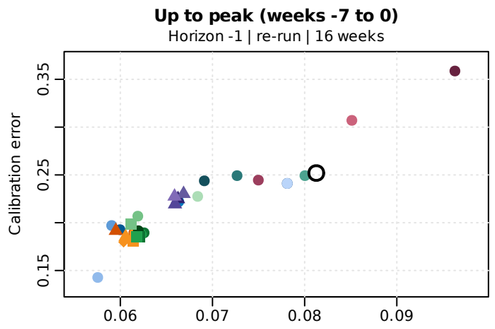


Feedback delay 0 + revisions (re-run)

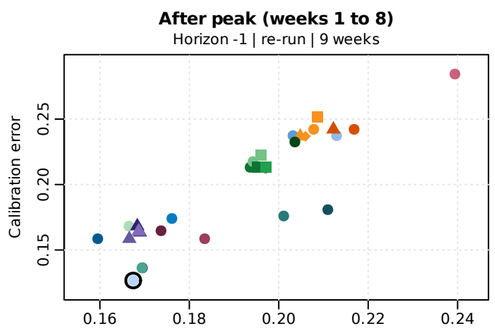
Jun 2025 to Sep 2026
Horizon -1 | re-run | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 16 weeks



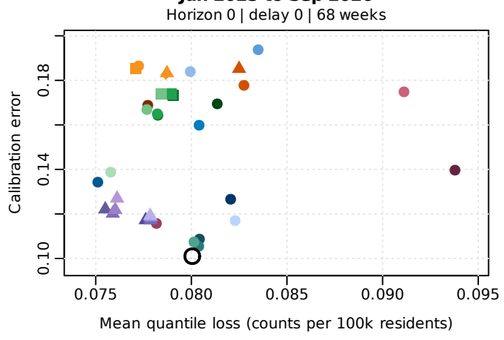
After peak (weeks 1 to 8)
Horizon -1 | re-run | 9 weeks



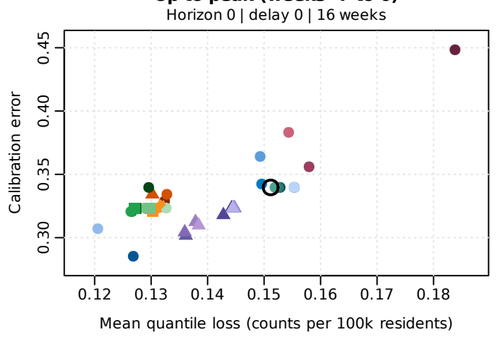
Horizon 0

Feedback delay 0

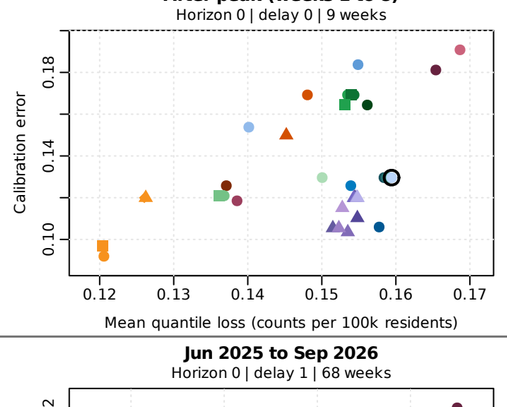
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 16 weeks

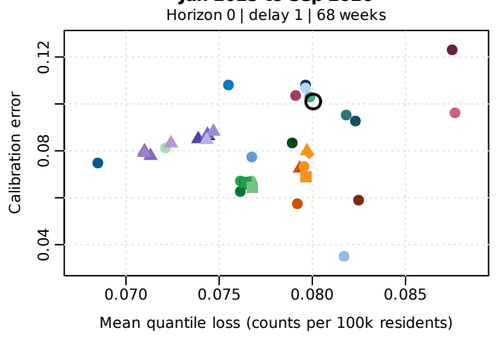


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 9 weeks

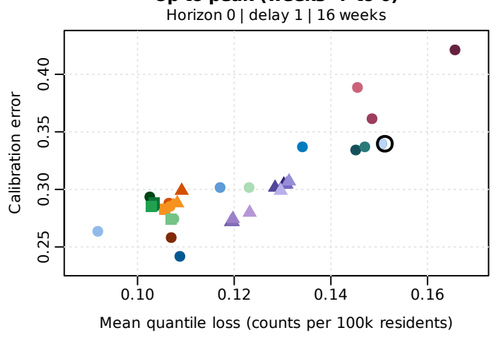


Feedback delay 1

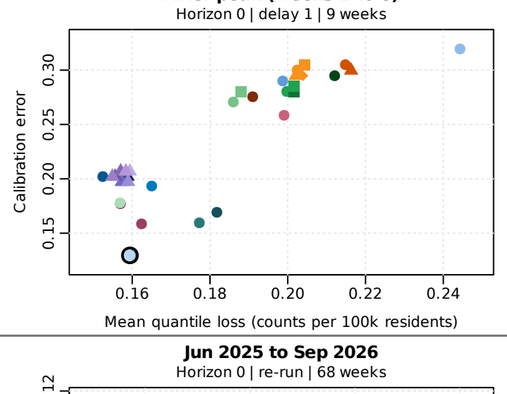
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 16 weeks

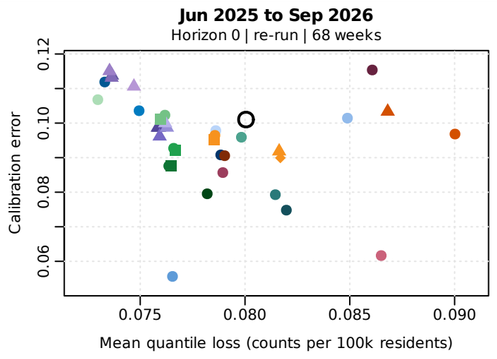


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 9 weeks

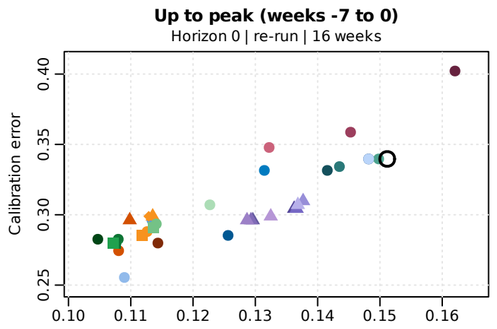


Feedback delay 0 + revisions (re-run)

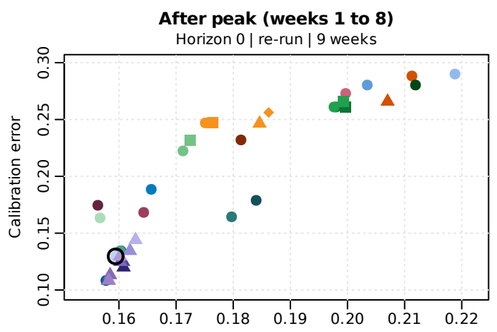
Jun 2025 to Sep 2026
Horizon 0 | re-run | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 16 weeks



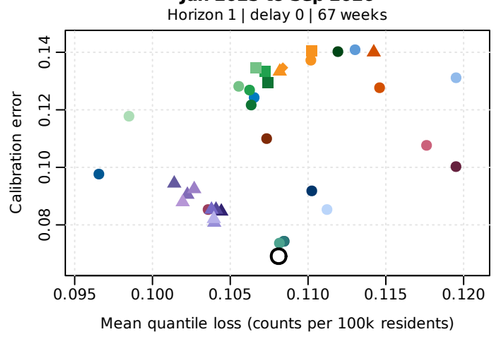
After peak (weeks 1 to 8)
Horizon 0 | re-run | 9 weeks



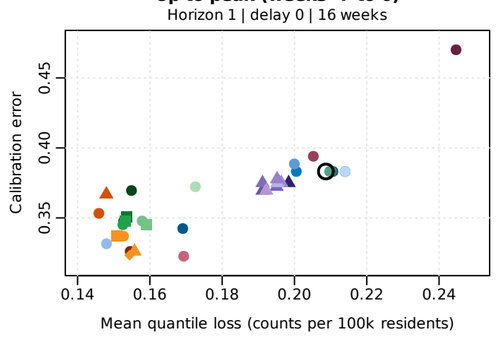
Horizon 1

Feedback delay 0

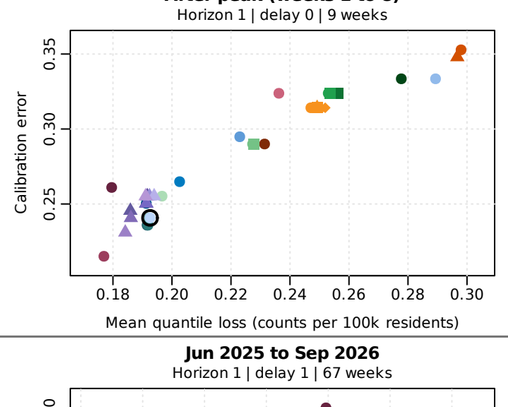
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 16 weeks

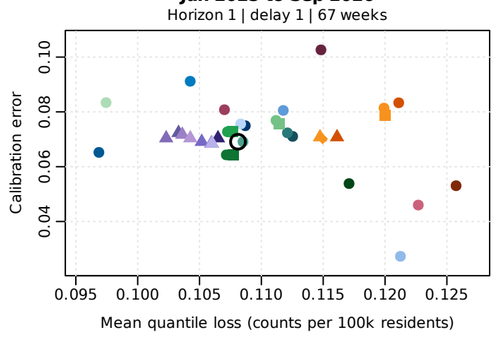


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 9 weeks

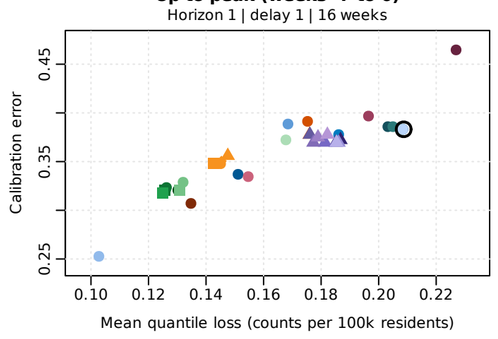


Feedback delay 1

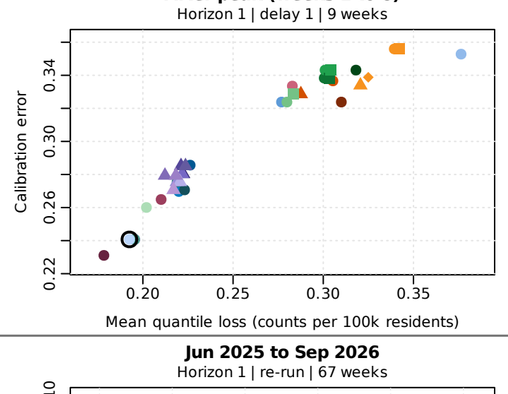
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 16 weeks

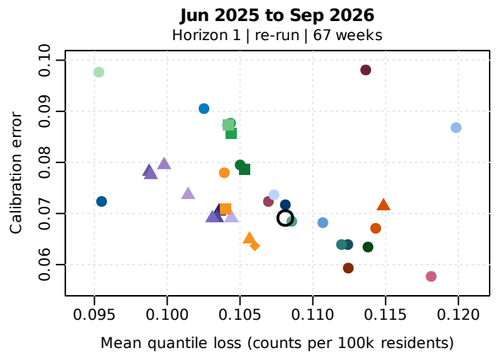


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 9 weeks

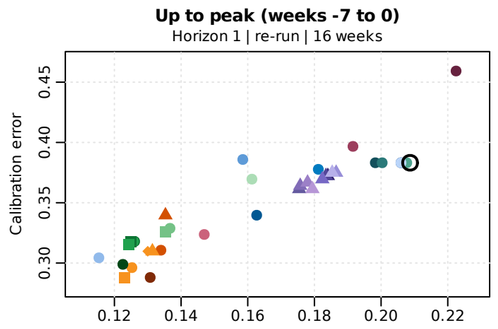


Feedback delay 0 + revisions (re-run)

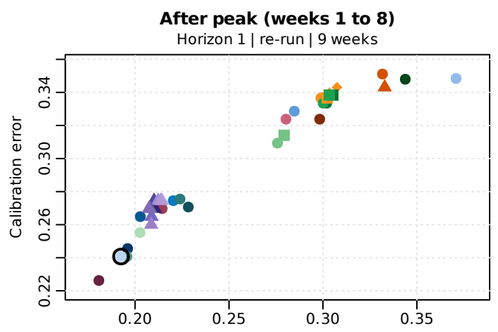
Jun 2025 to Sep 2026
Horizon 1 | re-run | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 16 weeks



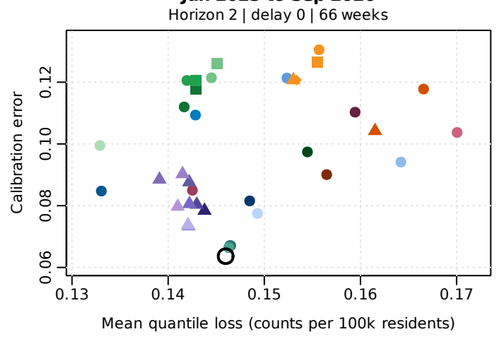
After peak (weeks 1 to 8)
Horizon 1 | re-run | 9 weeks



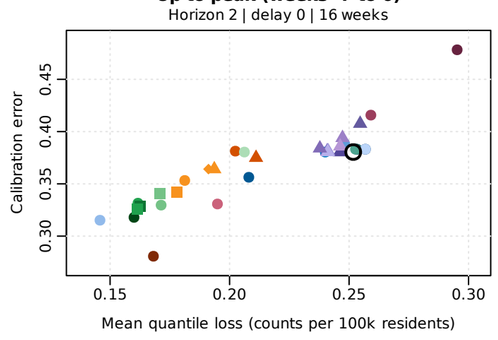
Horizon 2

Feedback delay 0

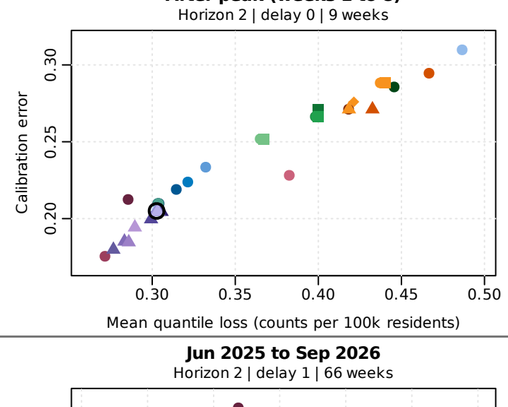
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 16 weeks

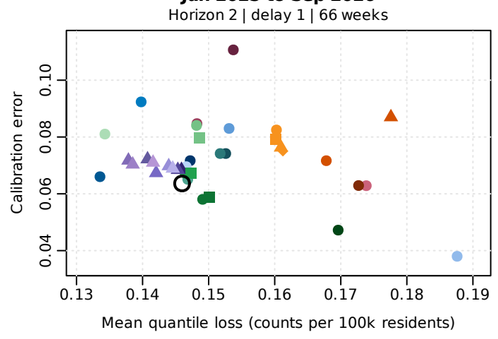


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 9 weeks

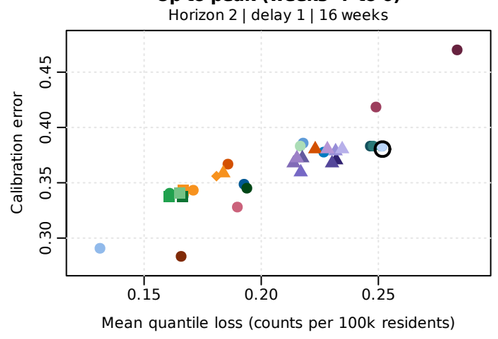


Feedback delay 1

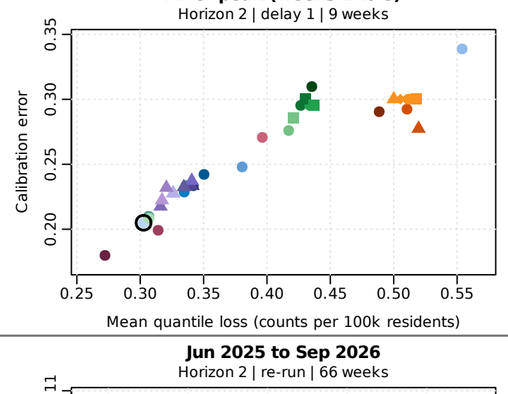
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 16 weeks

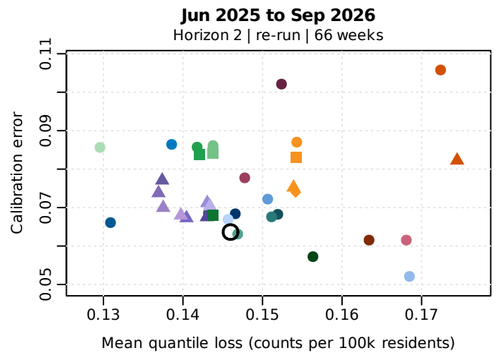


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 9 weeks

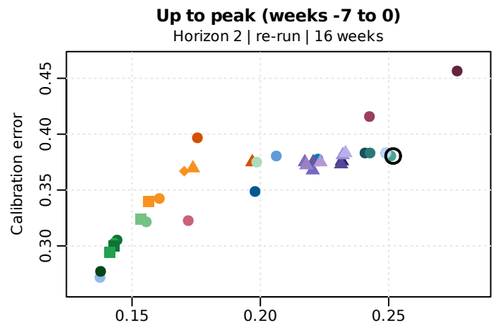


Feedback delay 0 + revisions (re-run)

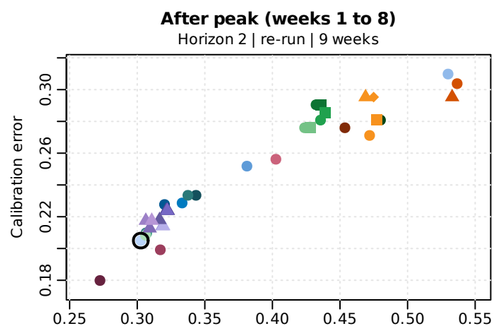
Jun 2025 to Sep 2026
Horizon 2 | re-run | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 16 weeks



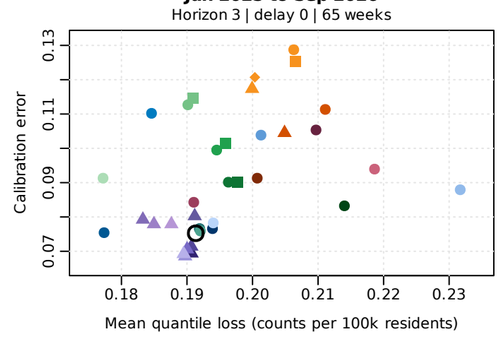
After peak (weeks 1 to 8)
Horizon 2 | re-run | 9 weeks



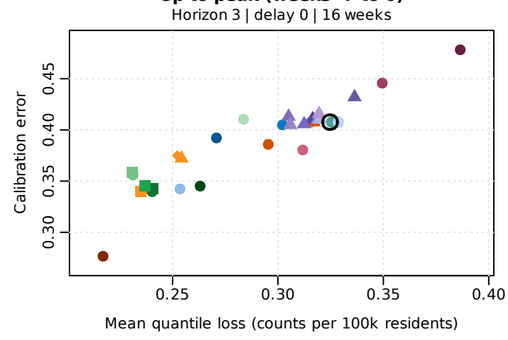
Horizon 3

Feedback delay 0

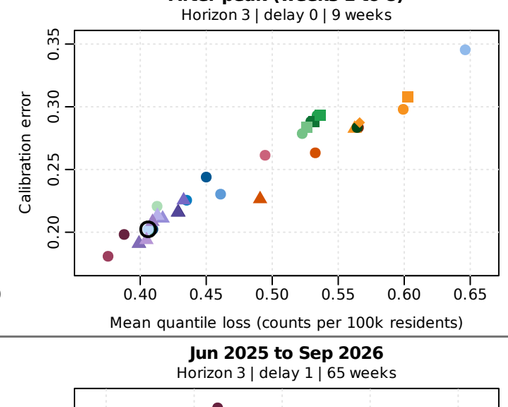
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 16 weeks

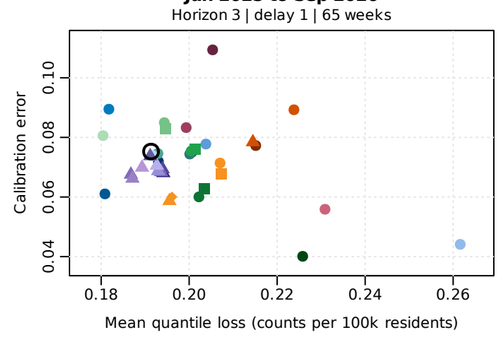


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 9 weeks

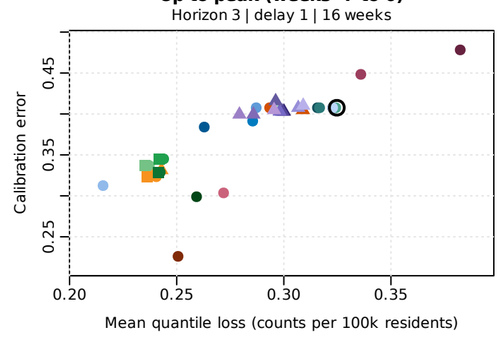


Feedback delay 1

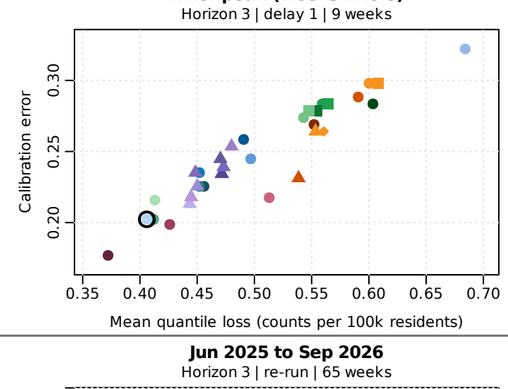
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 16 weeks

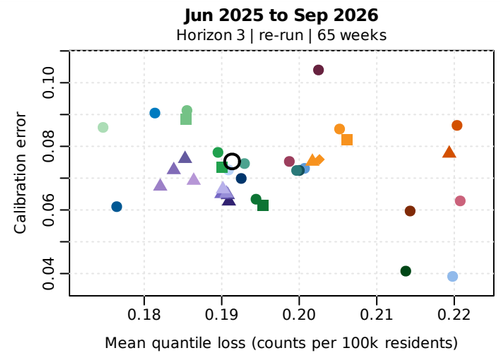


After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 9 weeks

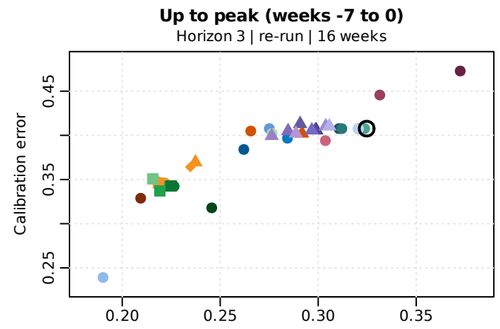


Feedback delay 0 + revisions (re-run)

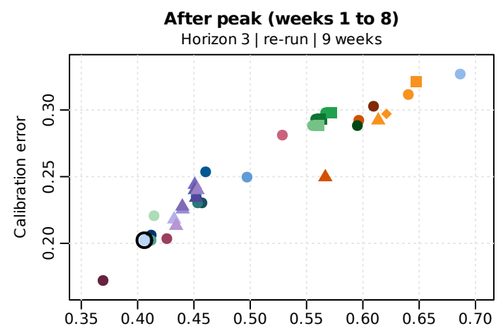
Jun 2025 to Sep 2026
Horizon 3 | re-run | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 16 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 9 weeks



Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=4)
- Rate grid: learned + sharing, common weights (H=4)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

w 0.003, leak 0.1)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Score-based experts
- Score experts: one vector (A, H=4, sharing 1/8)
- Score experts: soft switch, three signals (B, strength=1, H=4, sharing 1/8)
- Score experts: soft switch, target signal (C, delta=0, H=4, sharing 1/8)
- Score experts: context vectors, blended (D, delta=0, H=4, sharing 1/8)
- Score experts: context vectors, hard (D, delta=0, H=4, sharing 1/8)

Score-based experts without sharing
- Score experts: one vector (A, no sharing)
- Score experts: soft switch, three signals (B, strength=1, no sharing)
- Score experts: soft switch, target signal (C, delta=0, no sharing)
- Score experts: context vectors, blended (D, delta=0, no sharing)

Studentized m
- Studentized me
- Studentized me
- Median, consta

Studentized fu
- Studentized ful
- Studentized ful
- Studentized ful

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
vectors, hard (D, delta=0, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: lambda x 1 in decline (E, delta=0.05, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: soft sw
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday information cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the 80% interval width of the base / 2.563 (a normal sd).
With residuals, the original base is used until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets are initially zero; level scales = normal sd of the central interval of the base at each level, or the RMS of the last 5 released residuals at each level (weighted isotonic projection)

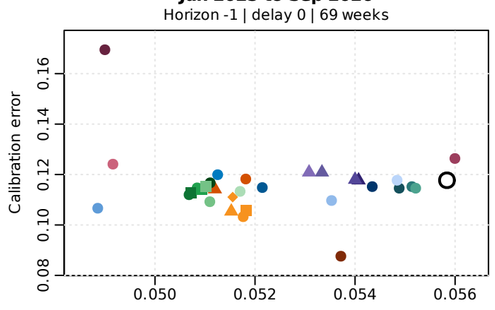
covid | Texas (48) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

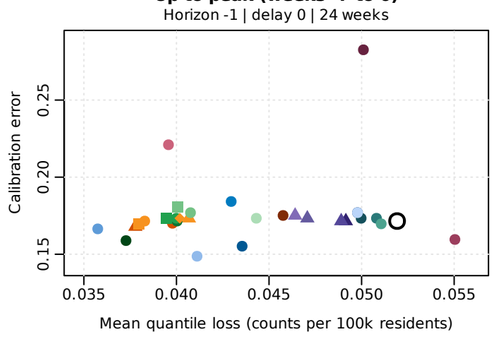
Horizon -1

Feedback delay 0

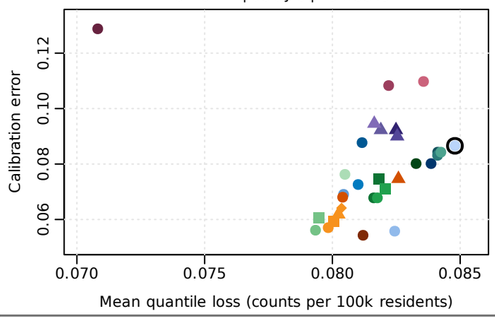
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 24 weeks

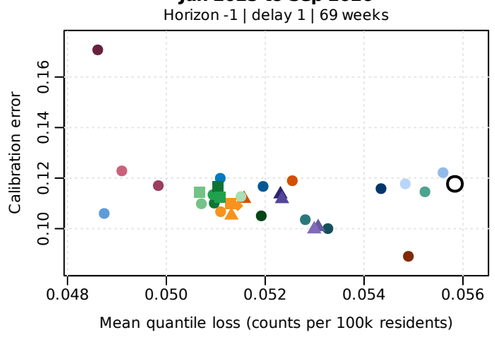


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 19 weeks

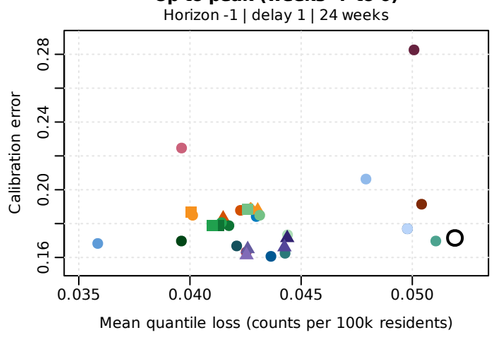


Feedback delay 1

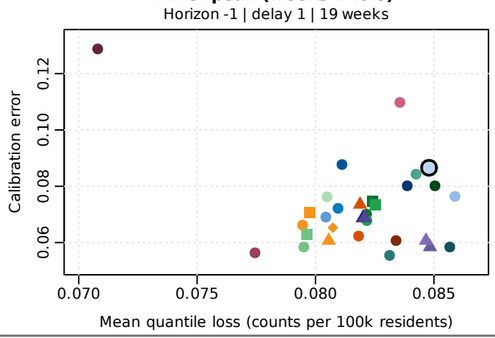
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 24 weeks

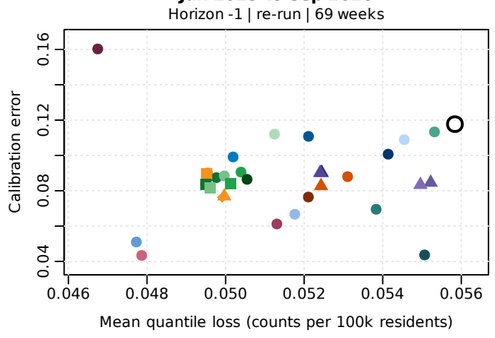


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 19 weeks

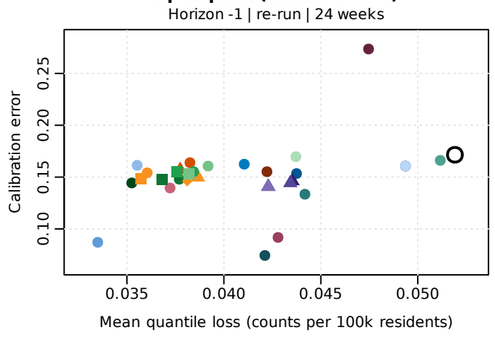


Feedback delay 0 + revisions (re-run)

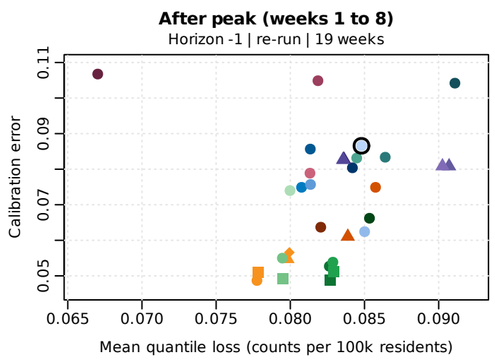
Jun 2025 to Sep 2026
Horizon -1 | re-run | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 24 weeks



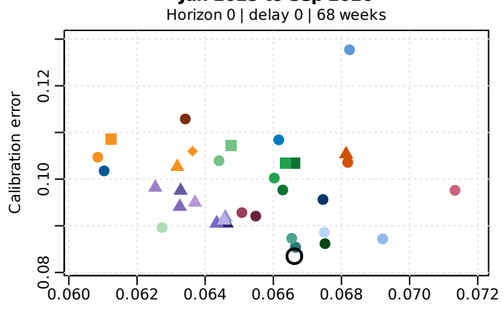
After peak (weeks 1 to 8)
Horizon -1 | re-run | 19 weeks



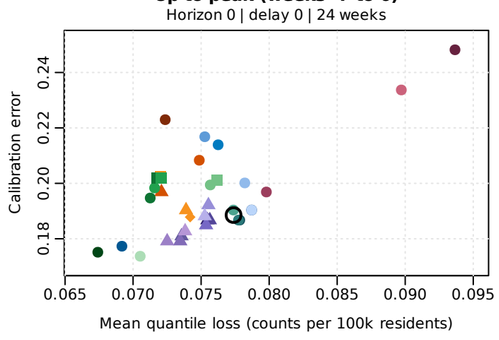
Horizon 0

Feedback delay 0

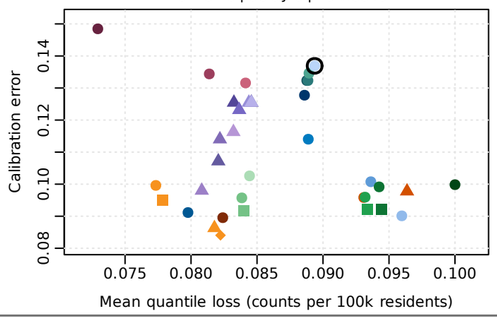
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 24 weeks

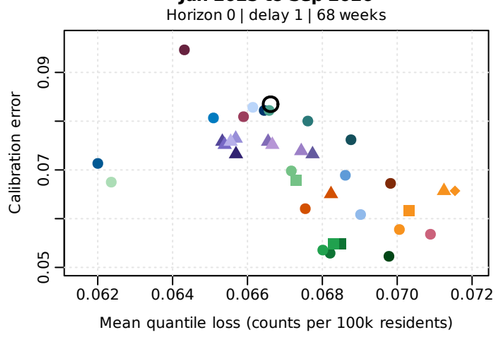


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 19 weeks

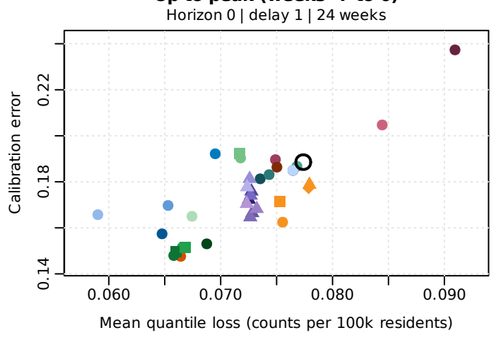


Feedback delay 1

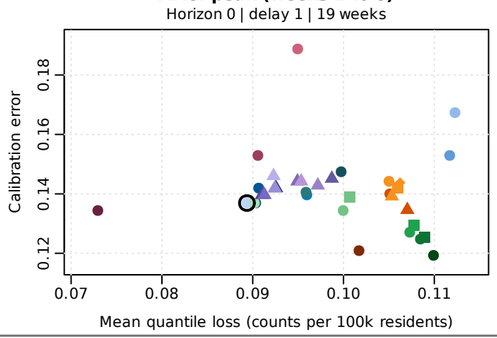
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 24 weeks

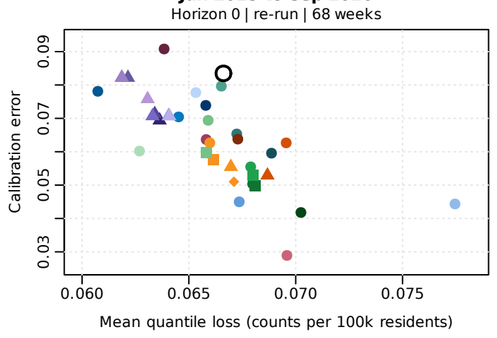


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 19 weeks

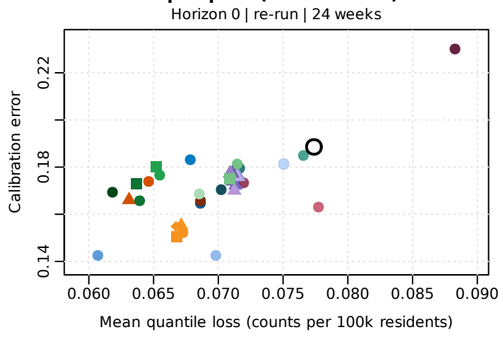


Feedback delay 0 + revisions (re-run)

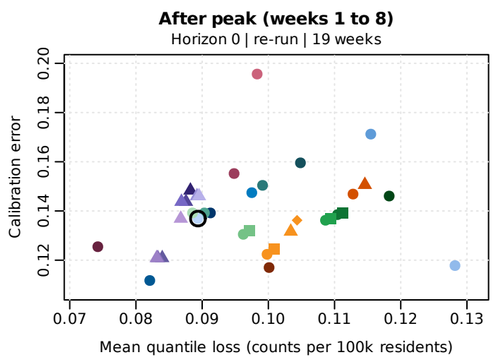
Jun 2025 to Sep 2026
Horizon 0 | re-run | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 24 weeks



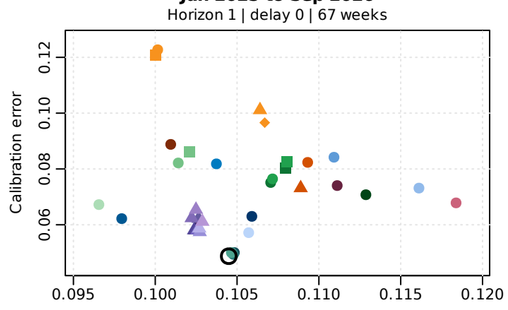
After peak (weeks 1 to 8)
Horizon 0 | re-run | 19 weeks



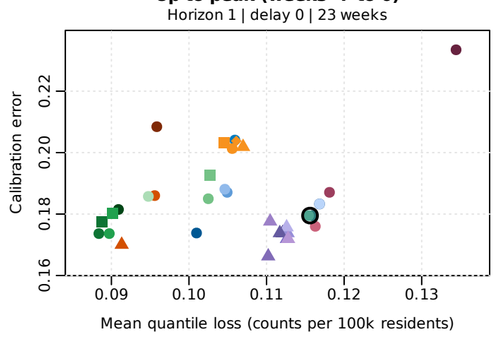
Horizon 1

Feedback delay 0

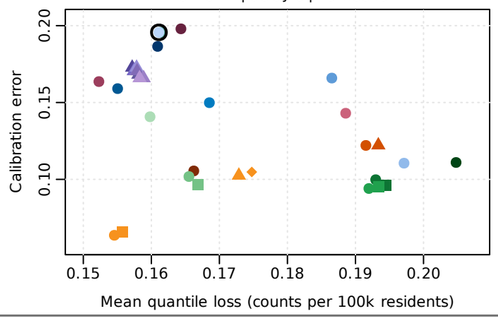
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 23 weeks

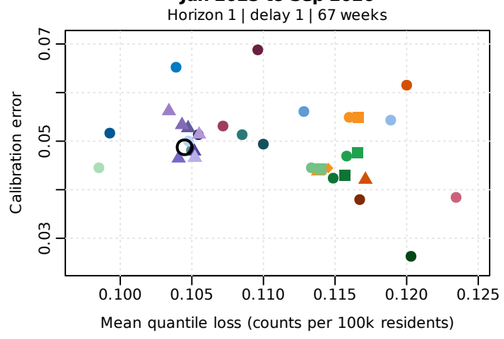


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 19 weeks

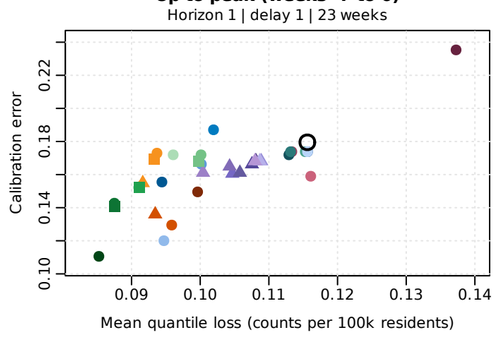


Feedback delay 1

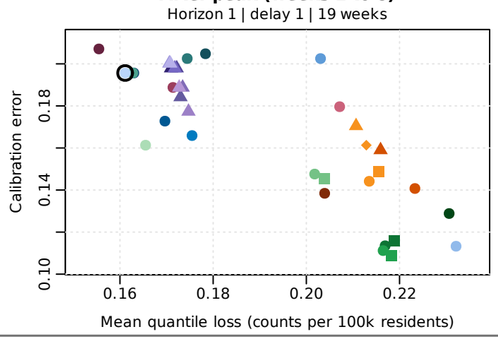
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 23 weeks

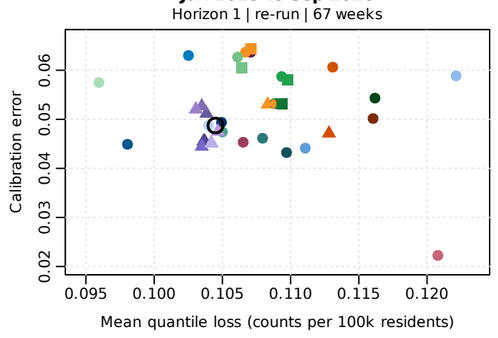


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 19 weeks

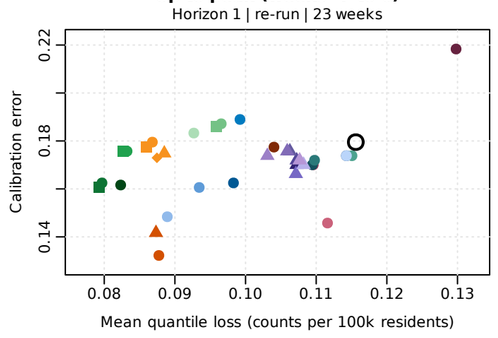


Feedback delay 0 + revisions (re-run)

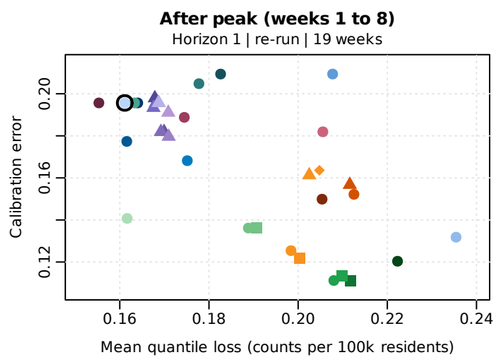
Jun 2025 to Sep 2026
Horizon 1 | re-run | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 23 weeks



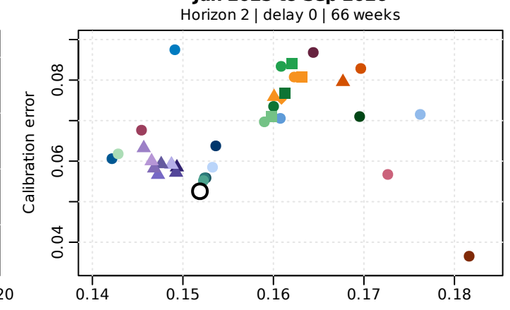
After peak (weeks 1 to 8)
Horizon 1 | re-run | 19 weeks



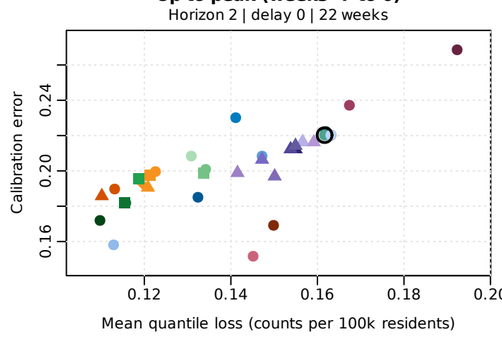
Horizon 2

Feedback delay 0

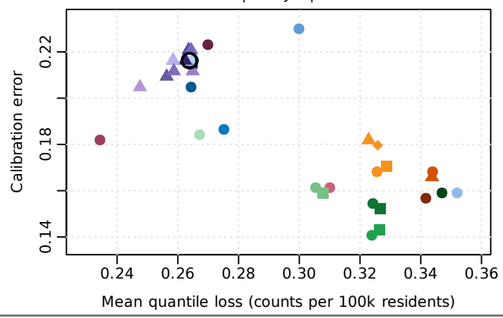
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 22 weeks

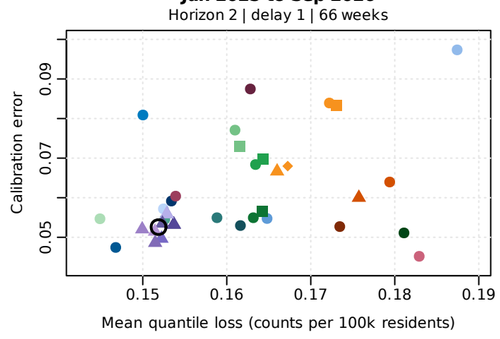


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 19 weeks

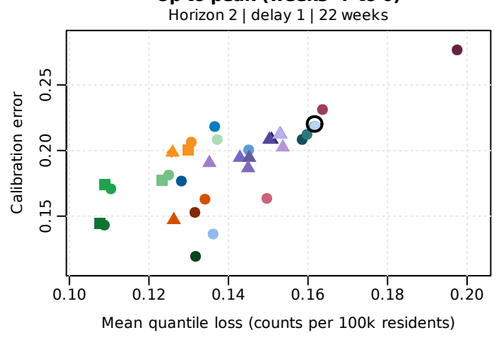


Feedback delay 1

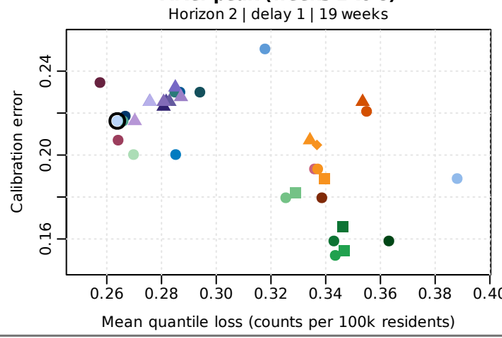
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 22 weeks

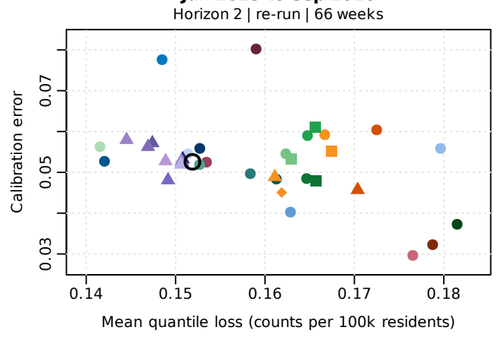


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 19 weeks

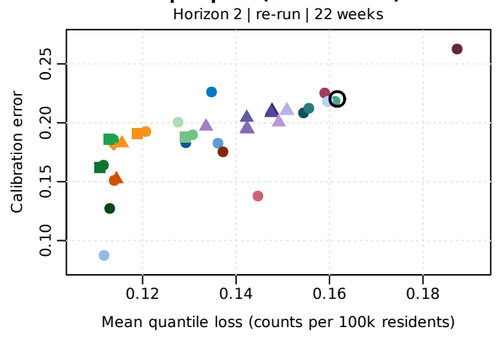


Feedback delay 0 + revisions (re-run)

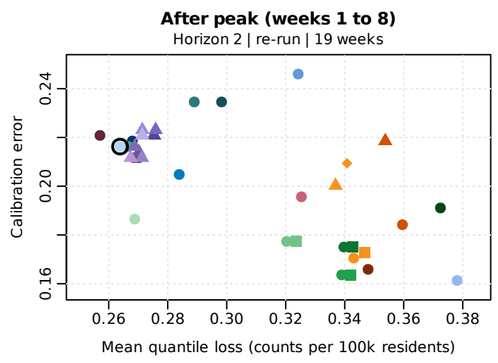
Jun 2025 to Sep 2026
Horizon 2 | re-run | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 22 weeks



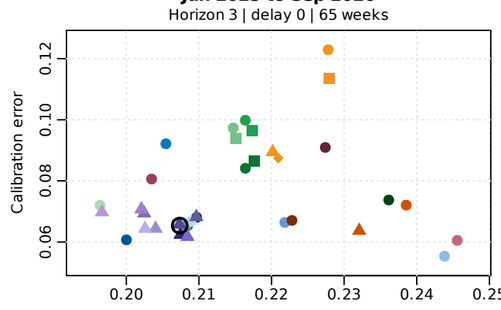
After peak (weeks 1 to 8)
Horizon 2 | re-run | 19 weeks



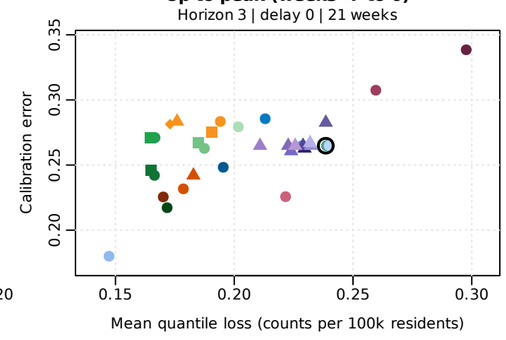
Horizon 3

Feedback delay 0

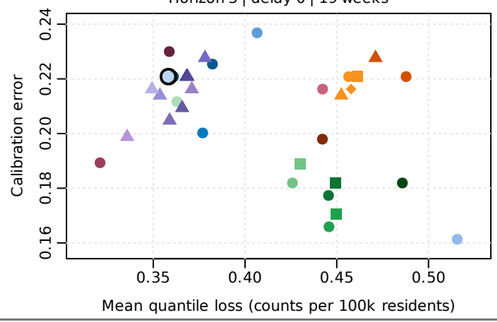
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 21 weeks

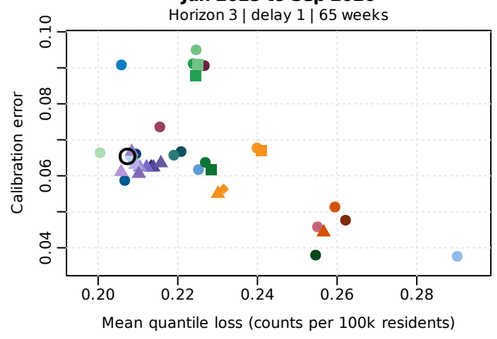


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 19 weeks

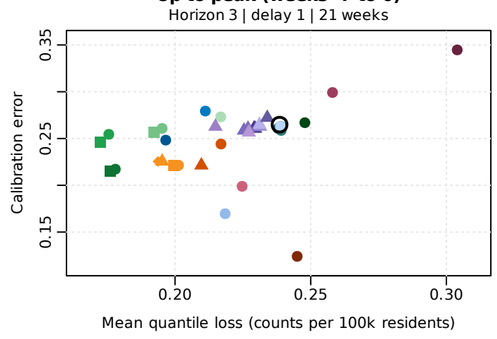


Feedback delay 1

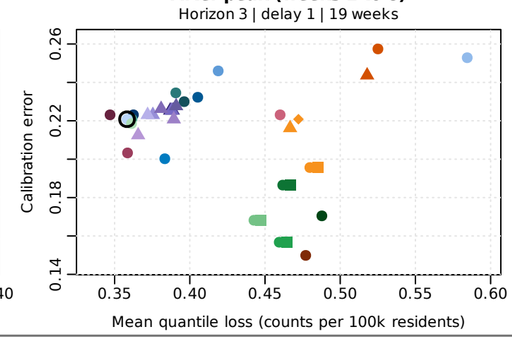
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 21 weeks

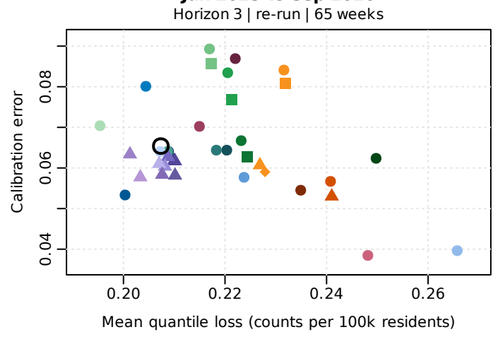


After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 19 weeks

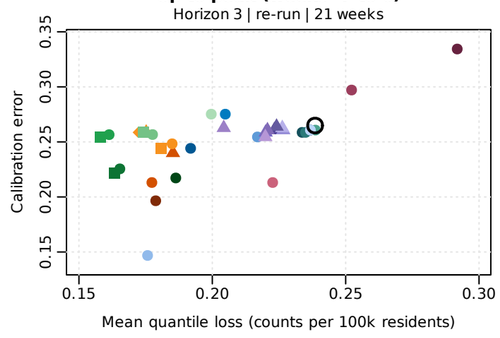


Feedback delay 0 + revisions (re-run)

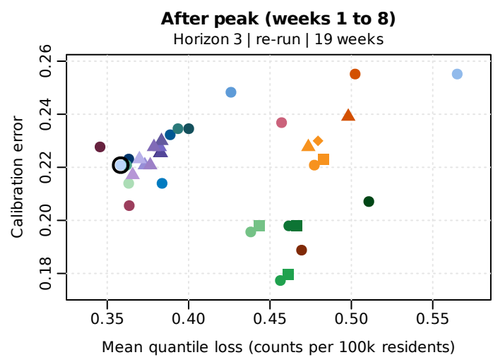
Jun 2025 to Sep 2026
Horizon 3 | re-run | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 21 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 19 weeks



Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=4)
- Rate grid: learned + sharing, common weights (H=4)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

w 0.003, leak 0.1)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Score-based experts
- Score experts: one vector (A, H=4, sharing 1/8)
- Score experts: soft switch, three signals (B, strength=1, H=4, sharing 1/8)
- Score experts: soft switch, target signal (C, delta=0, H=4, sharing 1/8)
- Score experts: context vectors, blended (D, delta=0, H=4, sharing 1/8)
- Score experts: context vectors, hard (D, delta=0, H=4, sharing 1/8)

Score-based experts without sharing
- Score experts: one vector (A, no sharing)
- Score experts: soft switch, three signals (B, strength=1, no sharing)
- Score experts: soft switch, target signal (C, delta=0, no sharing)
- Score experts: context vectors, blended (D, delta=0, no sharing)

Studentized m
- Studentized me
- Studentized me
- Median, consta

Studentized fu
- Studentized ful
- Studentized ful
- Studentized ful

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
vectors, hard (D, delta=0, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: lambda x 1 in decline (E, delta=0.05, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: soft sw
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday information cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the 80% interval width of the base / 2.563 (a normal sd).
With residuals, the original base is used until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets are initially zero; level scales = normal sd of the central interval of the base at each level, or the RMS of the last 5 released residuals at each level (weighted isotonic projection)

covid | New York (36) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday information cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the 80% interval width of the base / 2.563 (a normal sd).
With residuals, the original base is used until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets are initially zero; level scales = normal sd of the central interval of the base at each level, or the RMS of the last 5 released residuals at each level (weighted isotonic projection)

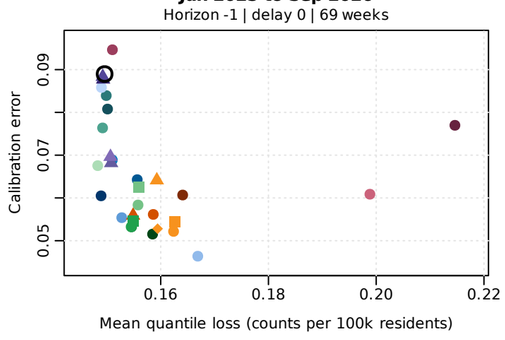
covid | Pennsylvania (42) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

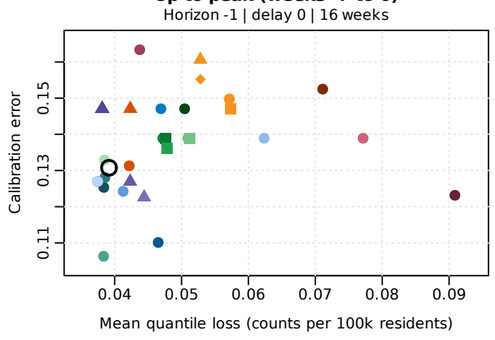
Horizon -1

Feedback delay 0

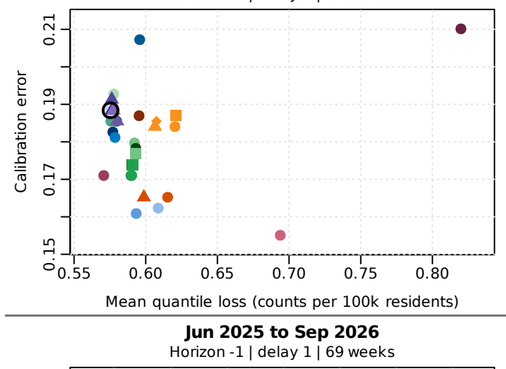
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 16 weeks

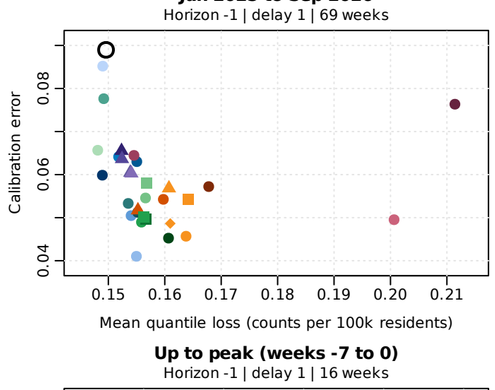


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 15 weeks

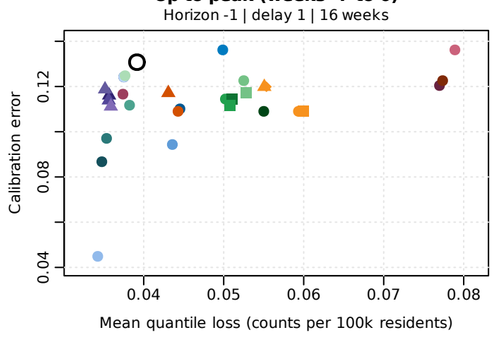


Feedback delay 1

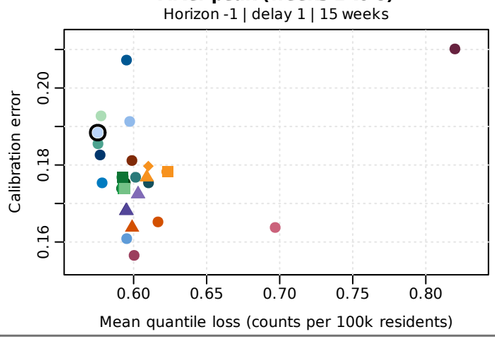
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 16 weeks

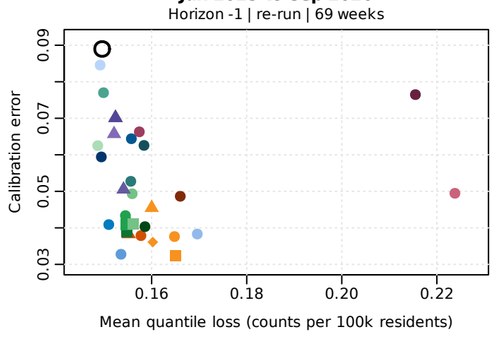


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 15 weeks

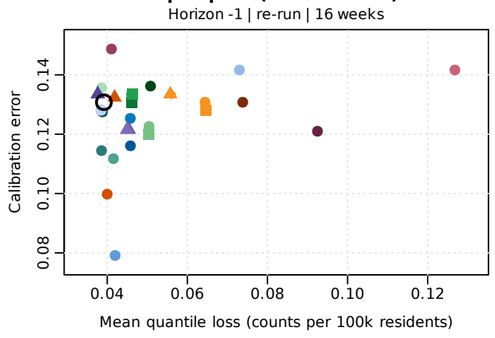


Feedback delay 0 + revisions (re-run)

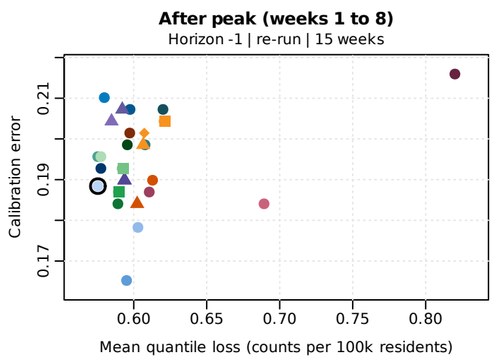
Jun 2025 to Sep 2026
Horizon -1 | re-run | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 16 weeks



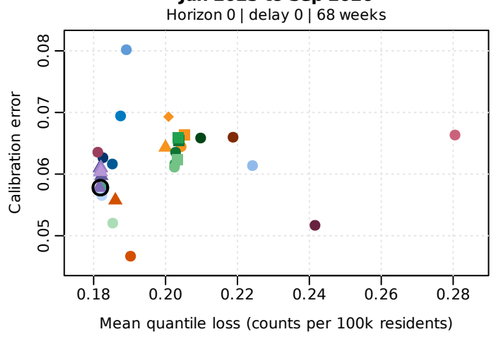
After peak (weeks 1 to 8)
Horizon -1 | re-run | 15 weeks



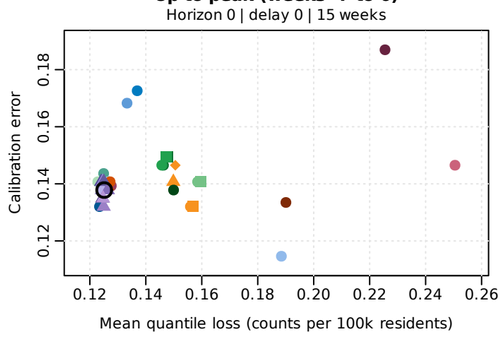
Horizon 0

Feedback delay 0

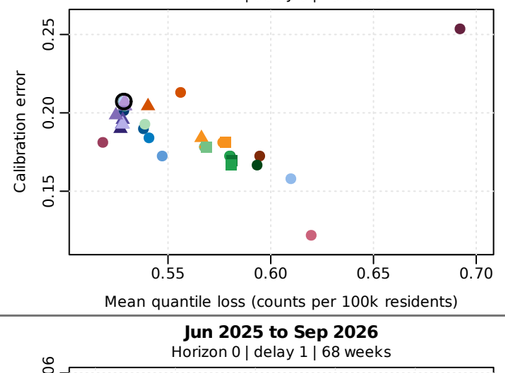
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 15 weeks

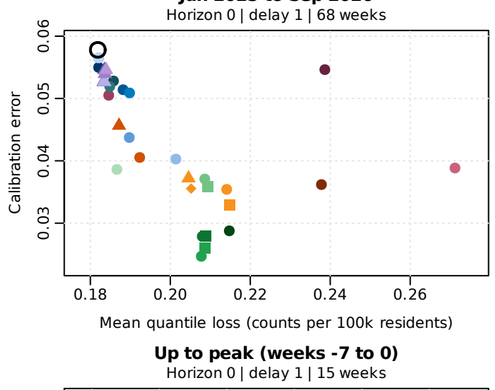


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 15 weeks

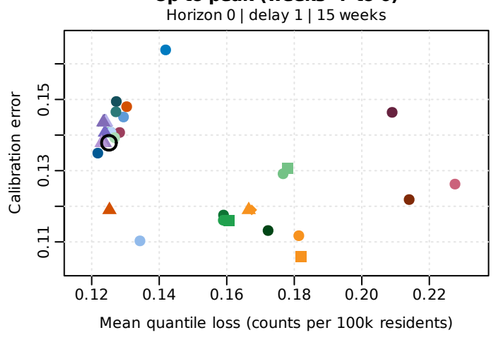


Feedback delay 1

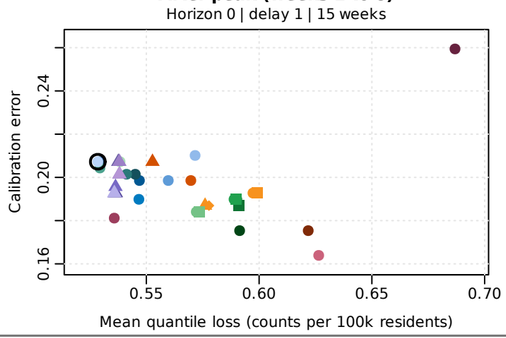
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 15 weeks

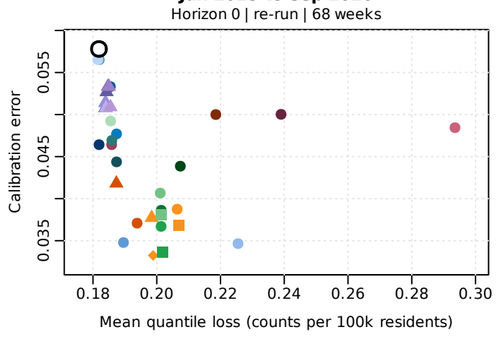


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 15 weeks

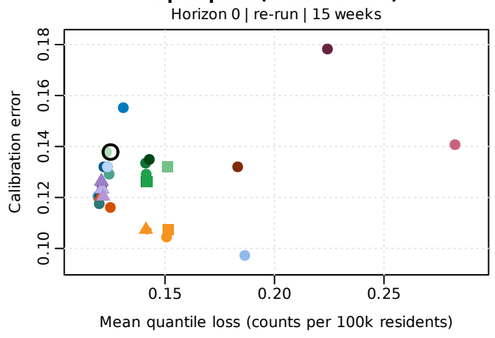


Feedback delay 0 + revisions (re-run)

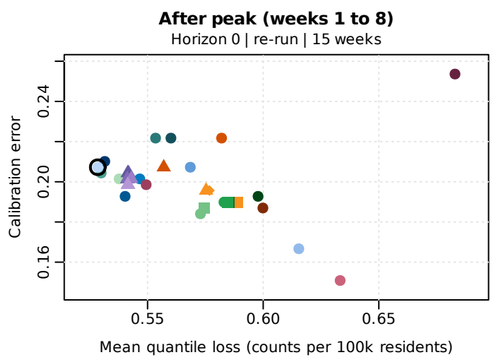
Jun 2025 to Sep 2026
Horizon 0 | re-run | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 15 weeks



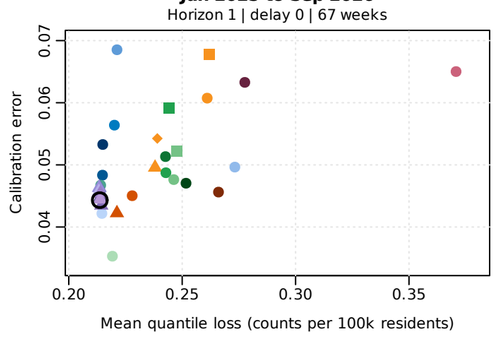
After peak (weeks 1 to 8)
Horizon 0 | re-run | 15 weeks



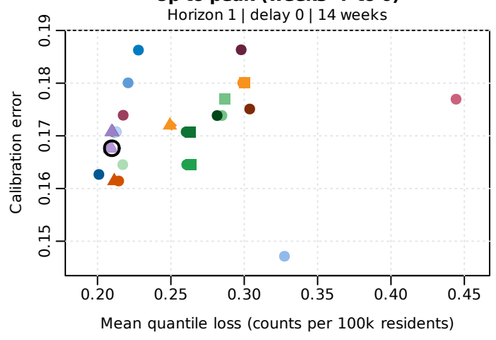
Horizon 1

Feedback delay 0

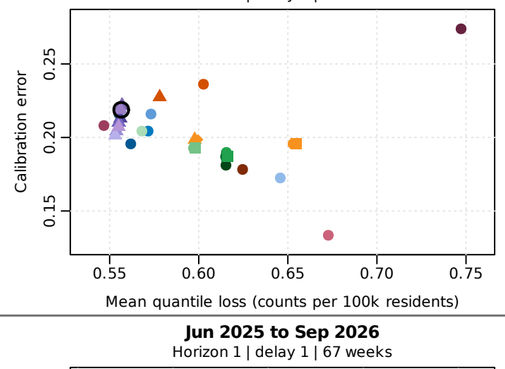
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 14 weeks

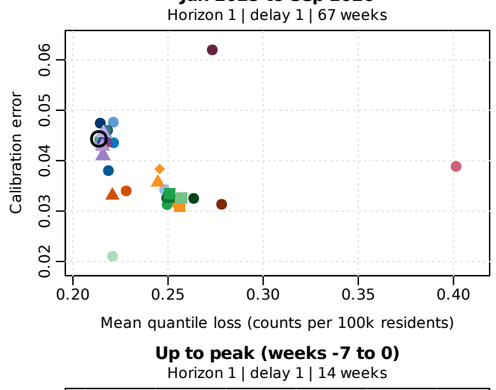


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 15 weeks

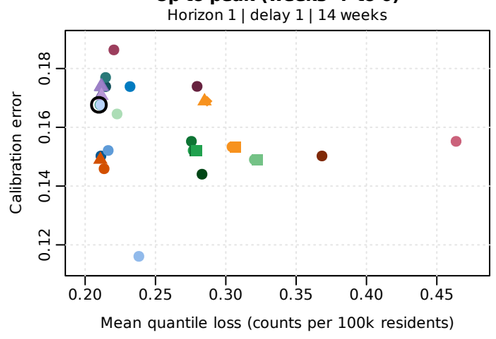


Feedback delay 1

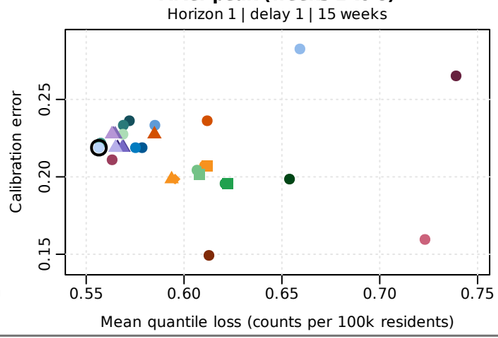
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 14 weeks

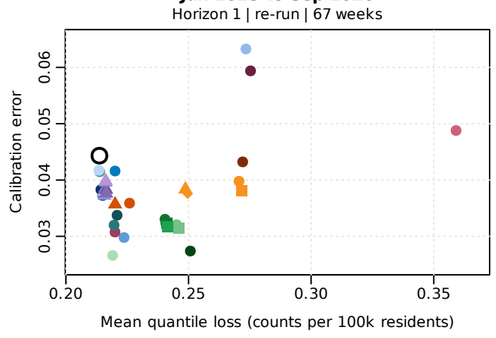


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 15 weeks

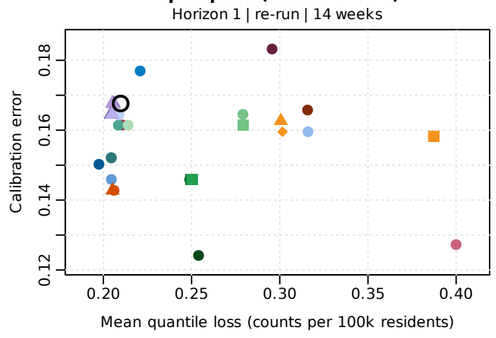


Feedback delay 0 + revisions (re-run)

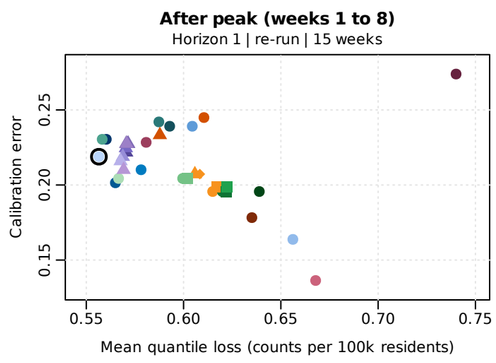
Jun 2025 to Sep 2026
Horizon 1 | re-run | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 14 weeks



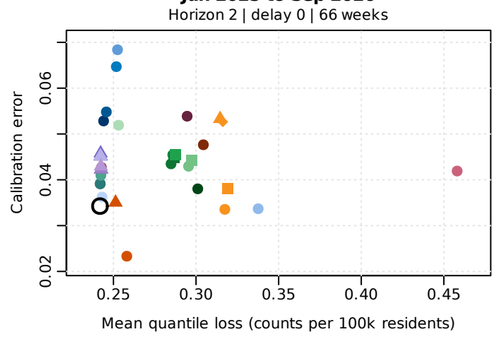
After peak (weeks 1 to 8)
Horizon 1 | re-run | 15 weeks



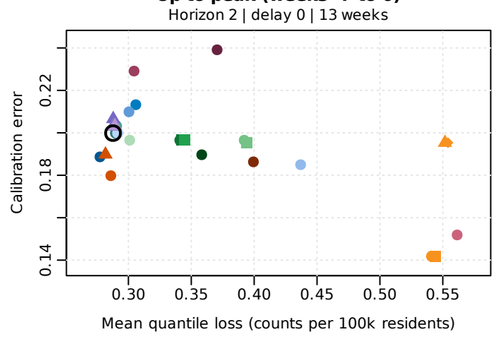
Horizon 2

Feedback delay 0

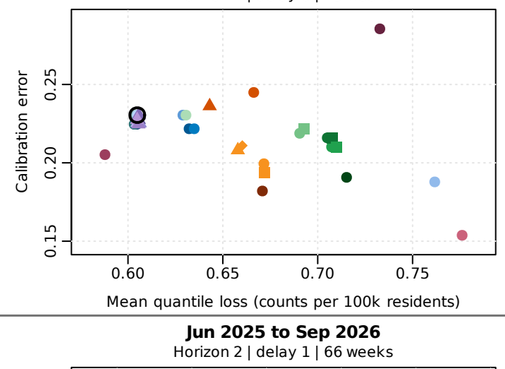
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 13 weeks

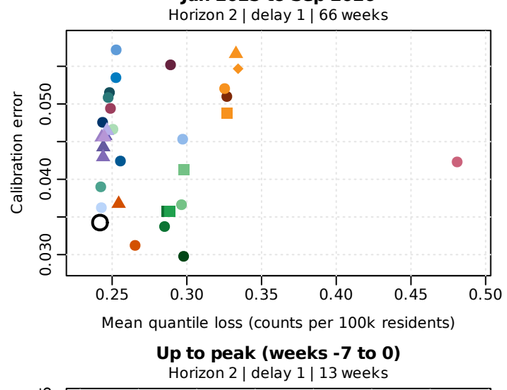


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 15 weeks

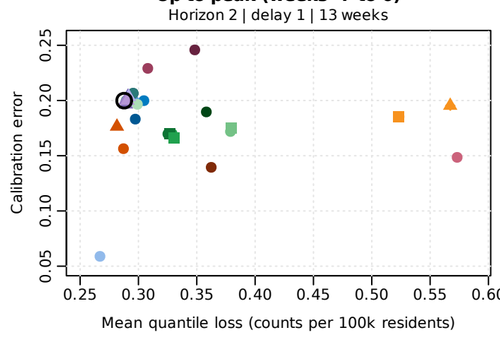


Feedback delay 1

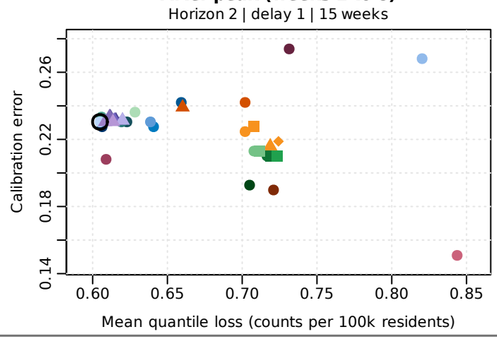
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 13 weeks

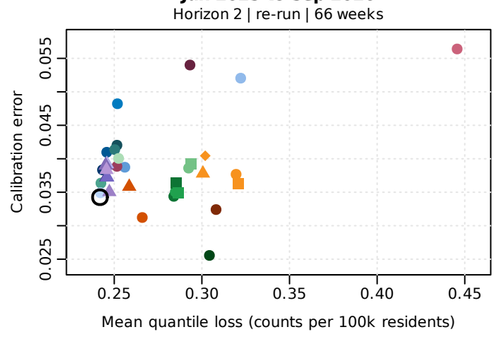


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 15 weeks

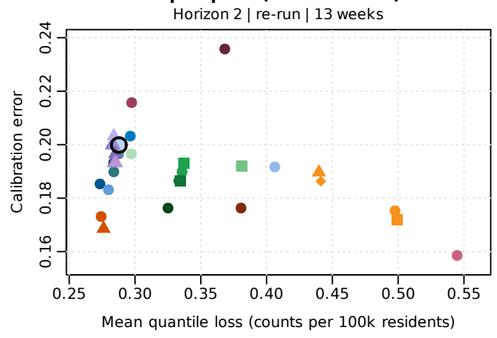


Feedback delay 0 + revisions (re-run)

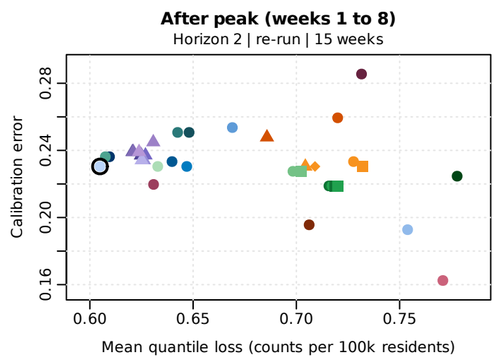
Jun 2025 to Sep 2026
Horizon 2 | re-run | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 13 weeks



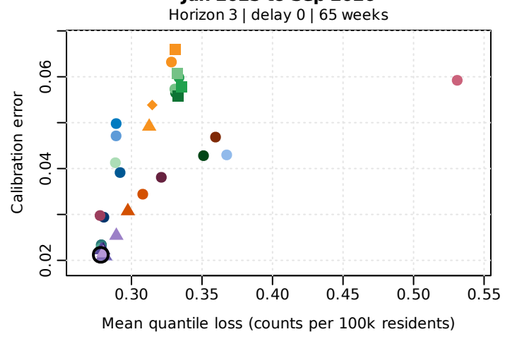
After peak (weeks 1 to 8)
Horizon 2 | re-run | 15 weeks



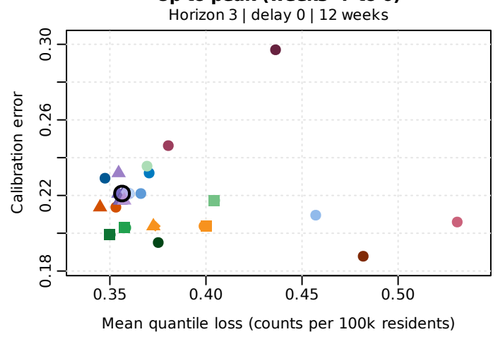
Horizon 3

Feedback delay 0

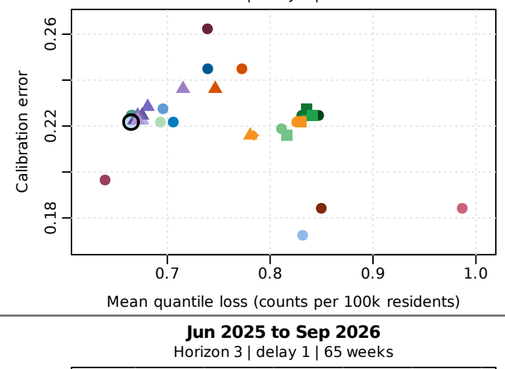
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 12 weeks

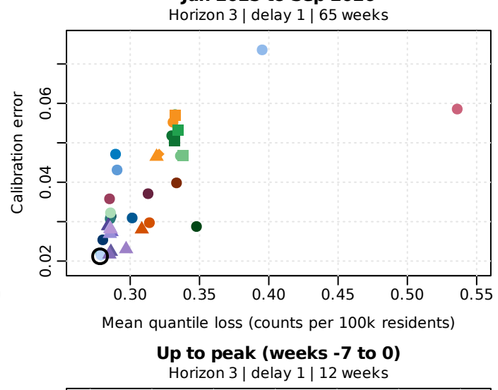


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 15 weeks

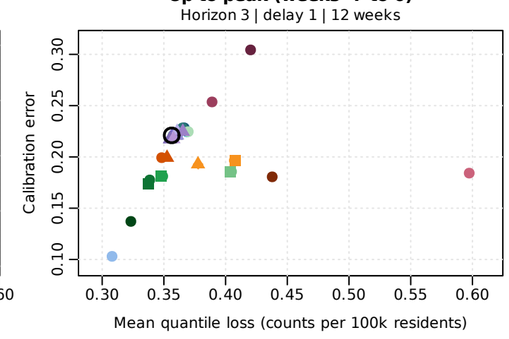


Feedback delay 1

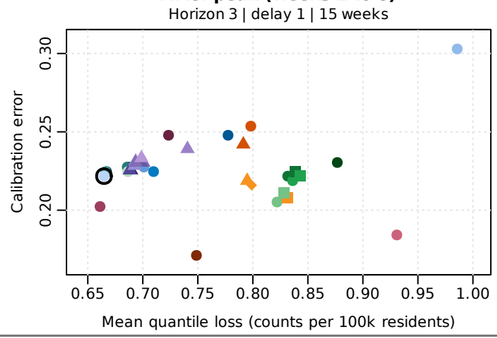
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 12 weeks

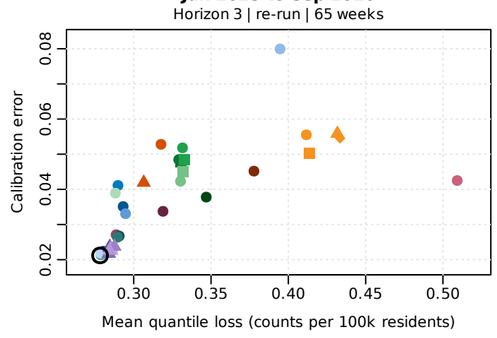


After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 15 weeks

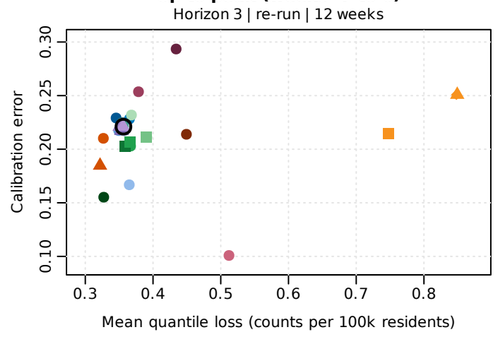


Feedback delay 0 + revisions (re-run)

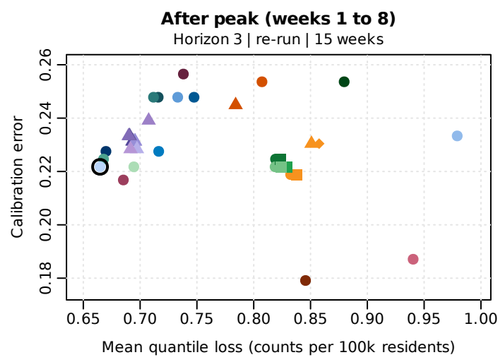
Jun 2025 to Sep 2026
Horizon 3 | re-run | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 12 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 15 weeks



Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=4)
- Rate grid: learned + sharing, common weights (H=4)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

w 0.003, leak 0.1)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Score-based experts
- Score experts: one vector (A, H=4, sharing 1/8)
- Score experts: soft switch, three signals (B, strength=1, H=4, sharing 1/8)
- Score experts: soft switch, target signal (C, delta=0, H=4, sharing 1/8)
- Score experts: context vectors, blended (D, delta=0, H=4, sharing 1/8)
- Score experts: context vectors, hard (D, delta=0, H=4, sharing 1/8)

Score-based experts without sharing
- Score experts: one vector (A, no sharing)
- Score experts: soft switch, three signals (B, strength=1, no sharing)
- Score experts: soft switch, target signal (C, delta=0, no sharing)
- Score experts: context vectors, blended (D, delta=0, no sharing)

Studentized m
- Studentized me
- Studentized me
- Median, consta

Studentized fu
- Studentized ful
- Studentized ful
- Studentized ful

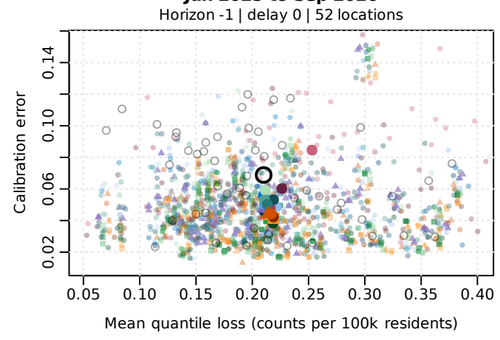
Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
vectors, hard (D, delta=0, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: lambda x 1 in decline (E, delta=0.05, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: soft sw
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday information cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the 80% interval width of the base / 2.563 (a normal sd).
With residuals, the original base is used until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets are initially zero; level scales = normal sd of the central interval of the base at each level, or the RMS of the last 5 released residuals at each level (weighted isotonic projection)

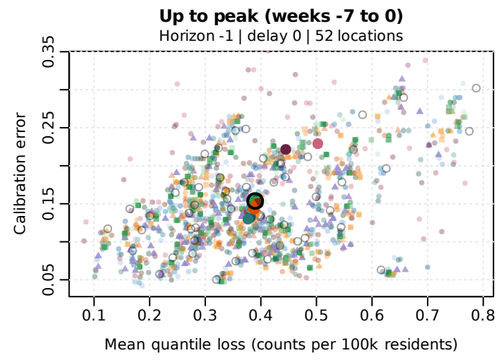
flu | all 52 locations | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

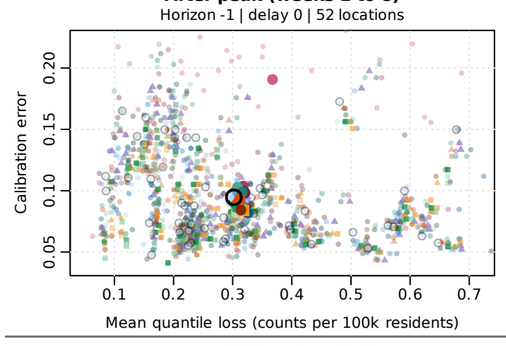
Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.
Small: single locations; large: means over locations; axes: 1st to 99th percentile of both.

Horizon -1

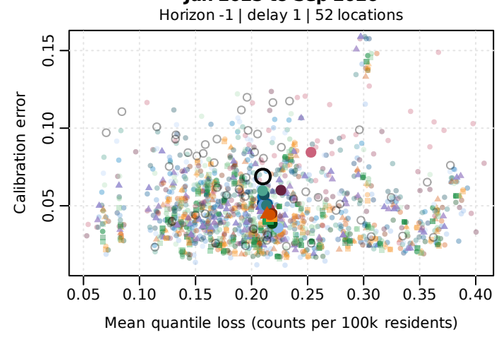
Feedback delay 0

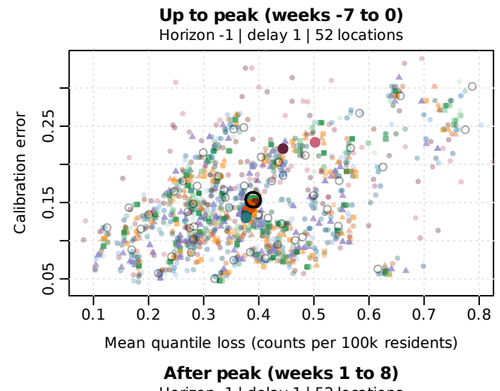


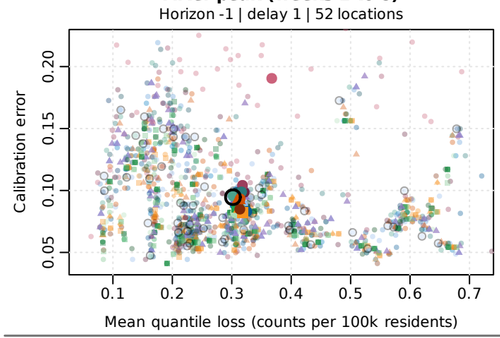




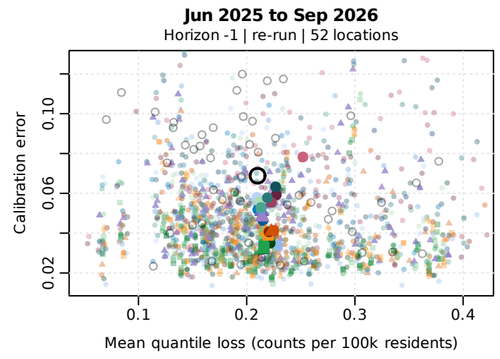
Feedback delay 1

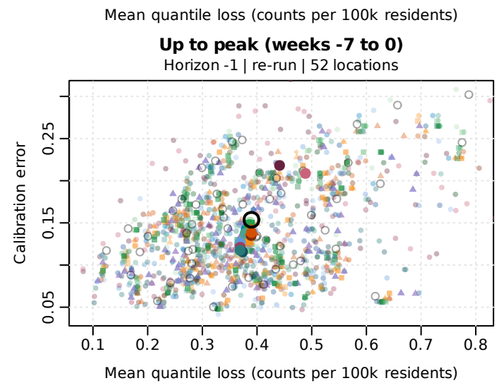


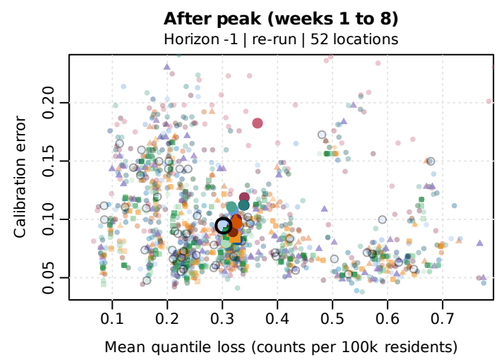




Feedback delay 0 + revisions (re-run)

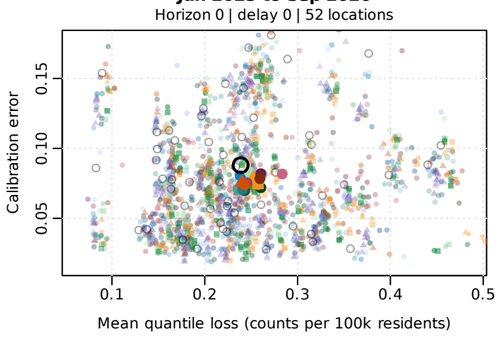


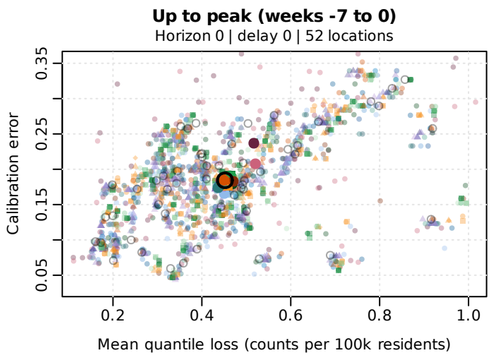


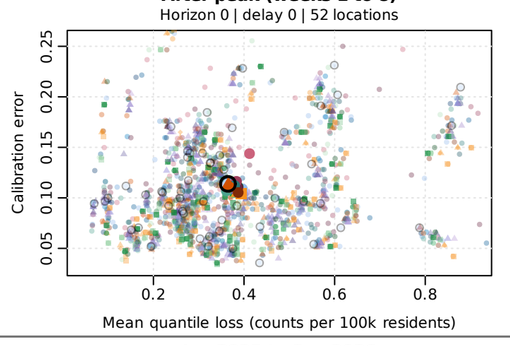


Horizon 0

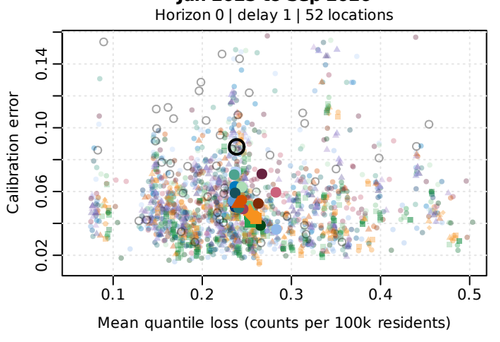
Feedback delay 0

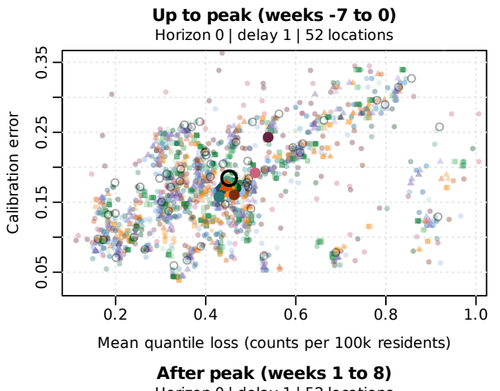


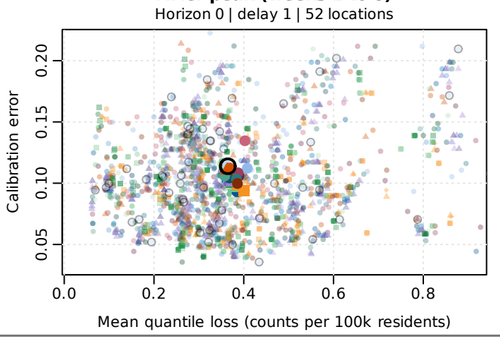




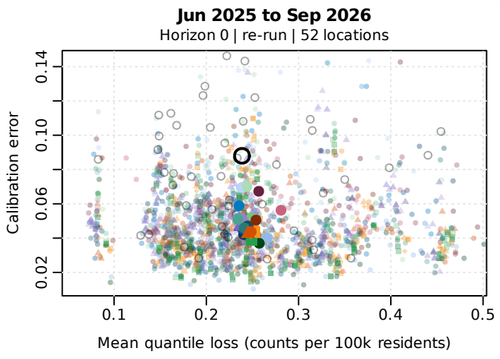
Feedback delay 1

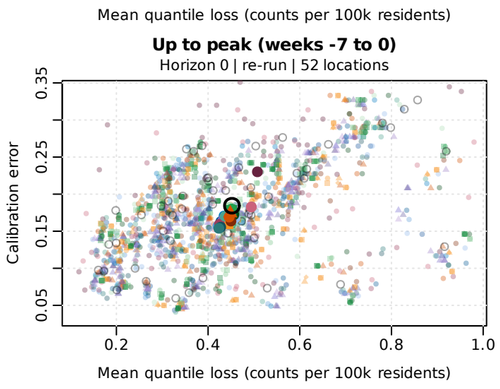


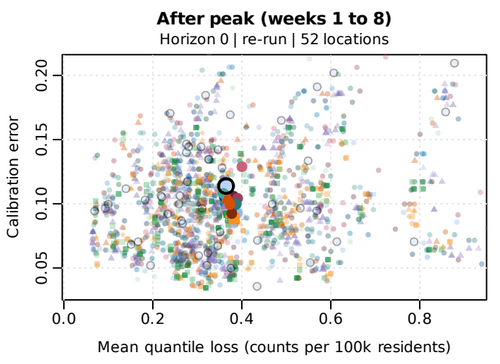




Feedback delay 0 + revisions (re-run)

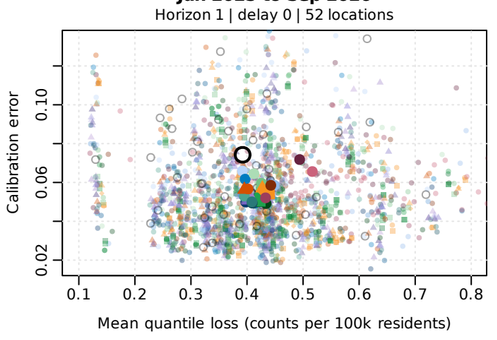


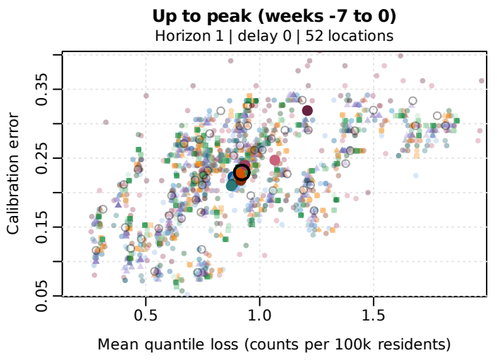


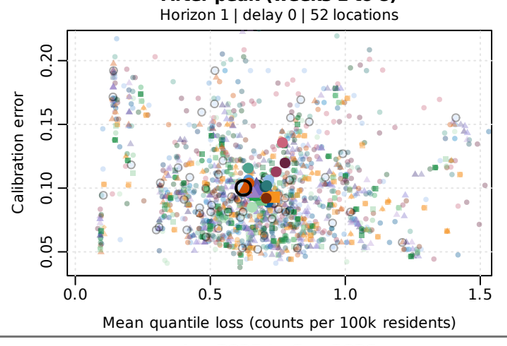


Horizon 1

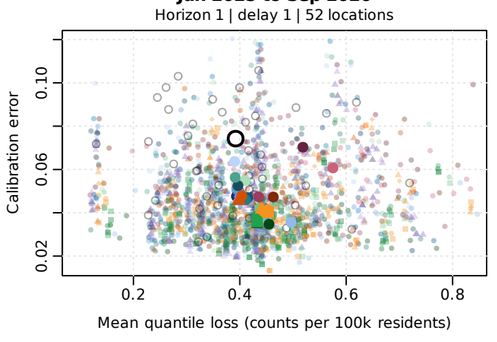
Feedback delay 0

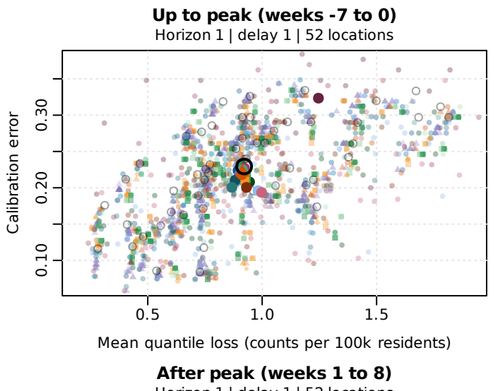


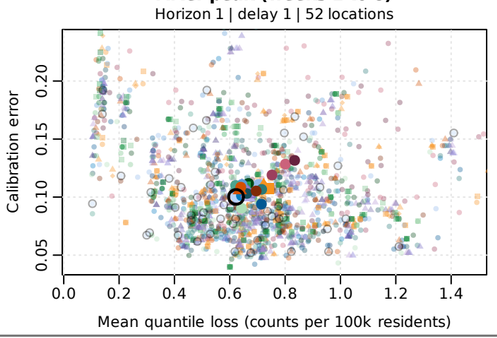




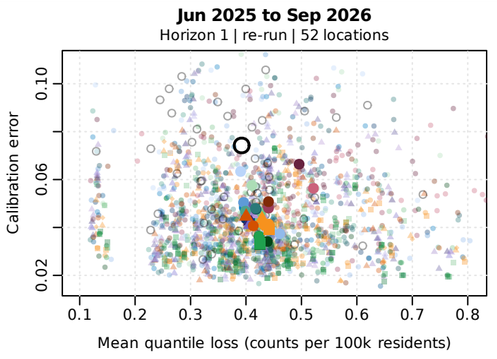
Feedback delay 1

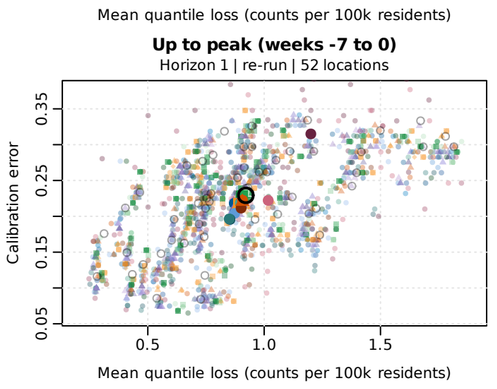


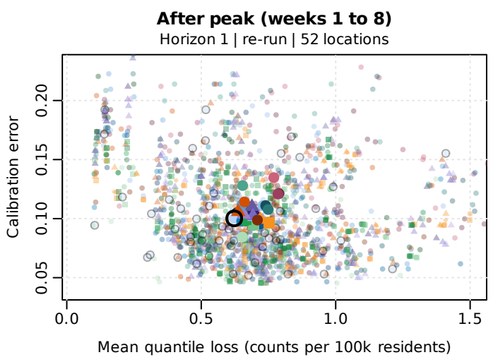




Feedback delay 0 + revisions (re-run)

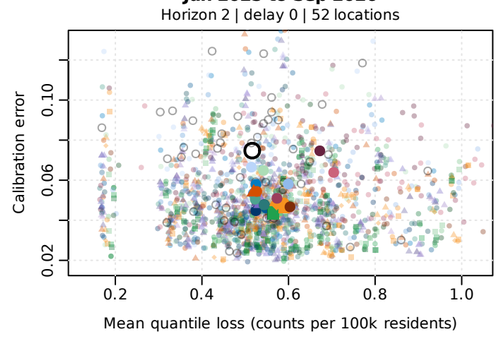


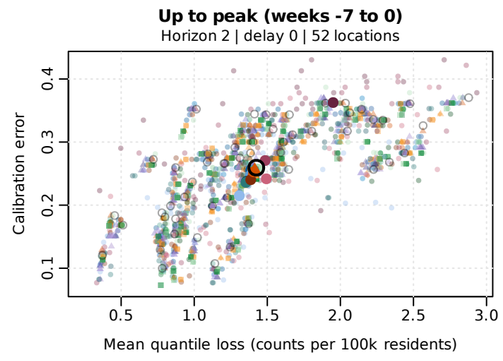


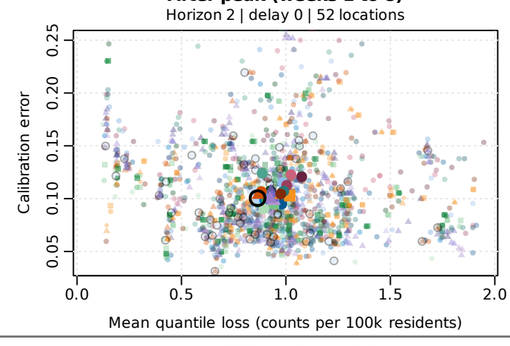


Horizon 2

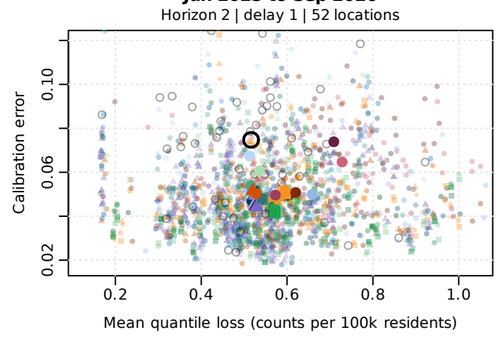
Feedback delay 0

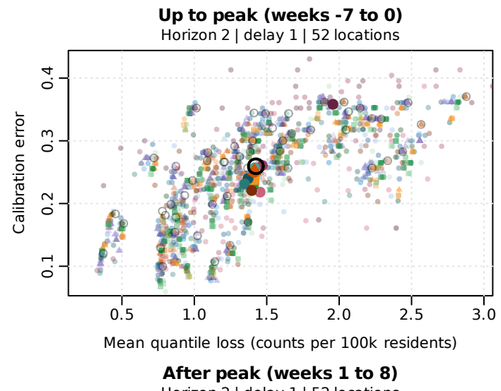


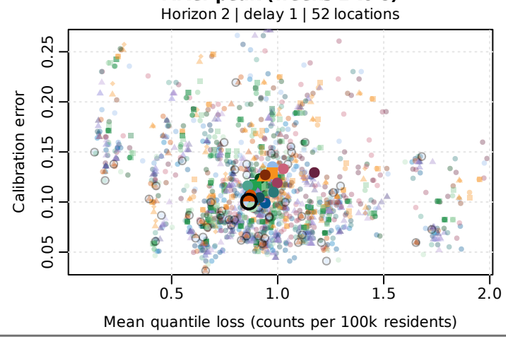




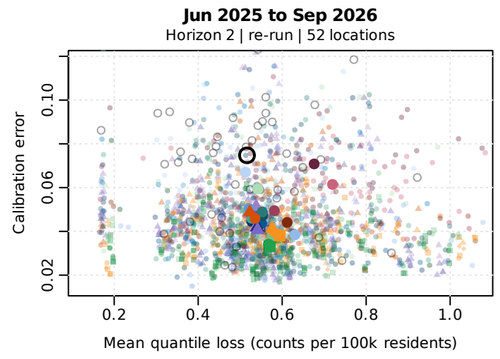
Feedback delay 1

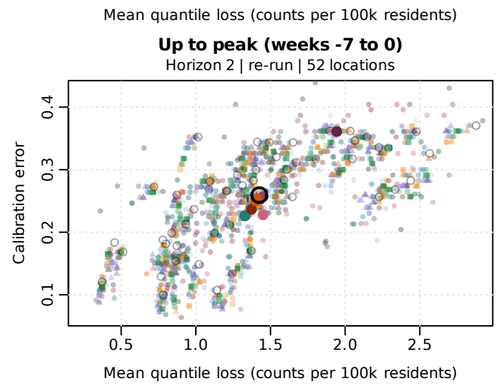


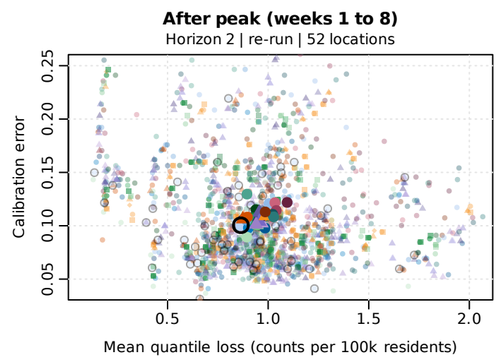




Feedback delay 0 + revisions (re-run)

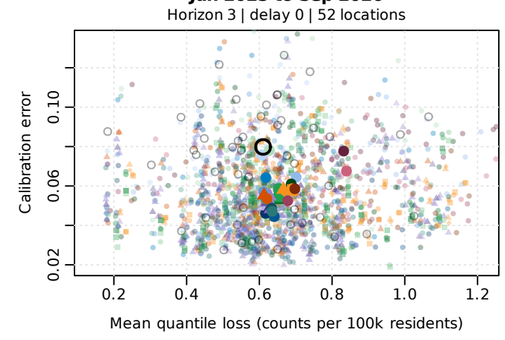


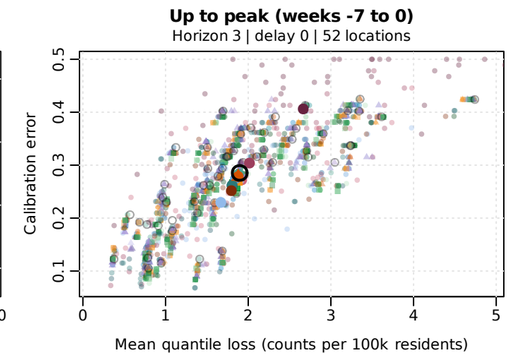


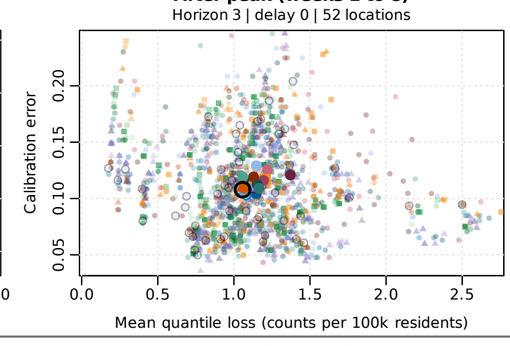


Horizon 3

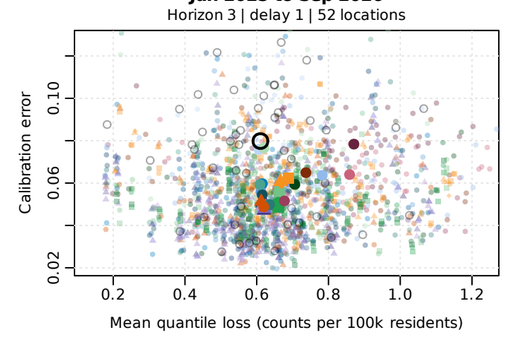
Feedback delay 0

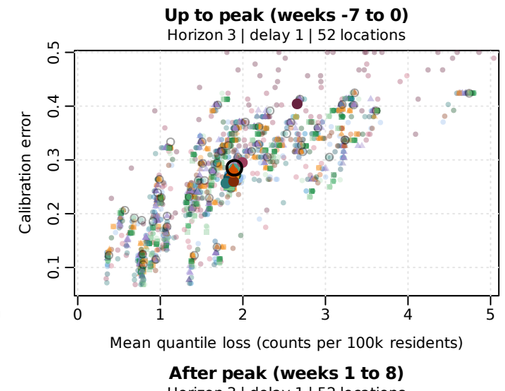


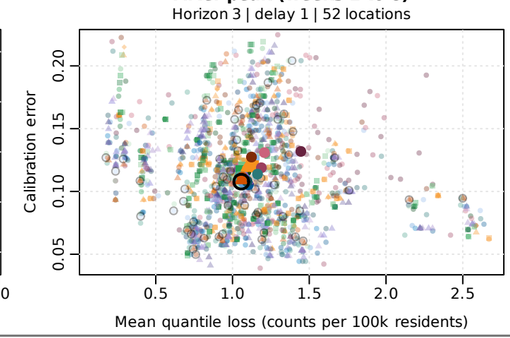




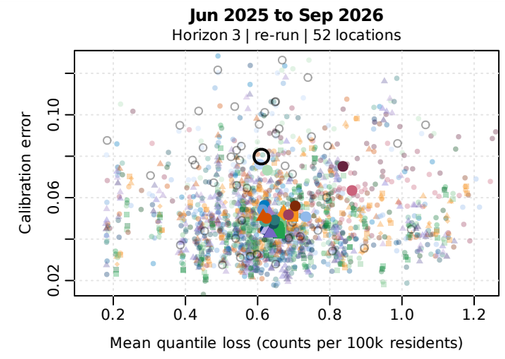
Feedback delay 1

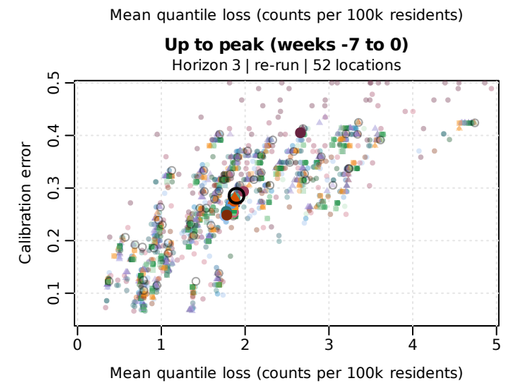


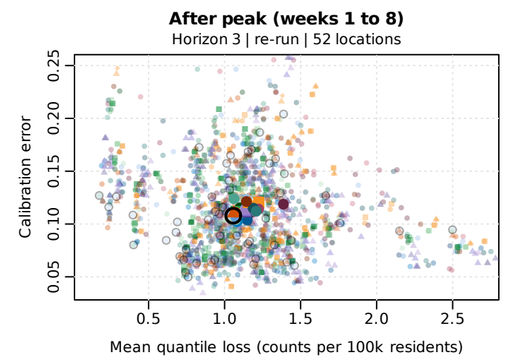




Feedback delay 0 + revisions (re-run)







Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=2)
- Rate grid: learned + sharing, common weights (H=2)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

slow 0.003, leak 0.1)
)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Score-based experts
- Score experts: one vector (A, H=4, sharing 1/8)
- Score experts: soft switch, target signal (C, delta=0.1, H=4, sharing 1/8)
- Score experts: context vectors, blended (D, delta=0, H=4, sharing 1/8)
- Score experts: context vectors, hard (D, delta=0, H=4, sharing 1/8)

Score-based experts without sharing
- Score experts: one vector (A, no sharing)
- Score experts: soft switch, target signal (C, delta=0.1, no sharing)
- Score experts: context vectors, blended (D, delta=0, no sharing)

Studentized m
- Studentized me
- Studentized me
- Median, constar

Studentized fu
- Studentized full
- Studentized full
- Studentized full

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
ntext vectors, hard (D, delta=0, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: lambda x 1 in decline (E, delta=0.05, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts:
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday information cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the 80% interval width of the base / 2.563 (a normal sd).
With residuals, the original base is used until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets are initially zero; level scales = normal sd of the central interval of the base at each level, or the RMS of the last 5 released residuals at each level (weighted isotonic projection)

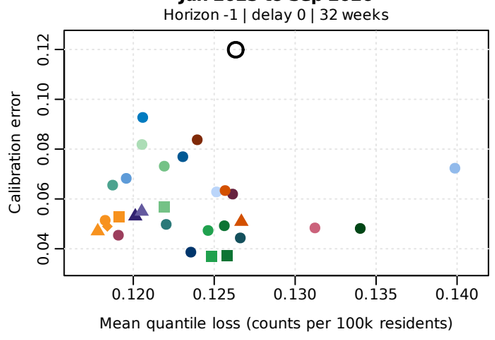
flu | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

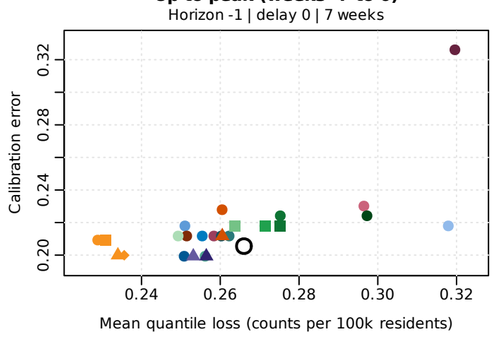
Horizon -1

Feedback delay 0

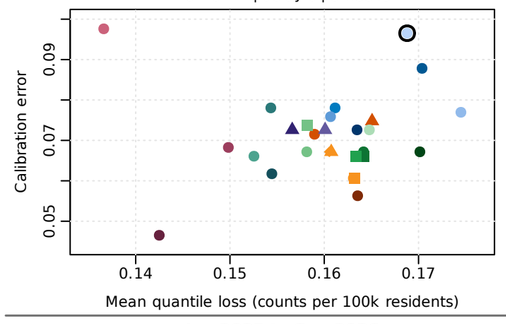
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 7 weeks

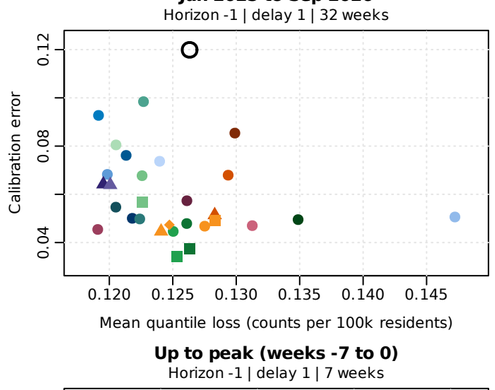


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 8 weeks

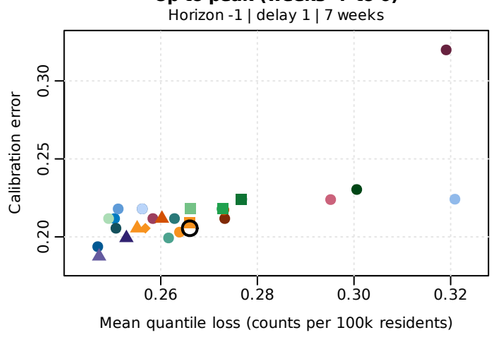


Feedback delay 1

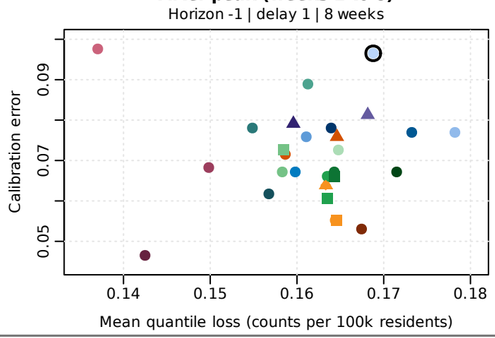
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 7 weeks

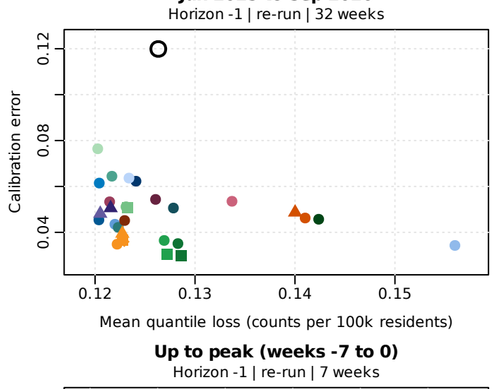


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 8 weeks

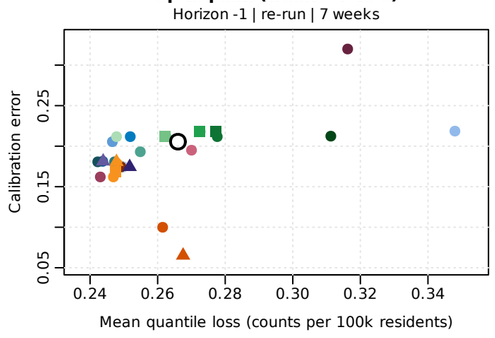


Feedback delay 0 + revisions (re-run)

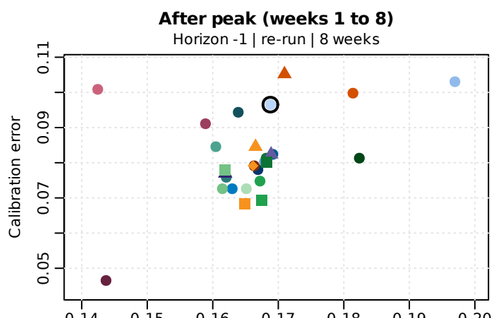
Jun 2025 to Sep 2026
Horizon -1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 7 weeks



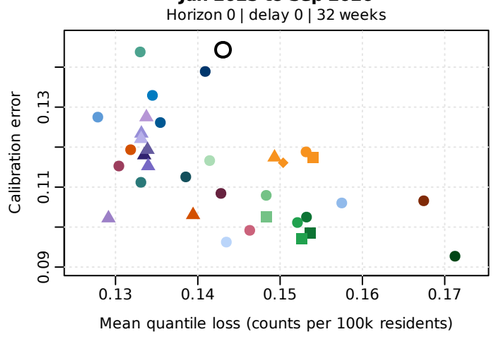
After peak (weeks 1 to 8)
Horizon -1 | re-run | 8 weeks



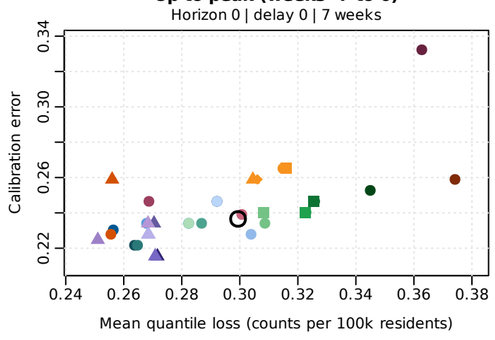
Horizon 0

Feedback delay 0

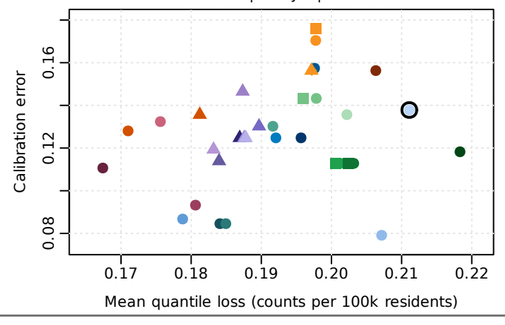
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 7 weeks

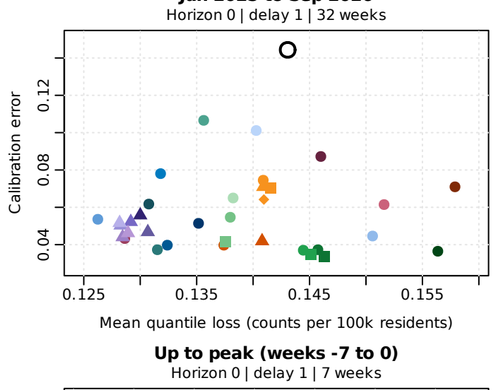


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 8 weeks

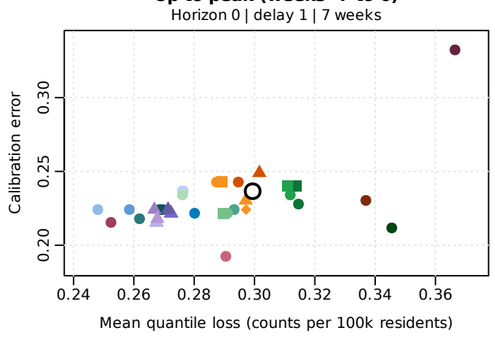


Feedback delay 1

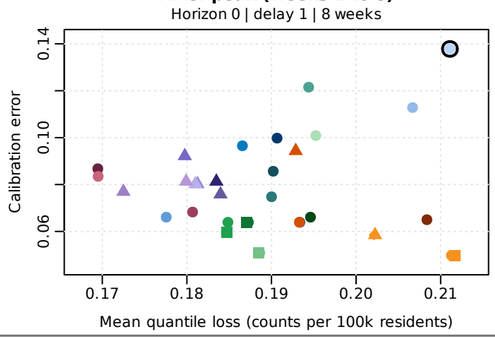
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 7 weeks

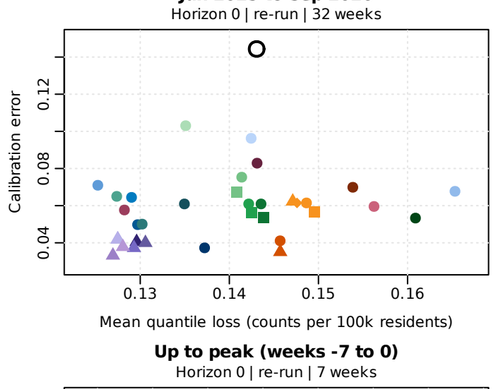


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 8 weeks

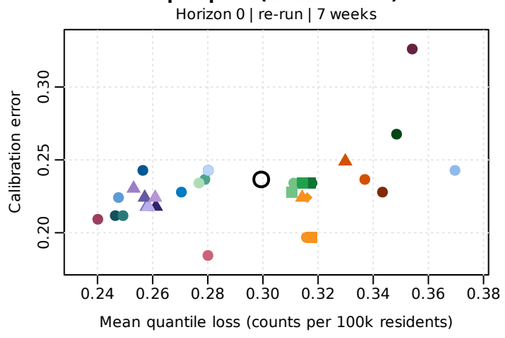


Feedback delay 0 + revisions (re-run)

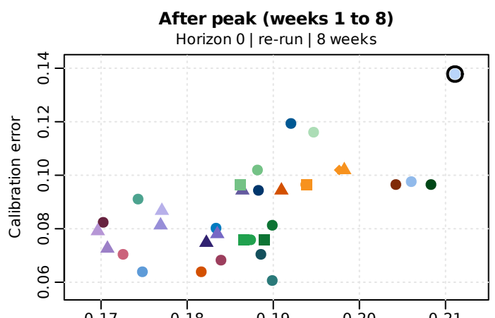
Jun 2025 to Sep 2026
Horizon 0 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 7 weeks



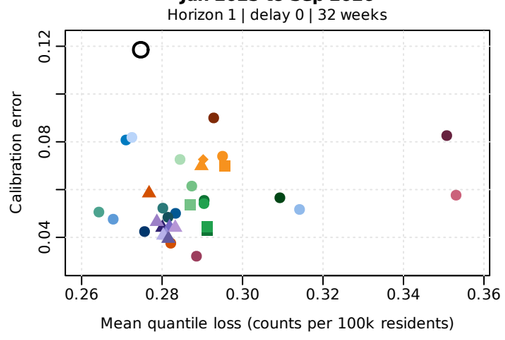
After peak (weeks 1 to 8)
Horizon 0 | re-run | 8 weeks



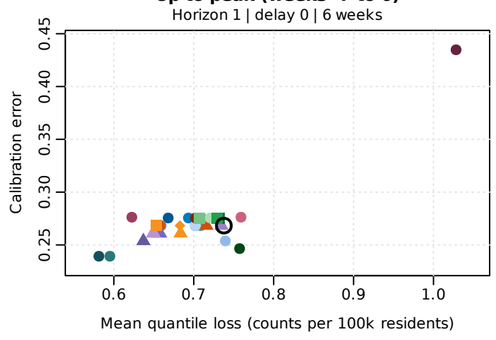
Horizon 1

Feedback delay 0

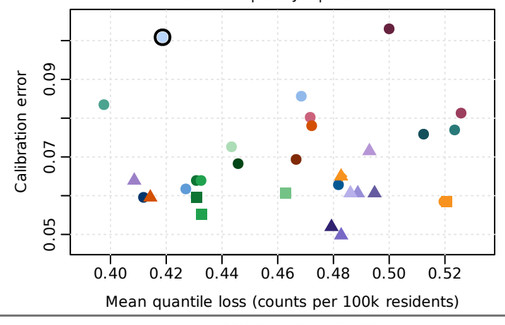
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 6 weeks

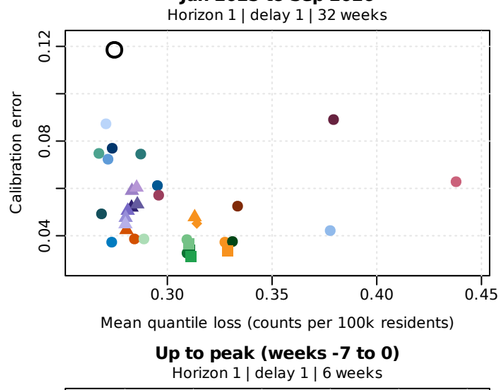


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 8 weeks

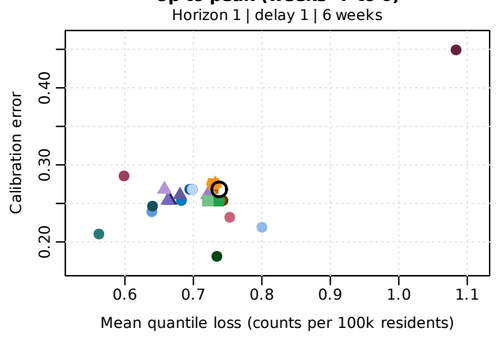


Feedback delay 1

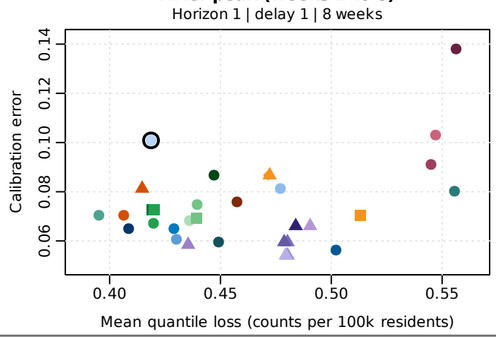
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 6 weeks

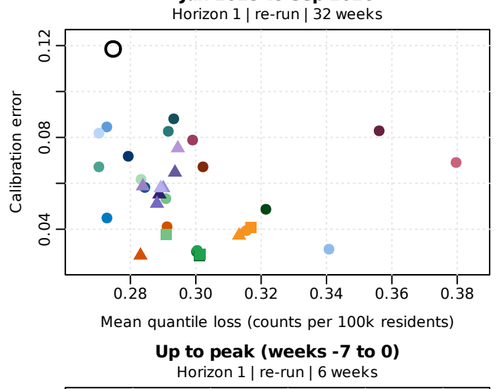


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 8 weeks

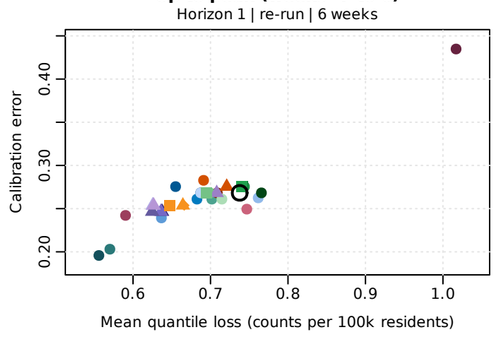


Feedback delay 0 + revisions (re-run)

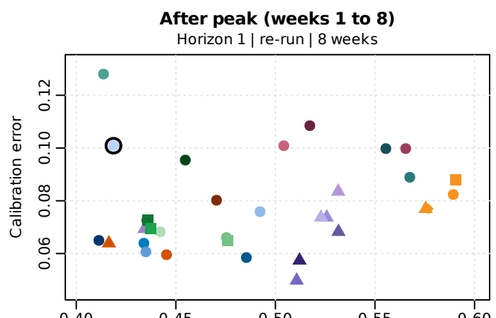
Jun 2025 to Sep 2026
Horizon 1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 6 weeks



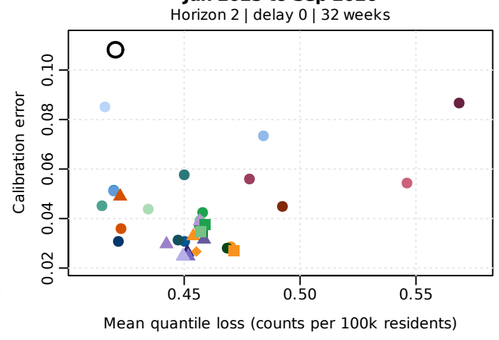
After peak (weeks 1 to 8)
Horizon 1 | re-run | 8 weeks



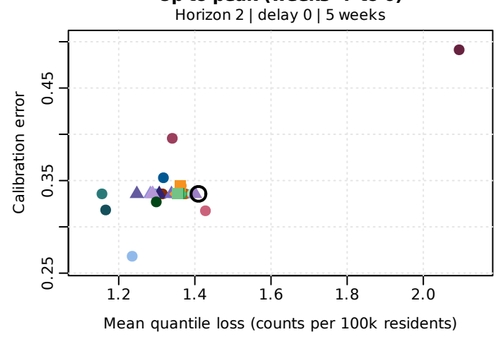
Horizon 2

Feedback delay 0

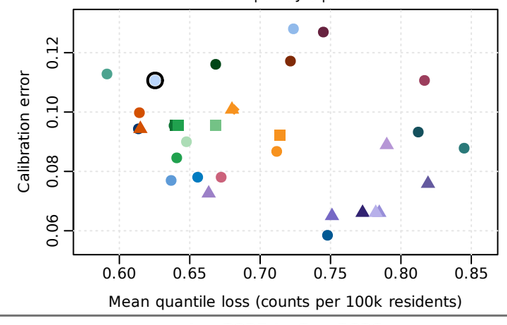
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 5 weeks

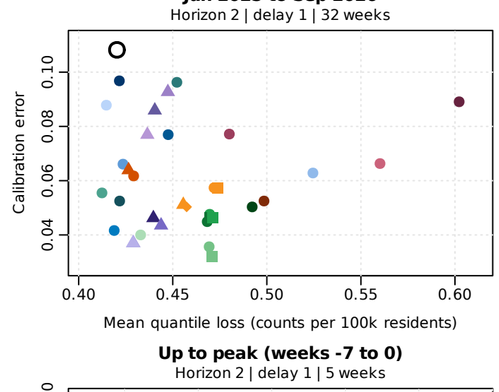


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 8 weeks

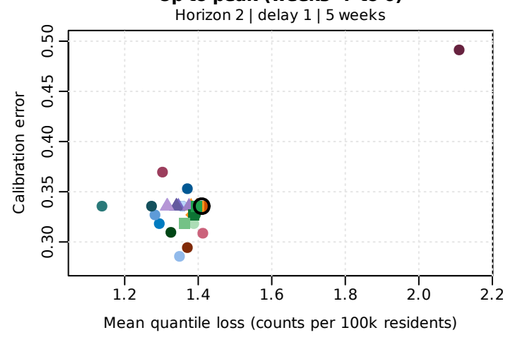


Feedback delay 1

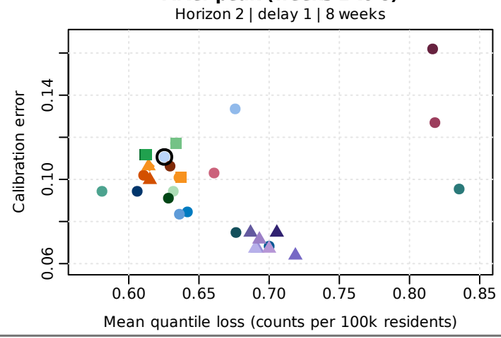
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 5 weeks

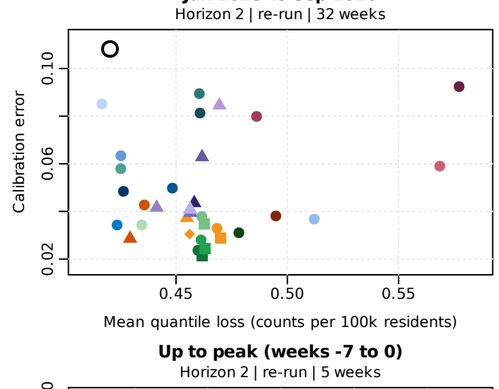


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 8 weeks

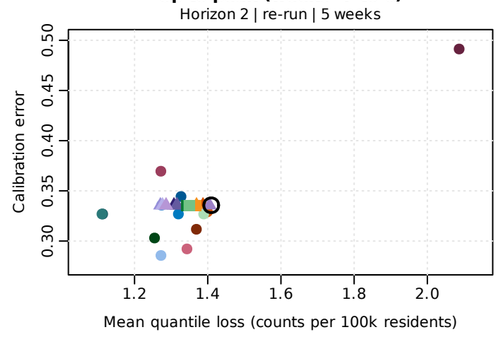


Feedback delay 0 + revisions (re-run)

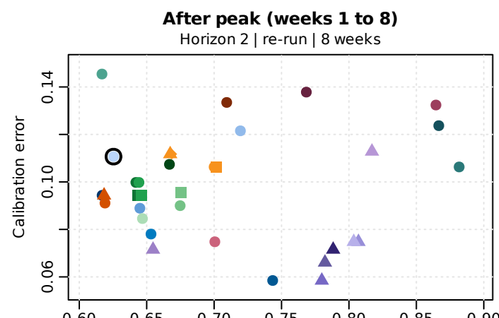
Jun 2025 to Sep 2026
Horizon 2 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 5 weeks



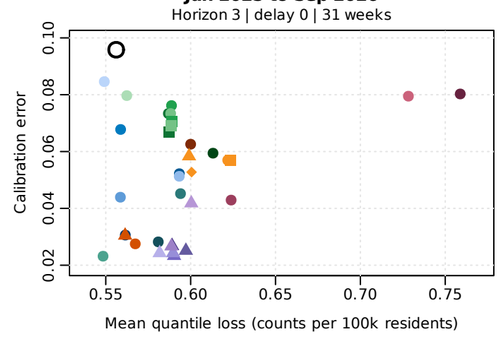
After peak (weeks 1 to 8)
Horizon 2 | re-run | 8 weeks



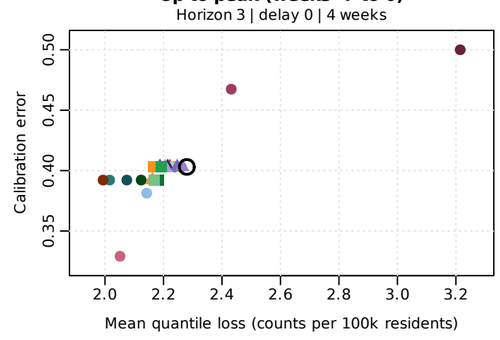
Horizon 3

Feedback delay 0

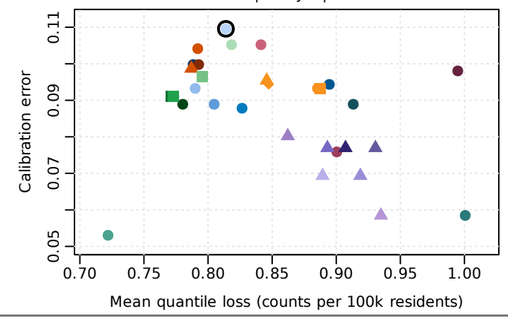
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 4 weeks

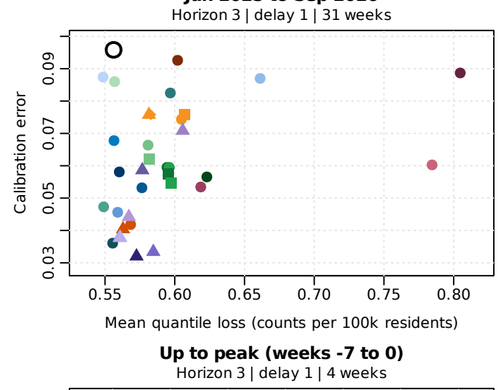


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 8 weeks

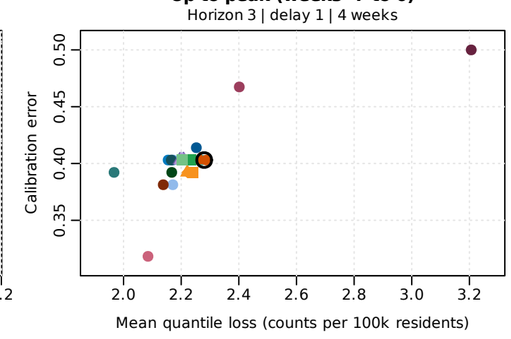


Feedback delay 1

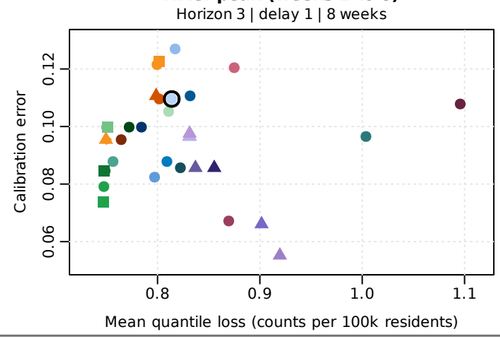
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 4 weeks

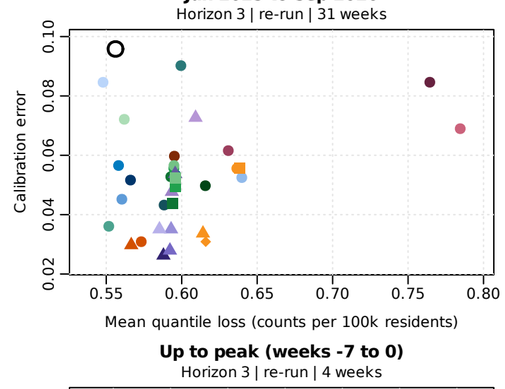


After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 8 weeks

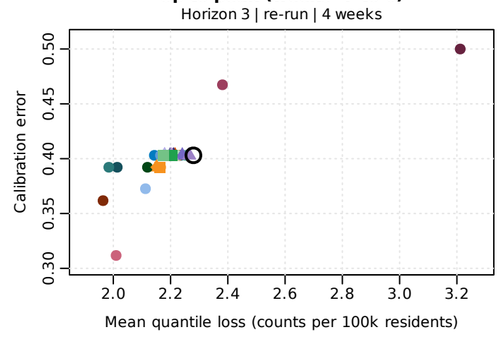


Feedback delay 0 + revisions (re-run)

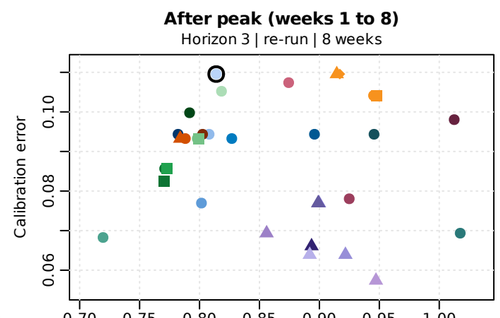
Jun 2025 to Sep 2026
Horizon 3 | re-run | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 4 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 8 weeks



Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=2)
- Rate grid: learned + sharing, common weights (H=2)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

slow 0.003, leak 0.1)
)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Score-based experts
- Score experts: one vector (A, H=4, sharing 1/8)
- Score experts: soft switch, target signal (C, delta=0.1, H=4, sharing 1/8)
- Score experts: context vectors, blended (D, delta=0, H=4, sharing 1/8)
- Score experts: context vectors, hard (D, delta=0, H=4, sharing 1/8)

Score-based experts without sharing
- Score experts: one vector (A, no sharing)
- Score experts: soft switch, target signal (C, delta=0.1, no sharing)
- Score experts: context vectors, blended (D, delta=0, no sharing)

Studentized m
- Studentized me
- Studentized me
- Median, constar

Studentized fu
- Studentized full
- Studentized full
- Studentized full

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
ntext vectors, hard (D, delta=0, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: lambda x 1 in decline (E, delta=0.05, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts:
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday information cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the 80% interval width of the base / 2.563 (a normal sd).
With residuals, the original base is used until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets are initially zero; level scales = normal sd of the central interval of the base at each level, or the RMS of the last 5 released residuals at each level (weighted isotonic projection)

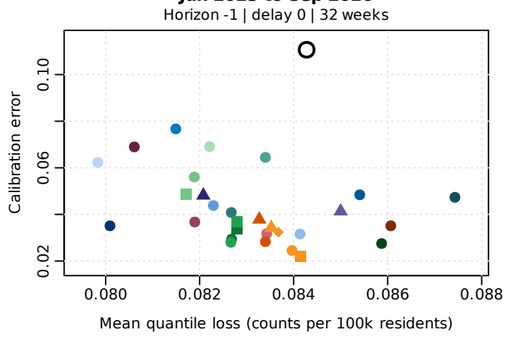
flu | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

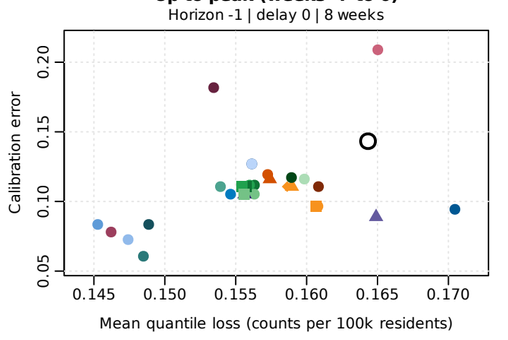
Horizon -1

Feedback delay 0

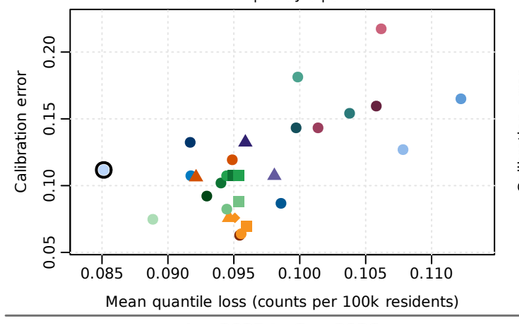
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 8 weeks

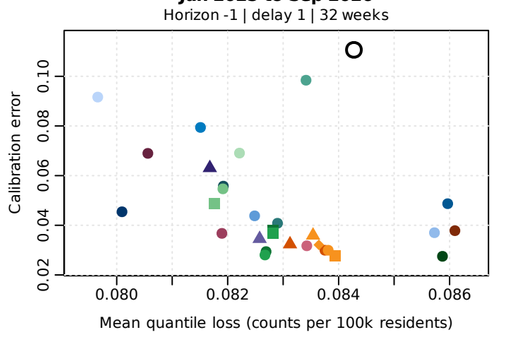


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 8 weeks

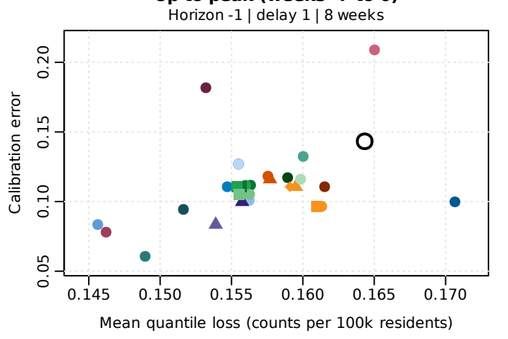


Feedback delay 1

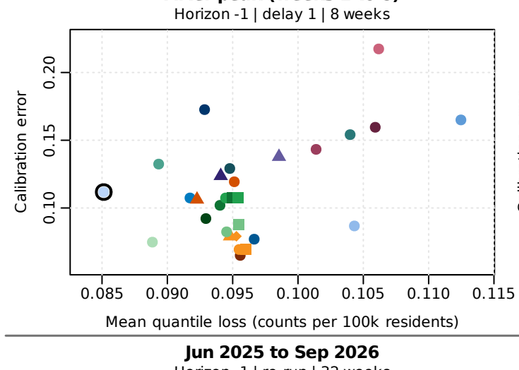
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 8 weeks

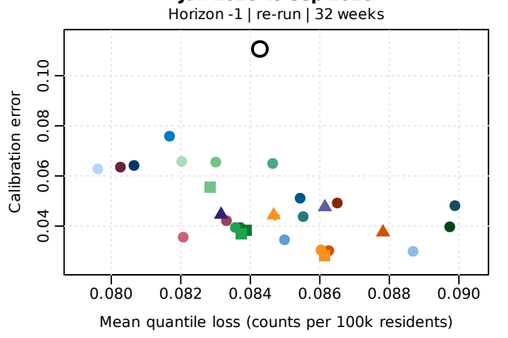


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 8 weeks

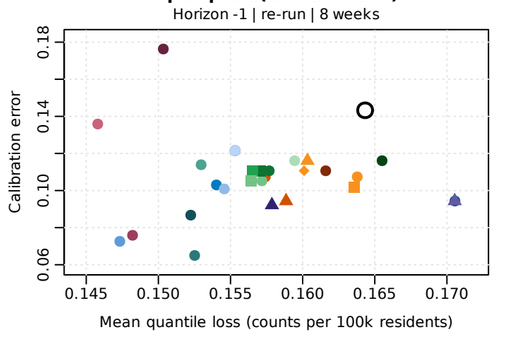


Feedback delay 0 + revisions (re-run)

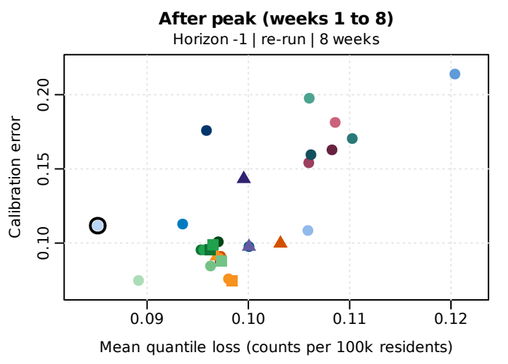
Jun 2025 to Sep 2026
Horizon -1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 8 weeks



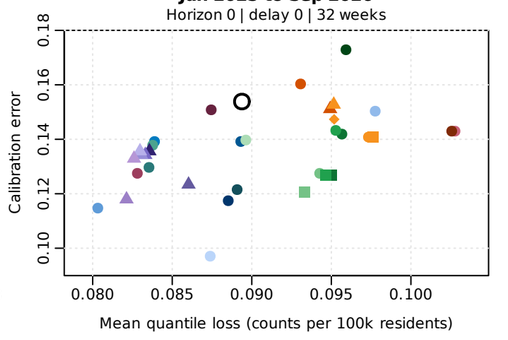
After peak (weeks 1 to 8)
Horizon -1 | re-run | 8 weeks



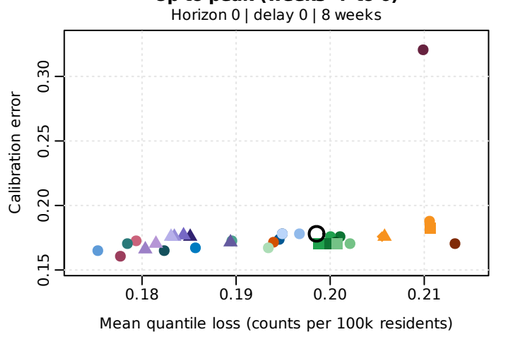
Horizon 0

Feedback delay 0

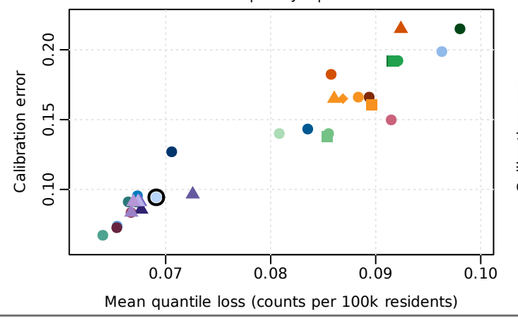
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 8 weeks

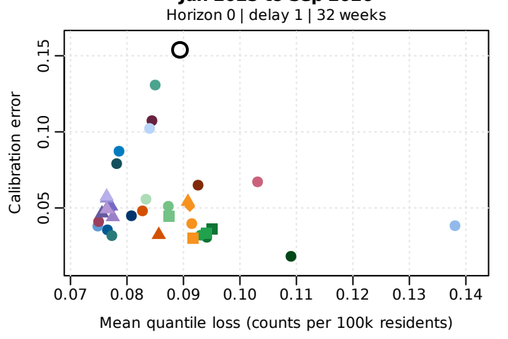


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 8 weeks

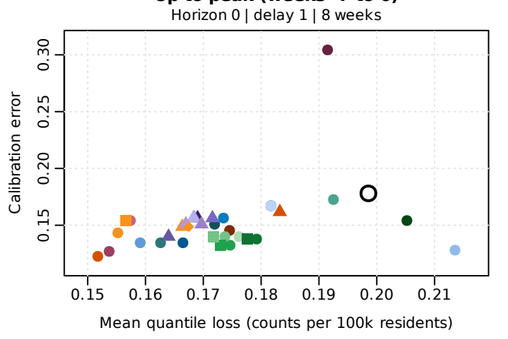


Feedback delay 1

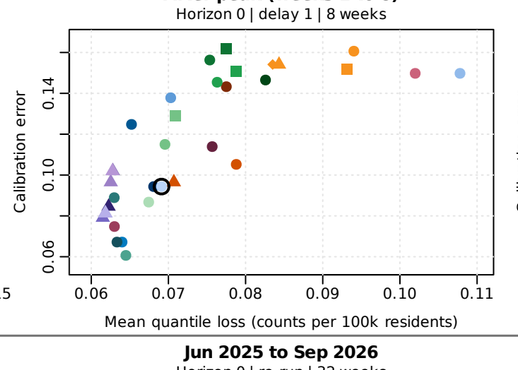
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 8 weeks

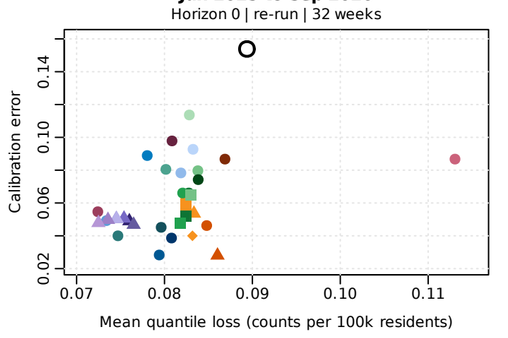


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 8 weeks

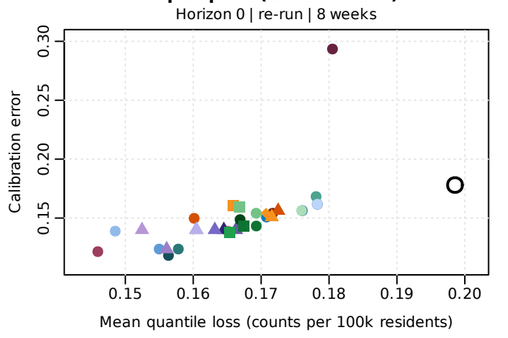


Feedback delay 0 + revisions (re-run)

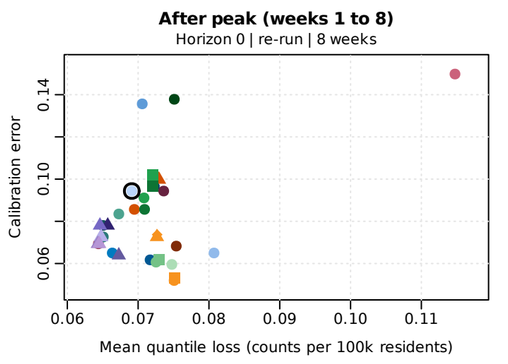
Jun 2025 to Sep 2026
Horizon 0 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 8 weeks



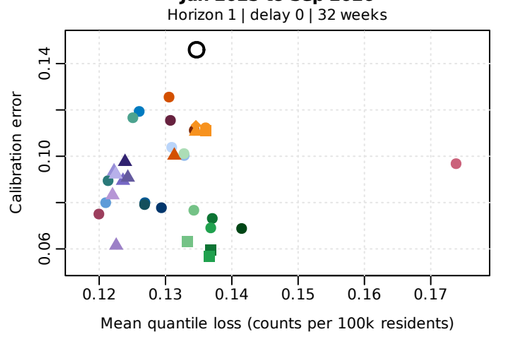
After peak (weeks 1 to 8)
Horizon 0 | re-run | 8 weeks



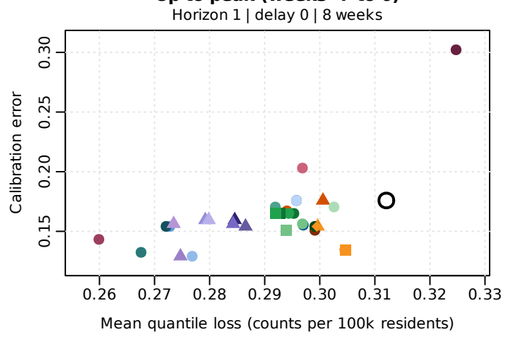
Horizon 1

Feedback delay 0

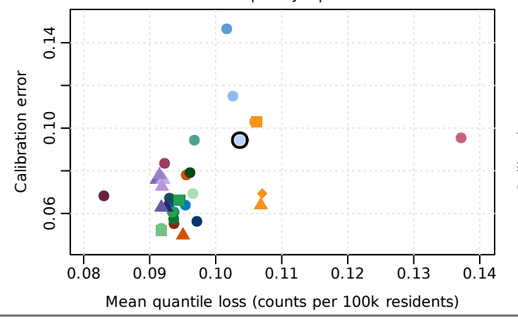
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 8 weeks

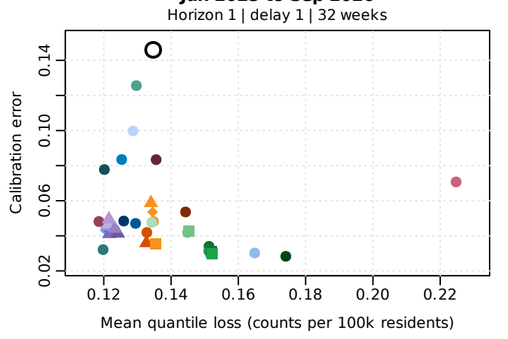


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 8 weeks

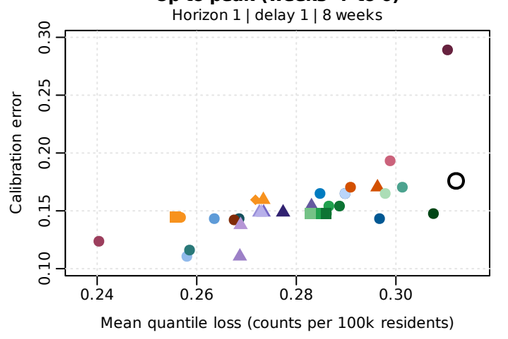


Feedback delay 1

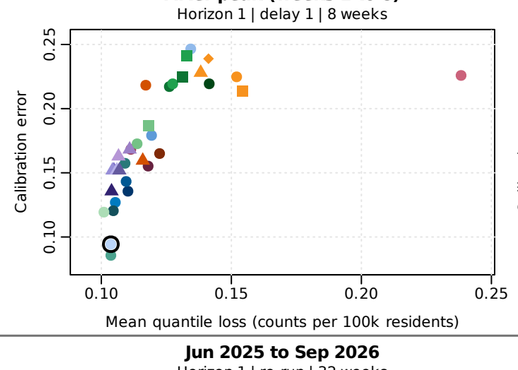
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 8 weeks

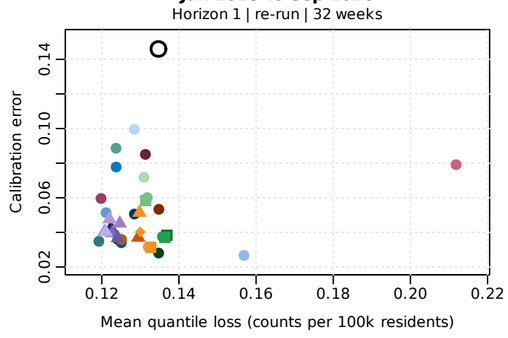


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 8 weeks

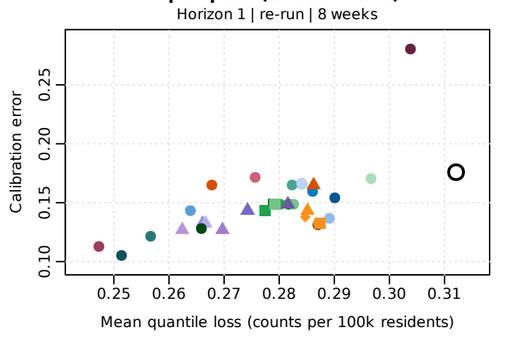


Feedback delay 0 + revisions (re-run)

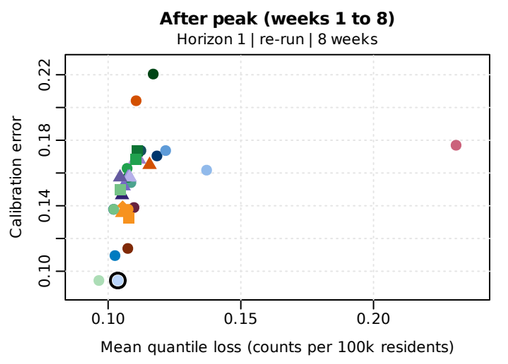
Jun 2025 to Sep 2026
Horizon 1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 8 weeks



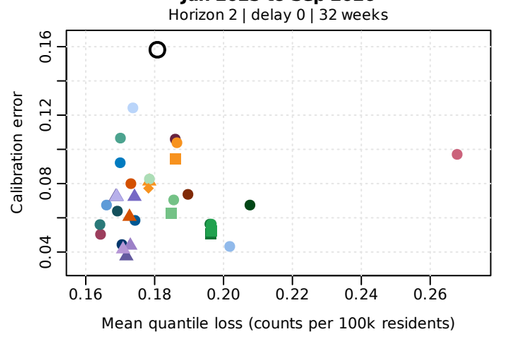
After peak (weeks 1 to 8)
Horizon 1 | re-run | 8 weeks



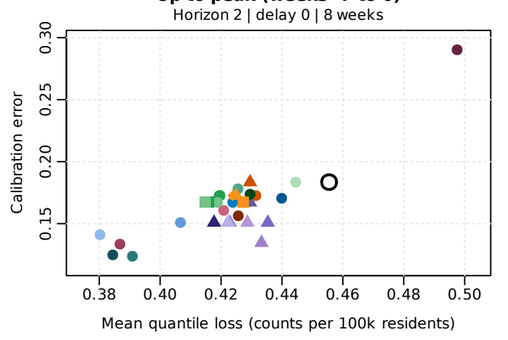
Horizon 2

Feedback delay 0

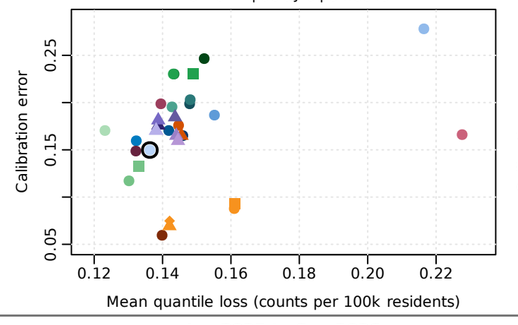
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 8 weeks

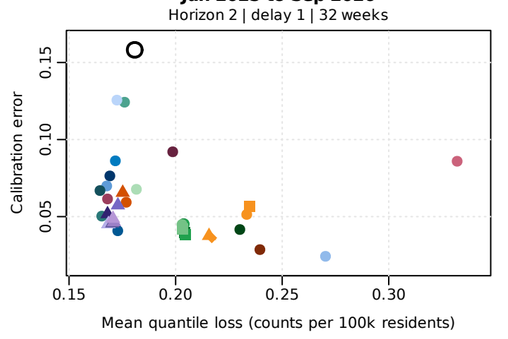


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 8 weeks

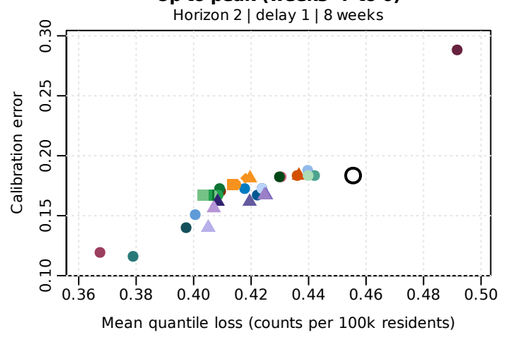


Feedback delay 1

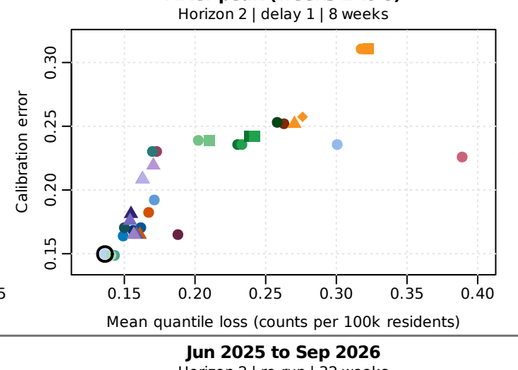
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 8 weeks

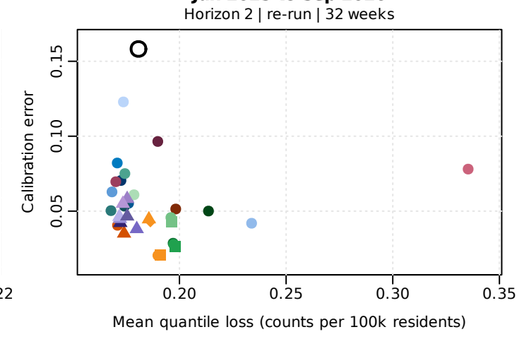


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 8 weeks

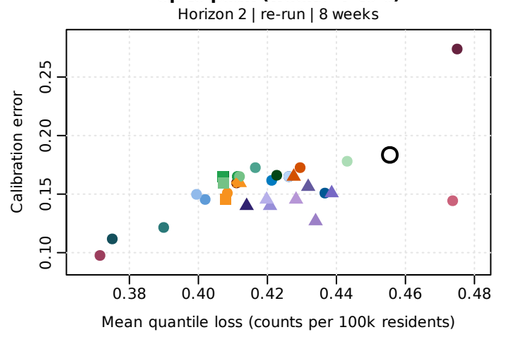


Feedback delay 0 + revisions (re-run)

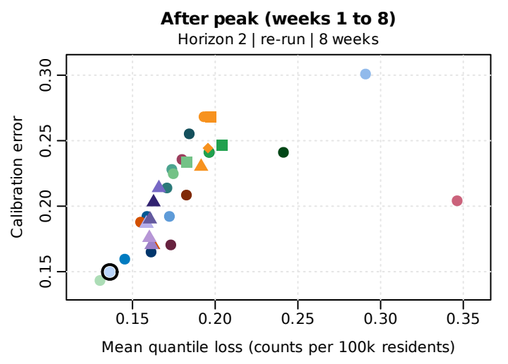
Jun 2025 to Sep 2026
Horizon 2 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 8 weeks



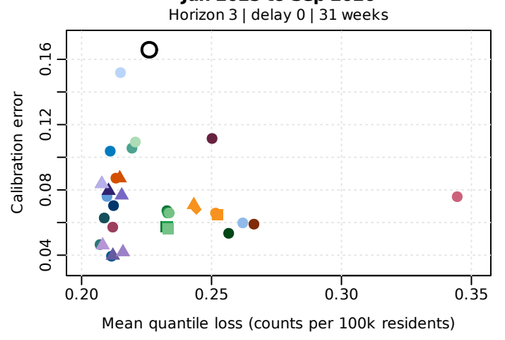
After peak (weeks 1 to 8)
Horizon 2 | re-run | 8 weeks



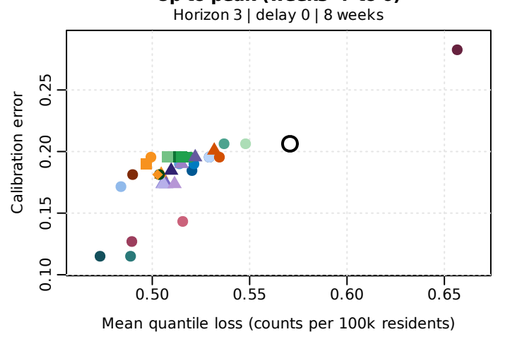
Horizon 3

Feedback delay 0

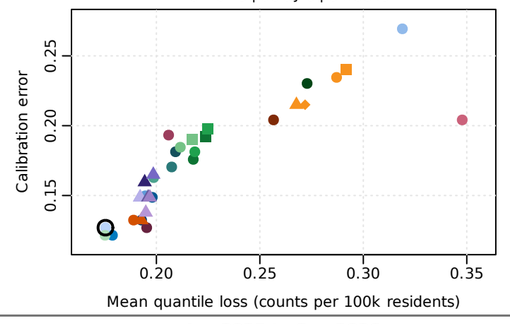
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 8 weeks

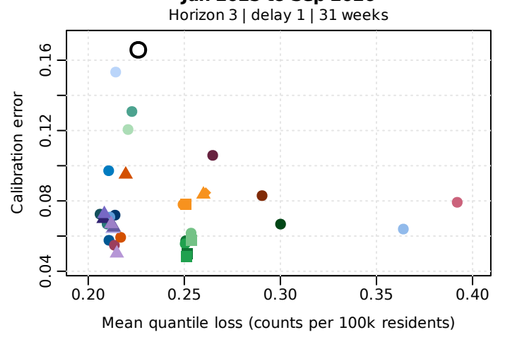


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 8 weeks

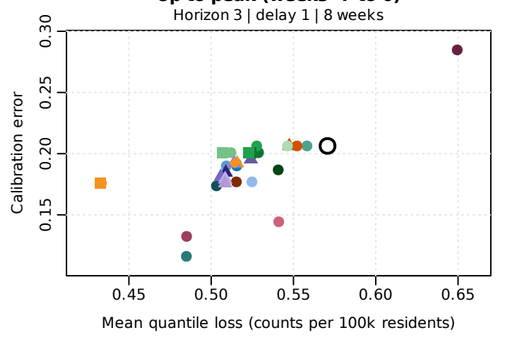


Feedback delay 1

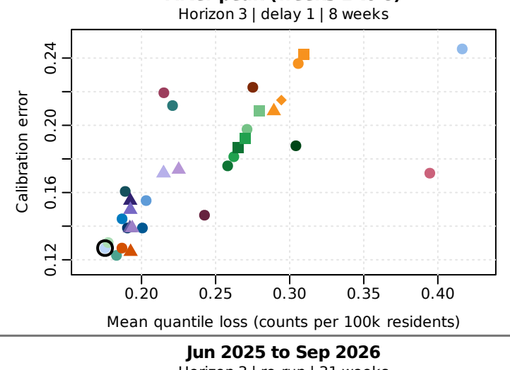
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 8 weeks

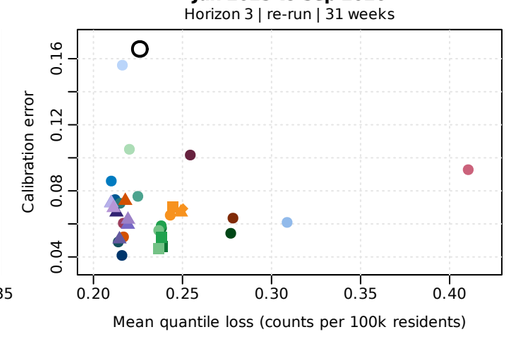


After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 8 weeks

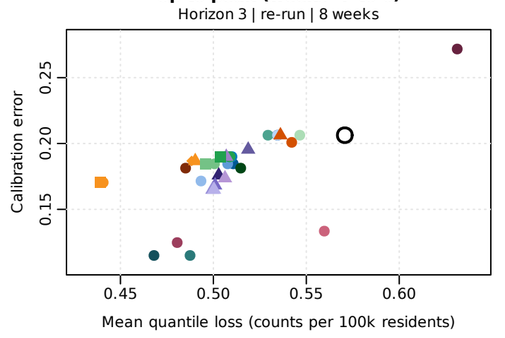


Feedback delay 0 + revisions (re-run)

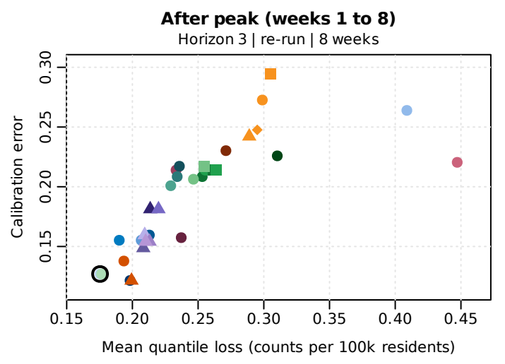
Jun 2025 to Sep 2026
Horizon 3 | re-run | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 8 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 8 weeks



Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=2)
- Rate grid: learned + sharing, common weights (H=2)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

slow 0.003, leak 0.1)
)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Score-based experts
- Score experts: one vector (A, H=4, sharing 1/8)
- Score experts: soft switch, target signal (C, delta=0.1, H=4, sharing 1/8)
- Score experts: context vectors, blended (D, delta=0, H=4, sharing 1/8)
- Score experts: context vectors, hard (D, delta=0, H=4, sharing 1/8)

Score-based experts without sharing
- Score experts: one vector (A, no sharing)
- Score experts: soft switch, target signal (C, delta=0.1, no sharing)
- Score experts: context vectors, blended (D, delta=0, no sharing)

Studentized m
- Studentized me
- Studentized me
- Median, constar

Studentized fu
- Studentized full
- Studentized full
- Studentized full

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
ntext vectors, hard (D, delta=0, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: lambda x 1 in decline (E, delta=0.05, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts:
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday information cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the 80% interval width of the base / 2.563 (a normal sd).
With residuals, the original base is used until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets are initially zero; level scales = normal sd of the central interval of the base at each level, or the RMS of the last 5 released residuals at each level (weighted isotonic projection)

flu | Texas (48) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

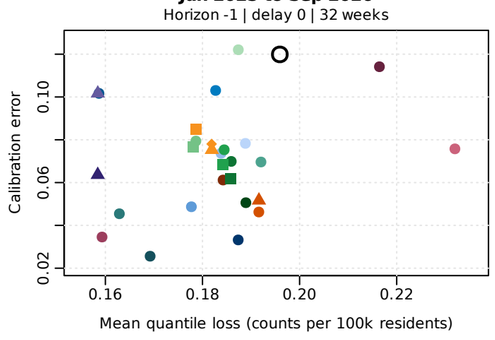
Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday information cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the 80% interval width of the base / 2.563 (a normal sd).
With residuals, the original base is used until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets are initially zero; level scales = normal sd of the central interval of the base at each level, or the RMS of the last 5 released residuals at each level (weighted isotonic projection)

flu | New York (36) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

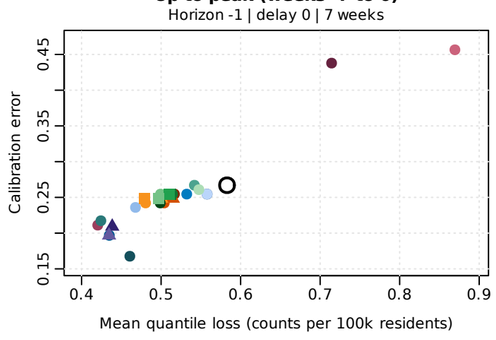
Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

Horizon -1

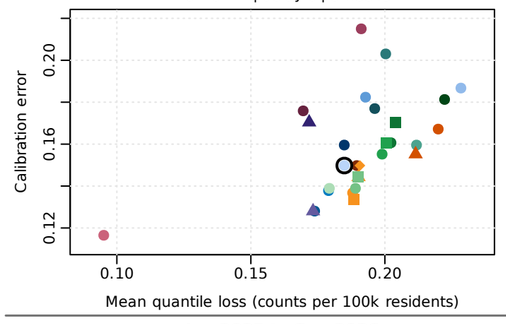
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 32 weeks



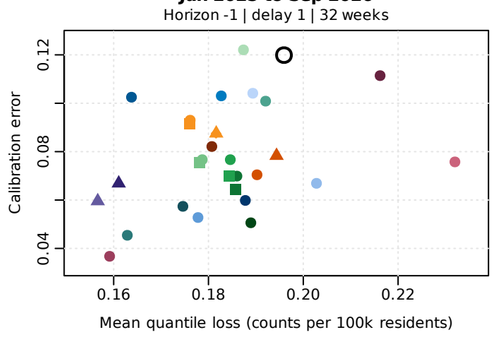
Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 7 weeks



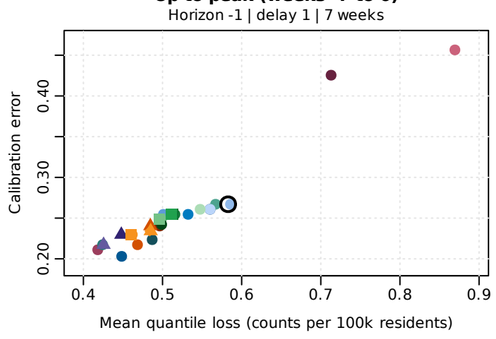
After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 8 weeks



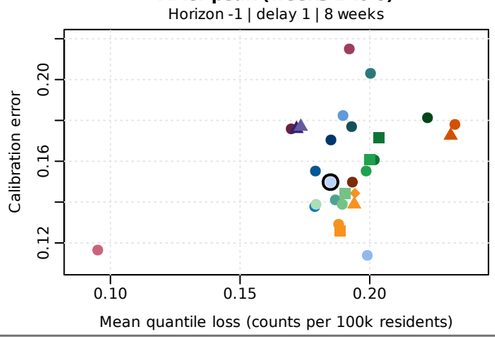
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 32 weeks



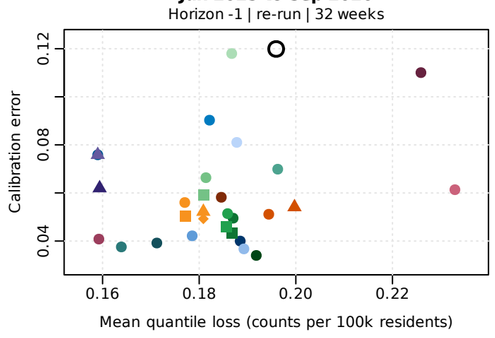
Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 7 weeks



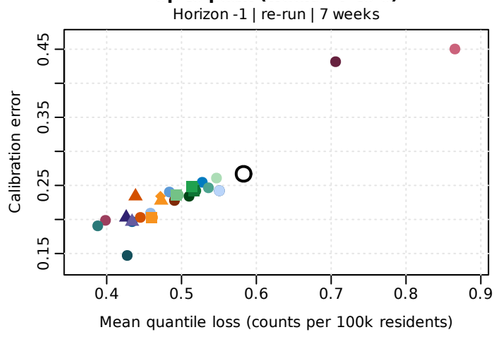
After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 8 weeks



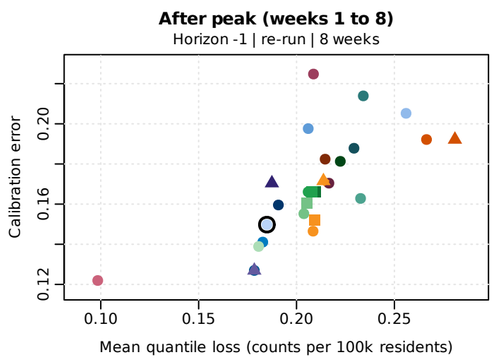
Jun 2025 to Sep 2026
Horizon -1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 7 weeks

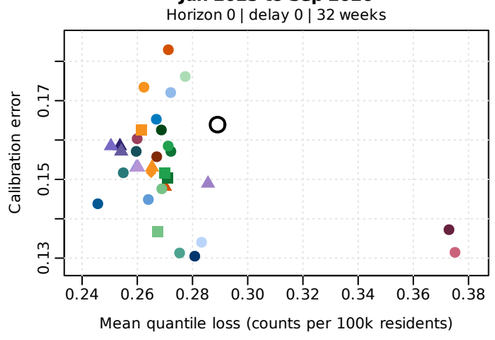


After peak (weeks 1 to 8)
Horizon -1 | re-run | 8 weeks

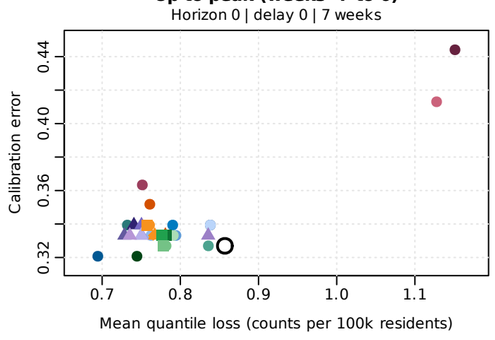


Horizon 0

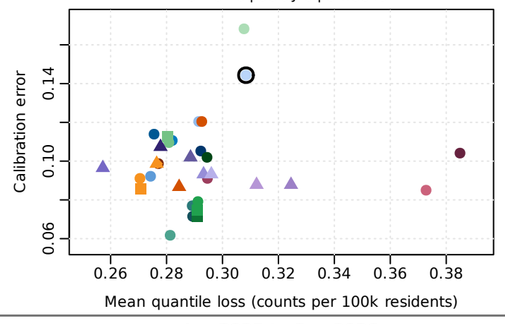
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 32 weeks



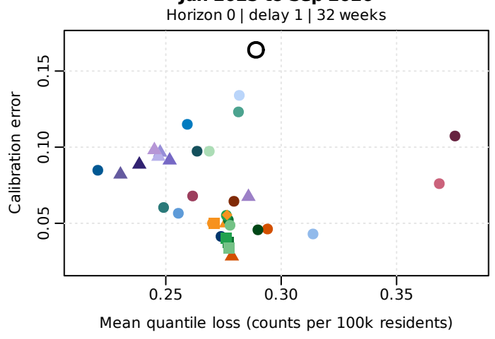
Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 7 weeks



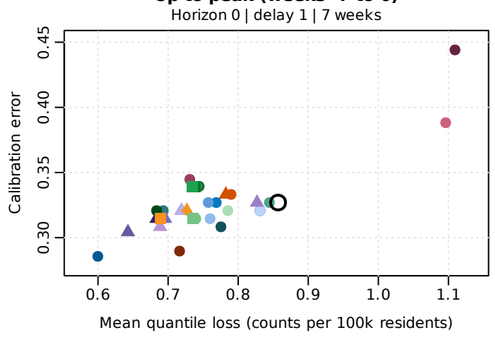
After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 8 weeks



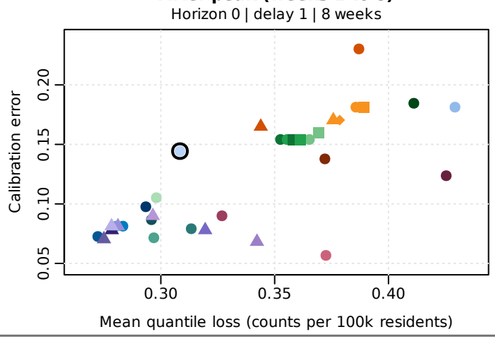
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 32 weeks



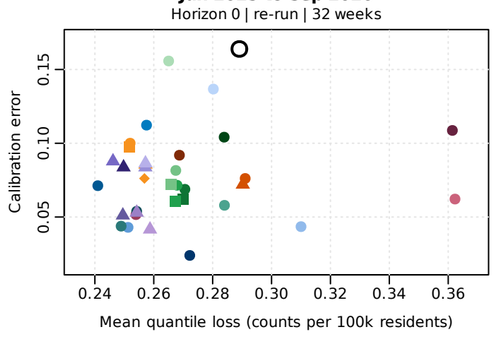
Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 7 weeks



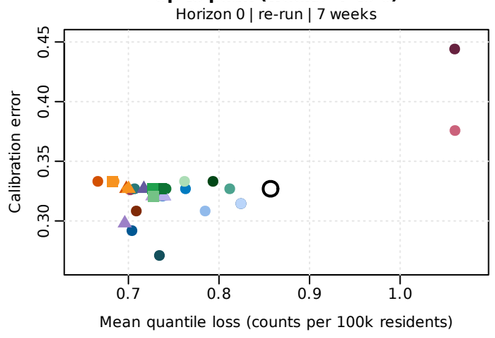
After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 8 weeks



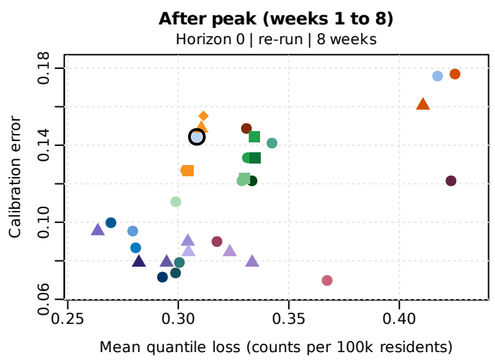
Jun 2025 to Sep 2026
Horizon 0 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 7 weeks

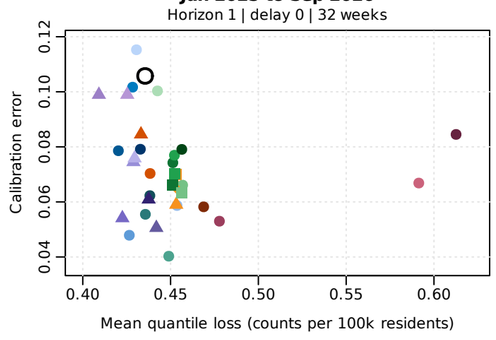


After peak (weeks 1 to 8)
Horizon 0 | re-run | 8 weeks

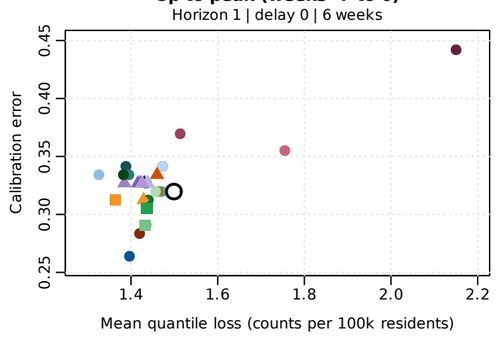


Horizon 1

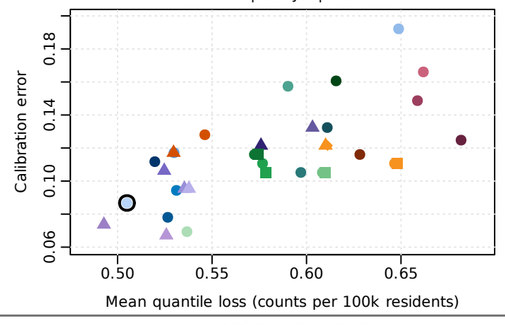
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 32 weeks



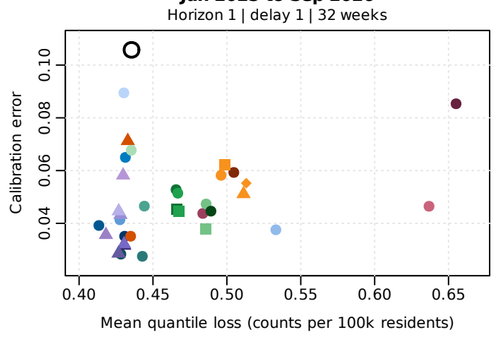
Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 6 weeks



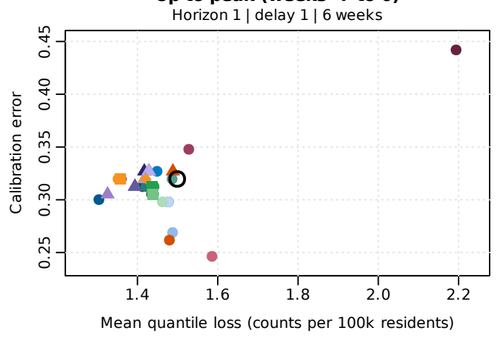
After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 8 weeks



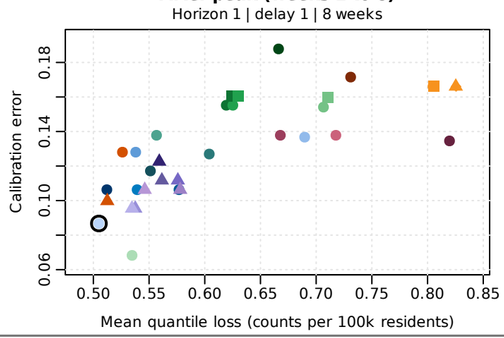
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 32 weeks



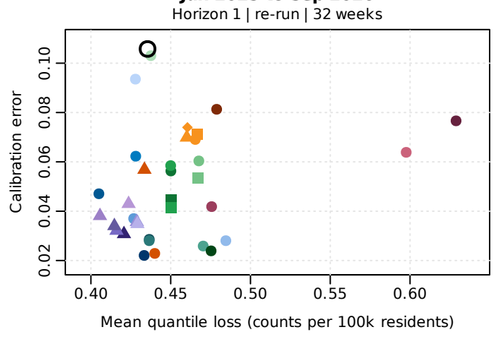
Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 6 weeks



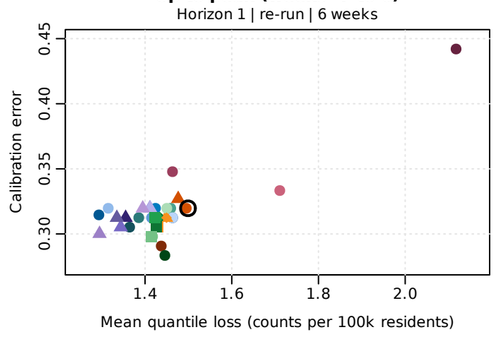
After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 8 weeks



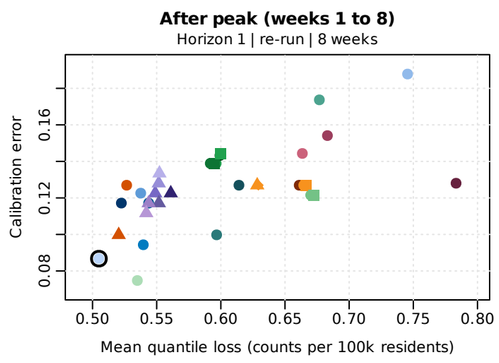
Jun 2025 to Sep 2026
Horizon 1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 6 weeks

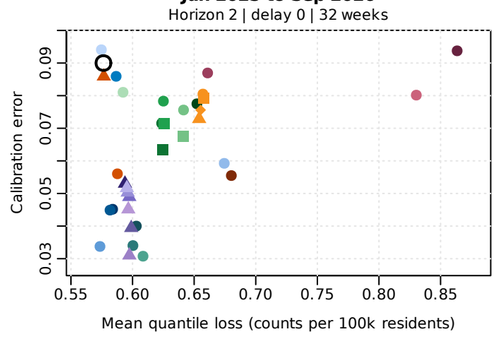


After peak (weeks 1 to 8)
Horizon 1 | re-run | 8 weeks

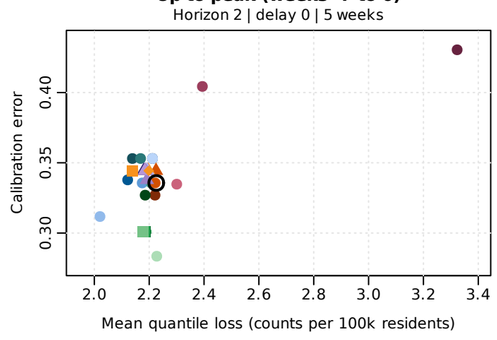


Horizon 2

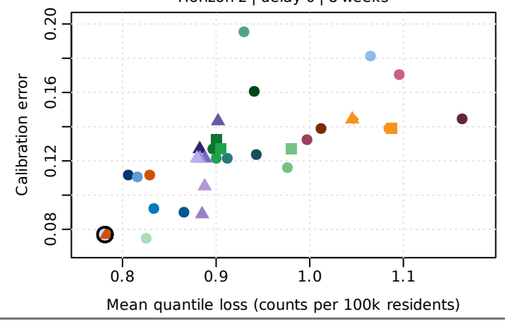
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 32 weeks



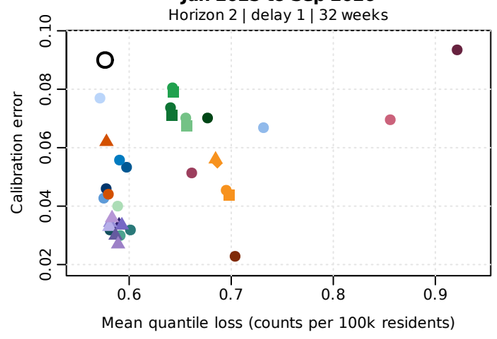
Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 5 weeks



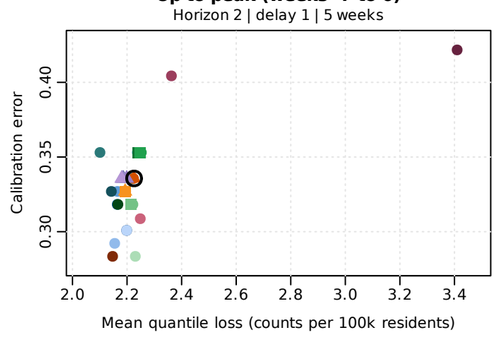
After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 8 weeks



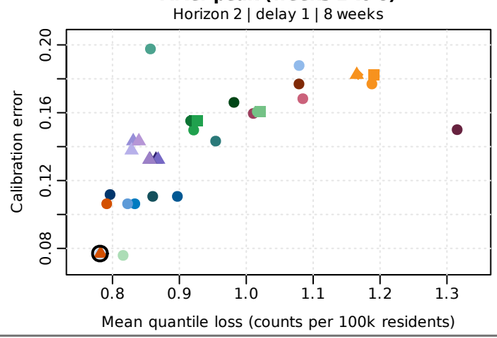
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 32 weeks



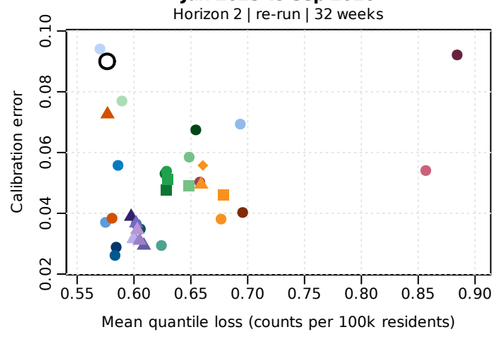
Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 5 weeks



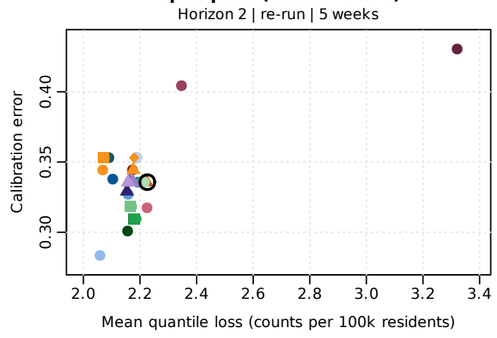
After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 8 weeks



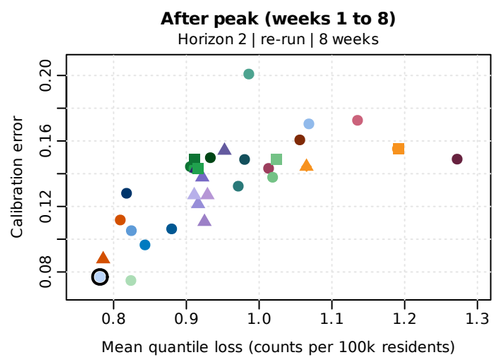
Jun 2025 to Sep 2026
Horizon 2 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 5 weeks

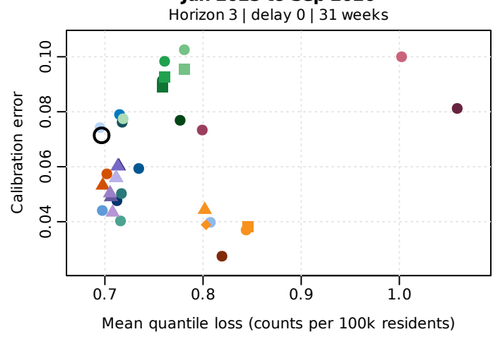


After peak (weeks 1 to 8)
Horizon 2 | re-run | 8 weeks

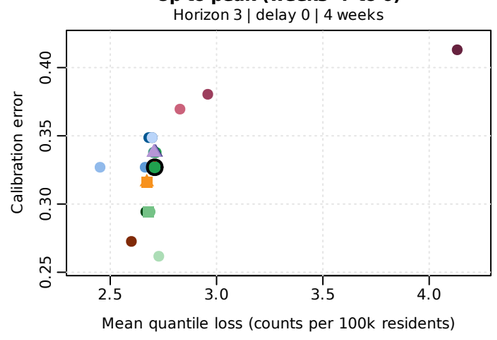


Horizon 3

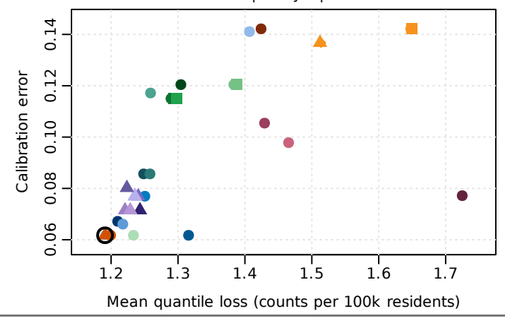
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 31 weeks



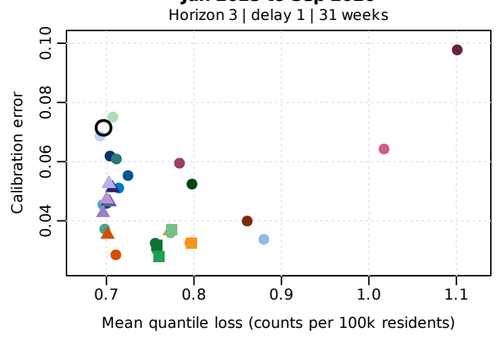
Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 4 weeks



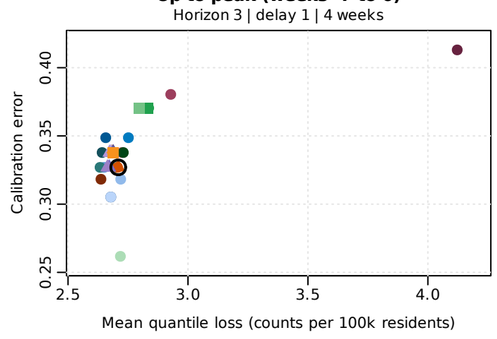
After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 8 weeks



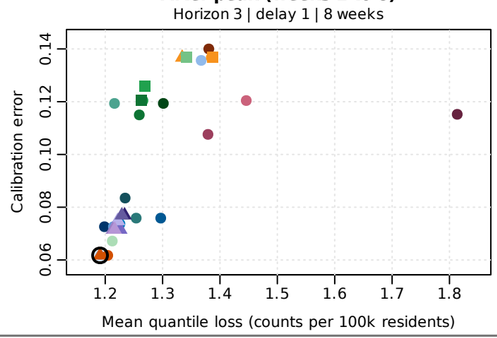
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 31 weeks



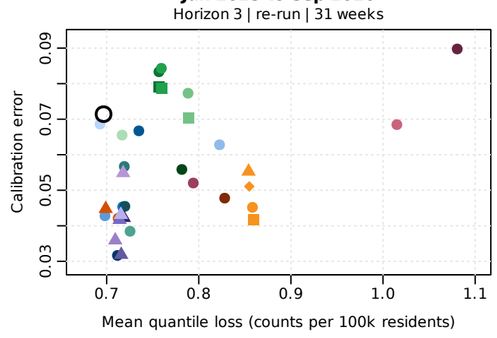
Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 4 weeks



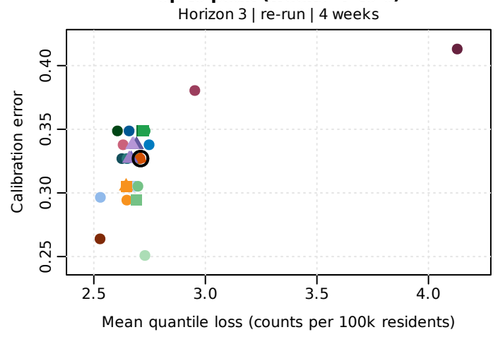
After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 8 weeks



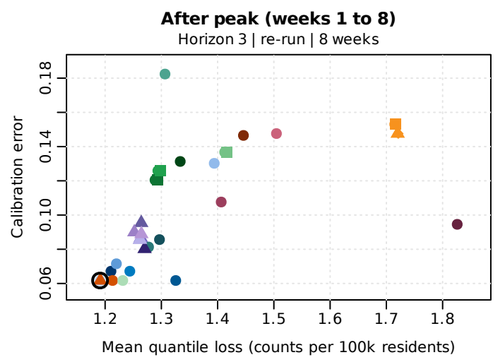
Jun 2025 to Sep 2026
Horizon 3 | re-run | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 4 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 8 weeks



Feedback delay 0

Feedback delay 1

Feedback delay 0 + revisions (re-run)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=2)
- Rate grid: learned + sharing, common weights (H=2)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

slow 0.003, leak 0.1)
)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Score-based experts
- Score experts: one vector (A, H=4, sharing 1/8)
- Score experts: soft switch, target signal (C, delta=0.1, H=4, sharing 1/8)
- Score experts: context vectors, blended (D, delta=0, H=4, sharing 1/8)
- Score experts: context vectors, hard (D, delta=0, H=4, sharing 1/8)

Score-based experts without sharing
- Score experts: one vector (A, no sharing)
- Score experts: soft switch, target signal (C, delta=0.1, no sharing)
- Score experts: context vectors, blended (D, delta=0, no sharing)

Studentized m
- Studentized me
- Studentized me
- Median, constar

Studentized fu
- Studentized full
- Studentized full
- Studentized full

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
ntext vectors, hard (D, delta=0, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: lambda x 1 in decline (E, delta=0.05, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts:
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday information cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the 80% interval width of the base / 2.563 (a normal sd).
With residuals, the original base is used until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets are initially zero; level scales = normal sd of the central interval of the base at each level, or the RMS of the last 5 released residuals at each level (weighted isotonic projection)

flu | Pennsylvania (42) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

slow 0.003, leak 0.1)
)

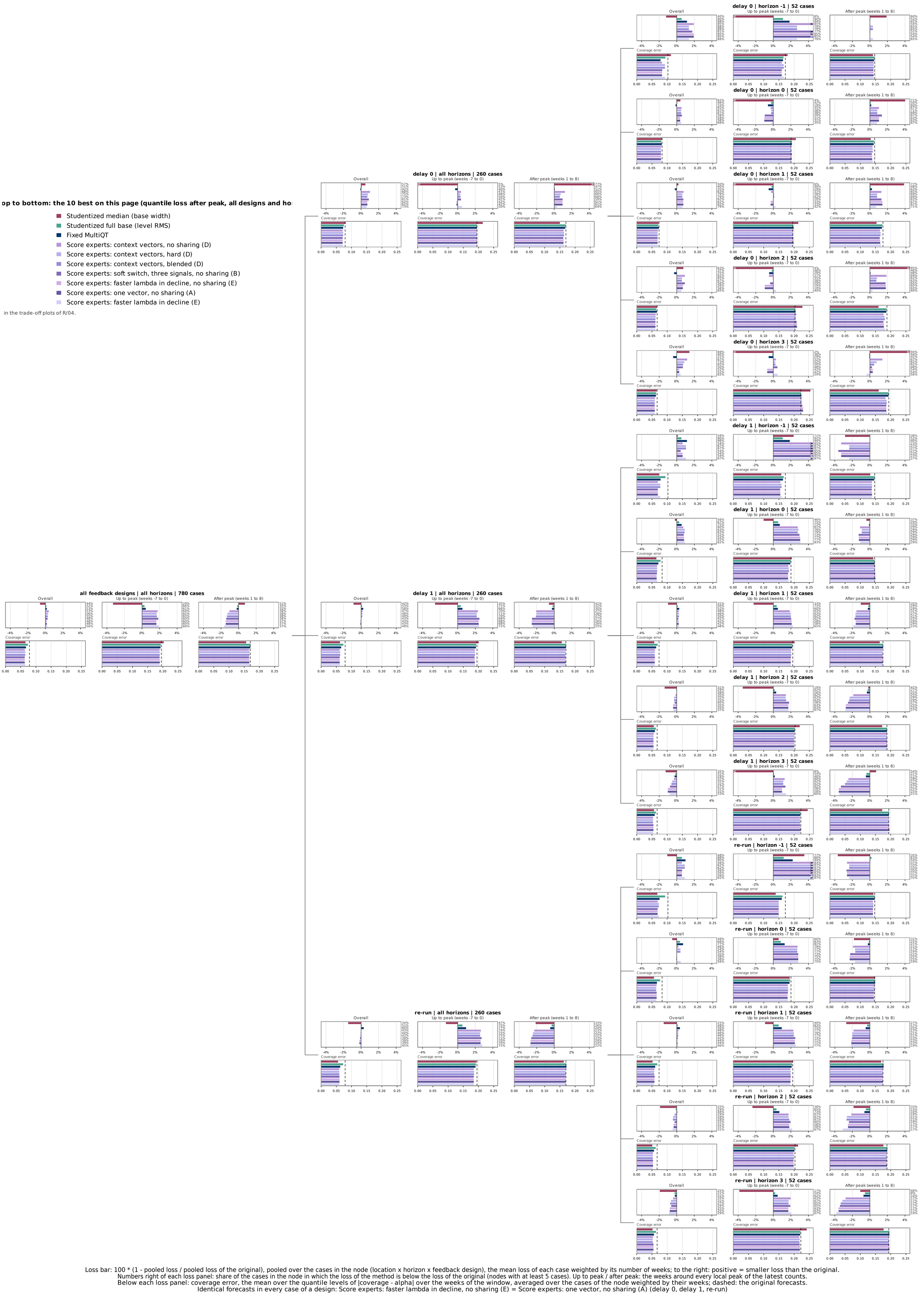
Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
ntext vectors, hard (D, delta=0, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: lambda x 1 in decline (E, delta=0.05, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts:
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday information cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the 80% interval width of the base / 2.563 (a normal sd).
With residuals, the original base is used until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets are initially zero; level scales = normal sd of the central interval of the base at each level, or the RMS of the last 5 released residuals at each level (weighted isotonic projection)

covid | all 52 locations pooled | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents

op to bottom: the 10 best on this page (quantile loss after peak, all designs and ho

- Studentized median (base width)
- Studentized full base (level RMS)
- Fixed MultiQT
- Score experts: context vectors, no sharing (D)
- Score experts: context vectors, hard (D)
- Score experts: context vectors, blended (D)
- Score experts: soft switch, three signals, no sharing (B)
- Score experts: faster lambda in decline, no sharing (E)
- Score experts: one vector, no sharing (A)
- Score experts: faster lambda in decline (E)

in the trade-off plots of R/04.



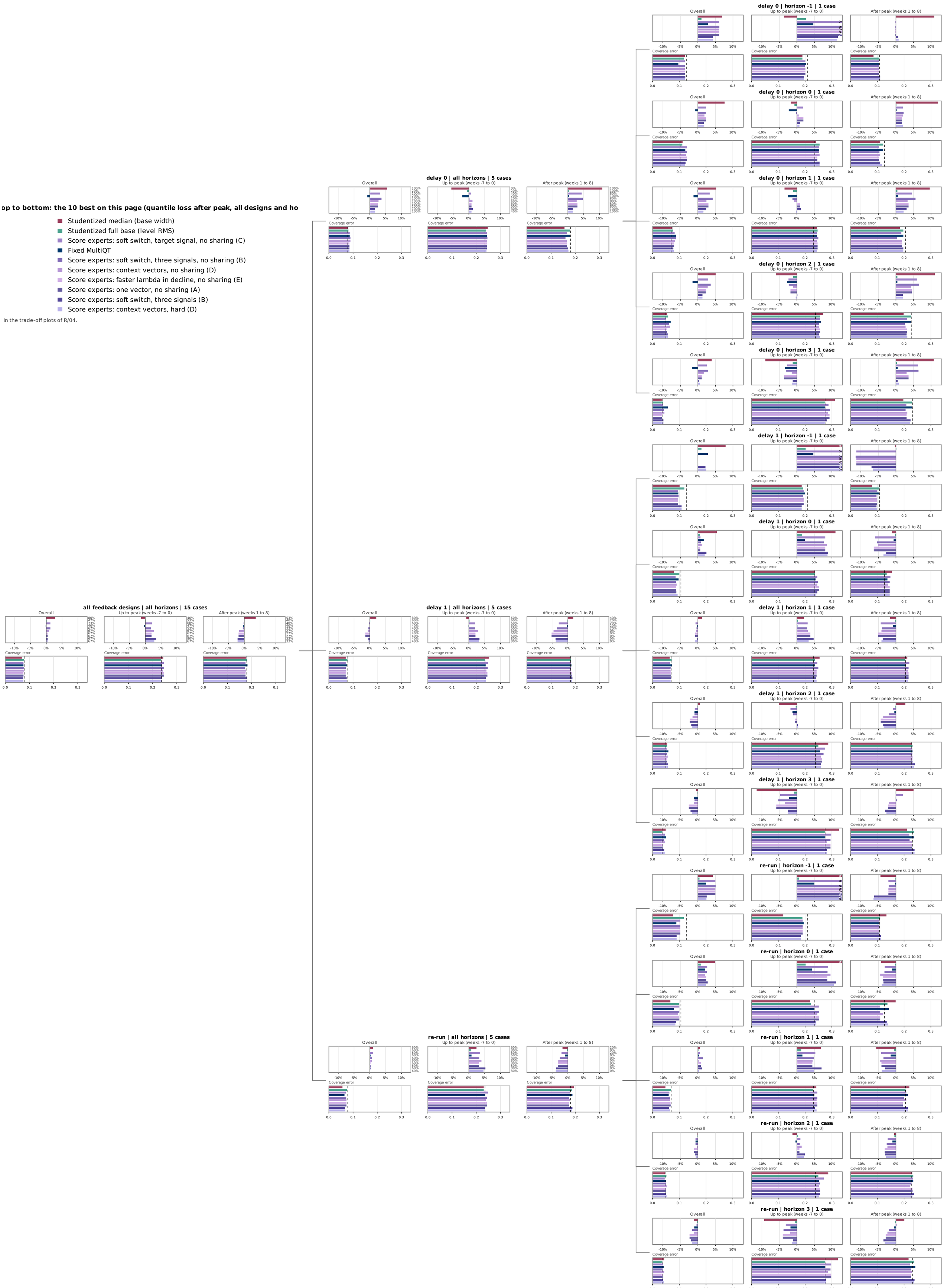
Loss bar: 100 * (1 - pooled loss / pooled loss of the original), pooled over the cases in the node (location x horizon x feedback design), the mean loss of each case weighted by its number of weeks; to the right: positive = smaller loss than the original.
Numbers right of each loss panel: share of the cases in the node in which the loss of the method is below the loss of the original (nodes with at least 5 cases). Up to peak / after peak: the weeks around every local peak of the latest counts.
Below each loss panel: coverage error, the mean over the quantile levels of |coverage - alpha| over the weeks of the window, averaged over the cases of the node weighted by their weeks; dashed: the original forecasts.
Identical forecasts in every case of a design: Score experts: faster lambda in decline, no sharing (E) = Score experts: one vector, no sharing (A) (delay 0, delay 1, re-run)

covid | United States (US) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents

op to bottom: the 10 best on this page (quantile loss after peak, all designs and ho

- Studentized median (base width)
- Studentized full base (level RMS)
- Score experts: soft switch, target signal, no sharing (C)
- Fixed MultiQT
- Score experts: soft switch, three signals, no sharing (B)
- Score experts: context vectors, no sharing (D)
- Score experts: faster lambda in decline, no sharing (E)
- Score experts: one vector, no sharing (A)
- Score experts: soft switch, three signals (B)
- Score experts: context vectors, hard (D)

in the trade-off plots of R/04.



Loss bar: 100 * (1 - pooled loss / pooled loss of the original), pooled over the cases in the node (location x horizon x feedback design), the mean loss of each case weighted by its number of weeks; to the right: positive = smaller loss than the original.
Numbers right of each loss panel: share of the cases in the node in which the loss of the method is below the loss of the original (nodes with at least 5 cases). Up to peak / after peak: the weeks around every local peak of the latest counts.
Below each loss panel: coverage error, the mean over the quantile levels of |coverage - alpha| over the weeks of the window, averaged over the cases of the node weighted by their weeks; dashed: the original forecasts.
Identical forecasts in every case of a design: Score experts: faster lambda in decline, no sharing (E) = Score experts: one vector, no sharing (A) (delay 0, delay 1, re-run)

covid | California (06) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents

op to bottom: the 10 best on this page (quantile loss after peak, all designs and ho

- Studentized median (residual RMS)
- Studentized median (base width)
- Studentized full base (level RMS)
- Fixed MultiQT
- Weekly starts: most recent, equal
- Score experts: soft switch, three signals, no sharing (B)
- Score experts: context vectors, no sharing (D)
- Score experts: faster lambda in decline, no sharing (E)
- Score experts: one vector, no sharing (A)
- Score experts: soft switch, target signal, no sharing (C)

in the trade-off plots of R/04.



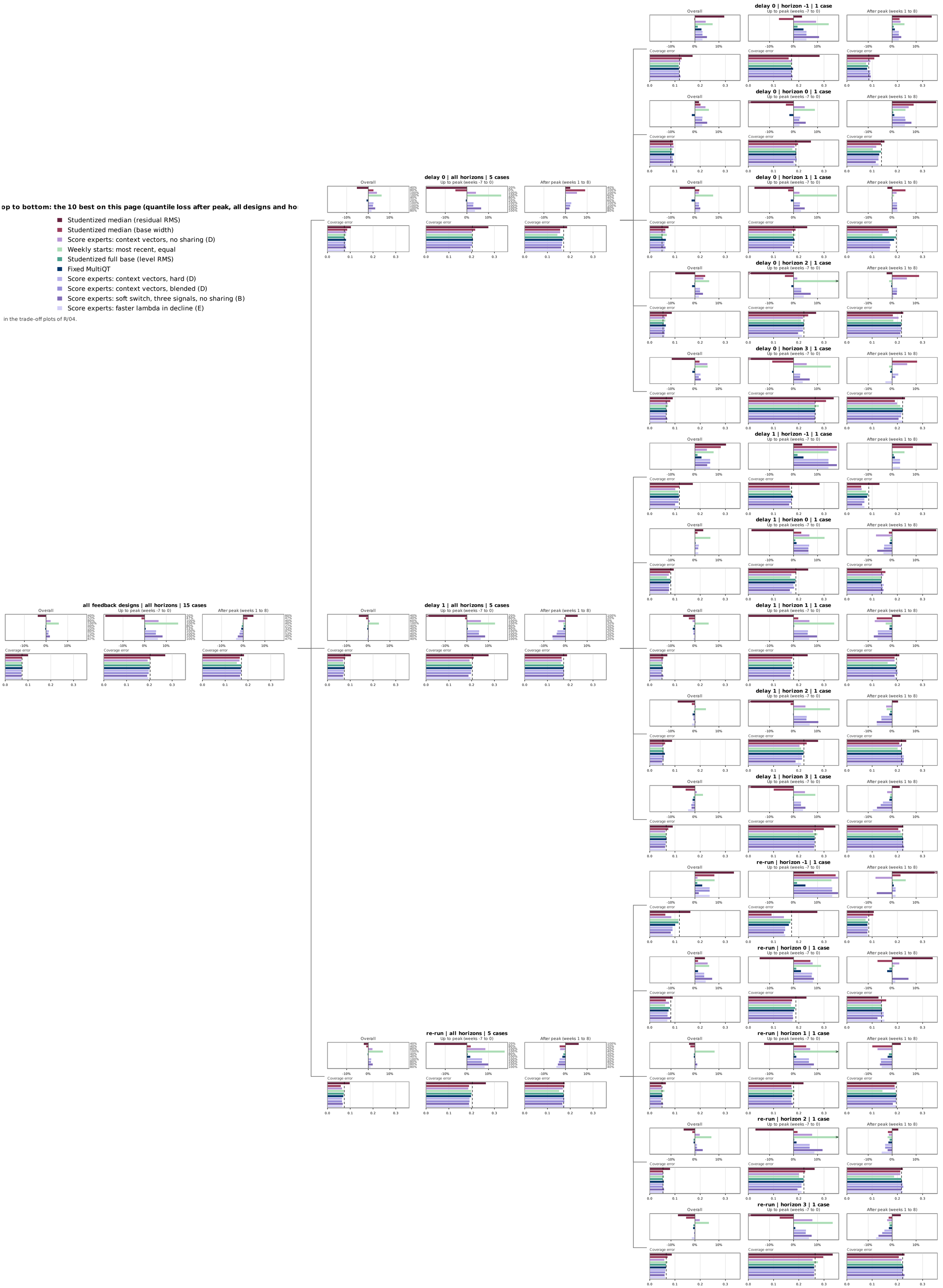
Loss bar: 100 * (1 - pooled loss / pooled loss of the original), pooled over the cases in the node (location x horizon x feedback design), the mean loss of each case weighted by its number of weeks; to the right: positive = smaller loss than the original.
Numbers right of each loss panel: share of the cases in the node in which the loss of the method is below the loss of the original (nodes with at least 5 cases). Up to peak / after peak: the weeks around every local peak of the latest counts.
Below each loss panel: coverage error, the mean over the quantile levels of |coverage - alpha| over the weeks of the window, averaged over the cases of the node weighted by their weeks; dashed: the original forecasts.
Identical forecasts in every case of a design: Score experts: faster lambda in decline, no sharing (E) = Score experts: one vector, no sharing (A) (delay 0, delay 1, re-run)

covid | Texas (48) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents

op to bottom: the 10 best on this page (quantile loss after peak, all designs and ho

- Studentized median (residual RMS)
- Studentized median (base width)
- Score experts: context vectors, no sharing (D)
- Weekly starts: most recent, equal
- Studentized full base (level RMS)
- Fixed MultiQT
- Score experts: context vectors, hard (D)
- Score experts: context vectors, blended (D)
- Score experts: soft switch, three signals, no sharing (B)
- Score experts: faster lambda in decline (E)

in the trade-off plots of R/04.



Loss bar: 100 * (1 - pooled loss / pooled loss of the original), pooled over the cases in the node (location x horizon x feedback design), the mean loss of each case weighted by its number of weeks; to the right: positive = smaller loss than the original.
Numbers right of each loss panel: share of the cases in the node in which the loss of the method is below the loss of the original (nodes with at least 5 cases). Up to peak / after peak: the weeks around every local peak of the latest counts.
Below each loss panel: coverage error, the mean over the quantile levels of |coverage - alpha| over the weeks of the window, averaged over the cases of the node weighted by their weeks; dashed: the original forecasts.
At horizon -1 (no target signal) the score mixtures C, D and E give the forecast of A (one vector).

covid | New York (36) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents

op to bottom: the 10 best on this page (quantile loss after peak, all designs and ho

- Studentized median (base width)
- Score experts: soft switch, target signal, no sharing (C)
- Score experts: soft switch, three signals, no sharing (B)
- Score experts: faster lambda in decline, no sharing (E)
- Score experts: one vector, no sharing (A)
- Score experts: context vectors, no sharing (D)
- Score experts: context vectors, hard (D)
- Score experts: context vectors, blended (D)
- Score experts: soft switch, three signals (B)
- Score experts: soft switch, target signal (C)

in the trade-off plots of R/04.



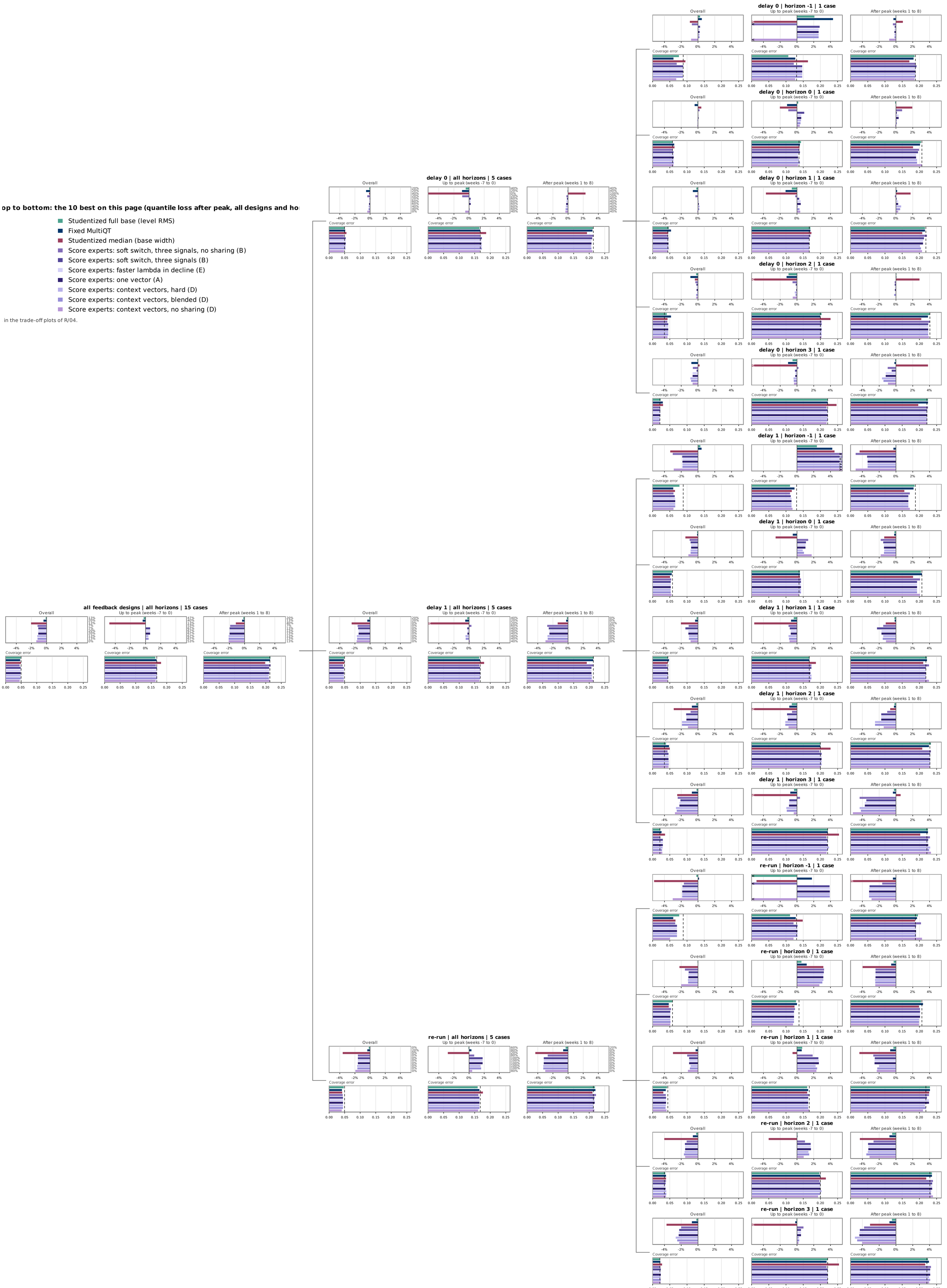
Loss bar: 100 * (1 - pooled loss / pooled loss of the original), pooled over the cases in the node (location x horizon x feedback design), the mean loss of each case weighted by its number of weeks; to the right: positive = smaller loss than the original.
Numbers right of each loss panel: share of the cases in the node in which the loss of the method is below the loss of the original (nodes with at least 5 cases). Up to peak / after peak: the weeks around every local peak of the latest counts.
Below each loss panel: coverage error, the mean over the quantile levels of |coverage - alpha| over the weeks of the window, averaged over the cases of the node weighted by their weeks; dashed: the original forecasts.
Identical forecasts in every case of a design: Score experts: faster lambda in decline, no sharing (E) = Score experts: one vector, no sharing (A) (delay 0, delay 1, re-run)

covid | Pennsylvania (42) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents

op to bottom: the 10 best on this page (quantile loss after peak, all designs and ho

- Studentized full base (level RMS)
- Fixed MultiQT
- Studentized median (base width)
- Score experts: soft switch, three signals, no sharing (B)
- Score experts: soft switch, three signals (B)
- Score experts: faster lambda in decline (E)
- Score experts: one vector (A)
- Score experts: context vectors, hard (D)
- Score experts: context vectors, blended (D)
- Score experts: context vectors, no sharing (D)

in the trade-off plots of R/04.



Loss bar: 100 * (1 - pooled loss / pooled loss of the original), pooled over the cases in the node (location x horizon x feedback design), the mean loss of each case weighted by its number of weeks; to the right: positive = smaller loss than the original.
Numbers right of each loss panel: share of the cases in the node in which the loss of the method is below the loss of the original (nodes with at least 5 cases). Up to peak / after peak: the weeks around every local peak of the latest counts.
Below each loss panel: coverage error, the mean over the quantile levels of |coverage - alpha| over the weeks of the window, averaged over the cases of the node weighted by their weeks; dashed: the original forecasts.
Identical forecasts in every case of a design: Score experts: faster lambda in decline (E) = Score experts: one vector (A) (delay 0, delay 1, re-run)

flu | all 52 locations pooled | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents

op to bottom: the 10 best on this page (quantile loss after peak, all designs and ho

- Fixed MultiQT
- Studentized full base (level RMS)
- Rate grid: learned, common weights
- Delphi sqrt 0.018 (flu prod)
- Weekly starts: most recent, equal
- Rate grid: learned, no sharing
- Delphi REF-op (covid prod)
- Weekly starts: equal
- Weekly starts: learned
- Starts equal + outer

in the trade-off plots of R/04.
non weights for all levels.
ne: + outer MultiQT.



Loss bar: 100 * (1 - pooled loss / pooled loss of the original), pooled over the cases in the node (location x horizon x feedback design), the mean loss of each case weighted by its number of weeks; to the right: positive = smaller loss than the original.
Numbers right of each loss panel: share of the cases in the node in which the loss of the method is below the loss of the original (nodes with at least 5 cases). Up to peak / after peak: the weeks around every local peak of the latest counts.
Below each loss panel: coverage error, the mean over the quantile levels of |coverage - alpha| over the weeks of the window, averaged over the cases of the node weighted by their weeks; dashed: the original forecasts.

flu | United States (US) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents

op to bottom: the 10 best on this page (quantile loss after peak, all designs and ho

- Studentized full base (level RMS)
- Fixed MultiQT
- Rate grid: learned, common weights
- Starts equal + outer
- Weekly starts: equal
- Weekly starts: learned
- Starts learned + outer
- Rate grid: learned, no sharing
- Delphi sqrt 0.018 (flu prod)
- Starts x rate grid: learned

in the trade-off plots of R/04.
non weights for all levels.
ne: + outer MultiQT.

Loss bar: 100 * (1 - pooled loss / pooled loss of the original), pooled over the cases in the node (location x horizon x feedback design), the mean loss of each case weighted by its number of weeks; to the right: positive = smaller loss than the original.
Numbers right of each loss panel: share of the cases in the node in which the loss of the method is below the loss of the original (nodes with at least 5 cases). Up to peak / after peak: the weeks around every local peak of the latest counts.
Below each loss panel: coverage error, the mean over the quantile levels of |coverage - alpha| over the weeks of the window, averaged over the cases of the node weighted by their weeks; dashed: the original forecasts.

flu | California (06) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents

op to bottom: the 10 best on this page (quantile loss after peak, all designs and ho

- Weekly starts: most recent, equal
- Delphi REF-op (covid prod)
- Score experts: faster lambda in decline (E)
- Score experts: one vector (A)
- Score experts: soft switch, three signals (B)
- Score experts: soft switch, target signal (C)
- Score experts: faster lambda in decline, no sharing (E)
- Score experts: one vector, no sharing (A)
- Score experts: soft switch, three signals, no sharing (B)
- Fixed MultiQT

in the trade-off plots of R/04.



Loss bar: 100 * (1 - pooled loss / pooled loss of the original), pooled over the cases in the node (location x horizon x feedback design), the mean loss of each case weighted by its number of weeks; to the right: positive = smaller loss than the original.
Numbers right of each loss panel: share of the cases in the node in which the loss of the method is below the loss of the original (nodes with at least 5 cases). Up to peak / after peak: the weeks around every local peak of the latest counts.
Below each loss panel: coverage error, the mean over the quantile levels of |coverage - alpha| over the weeks of the window, averaged over the cases of the node weighted by their weeks; dashed: the original forecasts.
nals (B) (delay 0, delay 1, re-run); Score experts: faster lambda in decline, no sharing (E) = Score experts: one vector, no sharing (A) (delay 0, delay 1, re-run); Score experts: faster lambda in decline, no sharing (E) = Score experts: soft switch, three signals, no sharing (B) (delay 0, delay 1, re-run);

flu | Texas (48) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents

op to bottom: the 10 best on this page (quantile loss after peak, all designs and ho

- Rate grid: equal
- Rate grid: learned + sharing, common weights
- Rate grid: sharing, common weights + outer
- Starts x rate grid + outer
- Starts x rate grid: learned
- Studentized full base (level RMS)
- Slope rate
- Rate grid: learned + sharing
- Rate grid: sharing + outer
- Starts learned + outer

in the trade-off plots of R/04.
non weights for all levels.
ne: + outer MultiQT.

all feedback designs | all horizons | 15 cases

delay 0 | all horizons | 5 cases

delay 1 | all horizons | 5 cases

re-run | all horizons | 5 cases

delay 0 | horizon -1 | 1 case

delay 0 | horizon 0 | 1 case

delay 0 | horizon 1 | 1 case

delay 0 | horizon 2 | 1 case

delay 0 | horizon 3 | 1 case

delay 1 | horizon -1 | 1 case

delay 1 | horizon 0 | 1 case

delay 1 | horizon 1 | 1 case

delay 1 | horizon 2 | 1 case

delay 1 | horizon 3 | 1 case

re-run | horizon -1 | 1 case

re-run | horizon 0 | 1 case

re-run | horizon 1 | 1 case

re-run | horizon 2 | 1 case

re-run | horizon 3 | 1 case

Overall | Up to peak (weeks -7 to 0) | After peak (weeks 1 to 8)

Coverage error

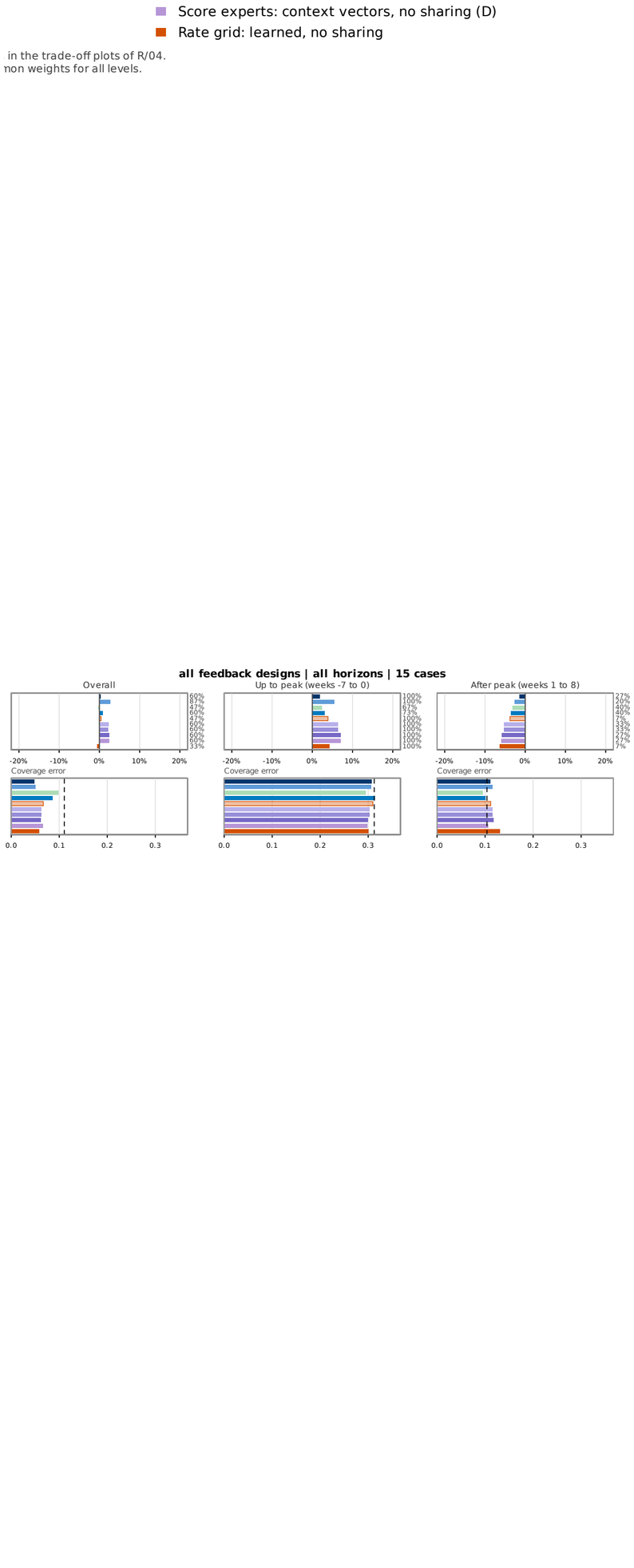
Loss bar: 100 * (1 - pooled loss / pooled loss of the original), pooled over the cases in the node (location x horizon x feedback design), the mean loss of each case weighted by its number of weeks; to the right: positive = smaller loss than the original.
Numbers right of each loss panel: share of the cases in the node in which the loss of the method is below the loss of the original (nodes with at least 5 cases). Up to peak / after peak: the weeks around every local peak of the latest counts.
Below each loss panel: coverage error, the mean over the quantile levels of |coverage - alpha| over the weeks of the window, averaged over the cases of the node weighted by their weeks; dashed: the original forecasts.

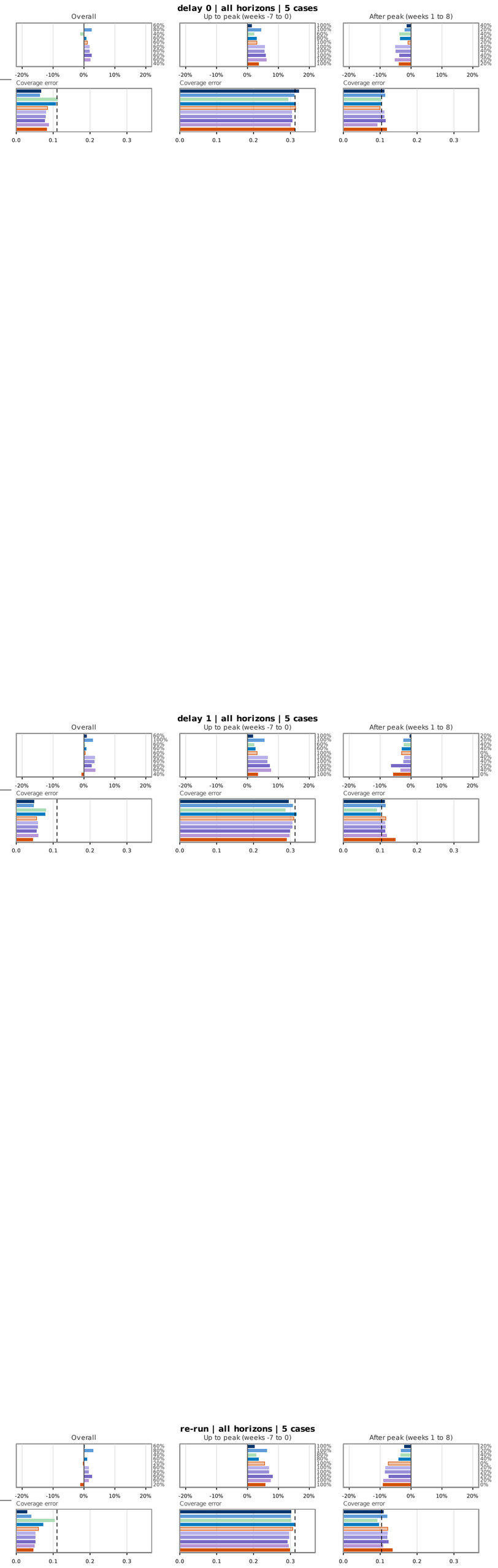
flu | New York (36) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents

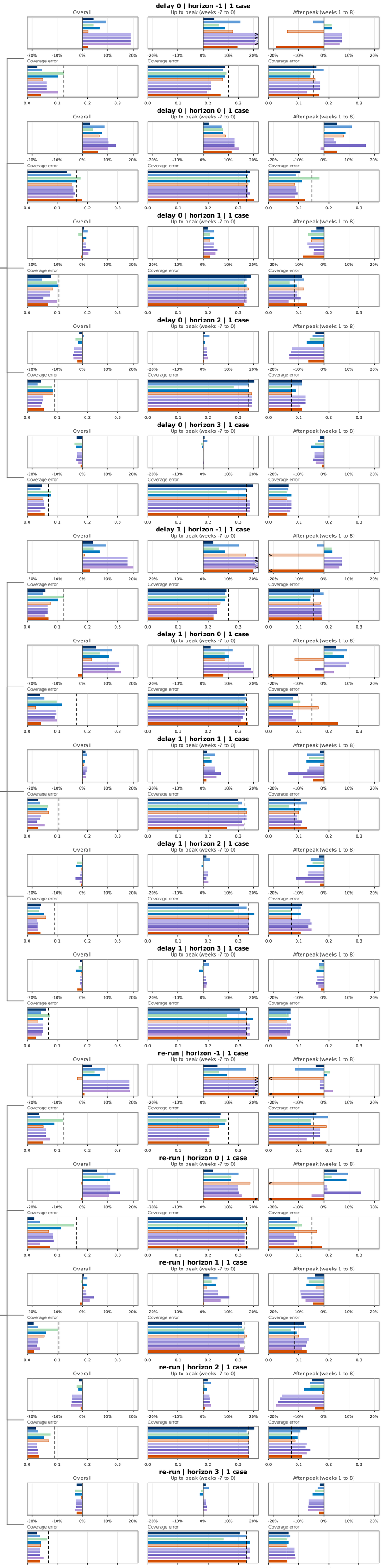
op to bottom: the 10 best on this page (quantile loss after peak, all designs and ho

- Fixed MultiQT
- Delphi sqrt 0.018 (flu prod)
- Weekly starts: most recent, equal
- Delphi REF-op (covid prod)
- Rate grid: learned, common weights
- Score experts: context vectors, hard (D)
- Score experts: context vectors, blended (D)
- Score experts: soft switch, target signal (C)
- Score experts: context vectors, no sharing (D)
- Rate grid: learned, no sharing

in the trade-off plots of R/04.
non weights for all levels.







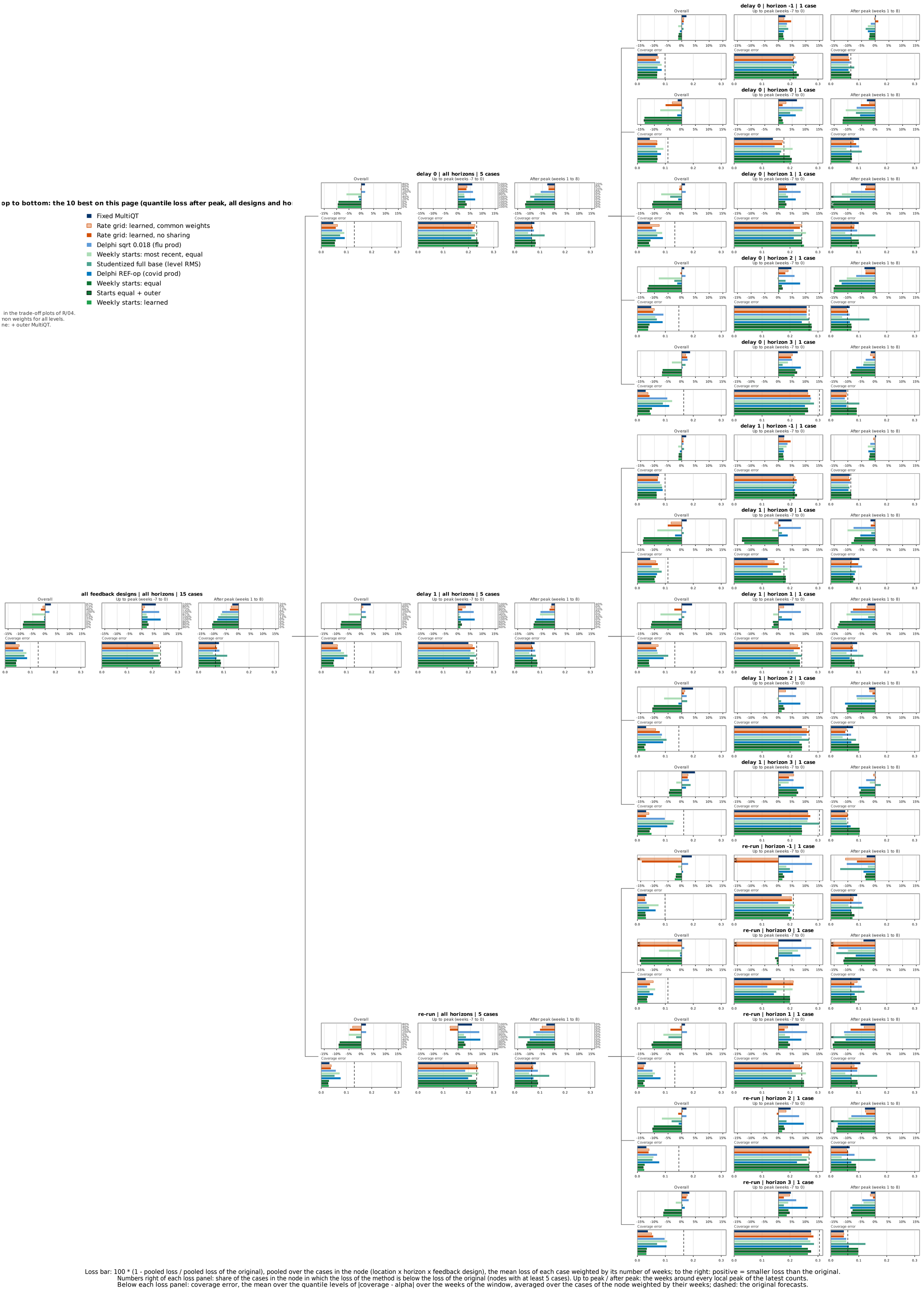
Loss bar: 100 * (1 - pooled loss / pooled loss of the original), pooled over the cases in the node (location x horizon x feedback design), the mean loss of each case weighted by its number of weeks; to the right: positive = smaller loss than the original.
Numbers right of each loss panel: share of the cases in the node in which the loss of the method is below the loss of the original (nodes with at least 5 cases). Up to peak / after peak: the weeks around every local peak of the latest counts.
Below each loss panel: coverage error, the mean over the quantile levels of |coverage - alpha| over the weeks of the window, averaged over the cases of the node weighted by their weeks; dashed: the original forecasts.
At horizon -1 (no target signal) the score mixtures C, D and E give the forecast of A (one vector).

flu | Pennsylvania (42) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents

op to bottom: the 10 best on this page (quantile loss after peak, all designs and ho

- Fixed MultiQT
- Rate grid: learned, common weights
- Rate grid: learned, no sharing
- Delphi sqrt 0.018 (flu prod)
- Weekly starts: most recent, equal
- Studentized full base (level RMS)
- Delphi REF-op (covid prod)
- Weekly starts: equal
- Starts equal + outer
- Weekly starts: learned

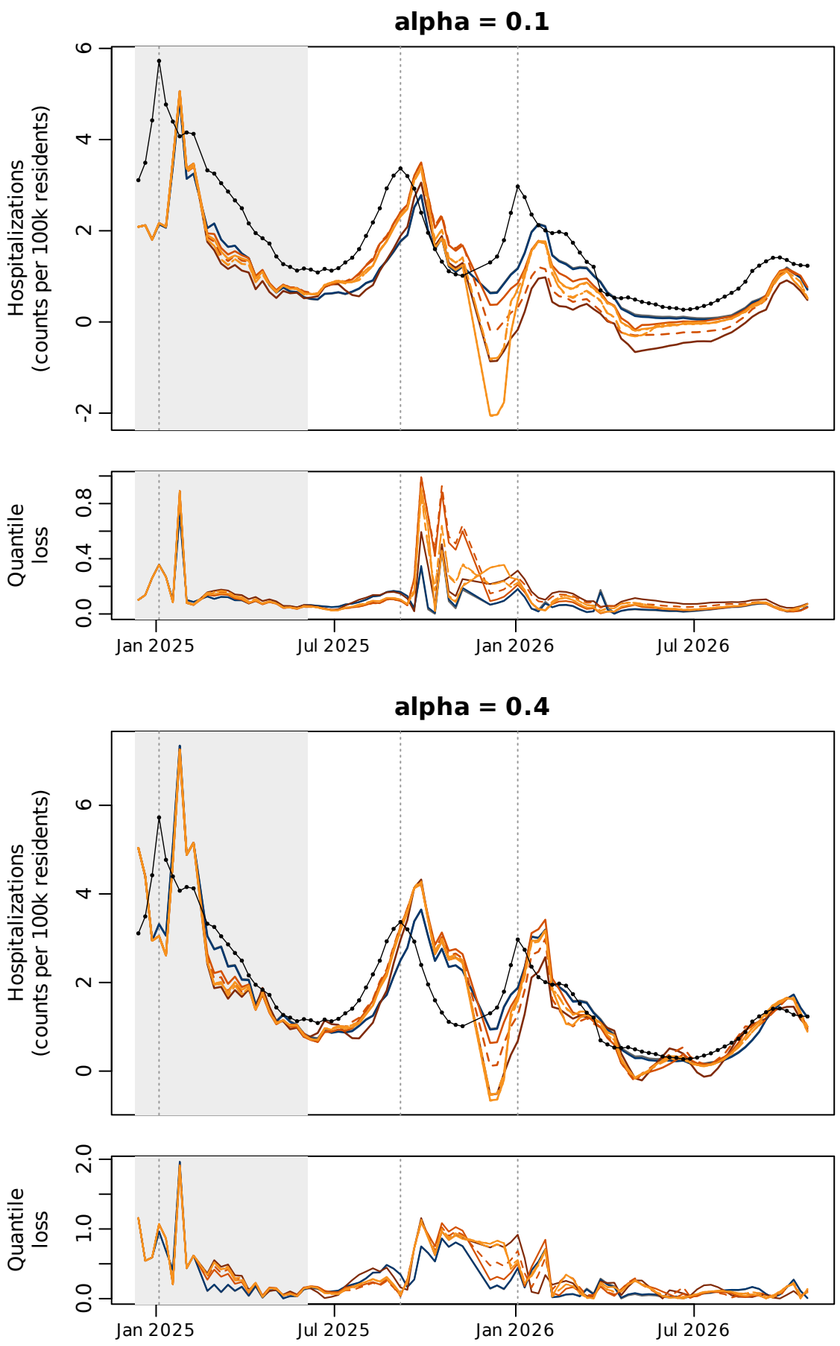
in the trade-off plots of R/04.
non weights for all levels.
ne: + outer MultiQT.

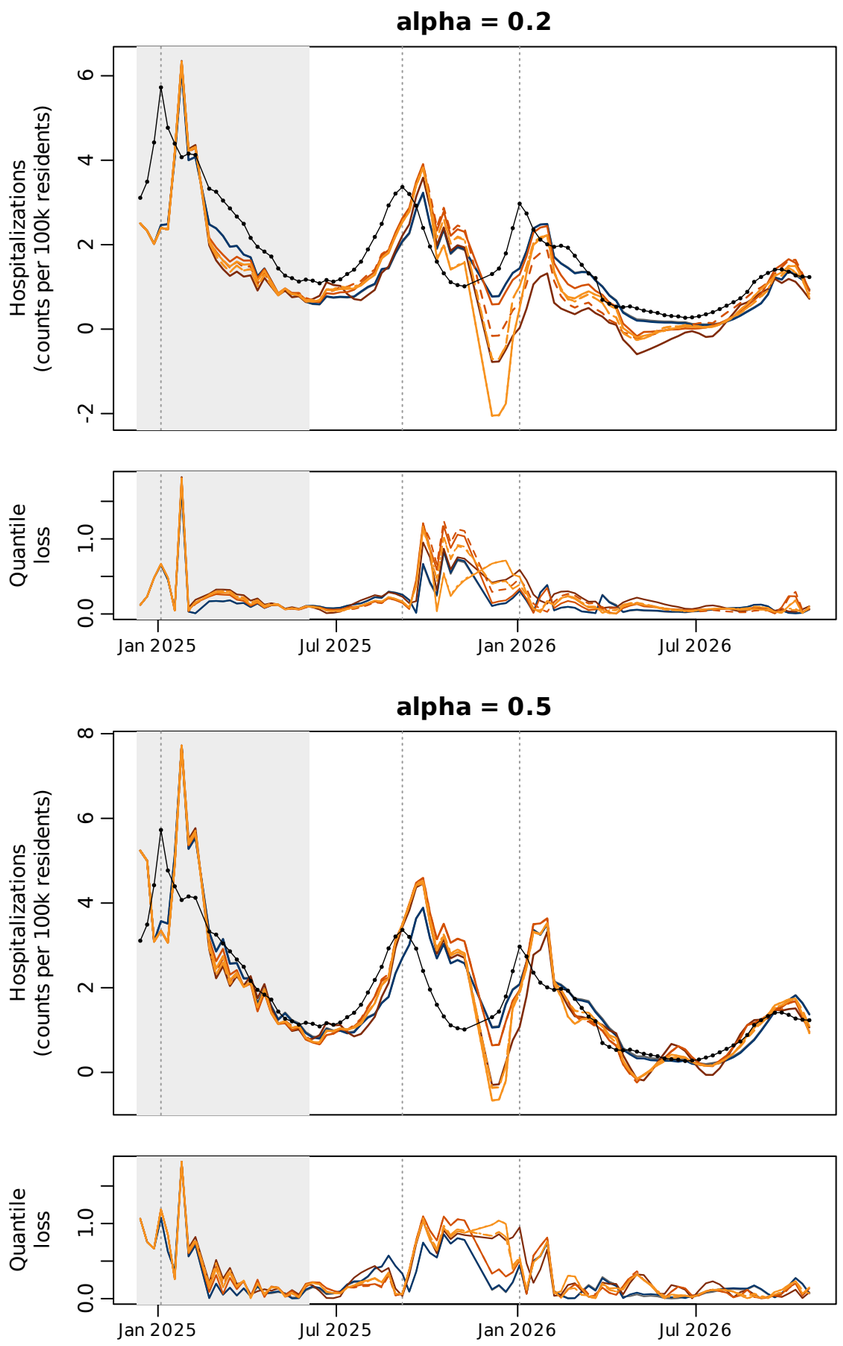


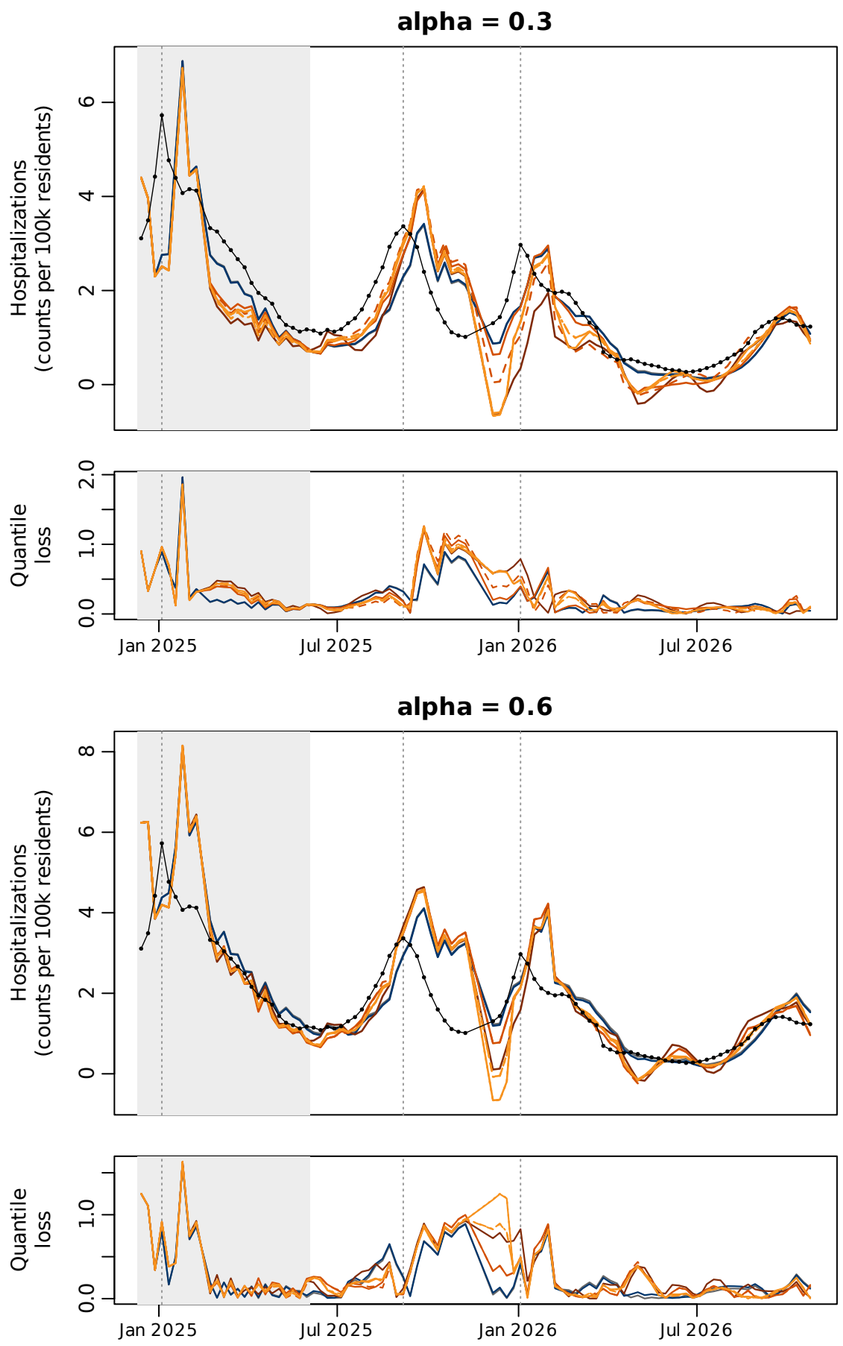
Loss bar: 100 * (1 - pooled loss / pooled loss of the original), pooled over the cases in the node (location x horizon x feedback design), the mean loss of each case weighted by its number of weeks; to the right: positive = smaller loss than the original.
Numbers right of each loss panel: share of the cases in the node in which the loss of the method is below the loss of the original (nodes with at least 5 cases). Up to peak / after peak: the weeks around every local peak of the latest counts.
Below each loss panel: coverage error, the mean over the quantile levels of |coverage - alpha| over the weeks of the window, averaged over the cases of the node weighted by their weeks; dashed: the original forecasts.

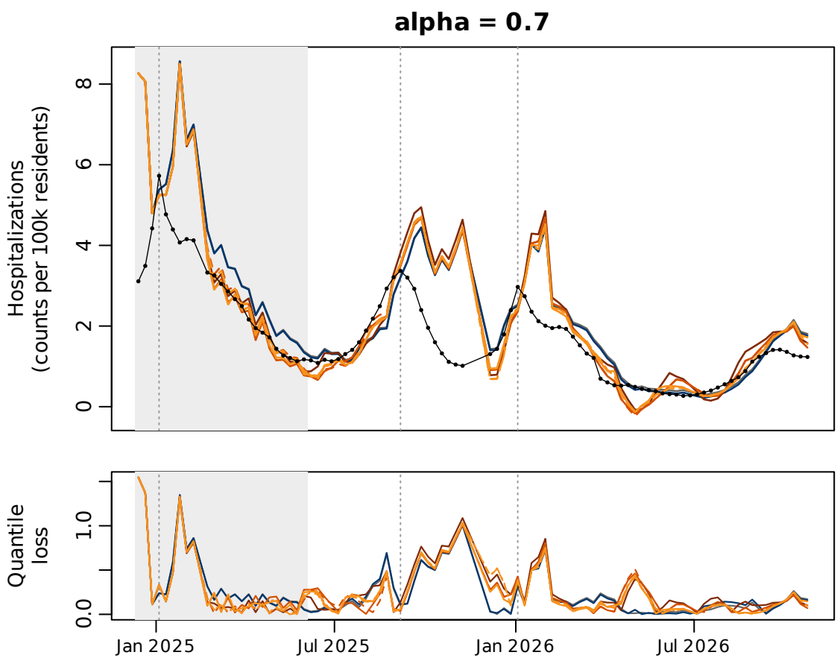
covid | horizon 2 | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0 | Rate-grid experts

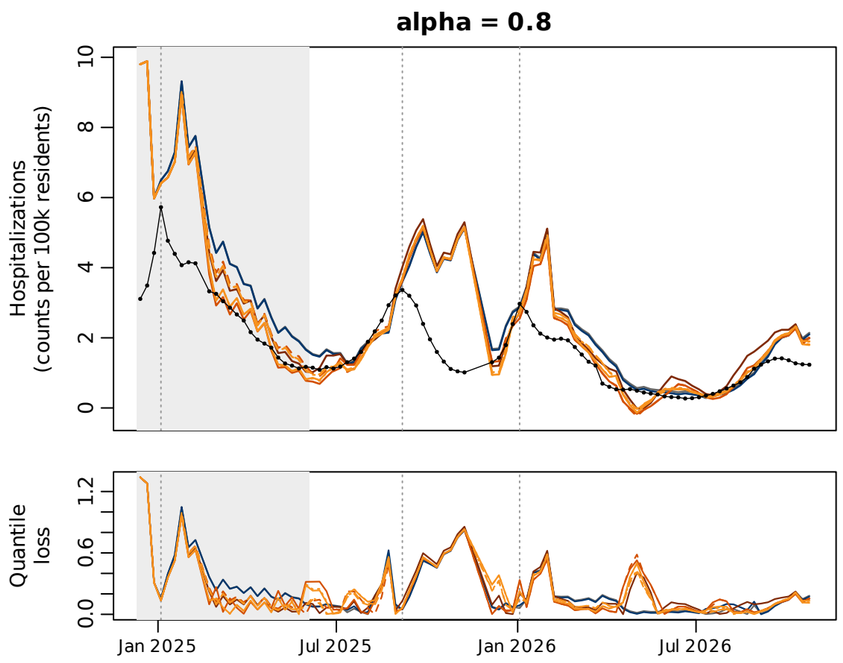
Alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Vertical dotted lines: peaks | Gray: tuning period (not scored)

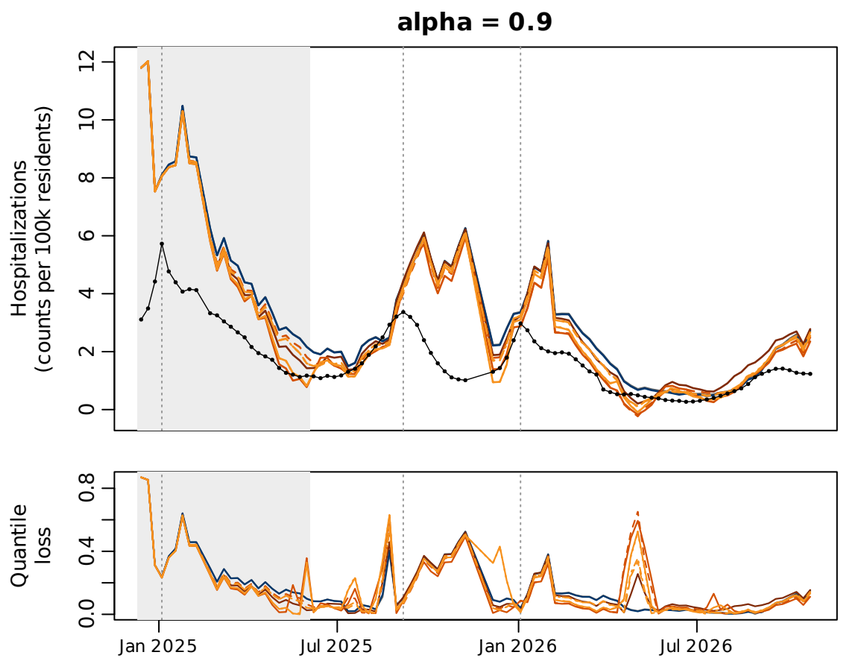








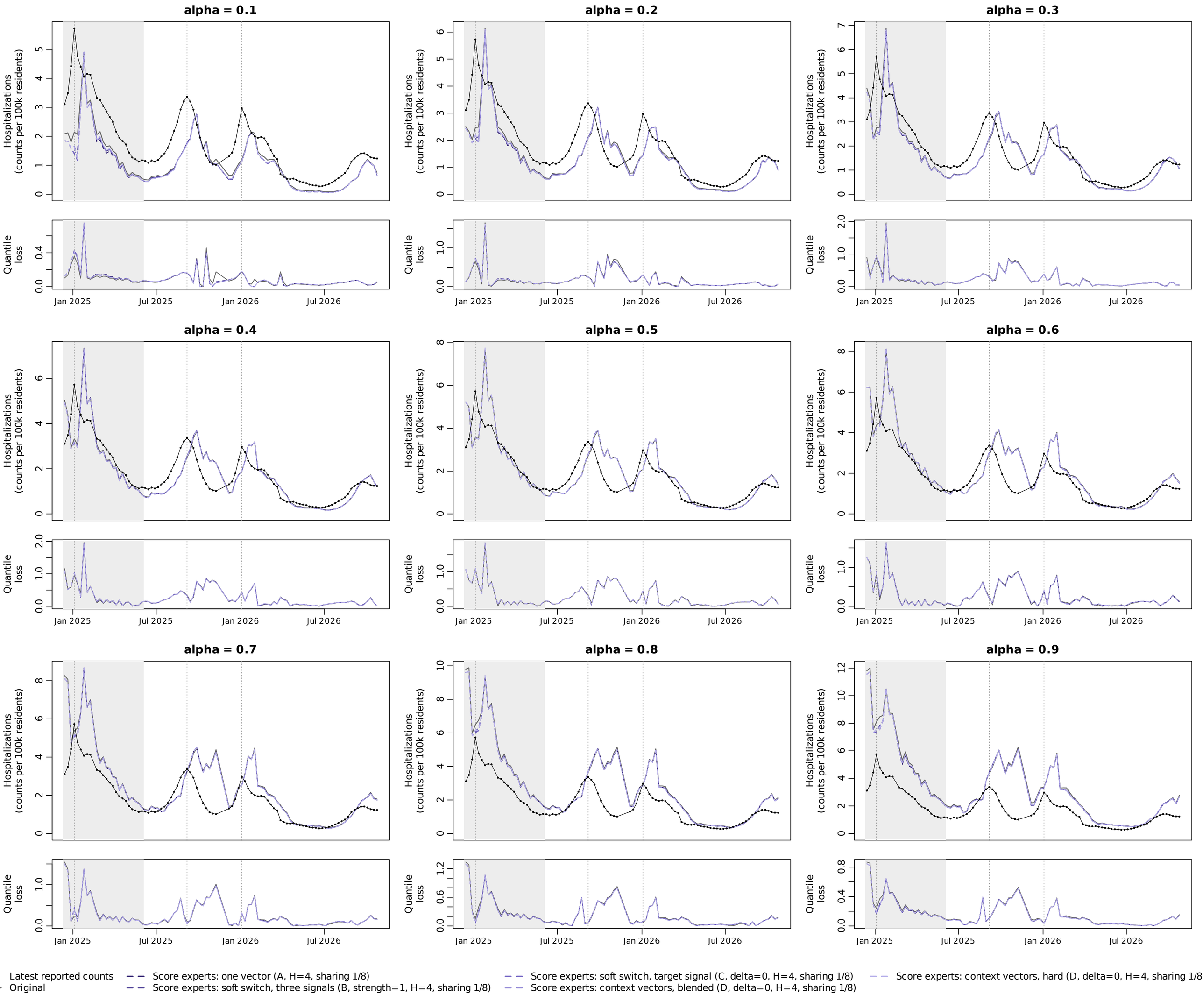




- Latest reported counts
- Original
- Fixed MultiQT (eta=0.001*S0)
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=4)
- Rate grid: learned + sharing, common weights (H=4)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

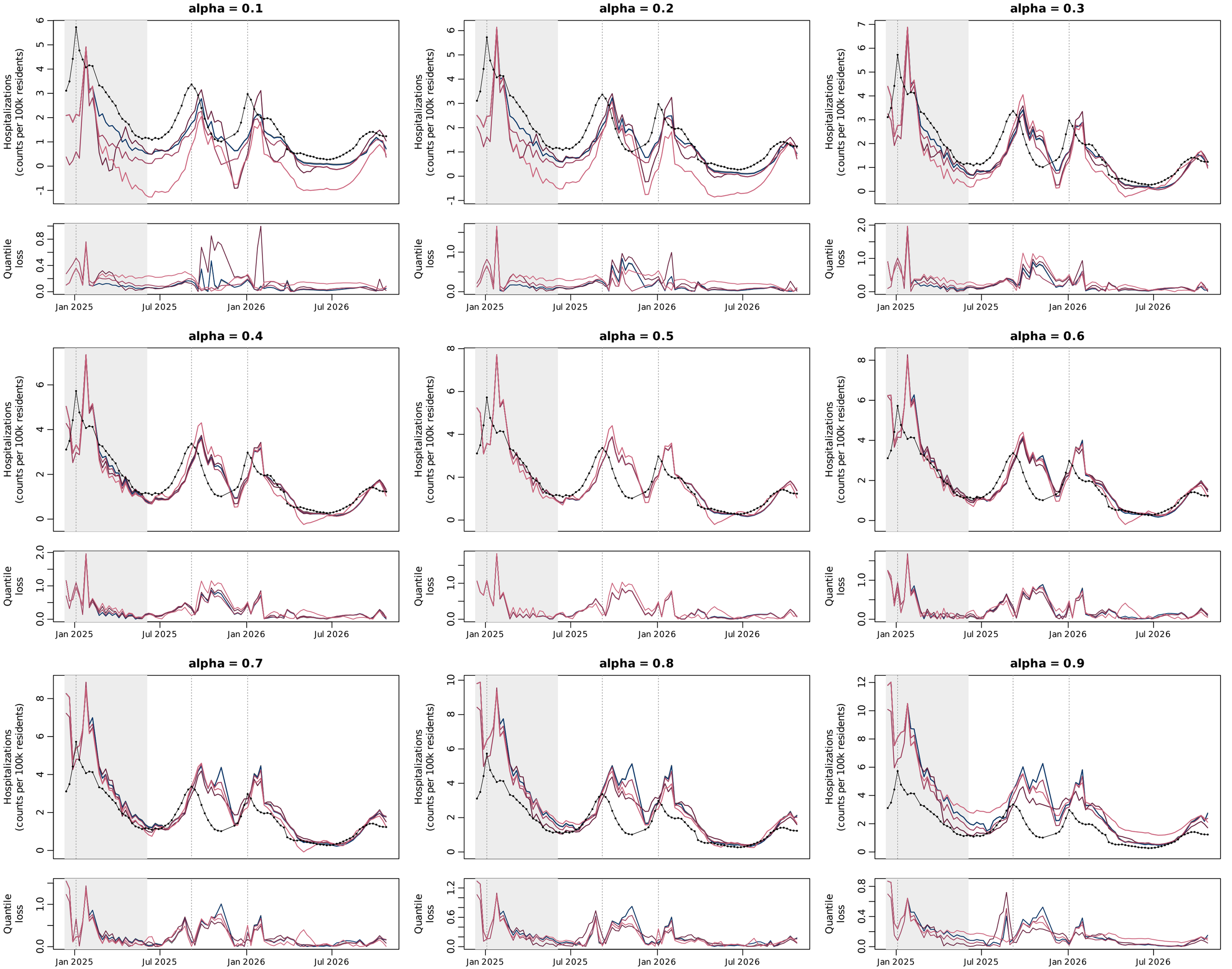
covid | horizon 2 | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0 | Score-based experts

Alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Vertical dotted lines: peaks | Gray: tuning period (not scored)



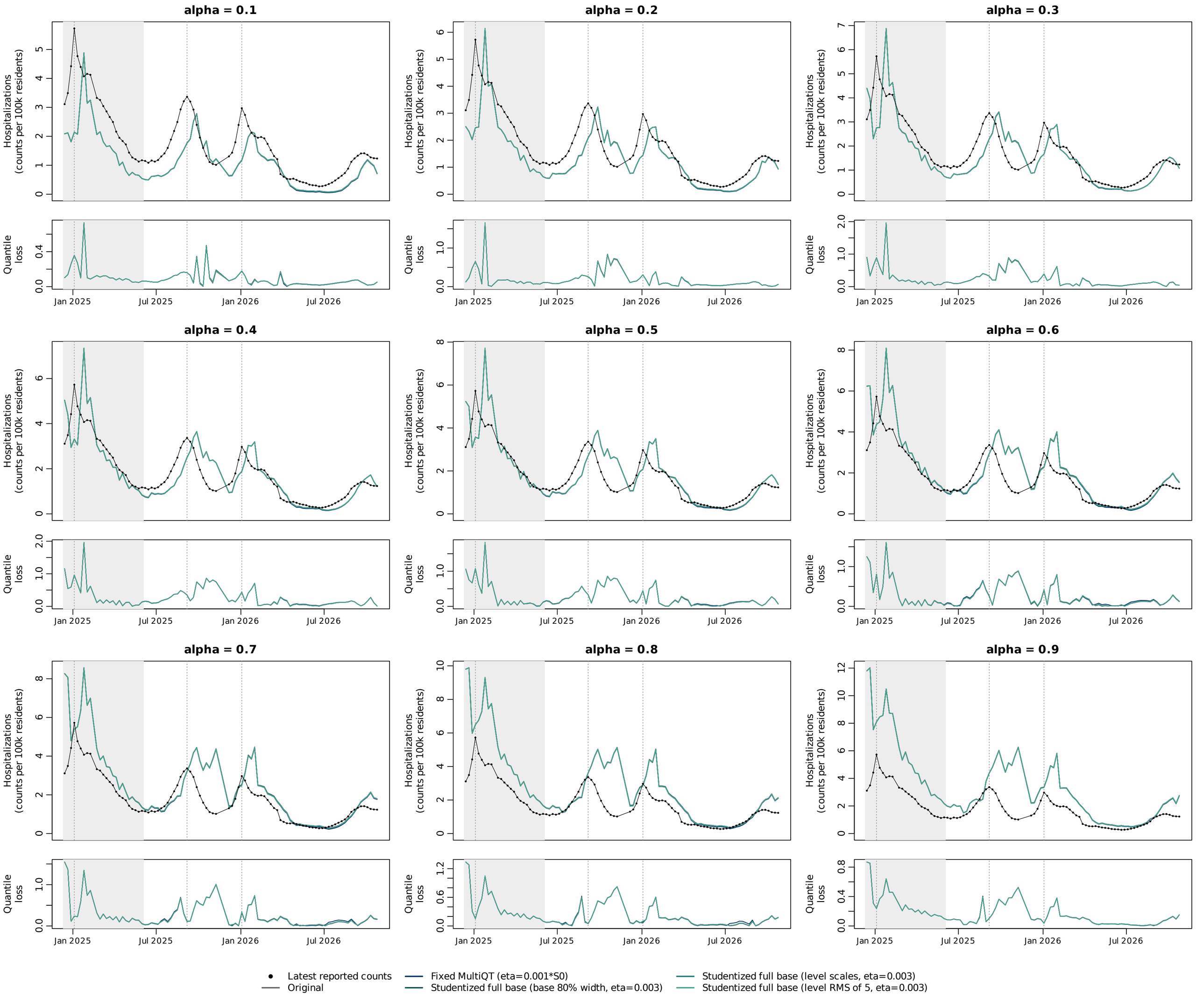
covid | horizon 2 | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0 | Studentized median

Alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Vertical dotted lines: peaks | Gray: tuning period (not scored)



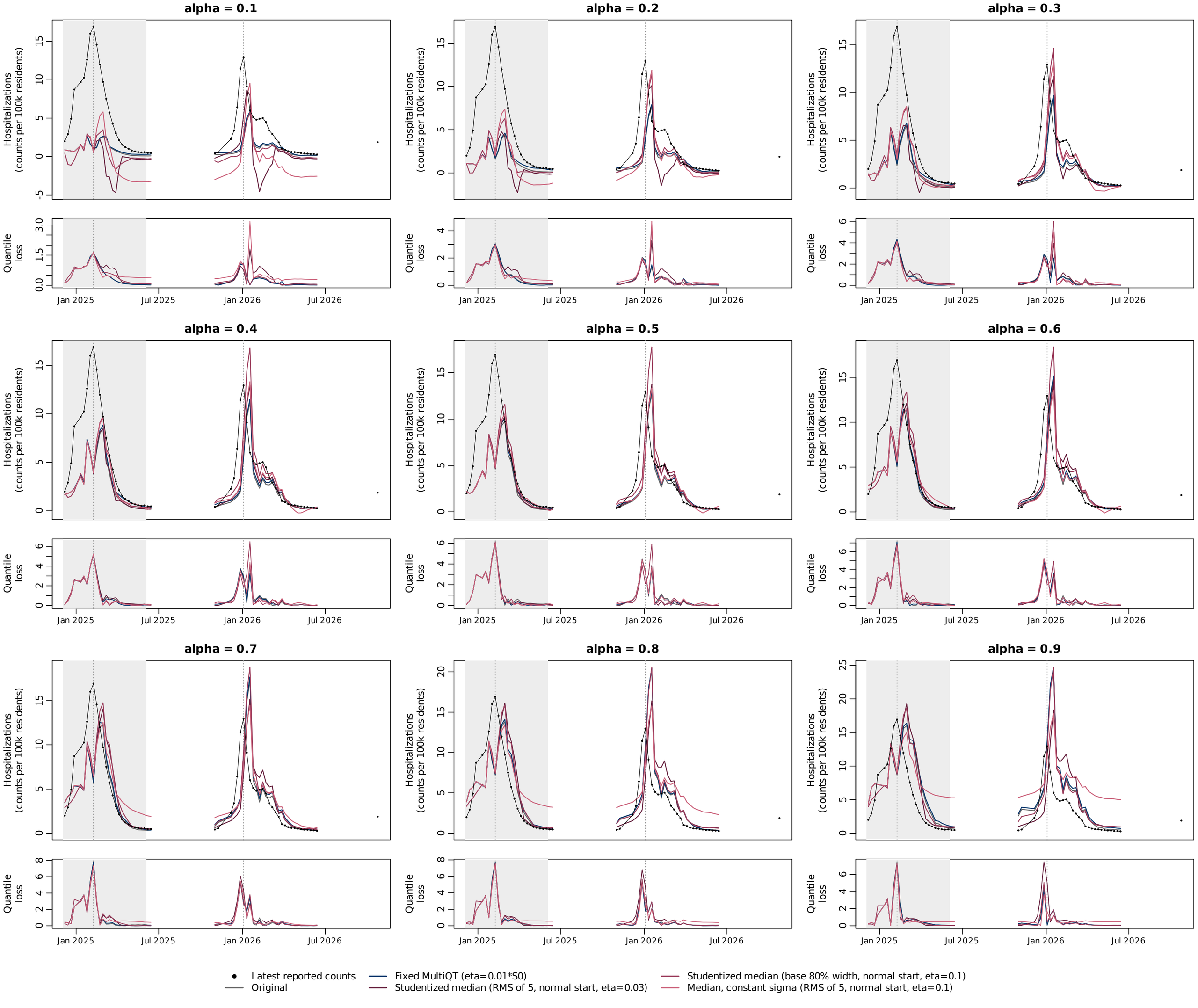
covid | horizon 2 | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0 | Studentized full base

Alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Vertical dotted lines: peaks | Gray: tuning period (not scored)



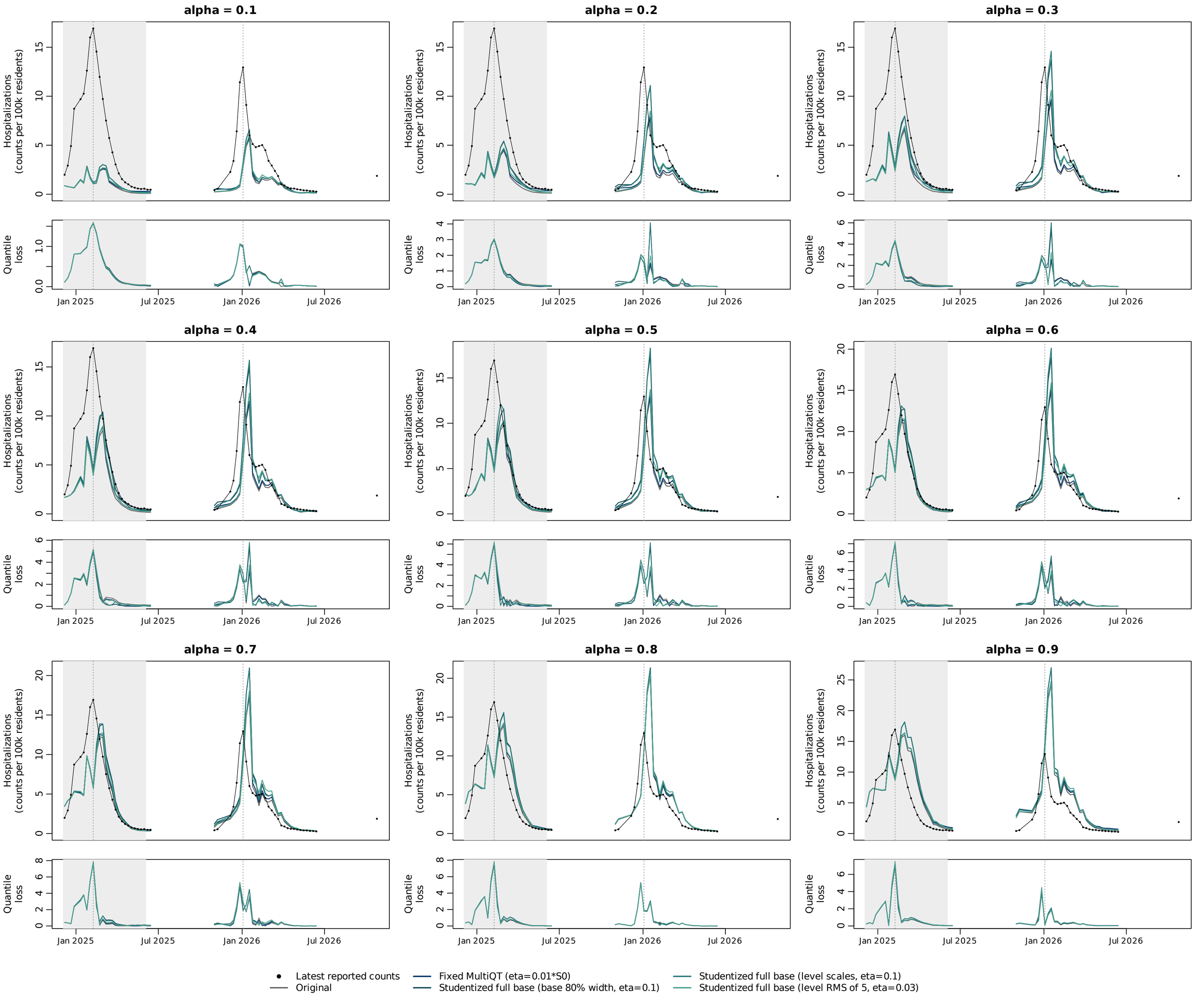
flu | horizon 2 | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0 | Studentized median

Alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Vertical dotted lines: peaks | Gray: tuning period (not scored)



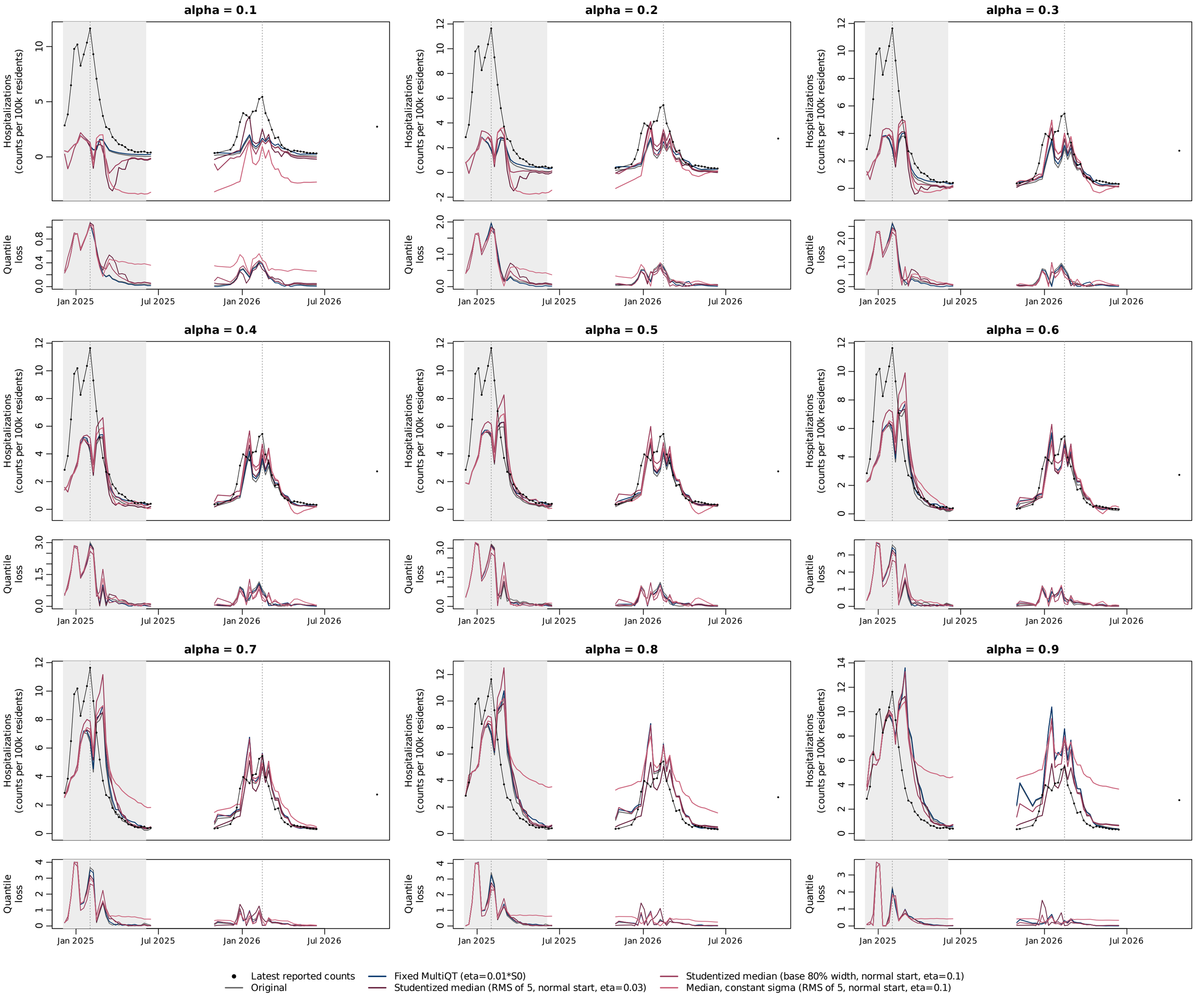
flu | horizon 2 | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0 | Studentized full base

Alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Vertical dotted lines: peaks | Gray: tuning period (not scored)



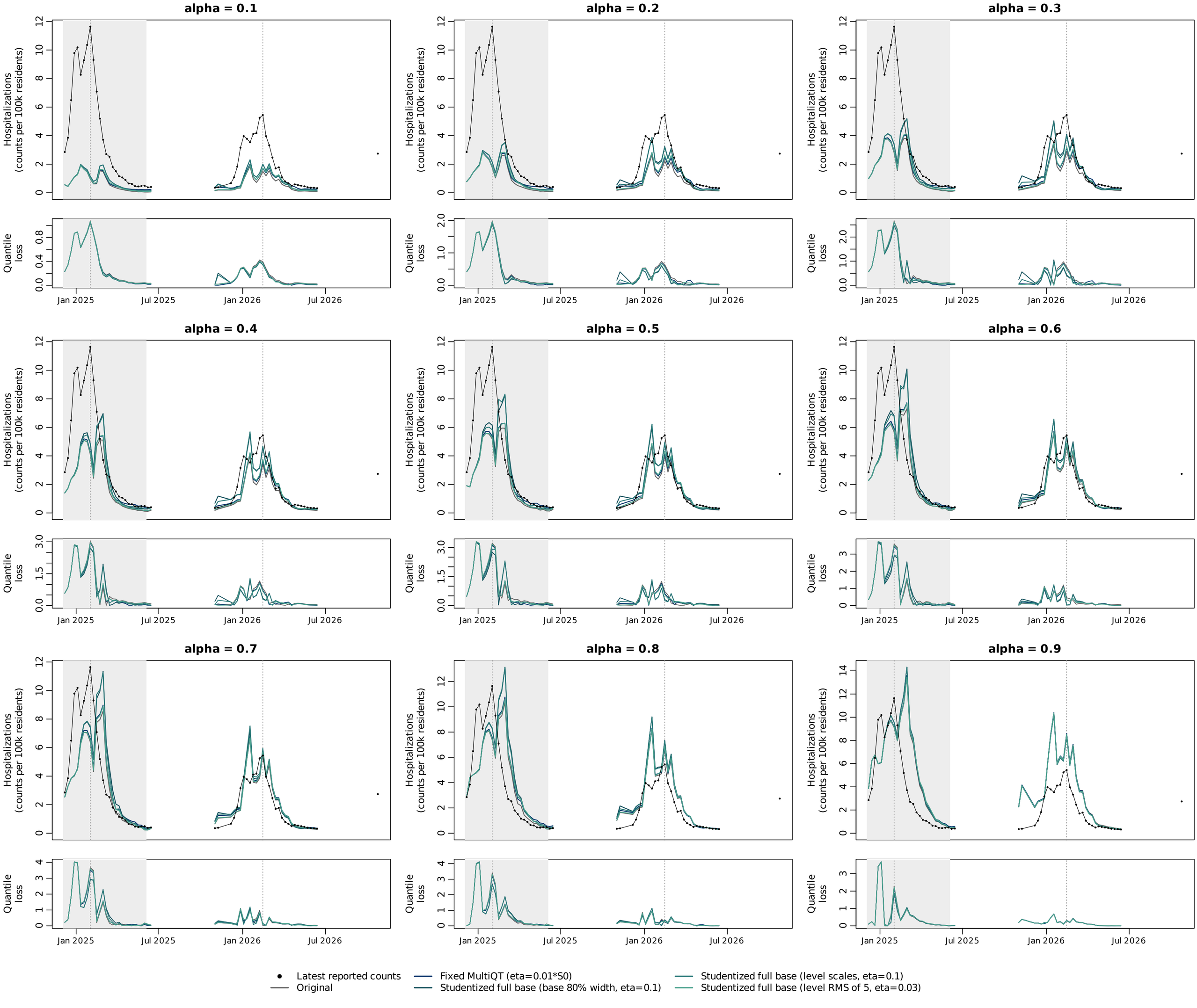
flu | horizon 2 | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0 | Studentized median

Alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Vertical dotted lines: peaks | Gray: tuning period (not scored)



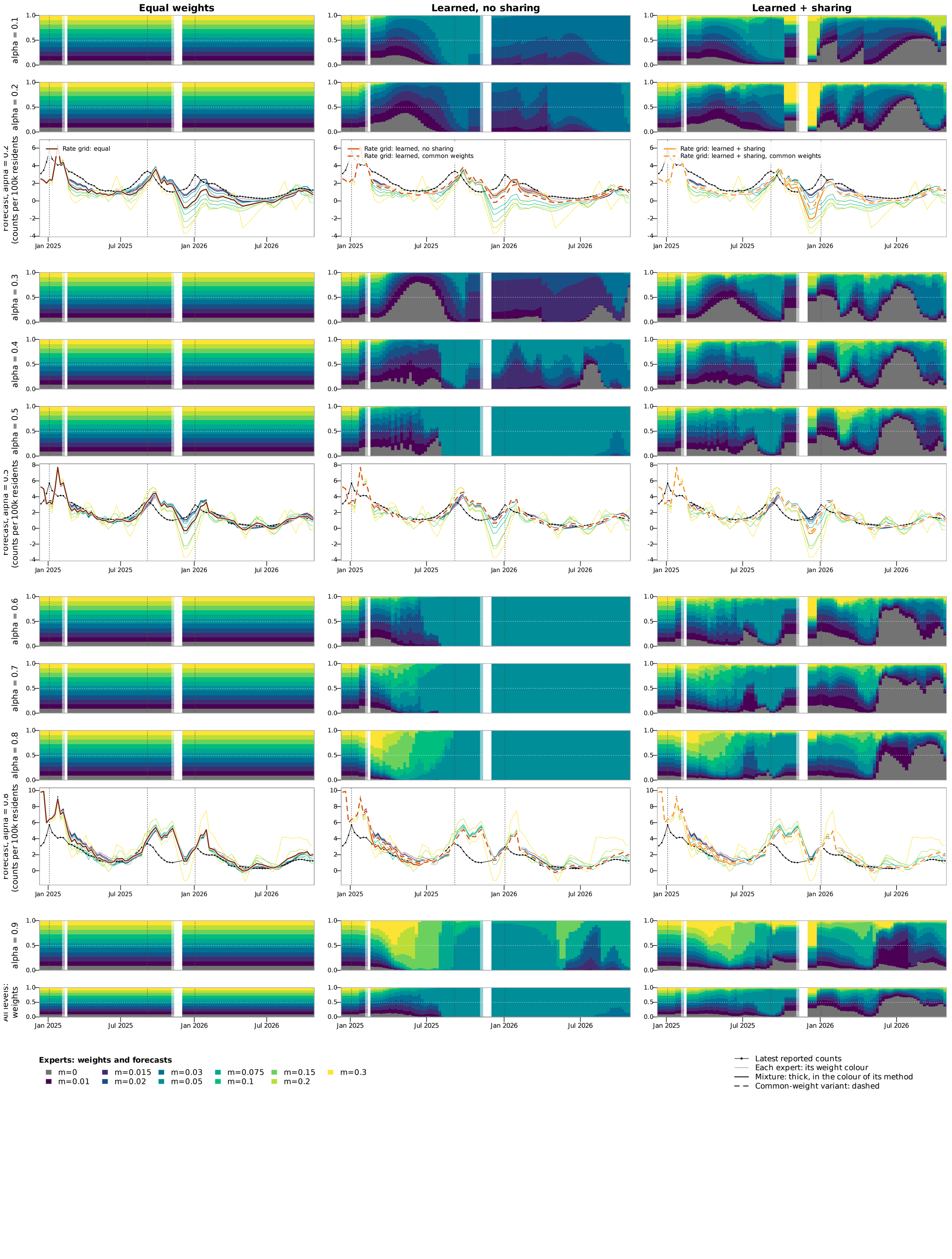
flu | horizon 2 | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0 | Studentized full base

Alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Vertical dotted lines: peaks | Gray: tuning period (not scored)



Frozen rate-grid weights

covid | horizon 2 | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0



Target date | Weights from the corresponding reference week | Vertical dotted lines: peaks

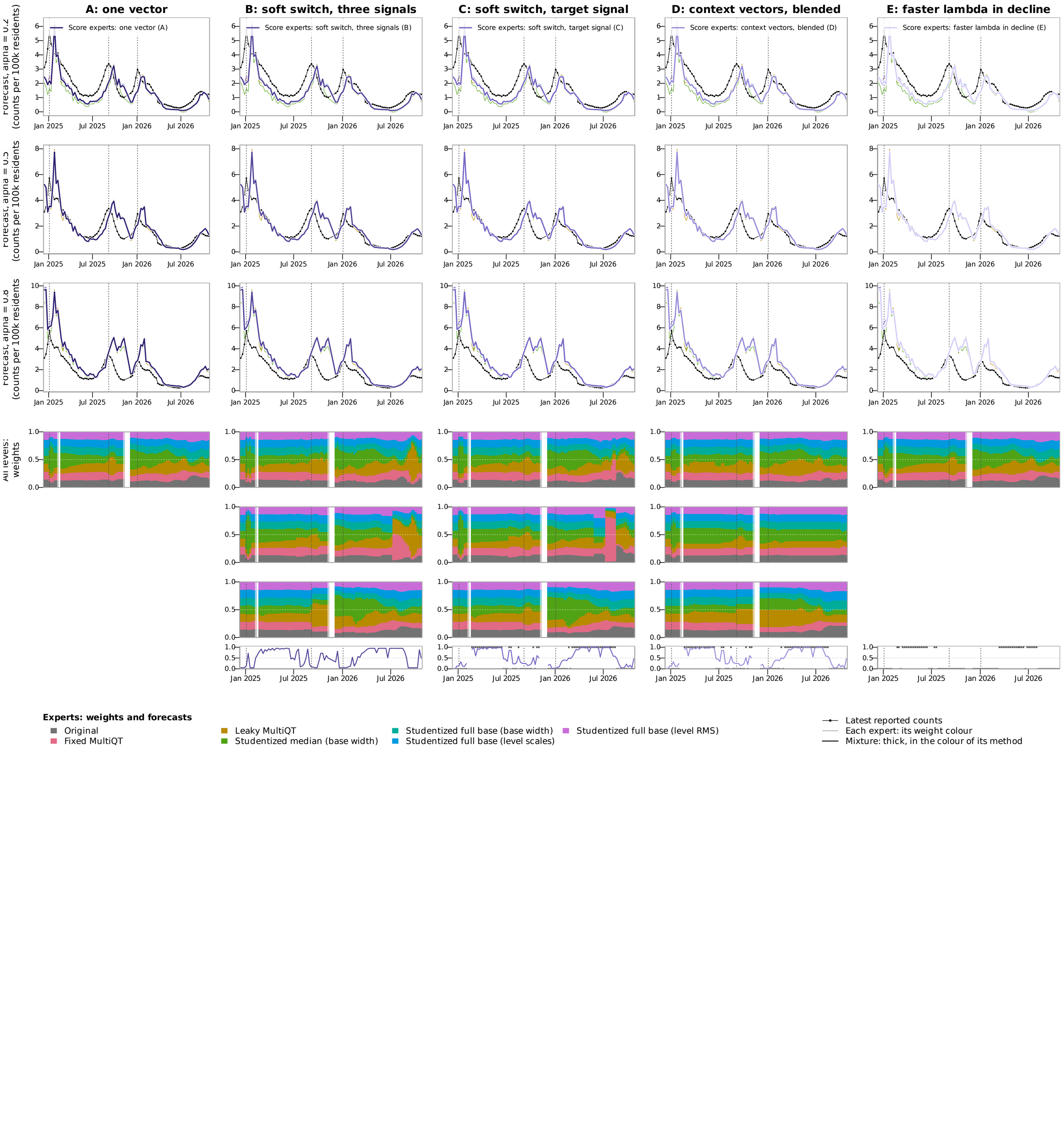
Rows alpha = 0.1-0.9 (mixtures with one weight vector per level): the weights of each level; below alpha = 0.2, 0.5 and 0.8 the forecasts at that level (per-level mixtures after crossing repair, before any outer correction).

Rows 'All levels': one weight vector for all levels (the mixture, or its common-weight variant), learned from the mean quantile loss over the levels; two-vector mixtures: baseline and declining vectors; context: G or the gate (line), c (points).

Lighter bars: rounds without a CMU forecast, filled with the latest earlier CMU forecast of the same target week.

Score-based expert weights

covid | horizon 2 | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0



Target date | Weights from the corresponding reference week | Vertical dotted lines: peaks

Rows alpha = 0.1-0.9 (mixtures with one weight vector per level): the weights of each level; below alpha = 0.2, 0.5 and 0.8 the forecasts at that level (per-level mixtures after crossing repair, before any outer correction).

Rows 'All levels': one weight vector for all levels (the mixture, or its common-weight variant), learned from the mean quantile loss over the levels; two-vector mixtures: baseline and declining vectors; context: G or the gate (line), c (points).

Lighter bars: rounds without a CMU forecast, filled with the latest earlier CMU forecast of the same target week.

Weekly-start weights

covid | horizon 2 | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0

Equal weights | **Learned** | **Starts x rate grid: learned**

alpha = 0.1 | alpha = 0.2 | alpha = 0.3 | alpha = 0.4 | alpha = 0.5 | alpha = 0.6 | alpha = 0.7 | alpha = 0.8 | alpha = 0.9

Forecast, alpha = 0.2 (counts per 100k residents)

Forecast, alpha = 0.5 (counts per 100k residents)

Forecast, alpha = 0.8 (counts per 100k residents)

Weekly starts: equal

Weekly starts: learned

Starts x rate grid: learned

Weekly starts: most recent, equal

Jan 2025 | Jul 2025 | Jan 2026 | Jul 2026

Weekly starts, m=0.01 (grid only) — first start Nov 2024 (dark) to last Oct 2026 (light)

Weekly starts, m=0.05 (all columns) — first start Nov 2024 (dark) to last Oct 2026 (light)

Weekly starts, m=0.2 (grid only) — first start Nov 2024 (dark) to last Oct 2026 (light)

Latest reported counts

Each expert: its weight colour

Mixture: thick, in the colour of its method

Target date | Weights from the corresponding reference week | Vertical dotted lines: peaks

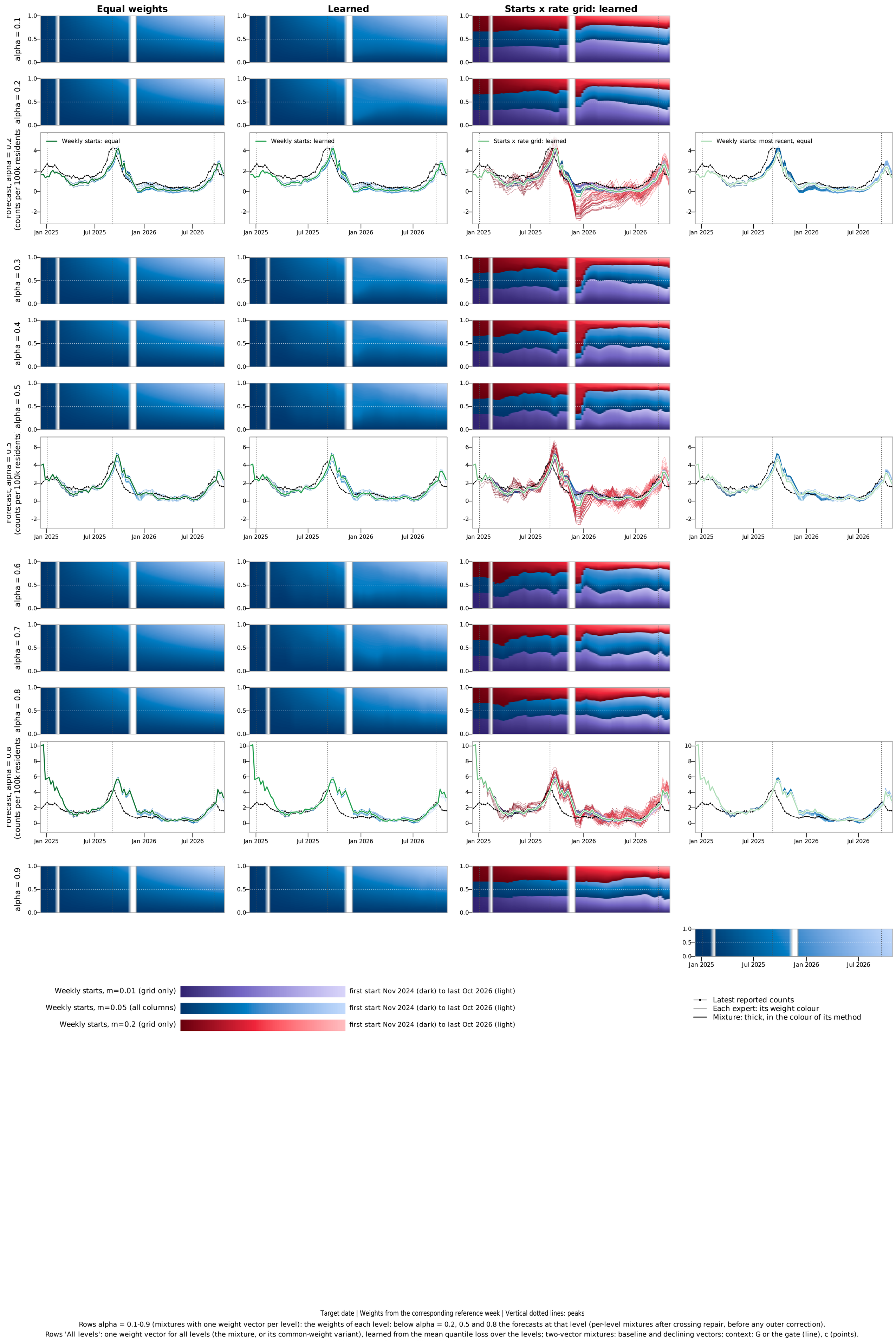
Rows alpha = 0.1-0.9 (mixtures with one weight vector per level): the weights of each level; below alpha = 0.2, 0.5 and 0.8 the forecasts at that level (per-level mixtures after crossing repair, before any outer correction).

Rows 'All levels': one weight vector for all levels (the mixture, or its common-weight variant), learned from the mean quantile loss over the levels; two-vector mixtures: baseline and declining vectors; context: G or the gate (line), c (points).

Lighter bars: rounds without a CMU forecast, filled with the latest earlier CMU forecast of the same target week.

Weekly-start weights

covid | horizon 2 | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0



Target date | Weights from the corresponding reference week | Vertical dotted lines: peaks

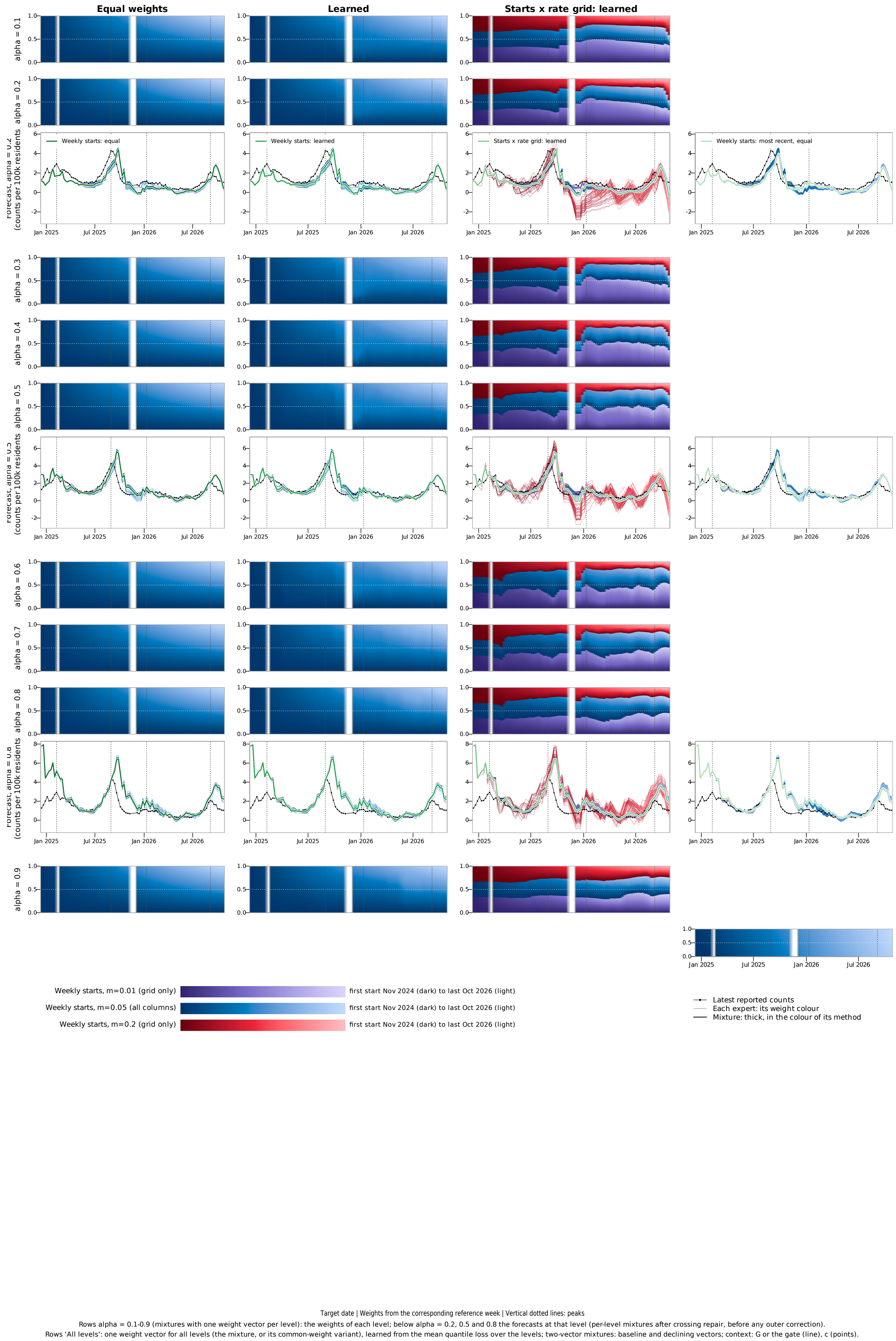
Rows alpha = 0.1-0.9 (mixtures with one weight vector per level): the weights of each level; below alpha = 0.2, 0.5 and 0.8 the forecasts at that level (per-level mixtures after crossing repair, before any outer correction).

Rows 'All levels': one weight vector for all levels (the mixture, or its common-weight variant), learned from the mean quantile loss over the levels; two-vector mixtures: baseline and declining vectors; context: G or the gate (line), c (points).

Lighter bars: rounds without a CMU forecast, filled with the latest earlier CMU forecast of the same target week.

Weekly-start weights

covid | horizon 2 | Texas (48) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0



Target date | Weights from the corresponding reference week | Vertical dotted lines: peaks

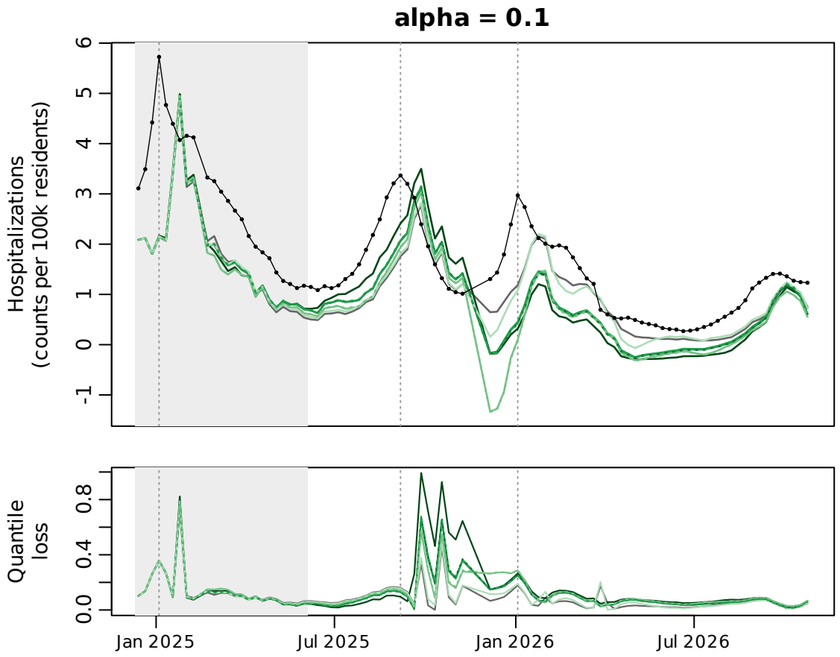
Rows alpha = 0.1-0.9 (mixtures with one weight vector per level): the weights of each level; below alpha = 0.2, 0.5 and 0.8 the forecasts at that level (per-level mixtures after crossing repair, before any outer correction).

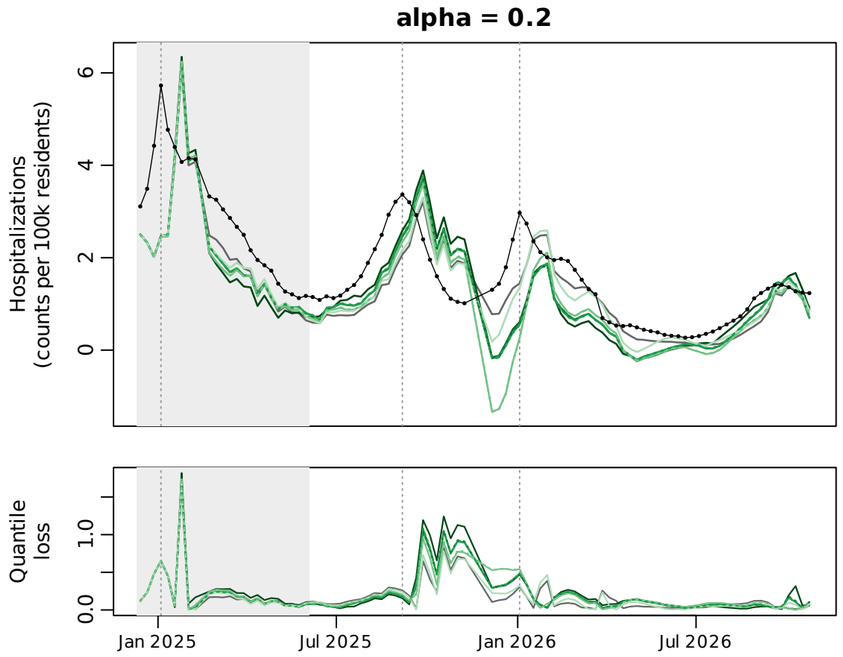
Rows 'All levels': one weight vector for all levels (the mixture, or its common-weight variant), learned from the mean quantile loss over the levels; two-vector mixtures: baseline and declining vectors; context: G or the gate (line), c (points).

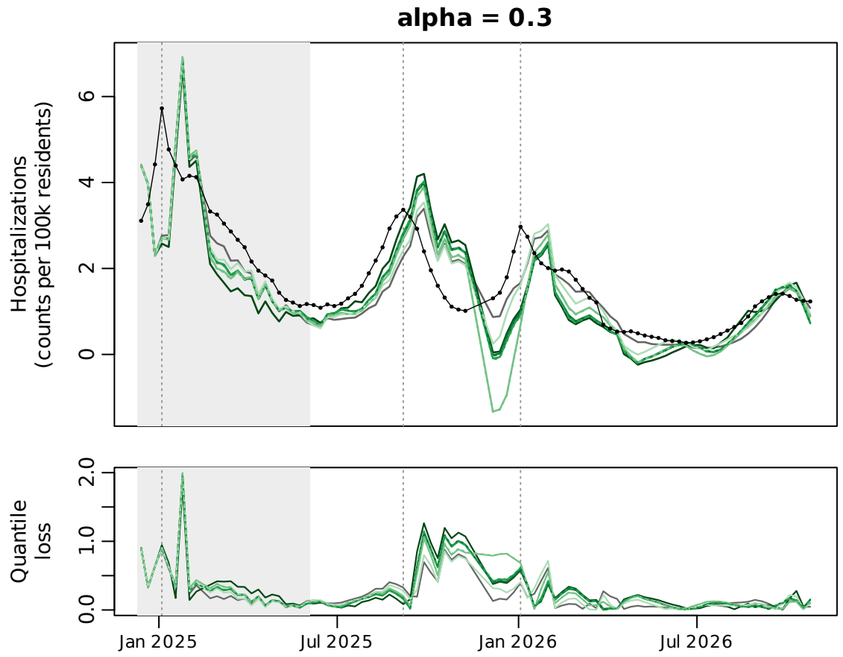
Lighter bars: rounds without a CMU forecast, filled with the latest earlier CMU forecast of the same target week.

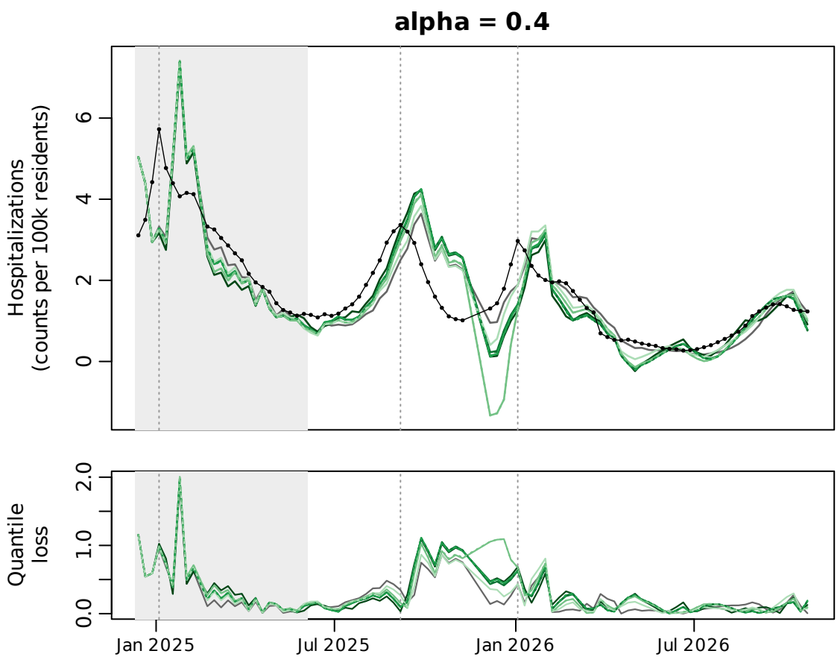
ovid | horizon 2 | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0 | Weekly starts: immediate admission

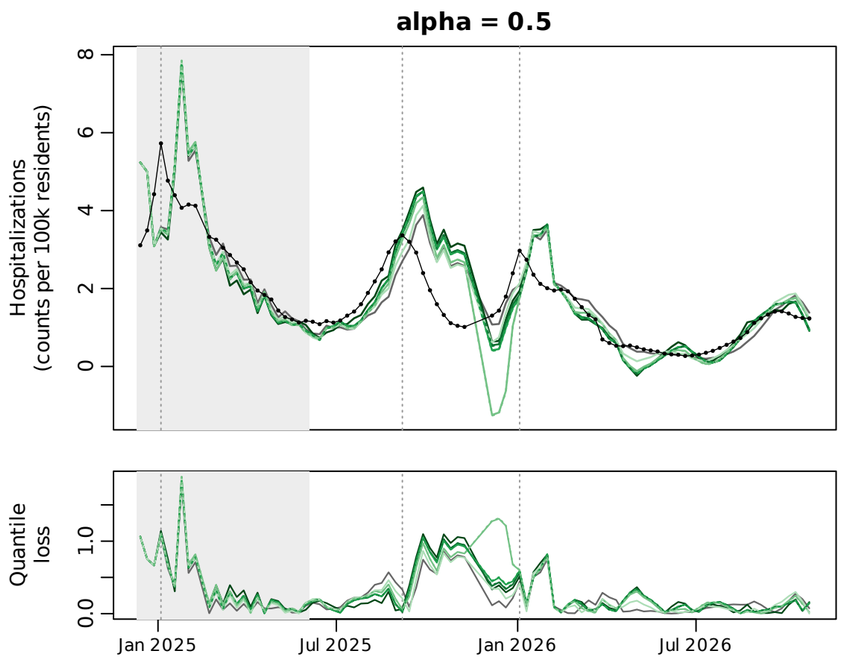
Alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Vertical dotted lines: peaks | Gray: tuning period (not scored)

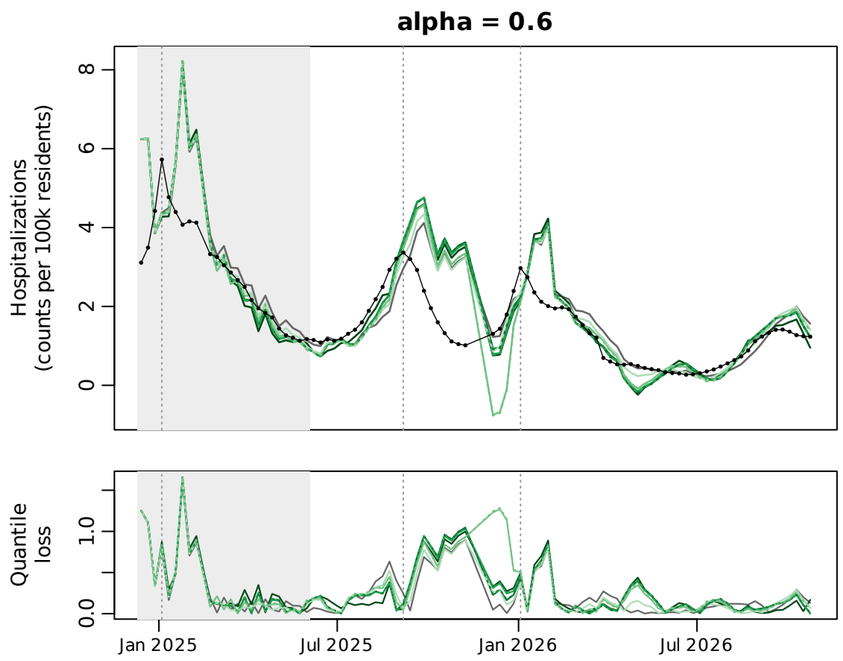


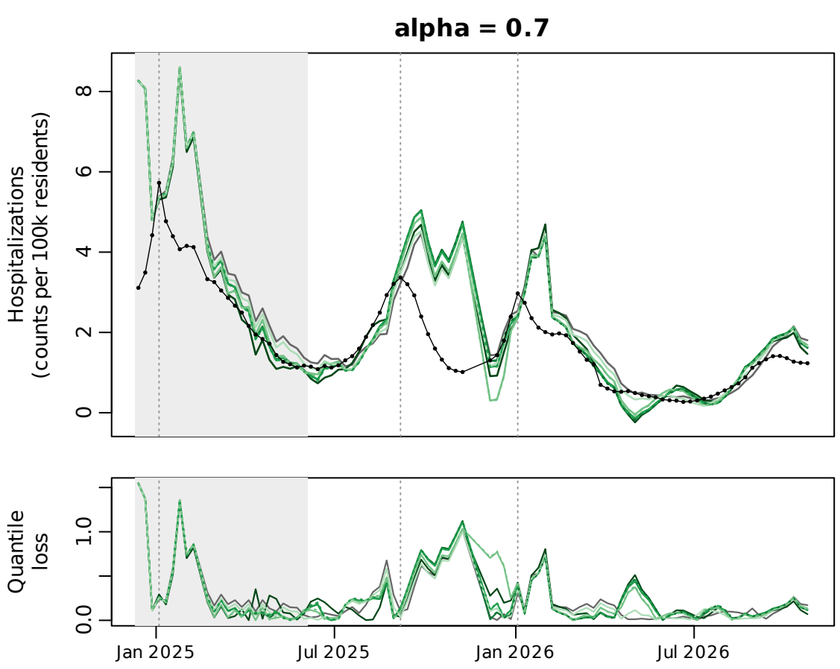


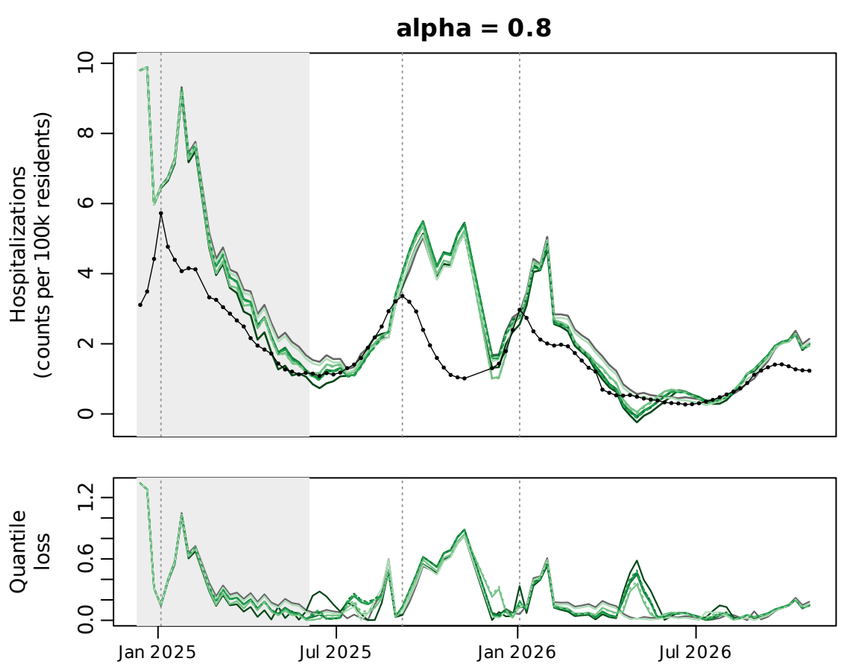


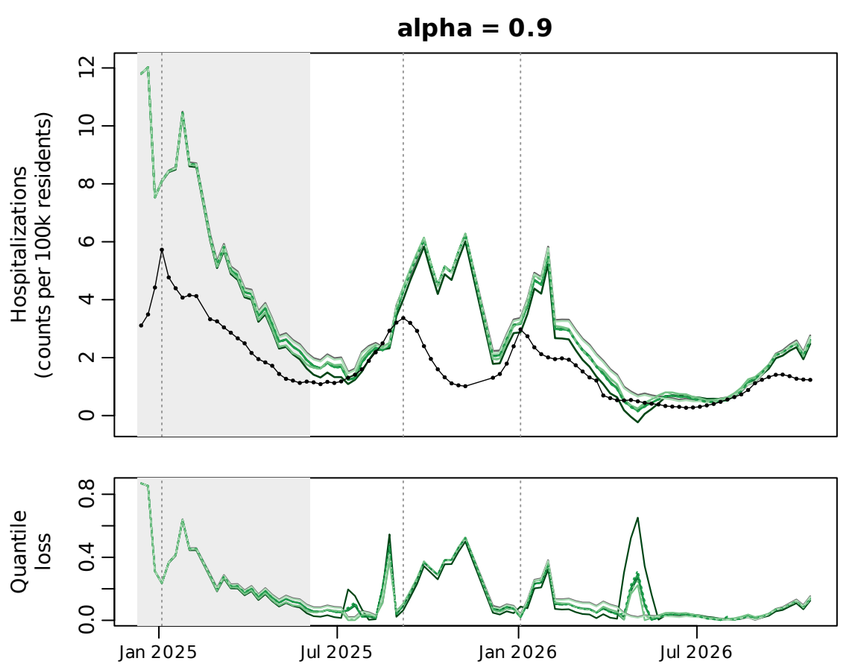








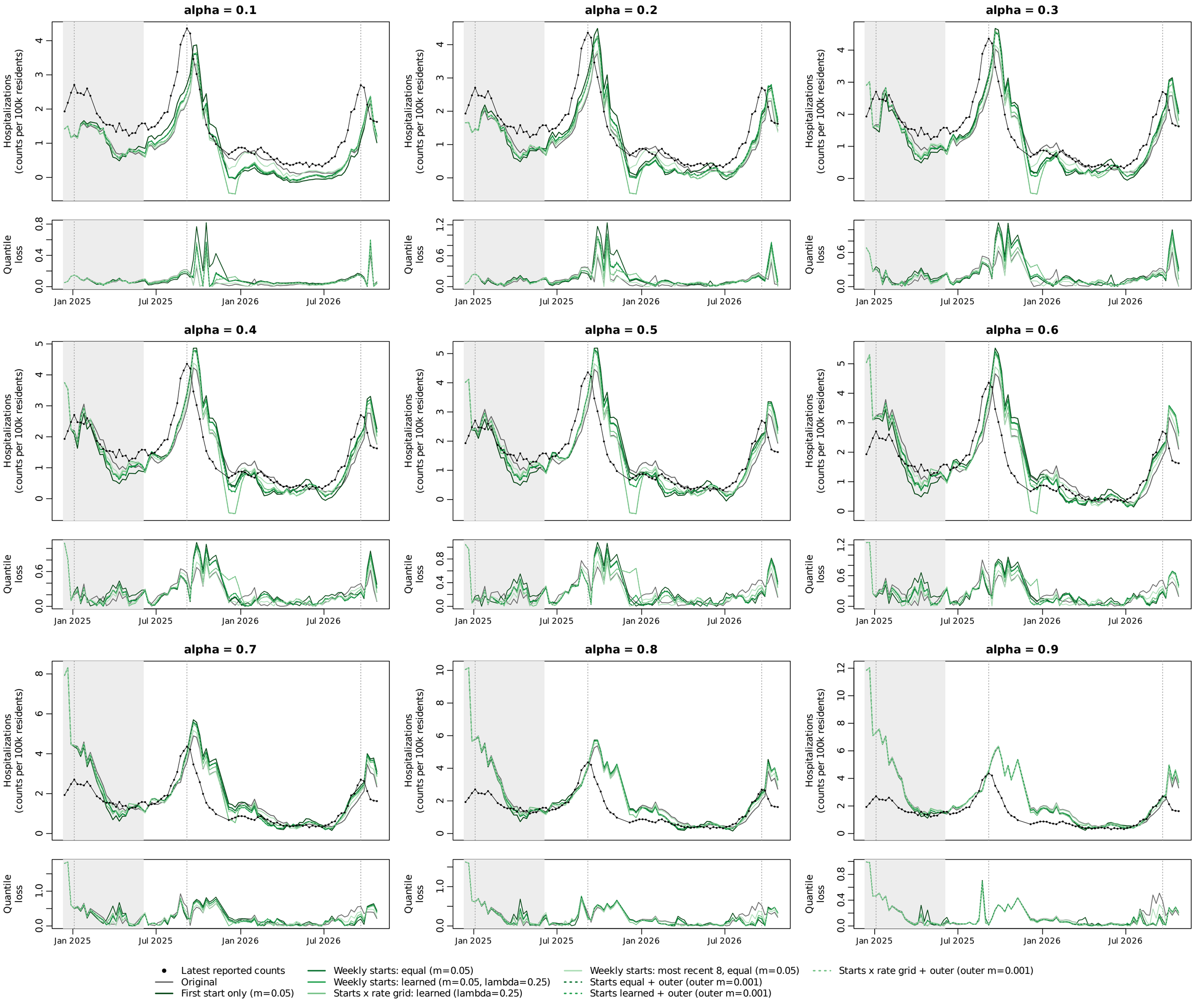




- Latest reported counts
- Original
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

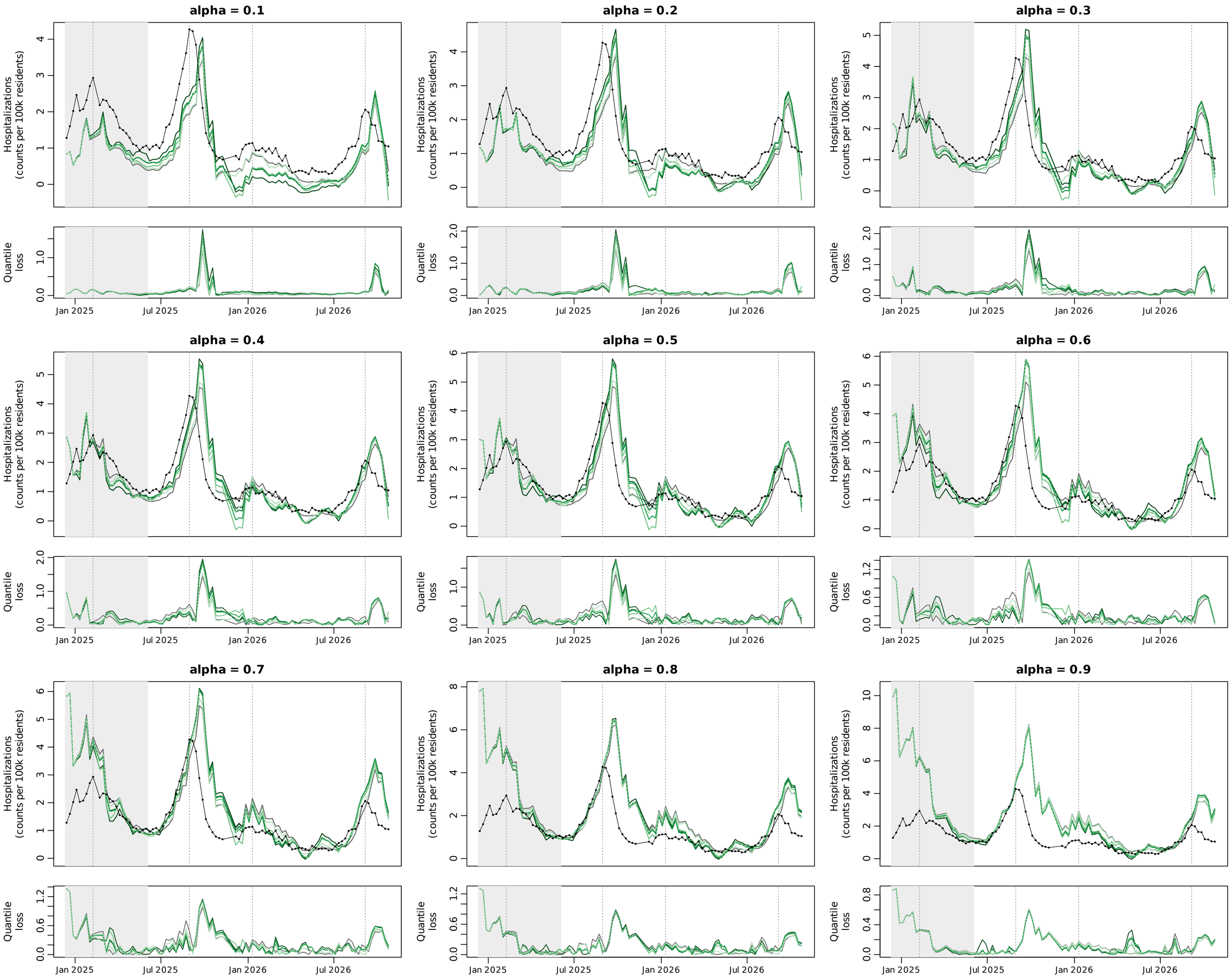
covid | horizon 2 | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0 | Weekly starts: immediate admission

Alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Vertical dotted lines: peaks | Gray: tuning period (not scored)



covid | horizon 2 | Texas (48) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0 | Weekly starts: immediate admission

Alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Vertical dotted lines: peaks | Gray: tuning period (not scored)

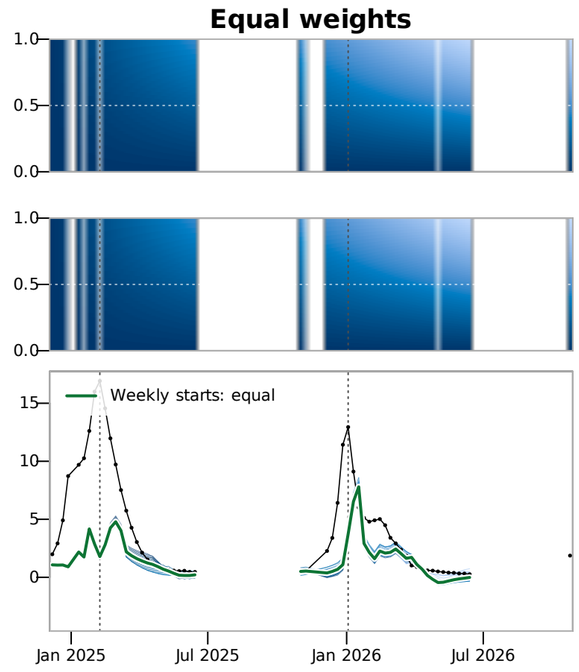


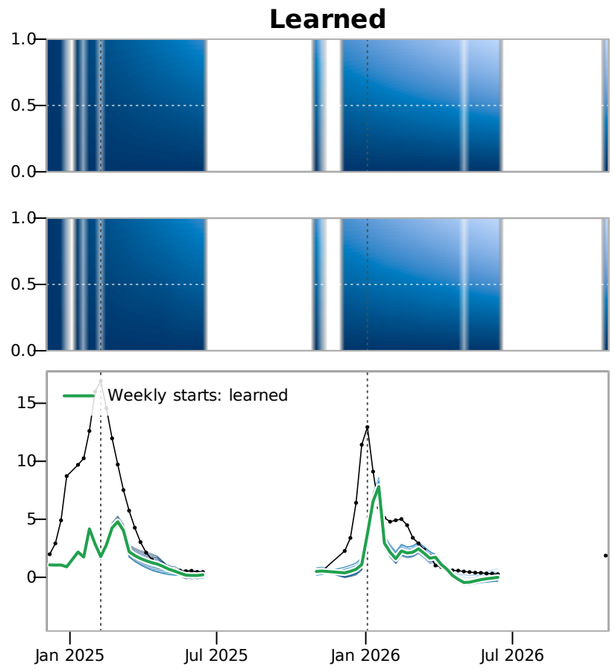
Weekly-start weights

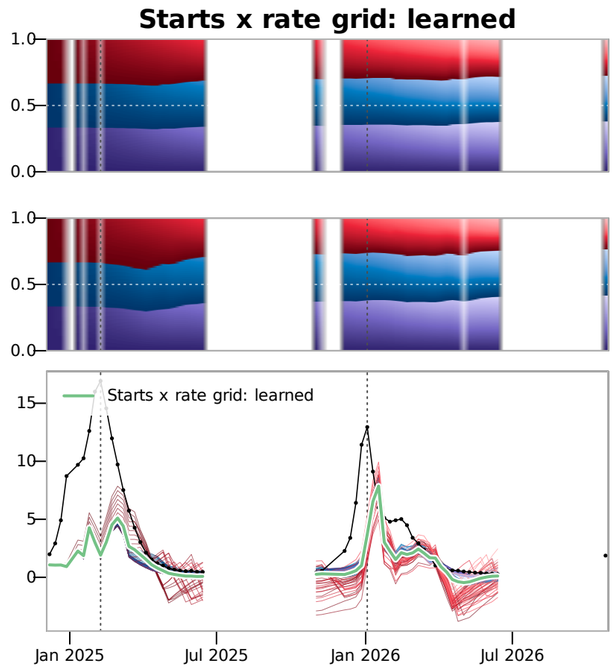
flu | horizon 2 | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0

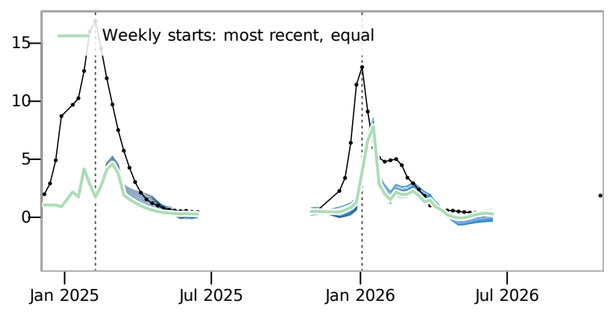
Equal weights | **Learned** | **Starts x rate grid: learned**

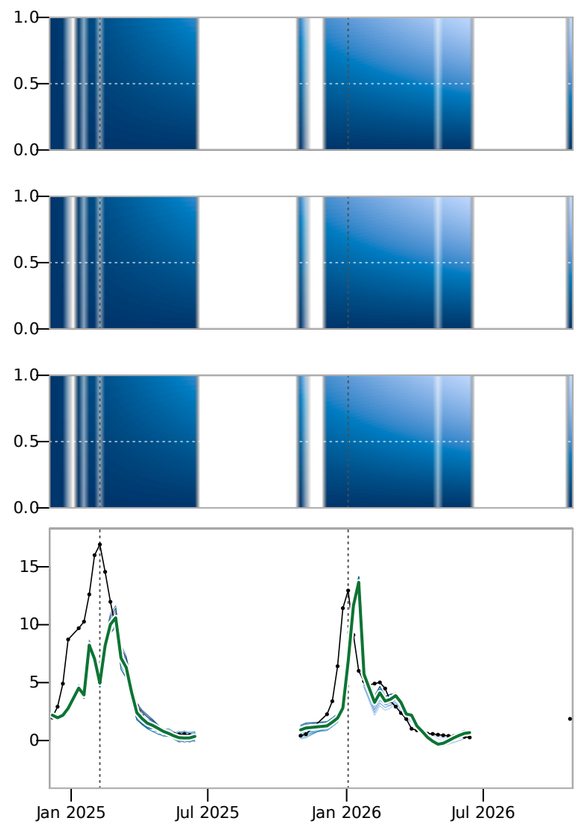
Forecast, alpha = 0.1 (counts per 100k residents)



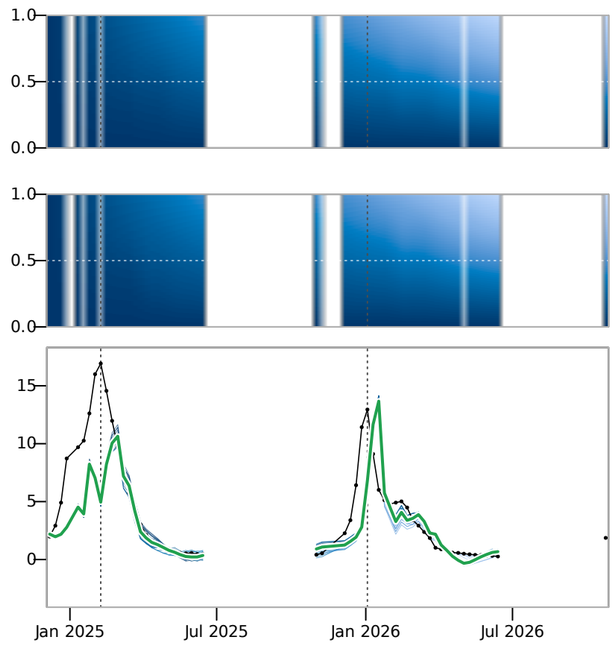


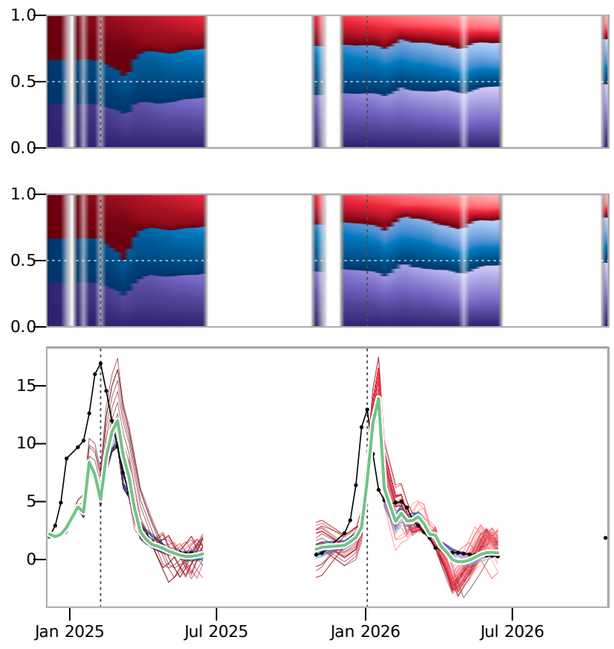


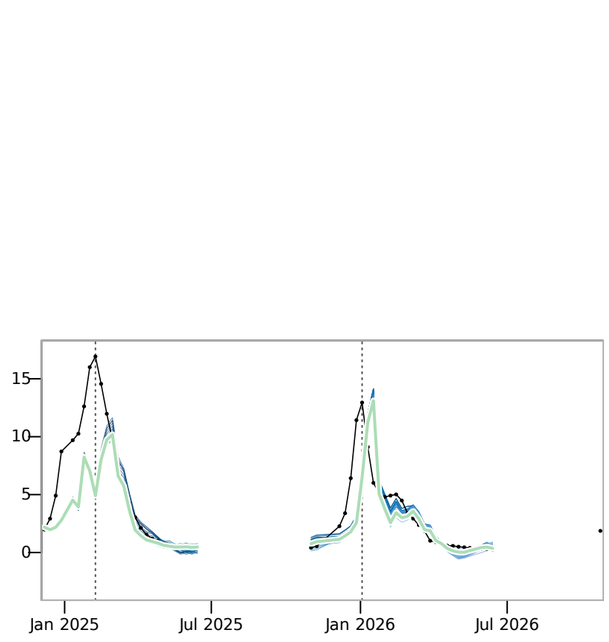


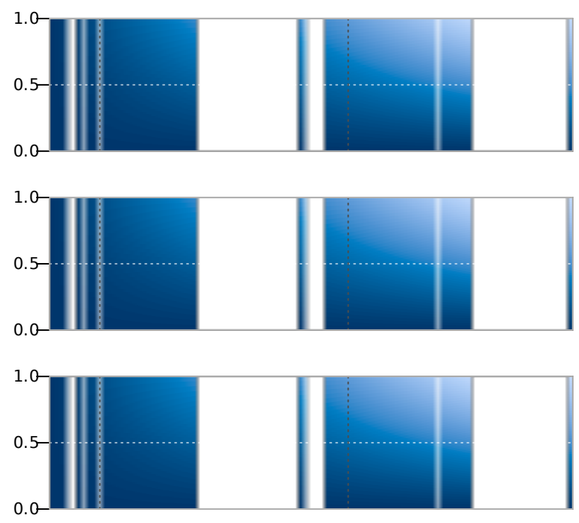


Forecast, alpha = 0.5 (counts per 100k residents)

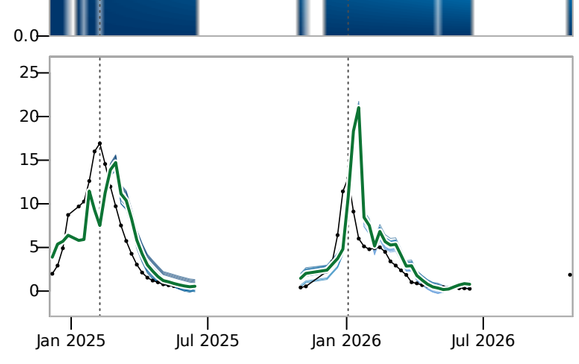


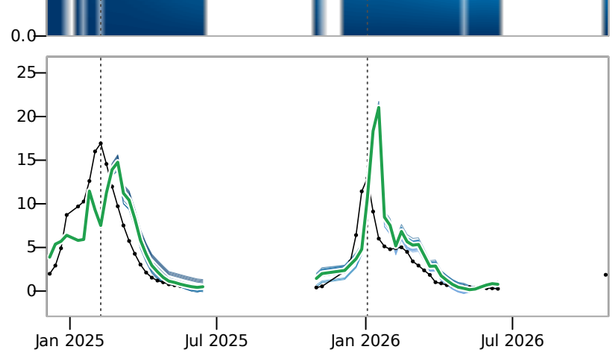


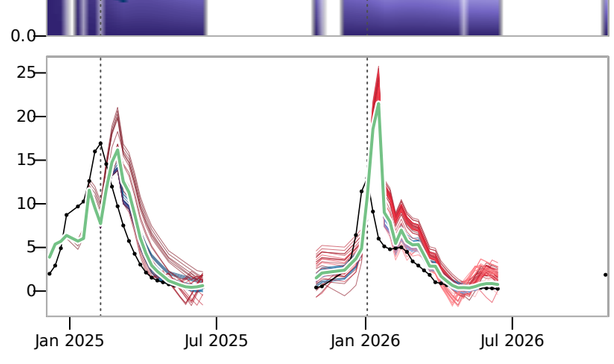


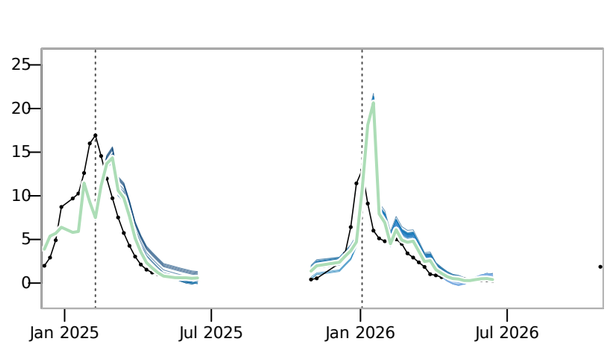


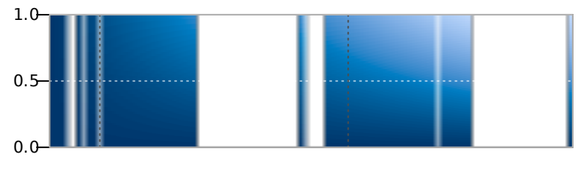
Forecast, alpha = 0.8 (counts per 100k residents)

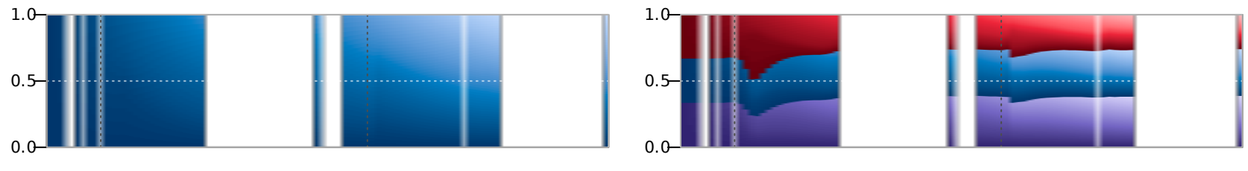


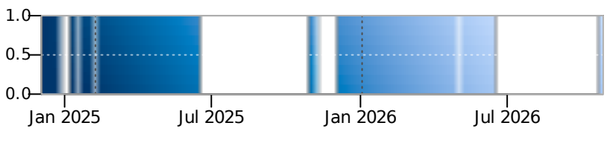


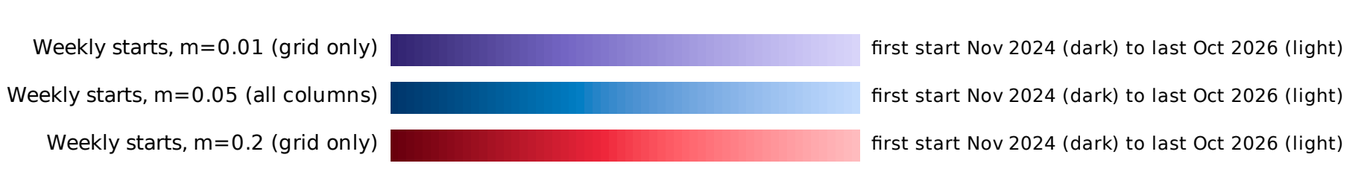


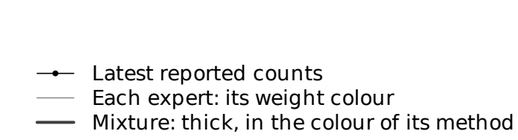












Target date | Weights from the corresponding reference week | Vertical dotted lines: peaks

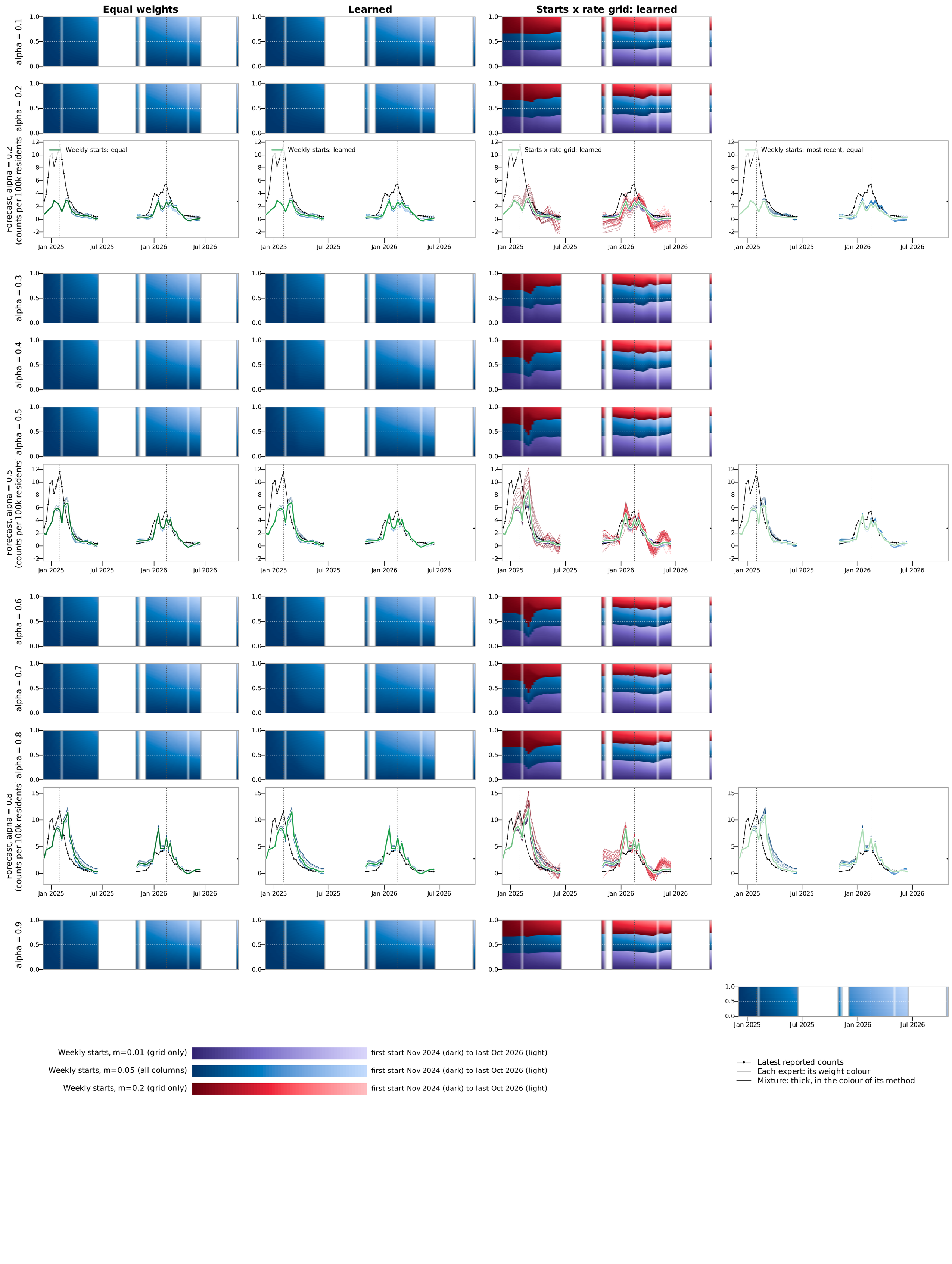
Rows alpha = 0.1-0.9 (mixtures with one weight vector per level): the weights of each level; below alpha = 0.2, 0.5 and 0.8 the forecasts at that level (per-level mixtures after crossing repair, before any outer correction).

Rows 'All levels': one weight vector for all levels (the mixture, or its common-weight variant), learned from the mean quantile loss over the levels; two-vector mixtures: baseline and declining vectors; context: G or the gate (line), c (points).

Lighter bars: rounds without a CMU forecast, filled with the latest earlier CMU forecast of the same target week.

Weekly-start weights

flu | horizon 2 | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0



Target date | Weights from the corresponding reference week | Vertical dotted lines: peaks

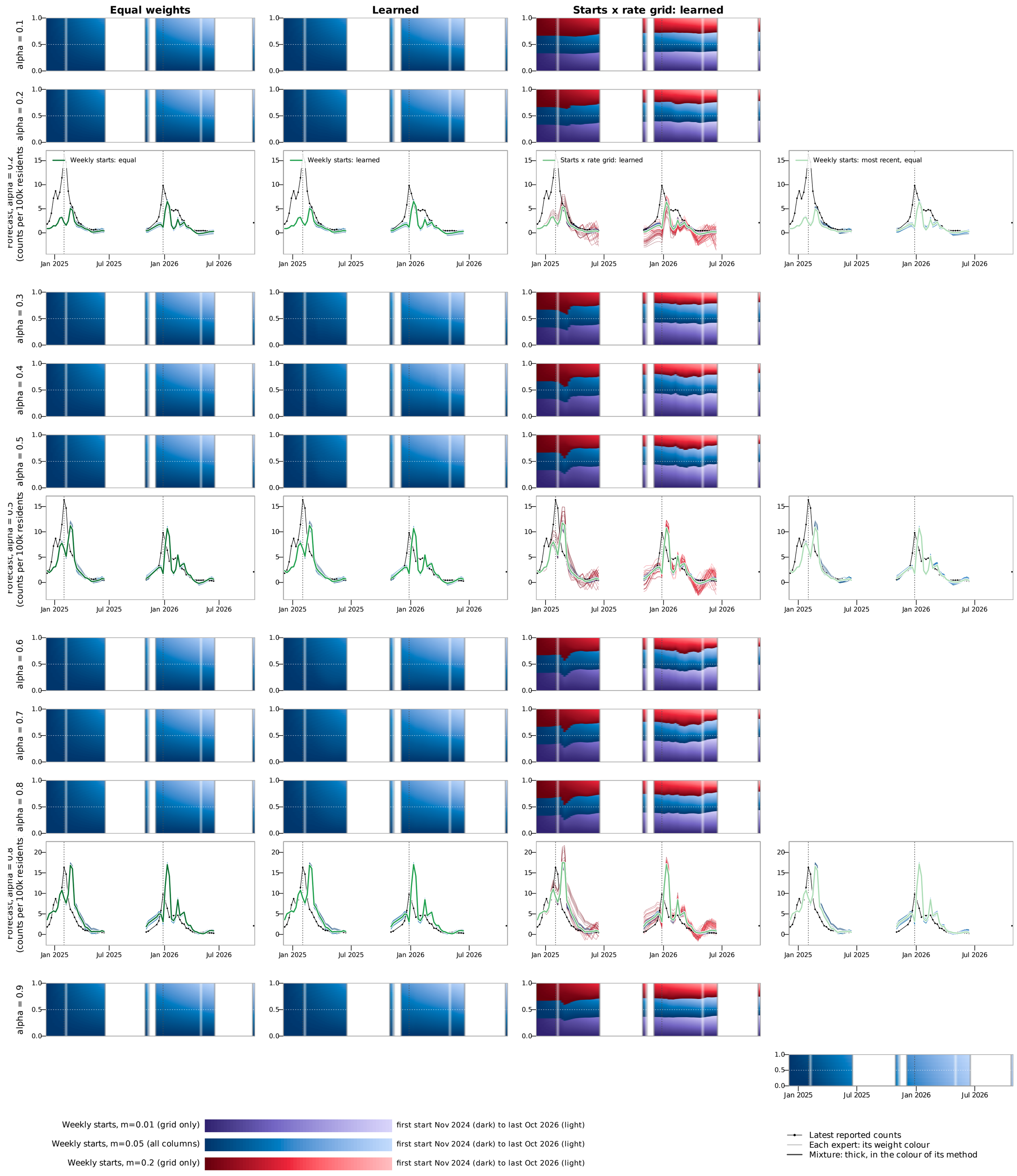
Rows alpha = 0.1-0.9 (mixtures with one weight vector per level): the weights of each level; below alpha = 0.2, 0.5 and 0.8 the forecasts at that level (per-level mixtures after crossing repair, before any outer correction).

Rows 'All levels': one weight vector for all levels (the mixture, or its common-weight variant), learned from the mean quantile loss over the levels; two-vector mixtures: baseline and declining vectors; context: G or the gate (line), c (points).

Lighter bars: rounds without a CMU forecast, filled with the latest earlier CMU forecast of the same target week.

Weekly-start weights

flu | horizon 2 | Texas (48) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0



Target date | Weights from the corresponding reference week | Vertical dotted lines: peaks

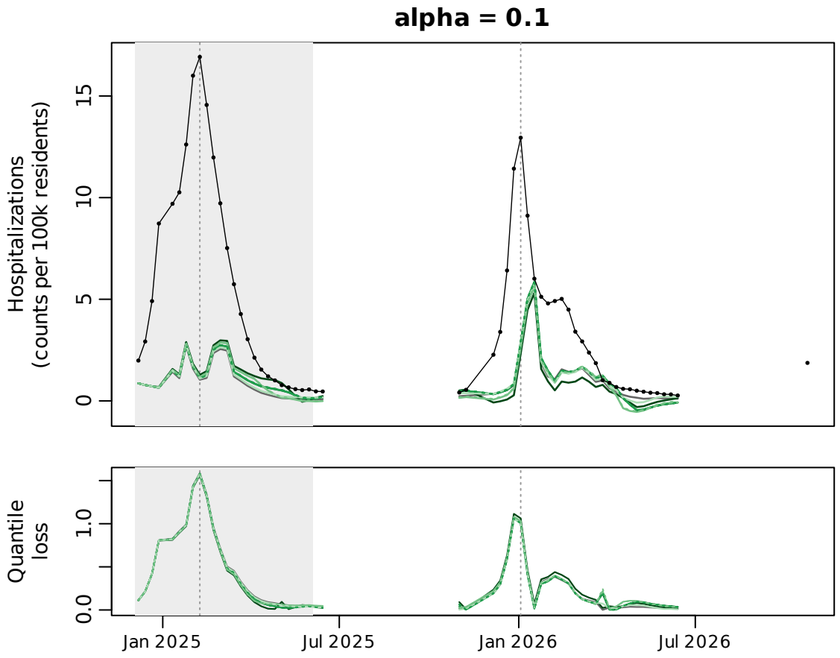
Rows alpha = 0.1-0.9 (mixtures with one weight vector per level): the weights of each level; below alpha = 0.2, 0.5 and 0.8 the forecasts at that level (per-level mixtures after crossing repair, before any outer correction).

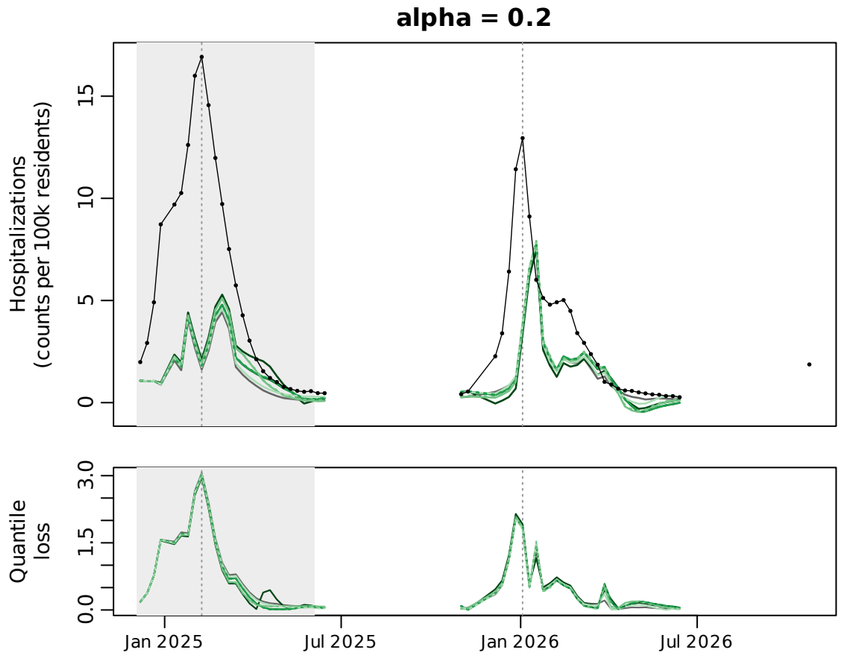
Rows 'All levels': one weight vector for all levels (the mixture, or its common-weight variant), learned from the mean quantile loss over the levels; two-vector mixtures: baseline and declining vectors; context: G or the gate (line), c (points).

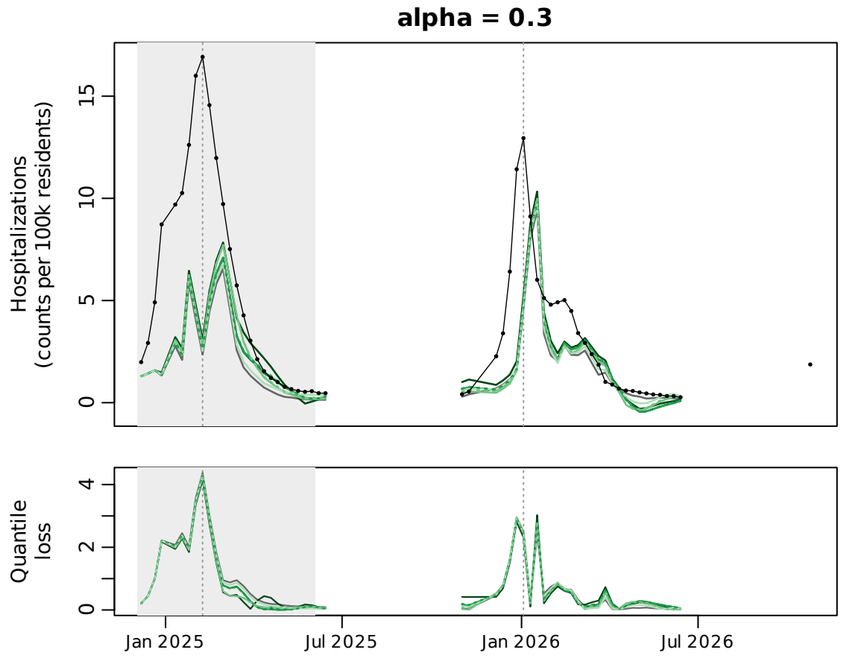
Lighter bars: rounds without a CMU forecast, filled with the latest earlier CMU forecast of the same target week.

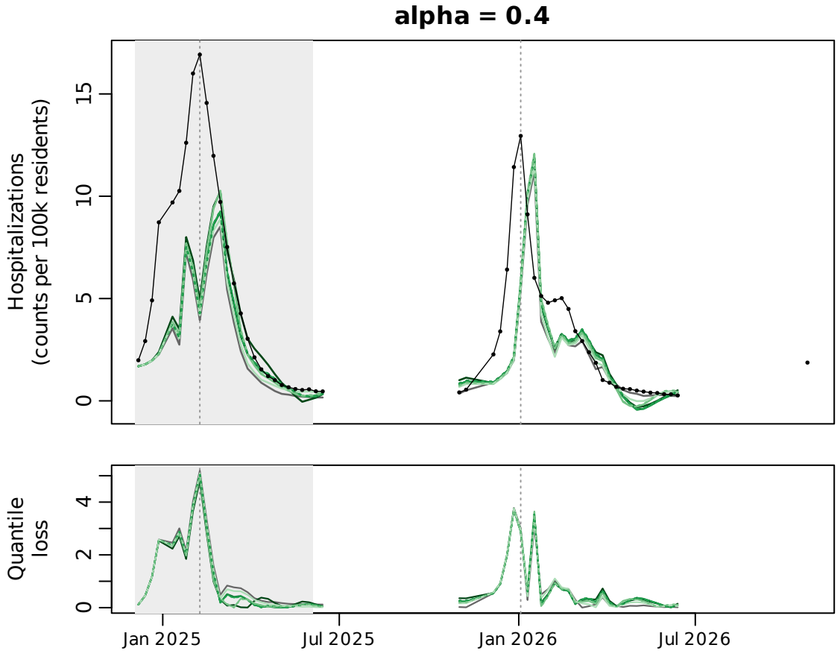
flu | horizon 2 | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0 | Weekly starts: immediate admission

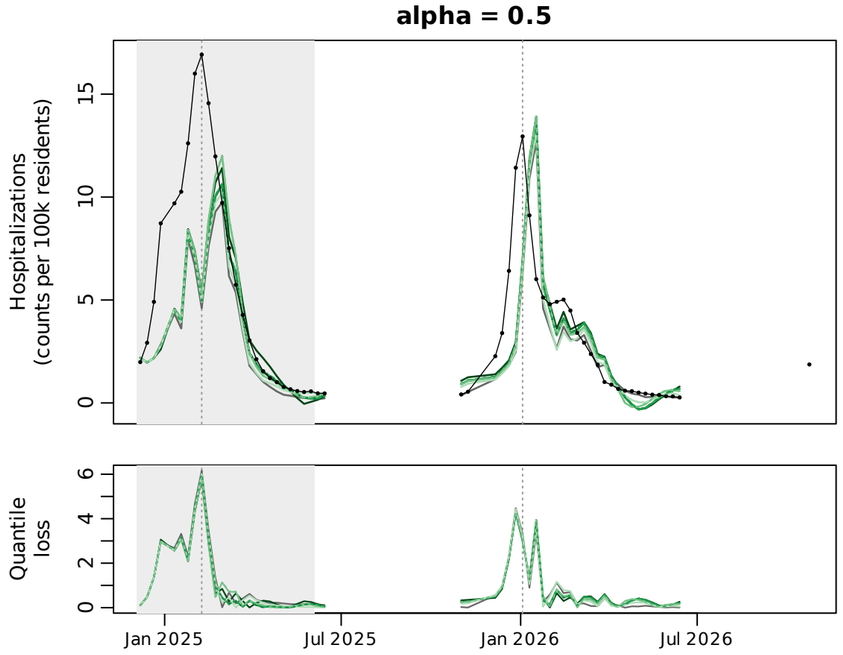
Alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Vertical dotted lines: peaks | Gray: tuning period (not scored)

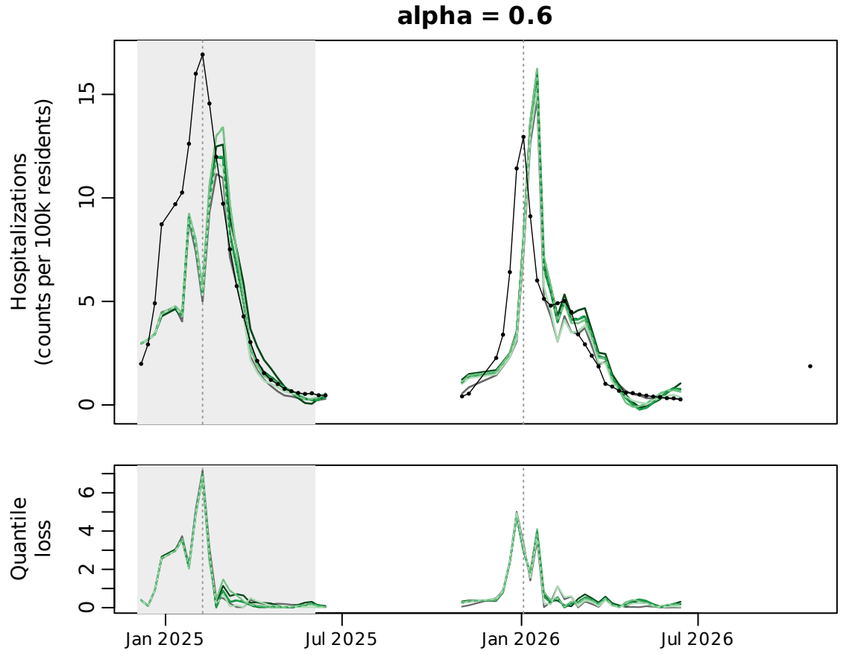


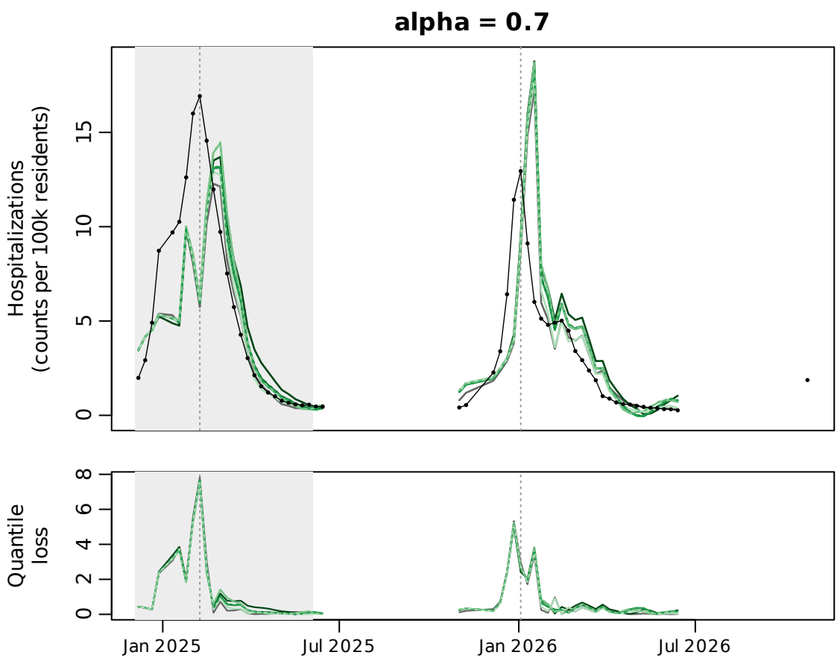


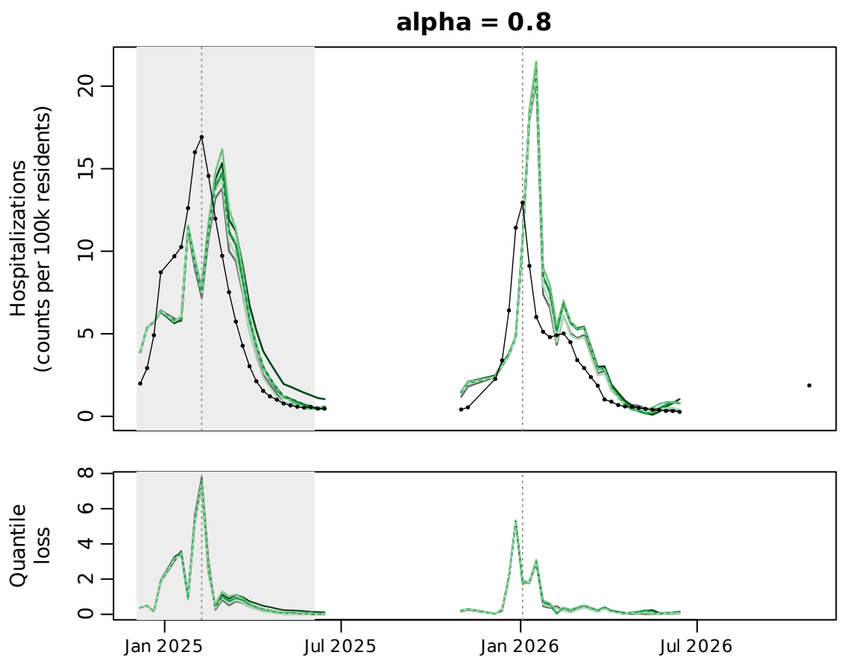


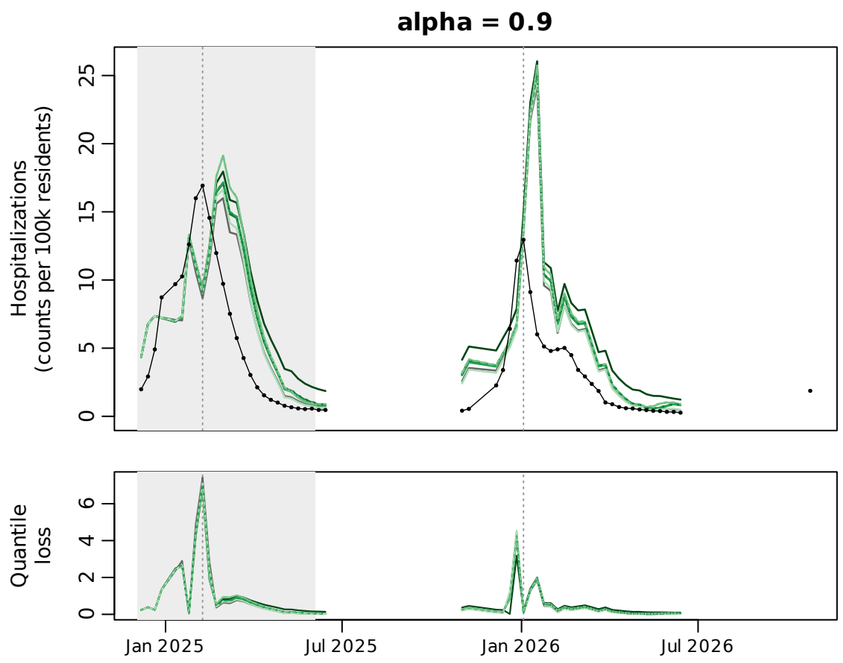








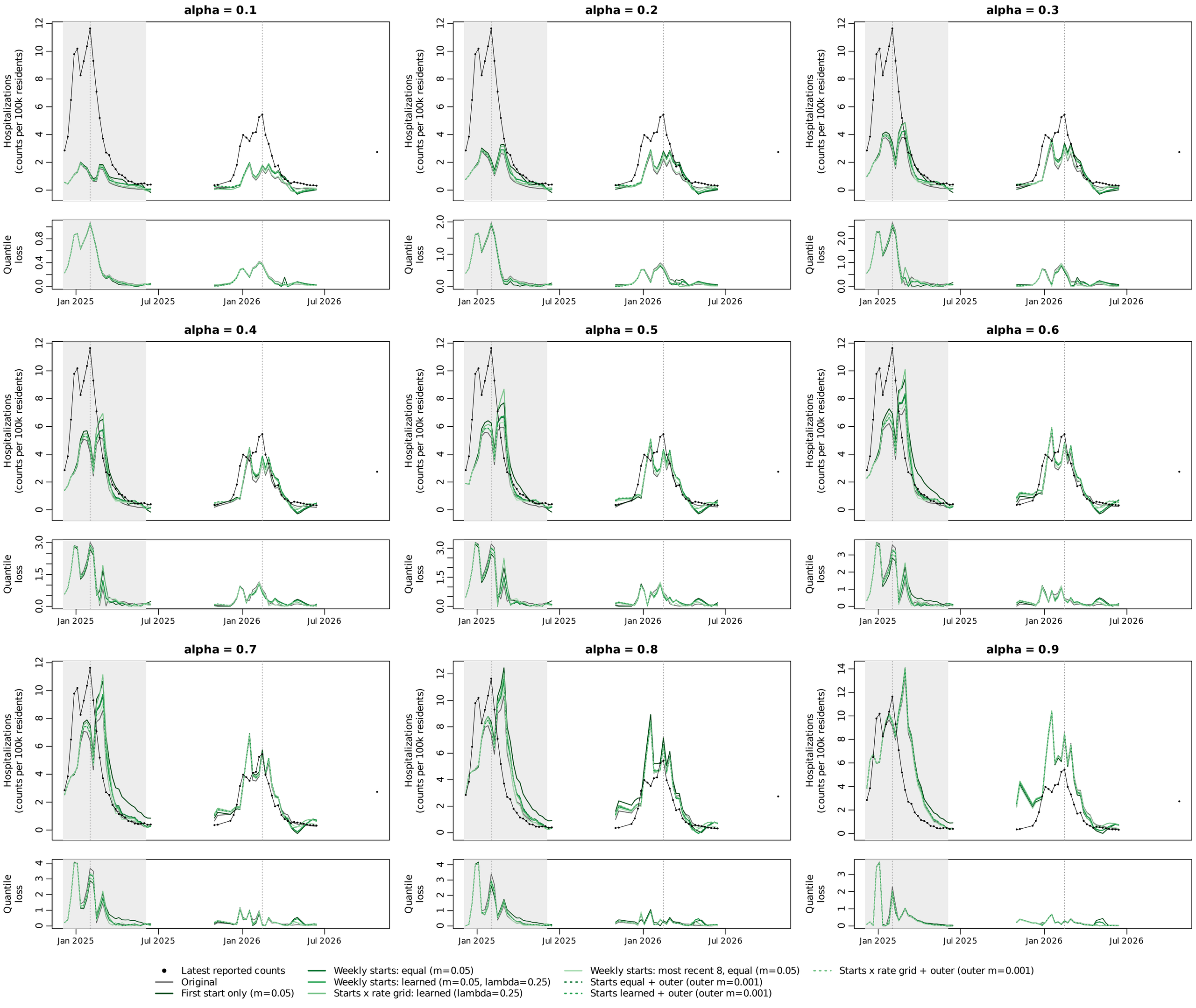




- Latest reported counts
- Original
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

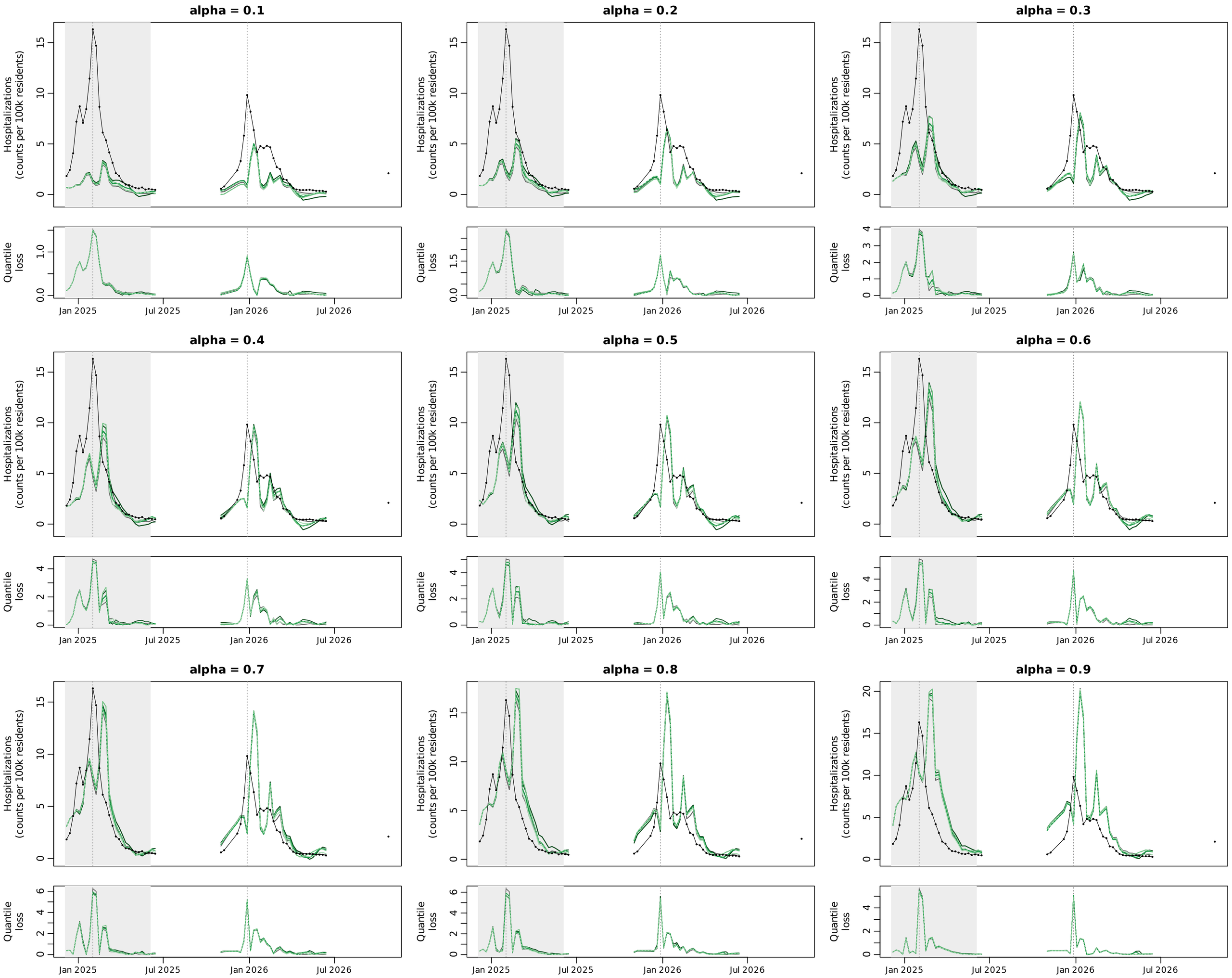
flu | horizon 2 | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0 | Weekly starts: immediate admission

Alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Vertical dotted lines: peaks | Gray: tuning period (not scored)



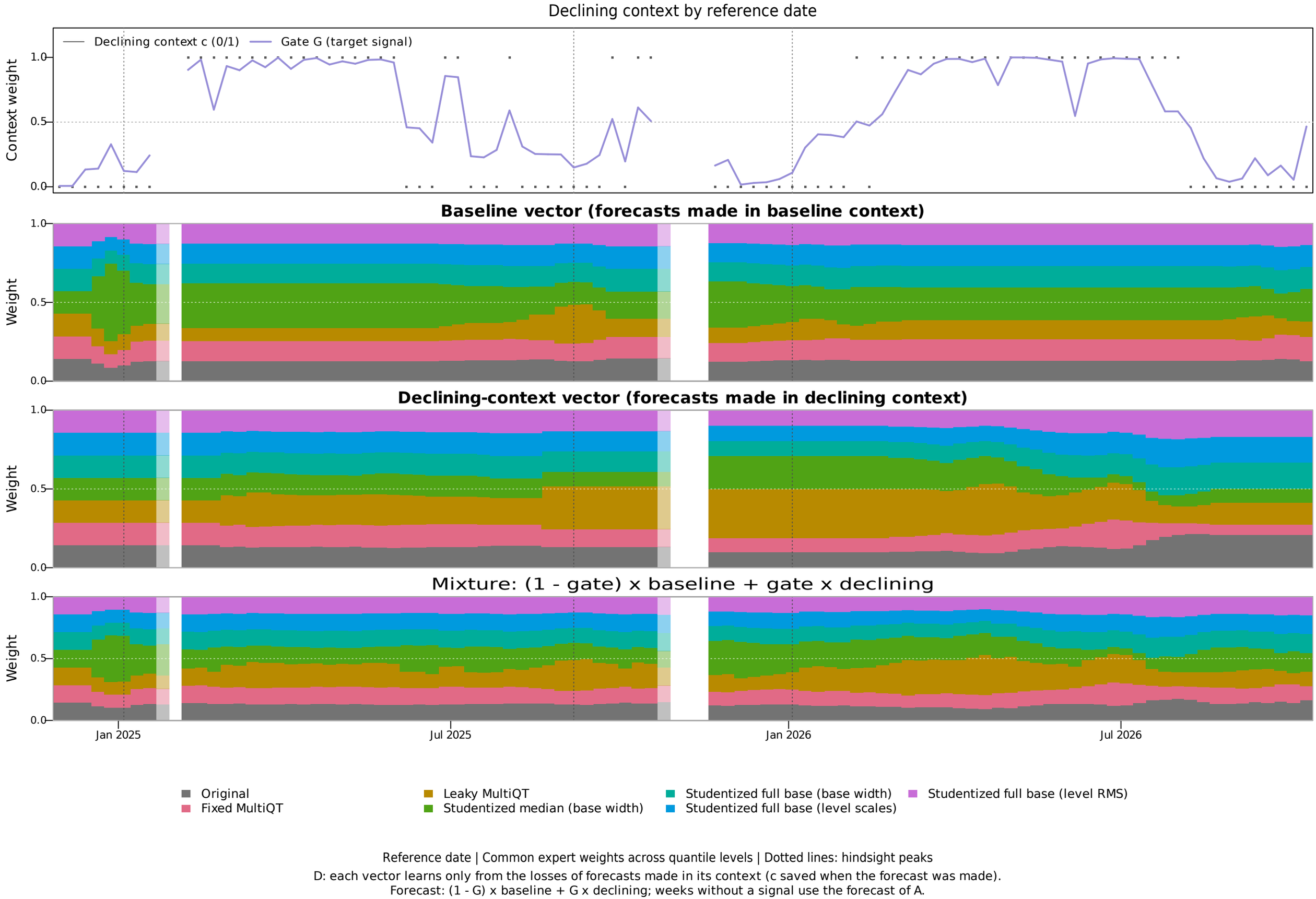
flu | horizon 2 | Texas (48) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0 | Weekly starts: immediate admission

Alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Vertical dotted lines: peaks | Gray: tuning period (not scored)



Score experts: context vectors, blended (D, delta=0, H=4, sharing 1/8)

covid | horizon 2 | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates | feedback delay 0



Reference date | Common expert weights across quantile levels | Dotted lines: hindsight peaks

D: each vector learns only from the losses of forecasts made in its context (c saved when the forecast was made).
Forecast: (1 - G) x baseline + G x declining; weeks without a signal use the forecast of A.
Lighter bars: filled forecast rounds.